

4
EDITION

Calculus

Early Transcendental Functions

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Chapter 2

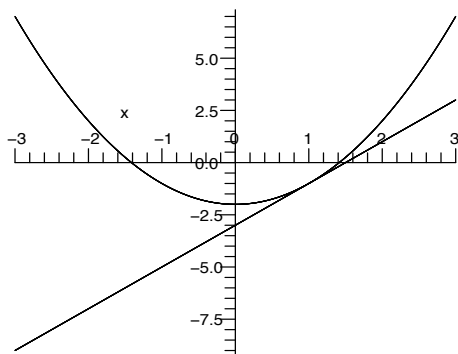
Differentiation

2.1 Tangent Line and Velocity

1. Slope is

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1+h)^2 - 2 - (-1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 + 2h}{h} \\ &= \lim_{h \rightarrow 0} (h + 2) = 2. \end{aligned}$$

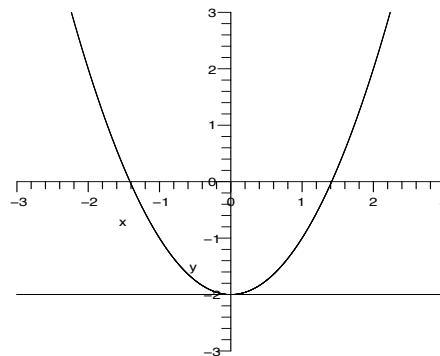
Tangent line is $y = 2(x-1) - 1$ or $y = 2x - 3$.



2. Slope is

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2}{h} = 0. \end{aligned}$$

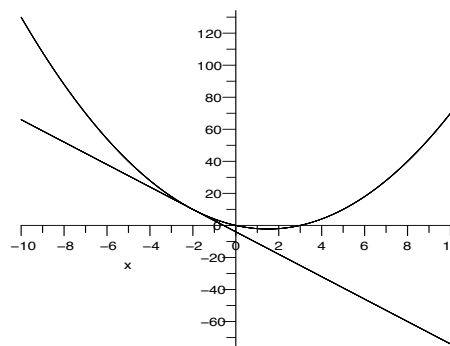
Tangent line is $y = -2$.



3. Slope is

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(-2+h) - f(-2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(-2+h)^2 - 3(-2+h) - (10)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-7h + h^2}{h} = -7. \end{aligned}$$

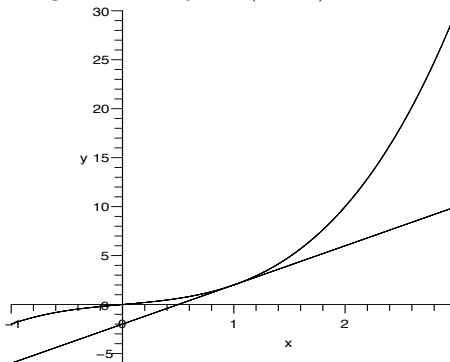
Tangent line is $y = -7(x+2) + 10$



4. Slope is

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1+3h+3h^2+h^3) + (1+h) - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4h+3h^2+h^3}{h} = \lim_{h \rightarrow 0} 4+3h+h^2 = 4. \end{aligned}$$

Tangent line is $y = 4(x-1) + 2$.

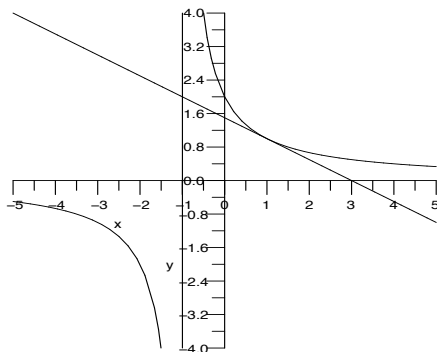


5. Slope is

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{2}{(1+h)+1} - \frac{2}{1+1}}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{2}{2+h} - 1}{h} = \lim_{h \rightarrow 0} \frac{\left(\frac{2-(2+h)}{2+h}\right)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\left(\frac{-h}{2+h}\right)}{h} = \lim_{h \rightarrow 0} \frac{-1}{2+h} = -\frac{1}{2}.
\end{aligned}$$

Tangent line is $y = -\frac{1}{2}(x-1) + 1$ or

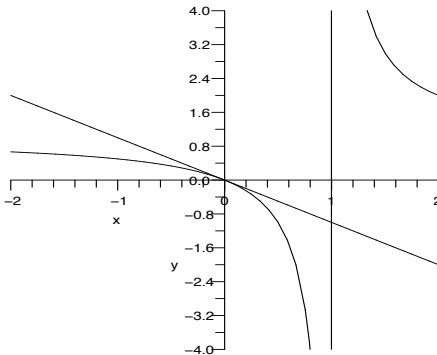
$$y = -\frac{x}{2} + \frac{3}{2}.$$



6. Slope is

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{h-1}{h} - 0}{h} \\
&= \lim_{h \rightarrow 0} \frac{1}{h-1} = -1
\end{aligned}$$

Tangent line is $y = -x$.

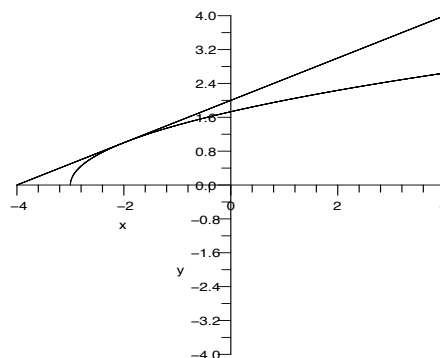


7. Slope is

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{f(-2+h) - f(-2)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sqrt{(-2+h)+3} - 1}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sqrt{h+1} - 1}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sqrt{h+1} - 1}{h} \cdot \frac{\sqrt{h+1} + 1}{\sqrt{h+1} + 1}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{(h+1) - 1}{h(\sqrt{h+1} + 1)} \\
&= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+1} + 1} = \frac{1}{2}.
\end{aligned}$$

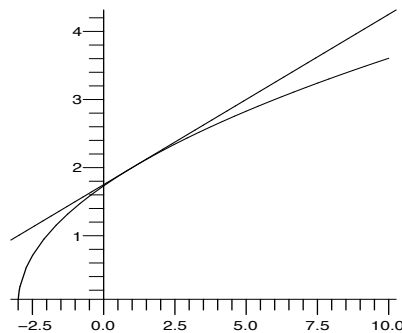
Tangent line is $y = \frac{1}{2}(x+2) + 1$ or $y = \frac{1}{2}x + 2$.



8. Slope is

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sqrt{(1+h)+3} - \sqrt{1+3}}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sqrt{h+4} - 2}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sqrt{h+4} - 2}{h} \cdot \frac{\sqrt{h+4} + 2}{\sqrt{h+4} + 2} \\
&= \lim_{h \rightarrow 0} \frac{h+4-4}{h} \cdot \frac{1}{\sqrt{h+4} + 2} \\
&= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+4} + 2} = \frac{1}{4}.
\end{aligned}$$

Tangent line is $y = \frac{1}{4}(x-1) + 2$.



9. $f(x) = x^3 - x$

No.	Points (x, y)	Slope
(a)	(1,0) and (2,6)	6
(b)	(2,6) and (3,24)	18
(c)	(1.5,1.875) and (2,6)	8.25
(d)	(2,6) and (2.5,13.125)	14.25
(e)	(1.9,4.959) and (2,6)	10.41
(f)	(2,6) and (2.1,7.161)	11.61

(g) Slope seems to be approximately 11.

10. $f(x) = \sqrt{x^2 + 1}$

No.	Points (x, y)	Slope
(a)	(1,1.414) and (2,2.236)	0.504
(b)	(2,2.236) and (3,3.162)	0.926
(c)	(1.5,1.803) and (2,2.236)	0.867
(d)	(2,2.236) and (2.5,2.269)	0.913
(e)	(1.9,2.147) and (2,2.236)	0.89
(f)	(2,2.236) and (2.1,2.325)	0.899

(g) Slope seems to be approximately 0.89.

11. $f(x) = \frac{x-1}{x+1}$

No.	Points (x, y)	Slope
(a)	(1,0) and (2,0.33)	0.33
(b)	(2,0.33) and (3,0.5)	0.17
(c)	(1.5,0.2) and (2,0.33)	0.26
(d)	(2,0.33) and (2.5,0.43)	0.2
(e)	(1.9,0.31) and (2,0.33)	0.2
(f)	(2,0.33) and (2.1,0.35)	0.2

(g) Slope seems to be approximately 0.2.

12. $f(x) = e^x$

No.	Points (x, y)	Slope
(a)	(1,2.718) and (2,7.389)	4.671
(b)	(2,7.389) and (3,20.085)	12.696
(c)	(1.5,4.481) and (2,7.389)	5.814
(d)	(2,7.389) and (2.5,12.182)	9.586
(e)	(1.9,6.686) and (2,7.389)	7.03
(f)	(2,7.389) and (2.1,8.166)	7.77

(g) Slope seems to be approximately 7.4

13. C, B, A, D. At the point labeled C, the slope is very steep and negative. At the point B, the slope is zero and at the point A, the slope is just more than zero. The slope of the line tangent to the point D is large and positive.

14. In order of increasing slope: D (large negative), C (small negative), B (small positive), and A (large positive).

15. (a) Velocity at time $t = 1$ is,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-4.9(1+h)^2 + 5 - (0.1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-4.9(1+2h+h^2) + 5 - (0.1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-9.8h - 4.9h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(-9.8 - 4.9h)}{h} = -9.8. \end{aligned}$$

- (b) Velocity at time $t = 2$ is,

$$\lim_{h \rightarrow 0} \frac{s(2+h) - s(2)}{h}$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{-4.9(2+h)^2 + 5 - (-14.6)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-4.9(4+4h+h^2) + 5 - (-14.6)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-19.6h - 4.9h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(-19.6 - 4.9h)}{h} = -19.6 \end{aligned}$$

16. (a) Velocity at time $t = 0$ is,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{s(0+h) - s(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{4h - 4.9h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(4 - 4.9h)}{h} \\ &= 4 - \lim_{h \rightarrow 0} 4.9h = 4. \end{aligned}$$

- (b) Velocity at time $t = 1$ is,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{4(1+h) - 4.9(1+h)^2 - (-0.9)}{h} \\ &= \lim_{h \rightarrow 0} \frac{4+4h-4.9-9.8h-4.9h^2+0.9}{h} \\ &= \lim_{h \rightarrow 0} \frac{-5.8h-4.9h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(-5.8-4.9h)}{h} = -5.8 \end{aligned}$$

17. (a) Velocity at time $t = 0$ is,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{s(0+h) - s(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{h+16} - 4}{h} \cdot \frac{\sqrt{h+16} + 4}{\sqrt{h+16} + 4} \\ &= \lim_{h \rightarrow 0} \frac{(h+16) - 16}{h(\sqrt{h+16} + 4)} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+16} + 4} = \frac{1}{8} \end{aligned}$$

- (b) Velocity at time $t = 2$ is,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{s(2+h) - s(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{18+h} - \sqrt{18}}{h} \\ & \text{Multiplying by } \frac{\sqrt{h+18} + \sqrt{18}}{\sqrt{h+18} + \sqrt{18}} \text{ gives} \\ &= \lim_{h \rightarrow 0} \frac{(h+18) - 18}{h(\sqrt{h+18} + \sqrt{18})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+18} + \sqrt{18}} = \frac{1}{2\sqrt{18}} \end{aligned}$$

18. (a) Velocity at time $t = 2$ is,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{s(2+h) - s(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{4}{(2+h)} - 2}{h} = \lim_{h \rightarrow 0} \frac{4-4-2h}{h(2+h)} \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{-2h}{h(2+h)} = \lim_{h \rightarrow 0} \frac{-2}{2+h} = -1.$$

- (b) Velocity at time $t = 4$ is,

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{s(4+h) - s(4)}{h} &= \lim_{h \rightarrow 0} \frac{\frac{4}{(4+h)} - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{4-1(4+h)}{(4+h)}}{h} = \lim_{h \rightarrow 0} \frac{\frac{4-4-h}{(4+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h(4+h)} = \lim_{h \rightarrow 0} \frac{-1}{4+h} = -\frac{1}{4} \end{aligned}$$

19. (a) Points: $(0, 10)$ and $(2, 74)$

$$\text{Average velocity: } \frac{74 - 10}{2} = 32$$

- (b) Second point: $(1, 26)$

$$\text{Average velocity: } \frac{74 - 26}{1} = 48$$

- (c) Second point: $(1.9, 67.76)$

$$\text{Average velocity: } \frac{74 - 67.76}{0.1} = 62.4$$

- (d) Second point: $(1.99, 73.3616)$

$$\text{Average velocity: } \frac{74 - 73.3616}{0.01} = 63.84$$

- (e) The instantaneous velocity seems to be 64.

20. (a) Points: $(0, 0)$ and $(2, 26)$

$$\text{Average velocity: } \frac{26 - 0}{2 - 0} = 13$$

- (b) Second point: $(1, 4)$

$$\text{Average velocity: } \frac{26 - 4}{2 - 1} = 22$$

- (c) Second point: $(1.9, 22.477)$

$$\text{Average velocity: } \frac{26 - 22.477}{2 - 1.9} = 35.23$$

- (d) Second point: $(1.99, 25.6318)$

$$\begin{aligned} \text{Average velocity:} \\ \frac{26 - 25.6318}{2 - 1.99} = 36.8203 \end{aligned}$$

- (e) The instantaneous velocity seems to be approaching 37.

21. (a) Points: $(0, 0)$ and $(2, \sqrt{20})$

$$\text{Average velocity: } \frac{\sqrt{20} - 0}{2 - 0} = 2.236068$$

- (b) Second point: $(1, 3)$

$$\text{Average velocity: } \frac{\sqrt{20} - 3}{2 - 1} = 1.472136$$

- (c) Second point: $(1.9, \sqrt{18.81})$

$$\begin{aligned} \text{Average velocity:} \\ \frac{\sqrt{20} - \sqrt{18.81}}{2 - 1.9} = 1.3508627 \end{aligned}$$

- (d) Second point: $(1.99, \sqrt{19.8801})$

$$\begin{aligned} \text{Average velocity:} \\ \frac{\sqrt{20} - \sqrt{19.88}}{2 - 1.99} = 1.3425375 \end{aligned}$$

- (e) One might conjecture that these numbers are approaching 1.34. The exact limit is $\frac{6}{\sqrt{20}} \approx 1.341641$.

22. (a) Points: $(0, -2.7279)$ and $(2, 0)$

$$\begin{aligned} \text{Average velocity:} \\ \frac{0 - (-2.7279)}{2 - 0} = 1.3639 \end{aligned}$$

- (b) Second point: $(1, -2.5244)$

$$\begin{aligned} \text{Average velocity:} \\ \frac{0 - (-2.5244)}{2 - 1} = 2.5244 \end{aligned}$$

- (c) Second point: $(1.9, -0.2995)$

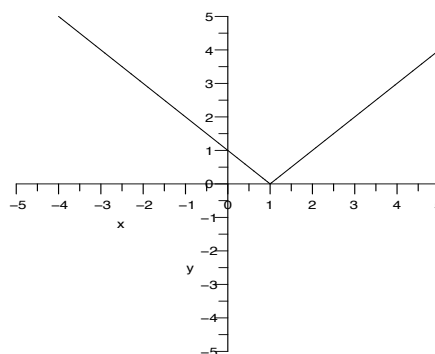
$$\begin{aligned} \text{Average velocity:} \\ \frac{0 - (-0.2995)}{2 - 1.9} = 2.995 \end{aligned}$$

- (d) Second point: $(1.99, -0.03)$

$$\text{Average velocity: } \frac{0 - (-0.03)}{2 - 1.99} = 3$$

- (e) The instantaneous velocity seems to be 3.

23. A graph makes it apparent that this function has a corner at $x = 1$.



Numerical evidence suggests that,

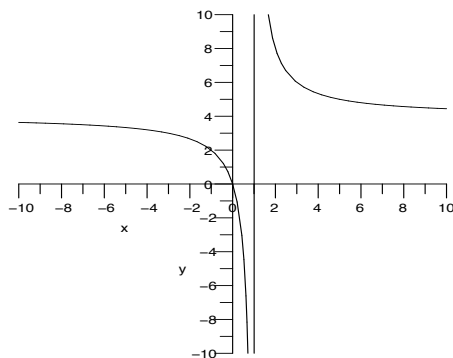
$$\lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = 1$$

$$\text{while } \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = -1.$$

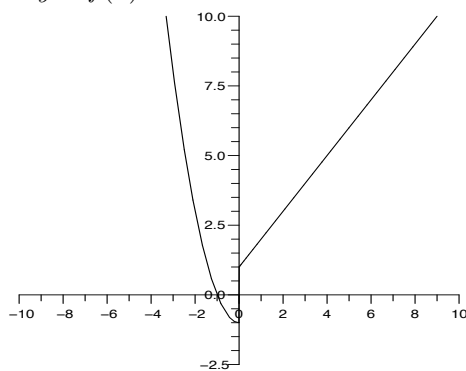
Since these are not equal, there is no tangent

line.

24. Tangent line does not exist at $x = 1$ because the function is not defined there.



25. From the graph it is clear that, curve is not continuous at $x = 0$ therefore tangent line at $y = f(x)$ at $x = 0$ does not exist.



Also,

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^-} \frac{h^2 - 1 - (-1)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h^2}{h} = \lim_{h \rightarrow 0^-} h = 0 \end{aligned}$$

Similarly,

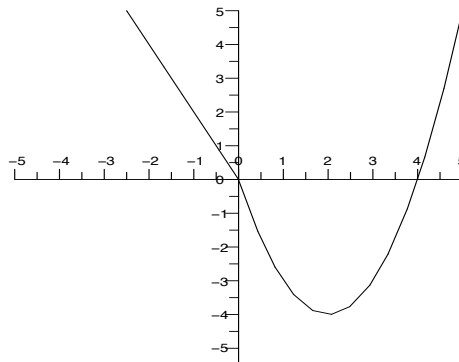
$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{h + 1 - (1)}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1. \end{aligned}$$

Numerical evidence suggest that,

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &\neq \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h}. \end{aligned}$$

Therefore tangent line does not exist at $x = 0$.

26. From the graph it is clear that, the curve of $y = f(x)$ is not smooth at $x = 0$ therefore tangent line at $x = 0$ does not exist.



Also,

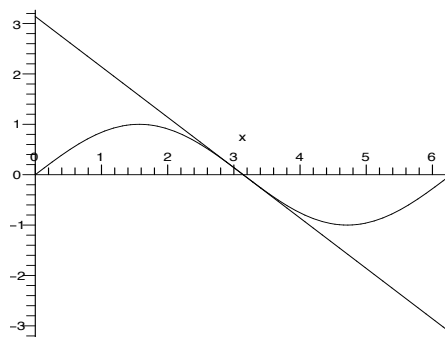
$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^-} \frac{-2h}{h} = -2 \\ \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} (h - 4) = -4. \end{aligned}$$

Numerical evidence suggest that,

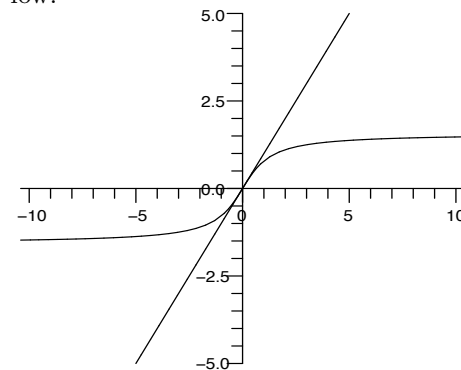
$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &\neq \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h}. \end{aligned}$$

Therefore tangent line does not exist at $x = 0$.

27. Tangent line at $x = \pi$ to $y = \sin x$ as below:

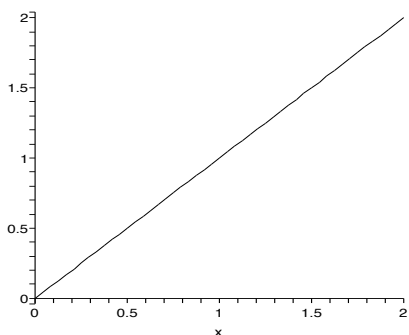


28. Tangent line at $x = 0$ to $y = \tan^{-1} x$ as below:



29. Since the graph has a corner at $x = 0$, tangent line does not exist.

30. The tangent line overlays the line:



31. (a) $\frac{f(4) - f(2)}{2} = 21,034$

Since $\frac{f(b)-f(a)}{b-a}$ is the average rate of change of function f between a and b . The expression tells us that the average rate of change of f between $a = 2$ to $b = 4$ is 21,034. That is the average rate of change in the bank balance between Jan. 1, 2002 and Jan. 1, 2004 was 21,034 (\$ per year).

(b) $2[f(4) - f(3.5)] = 25,036$

Note that $2[f(4) - f(3.5)] = f(4) - f(3.5)/2$. The expression says that the average rate of change in balance between July 1, 2003 and Jan. 1, 2004 was 25,036 (\$ per year).

(c) $\lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h} = 30,000$

The expression says that the instantaneous rate of change in the balance on Jan. 1, 2004 was 30,000 (\$ per year).

32. (a) $\frac{f(40) - f(38)}{2} = -2103$

Since $\frac{f(b)-f(a)}{b-a}$ is the average rate of change of function between a and b . The expression tells us that the average rate of change of f between $a = 38$ to $b = 40$ is -2103 . That is the average rate of depreciation between 38 and 40 thousand miles.

(b) $f(40) - f(39) = -2040$

The expression says that the average rate of depreciation between 39 and 40 thousand miles is -2040 .

(c) $\lim_{h \rightarrow 0} \frac{f(40+h) - f(40)}{h} = -2000$

The expression says that the instantaneous rate of depreciation in the value of the car when it has 40 thousand miles is -2000 .

$$\begin{aligned} 33. \quad v_{avg} &= \frac{f(s) - f(r)}{s - r} \\ v_{avg} &= \frac{f(s) - f(r)}{s - r} \\ &= \frac{as^2 + bs + c - (ar^2 + br + c)}{s - r} \\ &= \frac{a(s^2 - r^2) + b(s - r)}{s - r} \\ &= \frac{a(s + r)(s - r) + b(s - r)}{s - r} \\ &= a(s + r) + b \end{aligned}$$

Let $v(r)$ be the velocity at $t = r$. We have, $v(r) =$

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(r+h) - f(r)}{h} &= \lim_{h \rightarrow 0} \frac{a(r+h)^2 + b(r+h) + c - (ar^2 + br + c)}{h} \\ &= \lim_{h \rightarrow 0} \frac{a(r^2 + 2rh + h^2) + bh - ar^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2ar + ah + b)}{h} \\ &= \lim_{h \rightarrow 0} (2ar + ah + b) = 2ar + b \end{aligned}$$

So, $v(r) = 2ar + b$.

The same argument shows that $v(s) = 2as + b$.

$$\begin{aligned} \text{Finally} \quad \frac{v(r) + v(s)}{2} &= \frac{(2ar + b) + (2as + b)}{2} \\ &= \frac{2a(s + r) + 2b}{2} \\ &= a(s + r) + b = v_{avg} \end{aligned}$$

34. $f(t) = t^3 - t$ works with $r = 0, s = 2$. The average velocity between r and s is, $\frac{6-0}{2-0} = 3$.

The instantaneous velocity at r is

$$\lim_{h \rightarrow 0} \frac{(0+h)^3 - (0+h) - 0}{h} = 0$$

and the instantaneous velocity at s is,

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{(2+h)^3 - (2+h) - 6}{h} &= \lim_{h \rightarrow 0} \frac{8 + 12h + 6h^2 + h^3 - 2 - h - 6}{h} \\ &= \lim_{h \rightarrow 0} 11 + 6h + h^2 = 11 \end{aligned}$$

so, the average between the instantaneous velocities is 5.5.

35. (a) $y = x^3 + 3x + 1$

$$y' = 3x^2 + 3$$

Since the slope needed to be 5, $y' = 5$.

$$3x^2 + 3 = 5$$

$$\Rightarrow 3x^2 = 5 - 3$$

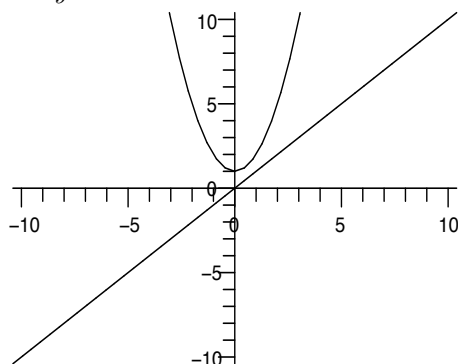
$$\Rightarrow x^2 = \frac{2}{3}$$

$$\Rightarrow x = \pm \sqrt{\frac{2}{3}}$$

Therefore, slope of tangent line at $x = \sqrt{\frac{2}{3}}$ and $x = -\sqrt{\frac{2}{3}}$ to $y = x^3 + 3x + 1$ equals 5.

- (b) Since the slope needed to be 1, $y' = 1$. $3x^2 + 3 = 1$ which has no real roots. Therefore slope of tangent line to $y = x^3 + 3x + 1$ cannot equal 1.

36. (a) From the graph it is clear that $y = x^2 + 1$ and $y = x$ do not intersect.



- (b) $y = x^2 + 1$ and $y = x$
 $y = x^2 + 1 \Rightarrow y' = 2x$
 $y = x \Rightarrow y' = 1$
 For, $y = x^2 + 1$
 $y' = 2x = 1$.
 $2x = 1$
 $\Rightarrow x = \frac{1}{2}$

Therefore, tangent line at $x = \frac{1}{2}$ to $y = x^2 + 1$ is parallel to the tangent lines to $y = x$.

37. (a) $y = x^3 + 3x + 1$
 $\lim_{h \rightarrow 0} \frac{f(1+h) - f(x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{(1+h)^3 + 3(1+h) + 1 - 5}{h}$
 $= \lim_{h \rightarrow 0} \frac{(1+3h+3h^2+h^3)+(3+3h)+1-5}{h}$
 $= \lim_{h \rightarrow 0} \frac{6h+3h^2+h^3}{h}$
 $= \lim_{h \rightarrow 0} \frac{h(6+3h+h^2)}{h} = 6$

The point corresponding to $x = 1$ is (1,5). So, line with slope 6 through point (1,5) has equation $y = 6(x-1)+5$ or $y = 6x - 1$.

- (b) From part (a) we have, equation of tan-

gent line is $y = 6x - 1$.

Given that $y = x^3 + 3x + 1$.

Therefore, we write

$$x^3 + 3x + 1 = 6x - 1$$

$$x^3 - 3x + 2 = 0$$

$$(x-1)(x^2+x-2) = 0$$

$$(x-1)(x-1)(x+2) = 0$$

$$(x-1)^2(x+2) = 0.$$

Therefore, tangent line intersects $y = x^3 + 3x + 1$ at more than one point that is at $x = 1$ and $x = -2$.

- (c) $y = x^2 + 1$

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(c+h)^2 + 1 - (c^2 + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(c^2 + 2ch + h^2) + 1 - (c^2 + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c^2 + 2ch + h^2 + 1 - c^2 - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{2ch + h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2c + h)}{h} = 2c \end{aligned}$$

The point corresponding to $x = c$ is $(c, c^2 + 1)$. So, line with slope $2c$ through point $(c, c^2 + 1)$ has equation $y = 2c(x - c) + c^2 + 1$ or $y = 2cx - c^2 + 1$. Given that $y = x^2 + 1$

Therefore,

$$x^2 + 1 = 2cx - c^2 + 1$$

$$x^2 - 2cx + c^2 = 0$$

$$(x - c)^2 = 0.$$

Therefore, tangent line intersects $y = x^2 + 1$ only at one point that is at $x = c$.

38. Let $x = h + a$. Then $h = x - a$ and clearly $\frac{f(a+h) - f(a)}{h} = \frac{f(x) - f(a)}{x-a}$.

It is also clear that, $x \rightarrow a$ if and only if $h \rightarrow 0$. Therefore, if one of the two limits exists, then so does the other and

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x-a}.$$

39. The slope of the tangent line at $p = 1$ is approximately

$$\frac{-20 - 0}{2 - 0} = -10$$

which means that at $p = 1$ the freezing temperature of water decreases by 10 degrees Celsius per 1 atm increase in pressure. The slope of the tangent line at $p = 3$ is approximately

$$\frac{-11 - (-20)}{4 - 2} = 4.5$$

which means that at $p = 3$ the freezing temperature of water increases by 4.5 degrees Celsius per 1 atm increase in pressure.

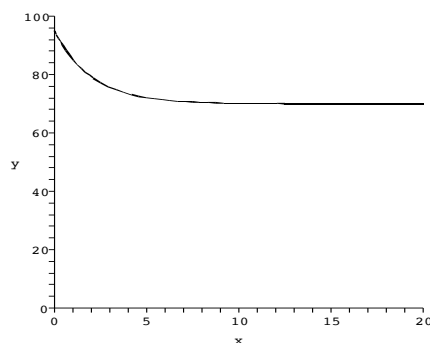
40. The slope of the tangent line at $v = 50$ is approximately $\frac{47 - 28}{60 - 40} = .95$.

This means that at an initial speed of 50mph the range of the soccer kick increases by .95 yards per 1 mph increase in initial speed.

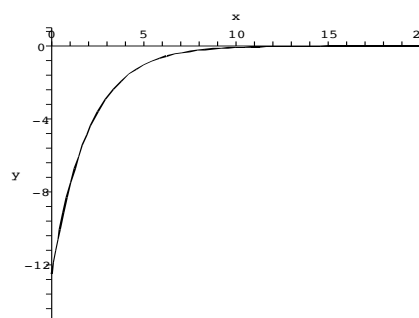
41. The hiker reached the top at the highest point on the graph (about 1.75 hours). The hiker was going the fastest on the way up at about 1.5 hours. The hiker was going the fastest on the way down at the point where the tangent line has the least (i.e. most negative) slope, at about 4 hours at the end of the hike. Where the graph is level the hiker was either resting or walking on flat ground.

42. The tank is the fullest at the first spike (at around 8 A.M.). The tank is the emptiest just before this at the lowest dip (around 7 A.M.). The tank is filling up the fastest where the graph has the steepest positive slope (in between 7 and 8 A.M.). The tank is emptying the fastest just after 8 A.M. where the graph has the steepest negative slope. The level portions most likely represent night when water usage is at a minimum.

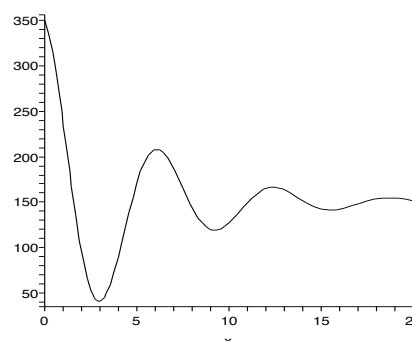
43. A possible graph of the temperature with respect to time:



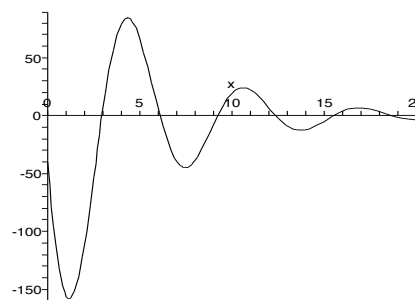
Graph of the rate of change of the temperature:



44. Possible graph of bungee-jumpers height:



A graph of the bungee-jumper's velocity:



2.2 The Derivative

1. Using (2.1):

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3(1+h) + 1 - (4)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3h}{h} = \lim_{h \rightarrow 0} 3 = 3 \end{aligned}$$

Using (2.2):

$$\begin{aligned} &\lim_{b \rightarrow 1} \frac{f(b) - f(1)}{b - 1} \\ &= \lim_{b \rightarrow 1} \frac{3b + 1 - (3 + 1)}{b - 1} \end{aligned}$$

$$\begin{aligned}
&= \lim_{b \rightarrow 1} \frac{3b - 3}{b - 1} \\
&= \lim_{b \rightarrow 1} \frac{3(b - 1)}{b - 1} \\
&= \lim_{b \rightarrow 1} 3 = 3
\end{aligned}$$

2. Using (2.1):

$$\begin{aligned}
f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \rightarrow 0} \frac{3(1+h)^2 + 1 - 4}{h} \\
&= \lim_{h \rightarrow 0} \frac{6h + 3h^2}{h} \\
&= \lim_{h \rightarrow 0} 6 + 3h = 6
\end{aligned}$$

Using (2.2):

$$\begin{aligned}
f'(1) &= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} \\
&= \lim_{x \rightarrow 1} \frac{(3x^2 + 1) - 4}{x - 1} \\
&= \lim_{x \rightarrow 1} \frac{3(x-1)(x+1)}{x - 1} \\
&= \lim_{x \rightarrow 1} 3(x+1) = 6
\end{aligned}$$

3. Using (2.1): Since

$$\begin{aligned}
&\frac{f(1+h) - f(1)}{h} \\
&= \frac{\sqrt{3(1+h)+1} - 2}{h} \\
&= \frac{\sqrt{4+3h} - 2}{h} \cdot \frac{\sqrt{4+3h} + 2}{\sqrt{4+3h} + 2} \\
&= \frac{h}{4+3h-4} \cdot \frac{\sqrt{4+3h} + 2}{\sqrt{4+3h} + 2} \\
&= \frac{h(\sqrt{4+3h} + 2)}{3h} = \frac{\sqrt{4+3h} + 2}{3} \\
&= \frac{\sqrt{4+3h} + 2}{3} = \frac{\sqrt{4+3h} + 2}{3},
\end{aligned}$$

we have

$$\begin{aligned}
f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\
&= \lim_{h \rightarrow 0} \frac{3}{\sqrt{4+3h} + 2} \\
&= \frac{3}{\sqrt{4+3(0)} + 2} = \frac{3}{4}
\end{aligned}$$

Using (2.2): Since

$$\begin{aligned}
&\frac{f(b) - f(1)}{b - 1} \\
&= \frac{\sqrt{3b+1} - 2}{b - 1} \\
&= \frac{(\sqrt{3b+1} - 2)(\sqrt{3b+1} + 2)}{(b-1)(\sqrt{3b+1} + 2)} \\
&= \frac{(b-1)\sqrt{3b+1} + 2}{(b-1)\sqrt{3b+1} + 2} \\
&= \frac{3}{\sqrt{3b+1} + 2},
\end{aligned}$$

we have

$$\begin{aligned}
f'(1) &= \lim_{b \rightarrow 1} \frac{f(b) - f(1)}{b - 1} \\
&= \lim_{b \rightarrow 1} \frac{3}{\sqrt{3b+1} + 2} \\
&= \frac{3}{\sqrt{4} + 2} = \frac{3}{4}
\end{aligned}$$

4. Using (2.1):

$$\begin{aligned}
f'(2) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{3}{(2+h)+1} - 1}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{3}{3+h} - \frac{3+3h}{3+h}}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{-h}{3+h}}{h} \\
&= \lim_{h \rightarrow 0} \frac{-1}{3+h} = -\frac{1}{3}
\end{aligned}$$

Using (2.2):

$$\begin{aligned}
f'(2) &= \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} \\
&= \lim_{x \rightarrow 2} \frac{\frac{3}{x+1} - 1}{x - 2} \\
&= \lim_{x \rightarrow 2} \frac{\frac{3}{x+1} - \frac{x+1}{x+1}}{x - 2} \\
&= \lim_{x \rightarrow 2} \frac{\frac{-(x-2)}{x+1}}{x - 2} \\
&= \lim_{x \rightarrow 2} \frac{-1}{x+1} = -\frac{1}{3}
\end{aligned}$$

5. $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{3(x+h)^2 + 1 - (3x)^2 + 1}{h} \\
&= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 + 1 - (3x^2 + 1)}{h} \\
&= \lim_{h \rightarrow 0} \frac{6xh + 3h^2}{h} \\
&= \lim_{h \rightarrow 0} (6x + 3h) = 6x
\end{aligned}$$

6. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) + 1 - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 2h}{h} \\
&= \lim_{h \rightarrow 0} \frac{h(2x + h - 2)}{h} = 2x - 2
\end{aligned}$$

7. $\lim_{b \rightarrow x} \frac{f(b) - f(x)}{b - x}$

$$\begin{aligned}
&= \lim_{b \rightarrow x} \frac{b^3 + 2b - 1 - (x^3 + 2x - 1)}{b - x}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{b \rightarrow x} \frac{(b-x)(b^2+bx+x^2+2)}{b-x} \\
&= \lim_{b \rightarrow x} b^2+bx+x^2+2 \\
&= 3x^2+2
\end{aligned}$$

$$\begin{aligned}
8. \quad f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{(x+h)^4 - 2(x+h)^2 + 1 - f(x)}{h} \\
&= \lim_{h \rightarrow 0} [4x^3 + 6x^2h + 4xh^2 + h^3 - 4x - 2h] \\
&= 4x^3 - 4x
\end{aligned}$$

$$\begin{aligned}
9. \quad \lim_{b \rightarrow x} \frac{f(b) - f(x)}{b-x} &= \lim_{b \rightarrow x} \frac{\frac{3}{b+1} - \frac{3}{x+1}}{b-x} \\
&= \lim_{b \rightarrow x} \frac{\frac{3(x+1) - 3(b+1)}{(b+1)(x+1)}}{b-x} \\
&= \lim_{b \rightarrow x} \frac{-3(b-x)}{(b+1)(x+1)(b-x)} \\
&= \lim_{b \rightarrow x} \frac{-3}{(b+1)(x+1)} \\
&= \frac{-3}{(x+1)^2}
\end{aligned}$$

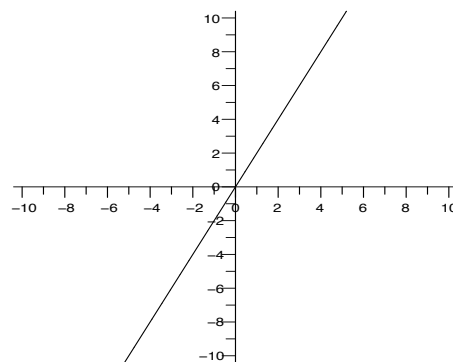
$$\begin{aligned}
10. \quad f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{2}{2(x+h)-1} - \frac{2}{2x-1}}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{2(2x-1) - 2(2x+2h-1)}{(2x+2h-1)(2x-1)}}{h} \\
&= \lim_{h \rightarrow 0} \frac{\frac{-4h}{(2x+2h-1)(2x-1)}}{h} \\
&= \lim_{h \rightarrow 0} \frac{-4}{(2x+2h-1)(2x-1)} \\
&= \frac{-4}{(2x-1)^2}
\end{aligned}$$

$$\begin{aligned}
11. \quad f(t) &= \sqrt{3t+1} \\
f'(t) &= \lim_{b \rightarrow t} \frac{f(b) - f(t)}{(b-t)} \\
&= \lim_{b \rightarrow t} \frac{\sqrt{3b+1} - \sqrt{3t+1}}{(b-t)} \\
&\text{Multiplying by } \frac{\sqrt{3b+1} + \sqrt{3t+1}}{\sqrt{3b+1} + \sqrt{3t+1}} \text{ gives}
\end{aligned}$$

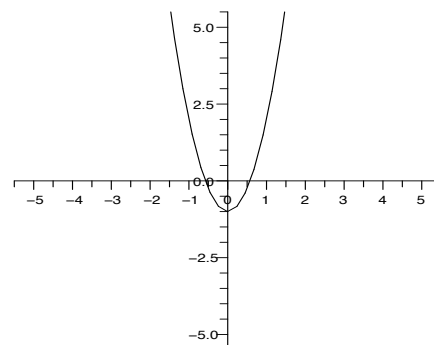
$$\begin{aligned}
f'(t) &= \lim_{b \rightarrow t} \frac{(3b+1) - (3t+1)}{(b-t)(\sqrt{3b+1} + \sqrt{3t+1})} \\
&= \lim_{b \rightarrow t} \frac{3(b-t)}{(b-t)(\sqrt{3b+1} + \sqrt{3t+1})} \\
&= \lim_{b \rightarrow t} \frac{3}{\sqrt{3b+1} + \sqrt{3t+1}} \\
&= \frac{3}{2\sqrt{3t+1}}
\end{aligned}$$

$$\begin{aligned}
12. \quad f(t) &= \sqrt{2t+4} \\
f'(t) &= \lim_{b \rightarrow t} \frac{f(b) - f(t)}{(b-t)} \\
&= \lim_{b \rightarrow t} \frac{\sqrt{2b+4} - \sqrt{2t+4}}{(b-t)} \\
&\text{Multiplying by } \frac{\sqrt{2b+4} + \sqrt{2t+4}}{\sqrt{2b+4} + \sqrt{2t+4}} \text{ gives} \\
f'(t) &= \lim_{b \rightarrow t} \frac{(2b+4) - (2t+4)}{(b-t)(\sqrt{2b+4} + \sqrt{2t+4})} \\
&= \lim_{b \rightarrow t} \frac{2(b-t)}{(b-t)(\sqrt{2b+4} + \sqrt{2t+4})} \\
&= \lim_{b \rightarrow t} \frac{2}{\sqrt{2b+4} + \sqrt{2t+4}} \\
&= \frac{2}{2\sqrt{2t+4}} = \frac{1}{\sqrt{2t+4}}
\end{aligned}$$

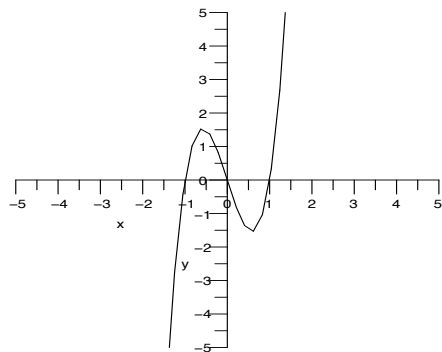
13. (a) The derivative should look like:



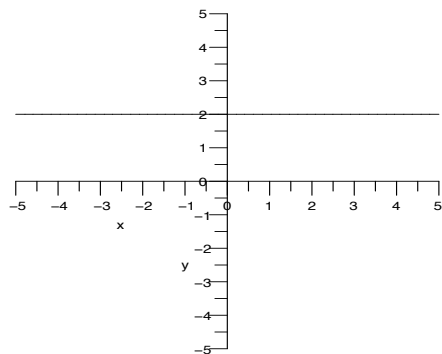
(b) The derivative should look like:



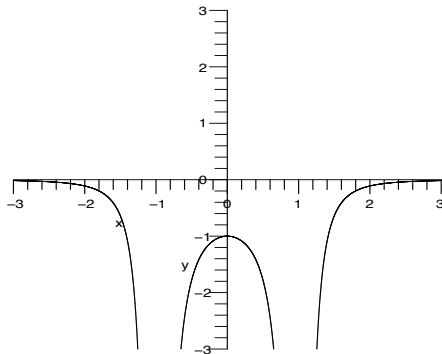
14. (a) The derivative should look like:



- (b) The derivative should look like:

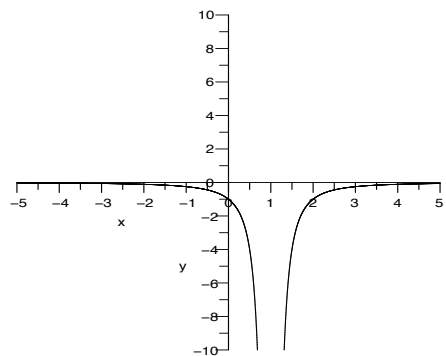


15. (a) The derivative should look like:

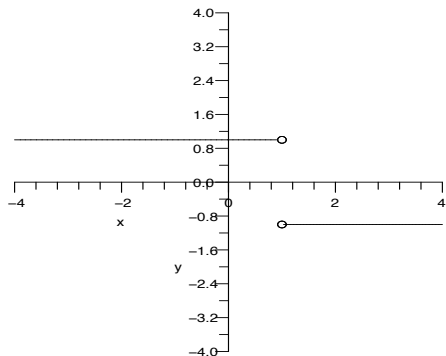


- (b) The derivative should look like:

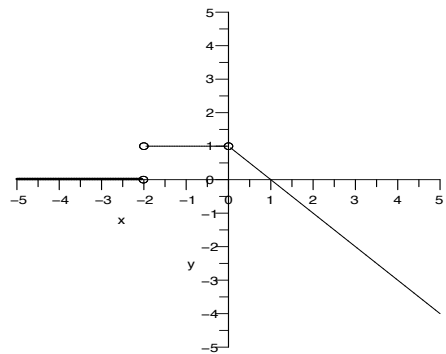
16. (a) The derivative should look like:

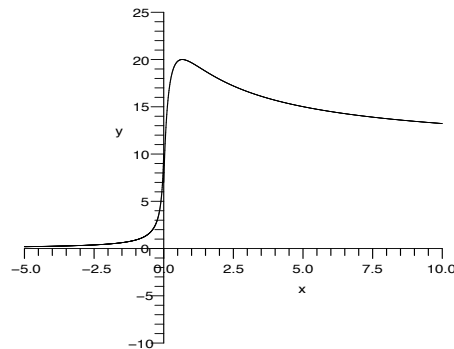
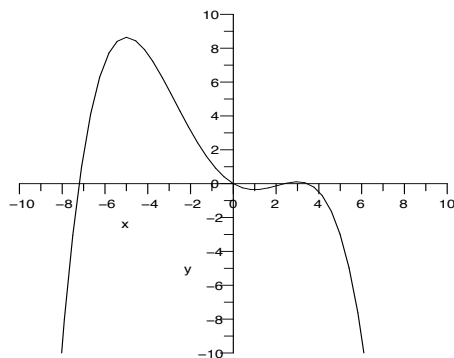


- (b) The derivative should look like:

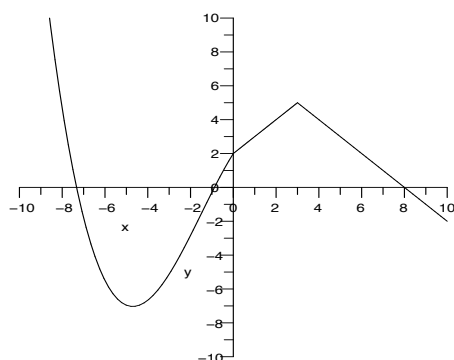


17. (a) The function should look like:





(b) The function should look like:



19. The left-hand derivative is

$$\begin{aligned} D_-f(0) &= \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{2h + 1 - 1}{h} = 2 \end{aligned}$$

The right-hand derivative is

$$\begin{aligned} D_+f(0) &= \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{3h + 1 - 1}{h} = 3 \end{aligned}$$

Since the one-sided limits do not agree ($2 \neq 3$), $f'(0)$ does not exist.

20. The left-hand derivative is

$$\begin{aligned} D_-f(0) &= \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{0 - 0}{h} = 0 \end{aligned}$$

The right-hand derivative is

$$\begin{aligned} D_+f(0) &= \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{2h}{h} = 2 \end{aligned}$$

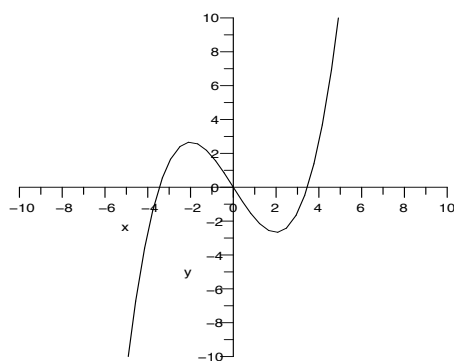
Since the one-sided limits do not agree ($0 \neq 2$), $f'(0)$ does not exist.

21. The left-hand derivative is

$$\begin{aligned} D_-f(0) &= \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h^2 - 0}{h} = 0 \end{aligned}$$

The right-hand derivative is

18. (a) The function should look like:



(b) The function should look like:

$$\begin{aligned}
 D_+f(0) &= \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{h^3 - 0}{h} = 0
 \end{aligned}$$

Since the one-sided limits are same ($0 = 0$), $f'(0)$ exist.

22. The left-hand derivative is

$$\begin{aligned}
 D_-f(0) &= \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} \\
 &= \lim_{h \rightarrow 0^-} \frac{2h}{h} = 2
 \end{aligned}$$

The right-hand derivative is

$$\begin{aligned}
 D_+f(0) &= \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{h^2 + 2h}{h} \\
 &= \lim_{h \rightarrow 0^+} \frac{h(h + 2)}{h} \\
 &= \lim_{h \rightarrow 0^+} h + 2 = 2
 \end{aligned}$$

Since the one-sided limits are same ($2 = 2$), $f'(0)$ exist.

23. $f(x) = \frac{x}{\sqrt{x^2 + 1}}$

x	$f(x)$	$\frac{f(x) - f(1)}{x - 1}$
1.1	0.7399401	0.3283329
1.01	0.7106159	0.3509150
1.001	0.7074601	0.3532884
1.0001	0.7071421	0.3535268
1.00001	0.7071103	0.3535507

The evidence of this table strongly suggests that the difference quotients (essentially) indistinguishable from the values (themselves) 0.353. If true, this would mean that $f'(1) \approx 0.353$.

24. $f(x) = xe^{x^2}$

x	$f(x)$	$\frac{f(x) - f(2)}{x - 2}$
1.1	172.7658734	635.6957329
1.01	114.2323717	503.6071639
1.001	109.6888867	492.5866054
1.0001	109.2454504	491.5034872
1.00001	109.201214	491.3953621
1.000001	109.1967915	491.3845515
1.0000001	109.1963492	491.3834702
1.00000001	109.1963050	491.3833622

The evidence of this table strongly suggests that the difference quotients (essentially) indistinguishable from the values (themselves) 491.383. If true, this would mean that $f'(2) \approx 491.383$.

25. $f(x) = \cos 3x$

x	$f(x)$	$\frac{f(x) - f(0)}{x - 0}$
0.1	0.9553365	-0.4466351
0.01	0.9995500	-0.0449966
0.001	0.9999955	-0.0045000
0.0001	1.0000000	-0.0004500
0.00001	1.0000000	-0.0000450

The evidence of this table strongly suggests that the difference quotients (essentially) indistinguishable from the values (themselves) 0. If true, this would mean that $f'(0) \approx 0$.

26. $f(x) = \ln 3x$

x	$f(x)$	$\frac{f(x) - f(2)}{x - 2}$
2.1	1.8405496	0.4879016
2.01	1.7967470	0.4987542
2.001	1.7922593	0.4998757
2.0001	1.7918095	0.4999875
2.00001	1.7917645	0.4999988
2.000001	1.7917600	0.4999999
2.0000001	1.7917595	0.5000000

The evidence of this table strongly suggests that the difference quotients (essentially) indistinguishable from the values (themselves) 0.5. If true, this would mean that $f'(2) \approx 0.5$.

27. Compute average velocities:

Time Interval	Average Velocity
(1.7, 2.0)	9.0
(1.8, 2.0)	9.5
(1.8, 2.0)	10.0
(2.0, 2.1)	10.0
(2.0, 2.2)	9.5
(2.0, 2.3)	9.0

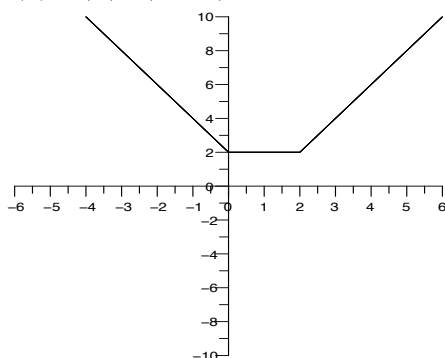
Our best estimate of velocity at $t = 2$ is 10.

28. Compute average velocities:

Time Interval	Average Velocity
(1.7, 2.0)	8
(1.8, 2.0)	8.5
(1.8, 2.0)	9.0
(2.0, 2.1)	8.0
(2.0, 2.2)	8.0
(2.0, 2.3)	7.67

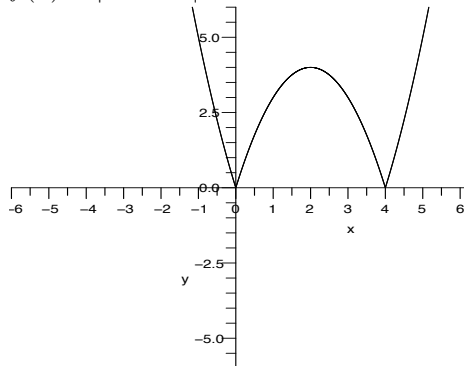
A velocity of between 8 and 9 seems to be a good guess.

29. (a) $f(x) = |x| + |x - 2|$



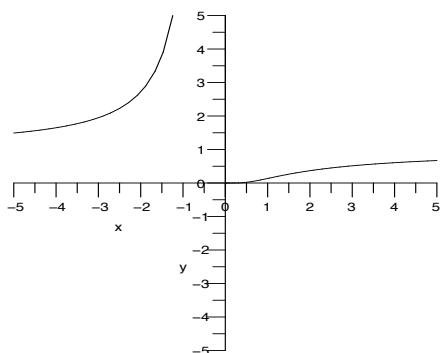
$f(x)$ is not differentiable at $x = 0$ and $x = 2$.

- (b) $f(x) = |x^2 - 4x|$



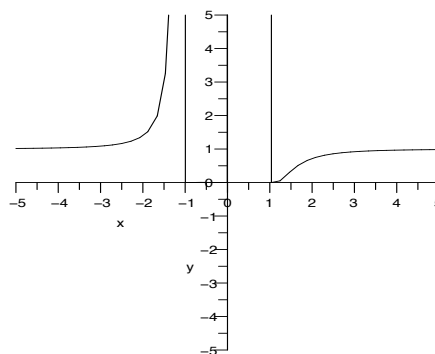
$f(x)$ is not differentiable at $x = 0$ and $x = 4$.

30. (a) $g(x) = e^{-2/x}$



$g(x)$ is not differentiable at $x = 0$.

- (b) $g(x) = e^{-2/(x^3-x)}$



$g(x)$ is not differentiable at $x = 0$ and $x = 1$.

31. $\lim_{h \rightarrow 0} \frac{(0+h)^p - 0^p}{h} = \lim_{h \rightarrow 0} \frac{h^p}{h} = \lim_{h \rightarrow 0} h^{p-1}$
 The last limit does not exist when $p < 1$, equals 1 when $p = 1$ and is 0 when $p > 1$. Thus $f(0)$ exists when $p \geq 1$.

32. $f(x) = \begin{cases} x^2 + 2x & x < 0 \\ ax + b & x \geq 0 \end{cases}$

For $h < 0$, $f(h) = h^2 + 2h$, $f(0) = b$

$$\begin{aligned} D_-f(0) &= \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{h^2 + 2h - b}{h} \end{aligned}$$

For f to be differentiable $D_-f(0)$ must exist.

$D_-f(0)$ exists if and only if $b = 0$.

Substituting $b = 0$, we get

$$D_-f(0) = \lim_{h \rightarrow 0^-} \frac{h^2 + 2h}{h} = \lim_{h \rightarrow 0^-} (h + 2) = 2$$

For $h > 0$, $f(h) = ah + b$, $f(0) = b$

$$\begin{aligned} D_+f(0) &= \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{ah + b - b}{h} \end{aligned}$$

$$= \lim_{h \rightarrow 0^+} \frac{ah}{h} = a$$

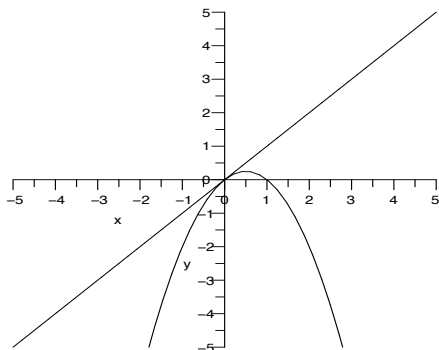
$D_+f(0) = 2$ if and only if $a = 2$.

33. Let $f(x) = -1 - x^2$ then for all, we have $f(x) \leq x$. But at $x = -1$, we find $f(-1) = -2$ and

$$\begin{aligned} f'(-1) &= \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1 - (-1+h)^2 - (-2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 - (1 - 2h + h^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2h - h^2}{h} = \lim_{h \rightarrow 0} (2 - h) = 2. \end{aligned}$$

So, $f'(x)$ is not always less than 1.

34. This is not always true. For example the function $f(x) = -x^2 + x$ satisfies the hypotheses but $f'(x) > 1$ for all $x < 0$ as the following graph shows.



$$\begin{aligned}
 35. \quad & \lim_{x \rightarrow a} \frac{[f(x)]^2 - [f(a)]^2}{x^2 - a^2} \\
 &= \lim_{x \rightarrow a} \frac{[f(x) - f(a)][f(x) + f(a)]}{(x - a)(x + a)} \\
 &= \left[\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right] \left[\lim_{x \rightarrow a} \frac{f(x) + f(a)}{x + a} \right] \\
 &= f'(a) \frac{2f(a)}{2a} = \frac{f(a)f'(a)}{a}
 \end{aligned}$$

36. Let $u = ch$ so $h = \frac{u}{c}$. Then we have

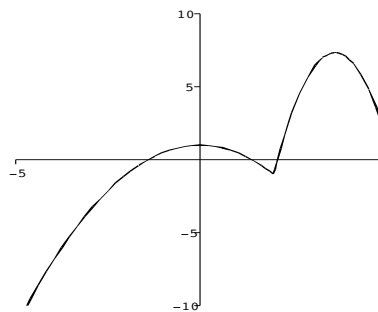
$$\begin{aligned}
 & \lim_{h \rightarrow 0} \frac{f(a + ch) - f(a)}{h} \\
 &= \lim_{\frac{u}{c} \rightarrow 0} \frac{f(a + u) - f(a)}{\frac{u}{c}} \\
 &= \lim_{u \rightarrow 0} \frac{f(a + u) - f(a)}{\frac{u}{c}} \\
 &= \lim_{u \rightarrow 0} c \left(\frac{f(a + u) - f(a)}{u} \right) \\
 &= c \lim_{u \rightarrow 0} \frac{f(a + u) - f(a)}{u} \\
 &= cf'(a)
 \end{aligned}$$

37. Because the curve appears to be bending upward, the slopes of these secant lines (based at $x = 1$ and with upper endpoint beyond 1) will increase with the upper endpoint. This has also the effect that any one of these slopes is greater than the actual derivative. Therefore $f'(1) < \frac{f(1.5) - f(1)}{0.5} < \frac{f(2) - f(1)}{1}$. As to where $f(1)$ fits in this list it seems necessary to read the graph and come up with estimates of $f(1)$ about 4, and $f(2)$ about 7. That would put the third number in the above list at about 3, comfortably less than $f(1)$.

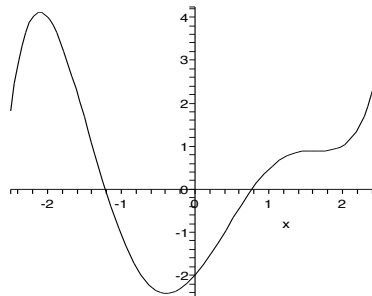
38. Note that $f(0) - f(-1)$ is the slope of the secant line from $x = -1$ to $x = 0$ (about),

and that $\frac{f(0) - f(-0.5)}{0.5}$ is the slope of the secant line from $x = -0.5$ to $x = 0$ (about-0.5). $f(0) = 3$ and $f'(0) = 0$. In increasing order, we have $f(0) - f(-1)$, $\frac{f(0) - f(-0.5)}{0.5}$, $f'(0)$, and $f'(0)$.

39. One possible such graph:



40. One possible such graph:



$$41. \quad \frac{d}{dx} (x^2) = 2x = 2x^1$$

$$\frac{d}{dx} (x^3) = 3x^2$$

$$\frac{d}{dx} (x^4) = 4x^3$$

In general

$$\frac{d}{dx} (x^n) = nx^{n-1}$$

$$42. \quad \frac{d}{dx} (x^n) = nx^{n-1}$$

$$\sqrt{x} = x^{1/2}$$

$$\frac{d}{dx} (\sqrt{x}) = \frac{d}{dx} (x^{1/2}) = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{x} = x^{-1}$$

$$\frac{d}{dx} \left(\frac{1}{x} \right) = \frac{d}{dx} (x^{-1}) = -1x^{-2} = \frac{-1}{x^2}$$

43. We estimate the derivative at $x = 2.5$ as follows $\frac{1.62 - 1.11}{2.7 - 2.39} = \frac{0.51}{0.31} = 1.64516$. For every increase of 1 meter in height of

serving point, there is an increase of 1.64516° in margin of error.

44. We estimate the derivative at $x = 2.854$ as follows

$$\frac{2.12 - 1.62}{3 - 2.7} = \frac{0.5}{0.3} = 1.66666.$$

For every increase of 1 meter in height of serving point, there is an increase of 1.66666° in margin of error.

45. We compile the rate of change in ton-MPG over each of the four two-year intervals for which data is given:

Intervals	Rate of change
(1992,1994)	0.4
(1994,1996)	0.4
(1996,1998)	0.4
(1998,2000)	0.2

These rates of change are measured in Ton-MPG per year. Either the first or second (they happen to agree) could be used as an estimate for the one-year “1994” while only the last is a promising estimate for the one-year interval “2000”. The mere fact that all these numbers are positive that efficiency is improving, the last number being smaller to suggest that the rate of improvement is slipping.

46. The average rate of change from 1992 to 1994 is 0.05, and from 1994 to 1996 is 0.1, so a good estimate of the rate of change in 1994 is 0.75. The average rate of change from 1998 to 2000 is -0.2 , and this is a good estimate for the rate of change in 2000. Comparing to exercise 35, we see that the MPG per ton increased, but the actual MPG for vehicles decreased. The weight of vehicles must have increased, if the weight remained then the actual MPG would have increased.

47. (a) meters per second
(b) items per dollar

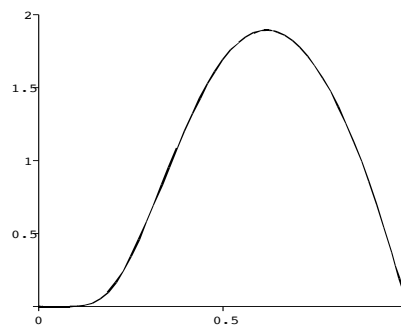
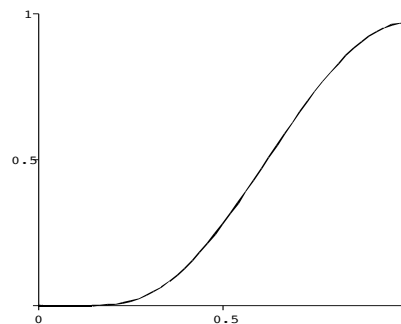
48. $c'(t)$ will represent the rate of change in amount of chemical and will be measured in grams per minute. $p'(x)$ will represent the rate of change of mass and will be measured in kg per meter.

49. If $f'(t) < 0$, the function is negatively sloped and decreasing, meaning the stock is losing value with the passing of time. This may be the basis for selling the stock if the current trend is expected to be a long term one.

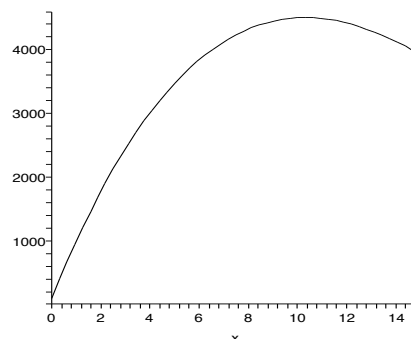
50. You should buy the stock with value $g(t)$. It is cheaper because $f(t) > g(t)$, and grow-

ing faster because $f'(t) < g'(t)$ (or possibly declining more slowly).

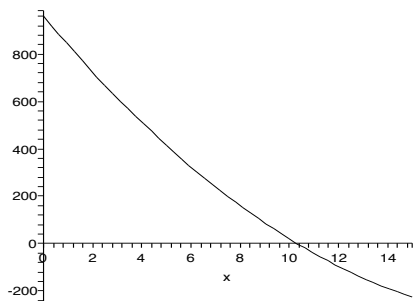
51. The following sketches are consistent with the hypotheses of infection $I'(t)$ rate rising, peaking and returning to zero. We started with the derivative (infection rate) and had to think backwards to construct the function $I(t)$. One can see in $I(t)$ the slope increasing up to the time of peak infection rate thereafter the slope decreasing but not the values. They merely level off.



52. One possible graph of the population $P(t)$:



Graph of $P'(t)$:



- 53.** Because the curve appears to be bending upward the slopes of the secant lines (based at $x = 1$ and with upper endpoint beyond 1) will increase with the upper endpoint. This has also the effect that any one of these slopes is greater than the actual derivative. Therefore

$$f'(1) < \frac{f(1.5) - f(1)}{0.5} < \frac{f(2) - f(1)}{1}.$$

As to where $f(1)$ fits in this list it seems necessary to read the graph and come up with estimates of $f(1)$ about 4, and $f(2)$ about 7. That would put the third number in the above list at about 3 comfortably less than $f(1)$.

54.
$$f(t) = \begin{cases} 0.1t & \text{if } 0 < t \leq 2 \cdot 10^4 \\ 2 \cdot 10^3 + (t - 2 \cdot 10^4)0.16 & \text{if } t > 2 \cdot 10^4 \end{cases}$$

This is another example of a piecewise linear function (this one is continuous), and although not differentiable at the income $x = 20000$, elsewhere we have

$$f'(x) = \begin{cases} 0.1 & 0 < t < 20000 \\ 0.16 & t > 20000 \end{cases}$$

2.3 Computation of Derivatives: The Power Rule

$$\begin{aligned} 1. \quad f'(x) &= \frac{d}{dx}(x^3) - \frac{d}{dx}(2x) + \frac{d}{dx}(1) \\ &= 3x^2 - 2\frac{d}{dx}(x) + 0 \\ &= 3x^2 - 2(1) \\ &= 3x^2 - 2 \end{aligned}$$

$$2. \quad f'(x) = 9x^8 - 15x^4 + 8x - 4$$

$$\begin{aligned} 3. \quad f'(t) &= \frac{d}{dt}(3t^3) - \frac{d}{dt}(2\sqrt{t}) \\ &= 3\frac{d}{dt}(t^3) - 2\frac{d}{dt}(t^{1/2}) \\ &= 3(3t^2) - 2\left(\frac{1}{2}t^{-1/2}\right) \\ &= 9t^2 - \frac{1}{\sqrt{t}} \end{aligned}$$

$$\begin{aligned} 4. \quad f(s) &= 5s^{1/2} - 4s^2 + 3 \\ f'(s) &= \frac{5}{2}s^{-1/2} - 8s \\ &= \frac{5}{2\sqrt{s}} - 8s \end{aligned}$$

$$\begin{aligned} 5. \quad f'(w) &= \frac{d}{dw}\left(\frac{3}{w}\right) - \frac{d}{dw}(8w) + \frac{d}{dw}(1) \\ &= 3\frac{d}{dw}(w^{-1}) - 8\frac{d}{dw}(w) + 0 \\ &= 3(-w^{-2}) - 8(1) \\ &= -\frac{3}{w^2} - 8 \end{aligned}$$

$$\begin{aligned} 6. \quad f'(y) &= \frac{d}{dy}\left(\frac{2}{y^4}\right) - \frac{d}{dy}(y^3) + \frac{d}{dy}(2) \\ &= 2\frac{d}{dy}(y^{-4}) - \frac{d}{dy}(y^3) + 0 \\ &= 2(-4y^{-5}) - 3(y^2) \\ &= -\frac{8}{y^5} - 3y^2 \end{aligned}$$

$$\begin{aligned} 7. \quad h'(x) &= \frac{d}{dx}\left(\frac{10}{\sqrt[3]{x}}\right) - \frac{d}{dx}(2x) + \frac{d}{dx}(\pi) \\ &= 10\frac{d}{dx}(x^{-1/3}) - 2\frac{d}{dx}(x) + 0 \\ &= 10\left(-\frac{1}{3}x^{-4/3}\right) - 2 \\ &= -\frac{10}{3x\sqrt[3]{x}} - 2 \end{aligned}$$

$$\begin{aligned} 8. \quad h'(x) &= \frac{d}{dx}(12x) - \frac{d}{dx}(x^2) - \frac{d}{dx}\left(\frac{3}{\sqrt[3]{x^2}}\right) \\ &= 12\frac{d}{dx}(x) - \frac{d}{dx}(x^2) - 3\frac{d}{dx}(x^{-2/3}) \\ &= 12 - 2x - 3\left(-\frac{2}{3}x^{-5/3}\right) \\ &= 12 - 2x + \frac{2}{x\sqrt[3]{x^2}} \end{aligned}$$

9. $f'(s) = \frac{d}{ds} 2s^{3/2} - \frac{d}{ds} (3s^{-1/3})$
 $= 2 \frac{d}{ds} (s^{3/2}) - 3 \frac{d}{ds} (s^{-1/3})$
 $= 2 \left(\frac{3}{2} s^{1/2} \right) - 3 \left(-\frac{1}{3} s^{-4/3} \right)$
 $= 3s^{1/2} + s^{-4/3}$
 $= 3\sqrt{s} + \frac{1}{\sqrt[3]{s^4}}$
10. $f'(t) = 3\pi t^{\pi-1} - 2.6t^{0.3}$
11. $f(x) = \frac{3x^2 - 3x + 1}{2x}$
 $= \frac{3x^2}{2x} - \frac{3x}{2x} + \frac{1}{2x}$
 $= \frac{3}{2}x - \frac{3}{2} + \frac{1}{2}x^{-1}$
 $f'(x) = \frac{d}{dx} \left(\frac{3}{2}x \right) - \frac{d}{dx} \left(\frac{3}{2} \right) + \frac{d}{dx} \left(\frac{1}{2}x^{-1} \right)$
 $= \frac{3}{2} \frac{d}{dx} (x) - 0 + \frac{1}{2} \frac{d}{dx} (x^{-1})$
 $= \frac{3}{2}(1) + \frac{1}{2}(-1x^{-2})$
 $= \frac{3}{2} - \frac{1}{2x^2}$
12. $f(x) = \frac{4x^2 - x + 3}{\sqrt{x}}$
 $= 4x^{3/2} - x^{1/2} + 3x^{-1/2}$
 $f'(x) = 6x^{1/2} - \frac{1}{2}x^{-1/2} - \frac{3}{2}x^{-3/2}$
13. $f(x) = x(3x^2 - \sqrt{x})$
 $= 3x^3 - x^{3/2}$
 $f'(x) = 3 \frac{d}{dx} (x^3) - \frac{d}{dx} (x^{3/2})$
 $= 3(3x^2) - \left(\frac{3}{2} x^{1/2} \right)$
 $= 9x^2 - \frac{3}{2} \sqrt{x}$
14. $f(x) = 3x^3 + 3x^2 - 4x - 4$,
 $f'(x) = 9x^2 + 6x - 4$
15. $f'(t) = \frac{d}{dt} (t^4 + 3t^2 - 2)$
 $= 4t^3 + 6t$
 $f''(t) = \frac{d}{dt} (4t^3 + 6t)$
 $= 12t^2 + 6$
16. $f(t) = 4t^2 - 12 + \frac{4}{t^2} = 4t^2 - 12 + 4t^{-2}$
- $f'(t) = \frac{d}{dt} (4t^2 - 12 + 4t^{-2})$
 $= 8t^2 - 0 + 4(-2t^{-3}) = 8t^2 - 8t^{-3}$
 $f''(t) = \frac{d}{dt} (8t^2 - 8t^{-3}) = 8 - 8(-3t^{-4})$
 $= 8 + 24t^{-4}$
 $f'''(t) = \frac{d}{dt} (8 + 24t^{-4}) = 0 + 24(-4t^{-5})$
 $= -96t^{-5} = -\frac{96}{t^5}$
17. $f(x) = 2x^4 - 3x^{-1/2}$
 $\frac{df}{dx} = 8x^3 + \frac{3}{2}x^{-3/2}$
 $\frac{d^2f}{dx^2} = 24x^2 - \frac{9}{4}x^{-5/2}$
18. $f(x) = x^6 - \sqrt{x} = x^6 - x^{1/2}$
 $\frac{df}{dx} = \frac{d}{dx} (x^6 - x^{1/2}) = 6x^5 - \frac{1}{2}x^{-1/2}$
 $\frac{d^2f}{dx^2} = \frac{d}{dx} \left(6x^5 - \frac{1}{2}x^{-1/2} \right)$
 $= 30x^4 - \frac{1}{2} \left(-\frac{1}{2}x^{-3/2} \right)$
 $= 30x^4 + \frac{1}{4}x^{-3/2}$
19. $f'(x) = \frac{d}{dx} \left(x^4 + 3x^2 - \frac{2}{\sqrt{x}} \right)$
 $= 4x^3 + 6x + x^{-3/2}$
 $f''(x) = \frac{d}{dx} (4x^3 + 6x + x^{-3/2})$
 $= 12x^2 + 6 - \frac{3}{2}x^{-5/2}$
 $f'''(x) = \frac{d}{dx} \left(12x^2 + 6 - \frac{3}{2}x^{-5/2} \right)$
 $= 24x + \frac{15}{4}x^{-7/2}$
 $f^4(x) = \frac{d}{dx} \left(24x + \frac{15}{4}x^{-7/2} \right)$
 $= 24 - \frac{105}{8}x^{-9/2}$
20. $f'(x) = 10x^9 - 12x^3 + 2$
 $f''(x) = 90x^8 - 36x^2$
 $f'''(x) = 720x^7 - 72x$
 $f^{(4)}(x) = 5040x^6 - 72$
 $f^{(5)}(x) = 30240x^5$
21. $s(t) = -16t^2 + 40t + 10$
 $v(t) = s'(t) = -32t + 40$
 $a(t) = v'(t) = s''(t) = -32$
22. $s(t) = -4.9t^2 + 12t - 3$
 $v(t) = s'(t) = -9.8t + 12$
 $a(t) = v'(t) = s''(t) = -9.8$

- 23.** $s(t) = \sqrt{t} + 2t^2 = t^{1/2} + 2t^2$
 $v(t) = s'(t) = \frac{1}{2}t^{-1/2} + 4t$
 $a(t) = v'(t) = s''(t) = -\frac{1}{4}t^{-3/2} + 4$
- 24.** $s(t) = 10 - 10t^{-1}$ $v(t) = s'(t) = 10t^{-2}$
 $a(t) = s''(t) = -20t^{-3}$
- 25.** $h(t) = -16t^2 + 40t + 5$
 $v(t) = h'(t) = -32t + 40$
 $a(t) = v'(t) = h''(t) = -32$

- (a) At time $t_0 = 1$
 $v(1) = 8$, object is going up.
 $a(1) = -32$, speed is decreasing.
- (b) At time $t_0 = 2$
 $v(2) = -24$, object is going down.
 $a(2) = -32$, speed is increasing.

- 26.** $h(t) = 10t^2 - 24t$
 $v(t) = h'(t) = 20t - 24$
 $a(t) = v'(t) = h''(t) = 20$

- (a) At time $t_0 = 2$
 $v(2) = 16$, object is going up.
 $a(2) = 20$, speed is increasing.
- (b) At time $t_0 = 1$
 $v(1) = -4$, object is going down.
 $a(1) = 20$, speed is decreasing.

- 27.** $f(x) = x^2 - 2$, $a = 2$, $f(2) = 2$,
 $f'(x) = 2x$, $f'(2) = 4$
The equation of the tangent line is
 $y = 4(x - 2) + 2$ or $y = 4x - 6$.

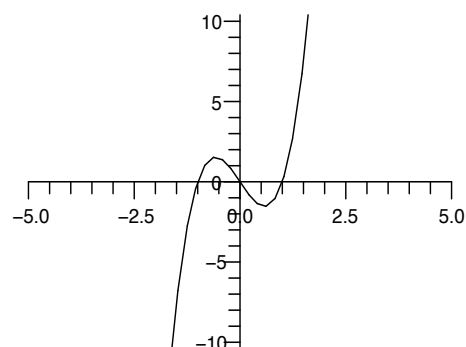
- 28.** $f(2) = 1$, $f'(x) = 2x - 2$, $f'(2) = 2$
Line through with slope 2 is
 $y = 2(x - 2) + 1$.

- 29.** $f(x) = 4\sqrt{x} - 2x$, $a = 4$
 $f(4) = 4\sqrt{4} - 2(4) = 0$.
 $f'(x) = \frac{d}{dx} (4x^{1/2} - 2x)$
 $= 2x^{-1/2} - 2 = \frac{2}{\sqrt{x}} - 2$
 $f'(4) = 1 - 2 = -1$
The equation of the tangent line is

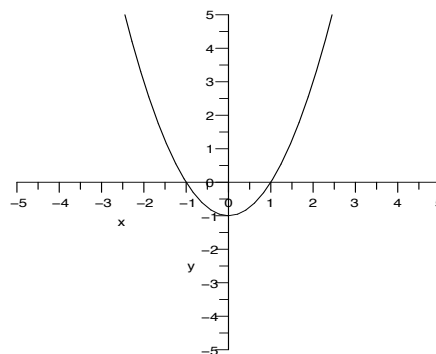
$$y = -1(x - 4) + 0 \text{ or } y = -x + 4.$$

- 30.** $f(x) = 3\sqrt{x} + 4$, $a = 2$
 $f(2) = 3\sqrt{2} + 4$
 $f'(x) = \frac{d}{dx} (3x^{1/2} + 4) = \frac{3}{2}x^{-1/2} = \frac{3}{2\sqrt{x}}$
 $f'(2) = \frac{3}{2\sqrt{2}}$
The equation of tangent line through
 $(2, 3\sqrt{2} + 4)$ with slope $\frac{3}{2\sqrt{2}}$ is
 $y = \frac{3}{2\sqrt{2}}(x - 2) + 3\sqrt{2} + 4.$

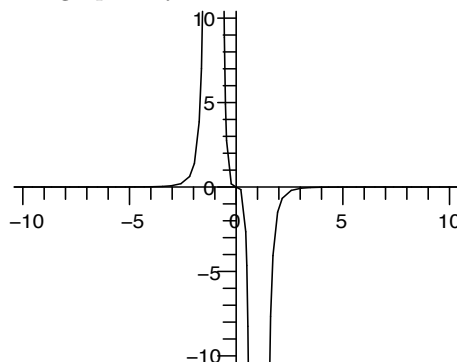
- 31. (a)** The graph of f' is as follows:



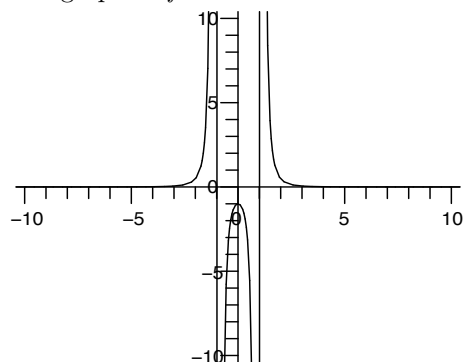
The graph of f'' is as follows.



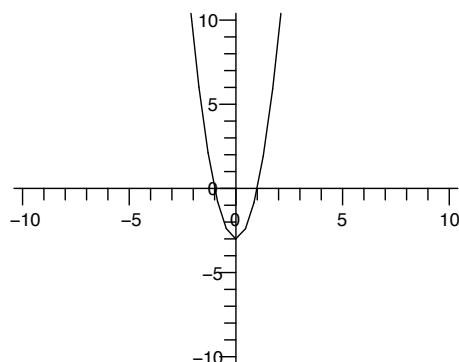
- (b) The graph of f' is as follows.



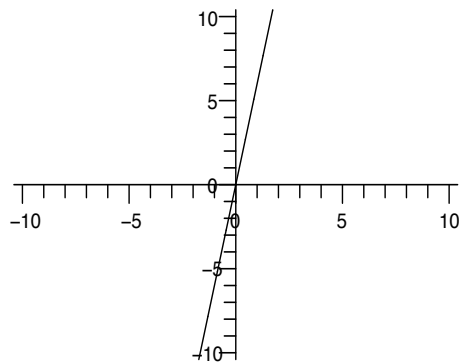
The graph of f'' is as follows.



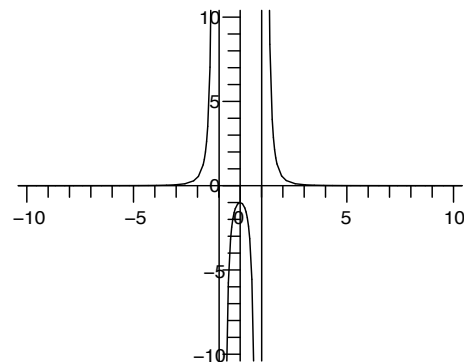
32. (a) The graph of f' is as follows.



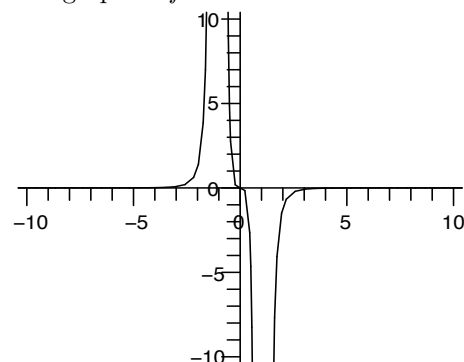
The graph of f'' is as follows



(b) The graph of f' is as follows.



The graph of f'' is as follows.



33. (a) $f(x) = x^3 - 3x + 1$

$$f'(x) = 3x^2 - 3$$

The tangent line to $y = f(x)$ is horizontal when

$$f'(x) = 0$$

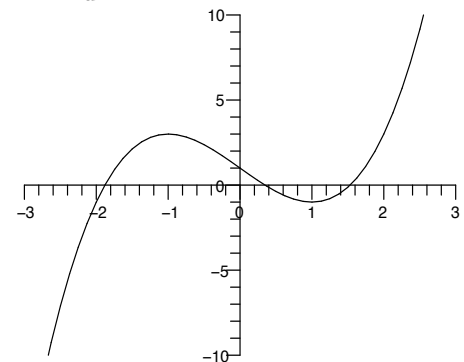
$$\Rightarrow 3x^2 - 3 = 0$$

$$\Rightarrow 3(x^2 - 1) = 0$$

$$\Rightarrow 3(x + 1)(x - 1) = 0$$

$$x = -1 \text{ or } x = 1.$$

(b) The graph shows that the first is a relative maximum, the second is a relative minimum.



(c) Now to determine the value(s) of x for

which the tangent line to $y = f(x)$ intersects the axis at 45° angle that is when $f'(x) = 1$.

$$3x^2 - 3 = 1$$

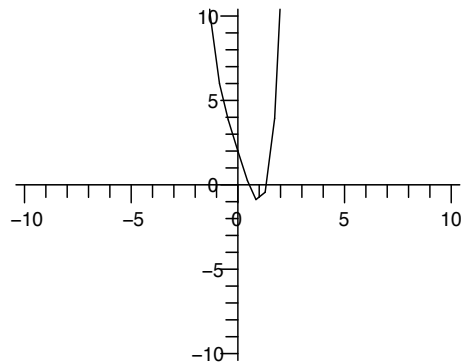
$$(x^2 - 1) = \frac{1}{3}$$

$$x^2 = \frac{4}{3}$$

$$x = \pm \frac{2}{\sqrt{3}}$$

34. (a) Now to determine the value(s) of x for which the tangent line to $y = f(x)$ intersects the axis at 45° angle that is when $f'(x) = 1$.
- $$3x^2 - 3 = 1$$
- $$3(x^2 - 1) = 1$$
- $$(x^2 - 1) = \frac{1}{3}$$
- $$x^2 = \frac{4}{3}$$
- $$x = \pm \frac{2}{\sqrt{3}}$$

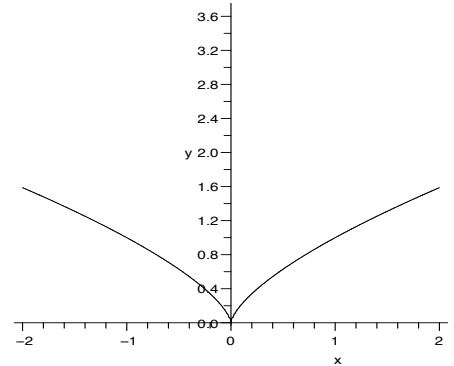
- (b) The graph shows that the function has global minimum at $(1, -1)$



- (c) Now to determine the value (s) of for which the tangent line to $y = f(x)$ intersects the axis at 45° angle that is when $f'(x) = 1$.
- $$4x^3 - 4 = 1$$
- $$(x^3 - 1) = \frac{1}{4}$$
- $$x^3 = \frac{5}{4} = \left(\frac{5}{2}\right)^{1/3}$$

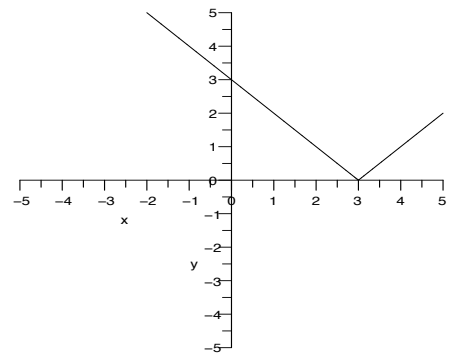
35. (a) $f(x) = x^{2/3}$
- $$f'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3\sqrt[3]{x}}$$

The slope of the tangent line to $y = f(x)$ does not exist where the derivative is undefined, which is only when $x=0$.



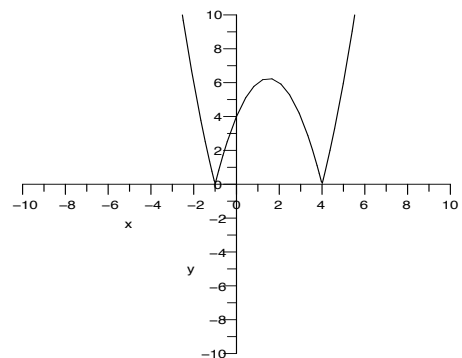
The graphical significance of this point is that there is vertical tangent here.

- (b) $f(x) = |x - 3|$
- $$f'(x) = \begin{cases} 1 & \text{when } x > 3 \\ -1 & \text{when } x < 3 \end{cases}$$
- $f'(x)$ is not defined at $x = 3$.



Though the graph of function is continuous at $x = 3$ tangent line does not exist as at this point there is sharp corner.

- (c) $f(x) = |x^2 - 3x - 4|$
- $$f'(x) = \begin{cases} 2x - 3 & \text{when } x > 4 \text{ or } x < -1 \\ -2x + 3 & \text{when } -1 < x < 4 \end{cases}$$
- $f'(x)$ is not defined at $x = -1, 4$.



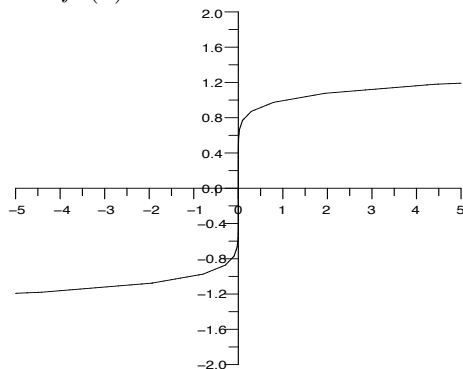
The graph shows that the function has global minima at $(-1, 0)$ and $(4, 0)$. The function has relative maximum at

$$\left(\frac{3}{2}, \frac{25}{4}\right).$$

36. (a) $f(x) = x^{1/9}$

$$f'(x) = \frac{1}{9}x^{-8/9} = \frac{1}{9x^{8/9}}$$

The $f'(x)$ is not defined at $x = 0$.

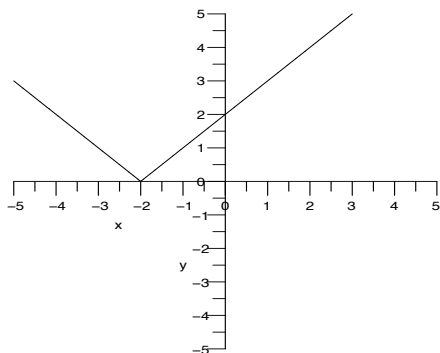


The graphical significance of this point is that there is vertical tangent here.

- (b) $f(x) = |x + 2|$

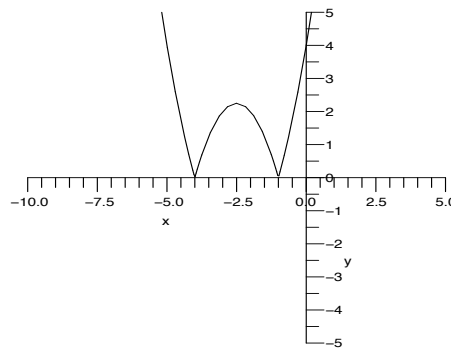
$$f'(x) = \begin{cases} 1 & \text{when } x > -2 \\ -1 & \text{when } x < -2 \end{cases}$$

The $f'(x)$ is not defined at $x = -2$.



Though the graph of function is continuous at $x = -2$, tangent line does not exist as at this point there is sharp corner.

- (c) $f(x) = |x^2 + 5x + 4| = |(x + 4)(x + 1)|$
 $f'(x) = \begin{cases} 2x + 5 & \text{when } x < -4 \text{ or } x > -1 \\ -2x - 5 & \text{when } -4 < x < -1 \end{cases}$
 The $f'(x)$ is not defined at $x = -4, -1$.



The graph shows that the function has global minima at $(-4, 0)$ and $(-1, 0)$. The function has relative maxima at $(-2.5, 2.25)$.

37. (a) $y = x^3 - 3x + 1$

$$y' = 3x^2 - 3 = 3(x^2 - 1)$$

The tangent line to $y = f(x)$ intersects the x -axis at a 45° angle when

$$f'(x) = 1$$

$$\Leftrightarrow 3(x^2 - 1) = 1$$

$$\Leftrightarrow x^2 = 1 + \frac{1}{3}$$

$$\Leftrightarrow x = \frac{2}{\sqrt{3}} \text{ or } x = -\frac{2}{\sqrt{3}}$$

- (b) The tangent line to $y = f(x)$ intersects the x -axis at a 30° angle when

$$f'(x) = \frac{1}{\sqrt{3}}.$$

$$\Leftrightarrow 3(x^2 - 1) = \frac{1}{\sqrt{3}}$$

$$\Leftrightarrow x^2 = 1 + \frac{1}{3\sqrt{3}}$$

$$\Leftrightarrow x = \left(1 + \frac{1}{3\sqrt{3}}\right)^{1/2} \text{ or }$$

$$x = -\left(1 + \frac{1}{3\sqrt{3}}\right)^{1/2}$$

38. Answers depend on CAS.

39. $f(x) = ax^2 + bx + c$, $f(0) = c$

$$f'(x) = 2ax + b, f'(0) = b$$

$$f''(x) = 2a, f''(0) = 2a$$

Given $f''(0) = 3$, we learn $2a = 3$, or $a = 3/2$. Given $f'(0) = 2$ we learn $2 = b$, and given $f(0) = -2$, we learn $c = -2$. In the end

$$f(x) = ax^2 + bx + c = \frac{3}{2}x^2 + 2x - 2$$

40. (a) $f(x) = \sqrt{x} = x^{1/2}$

$$f'(x) = \frac{1}{2}x^{-1/2}$$

$$f''(x) = \frac{1}{2} \left(-\frac{1}{2}\right) x^{-3/2}$$

$$\begin{aligned}
 f'''(x) &= \left(\frac{1}{2}\right) \left(\frac{-1}{2}\right) \left(\frac{-3}{2}\right) x^{-5/2} \\
 f^{(n)}(x) &= (-1)^{n-1} \frac{1 \cdot 3 \cdot 5 \cdots (2n-3)}{2^n} x^{-(2n-1)/2} \\
 &= (-1)^{n-1} \cdot \frac{1 \cdot 2 \cdot 3 \cdots (2n-2)}{2^n \cdot 2 \cdot 4 \cdots (2n-2)} x^{-(2n-1)/2} \\
 &= (-1)^{n-1} \cdot \frac{(2n-2)!}{2^{2n-1} (n-1)!} x^{-(2n-1)/2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad f'(x) &= -2x^{-3} \\
 f''(x) &= 6x^{-4} \\
 f'''(x) &= -24x^{-5}
 \end{aligned}$$

The pattern is

$$f^{(n)}(x) = (-1)^n (n+1)! x^{-n-2}$$

41. For $y = \frac{1}{x}$, we have $y' = -\frac{1}{x^2}$. Thus, the slope of the tangent line at $x = a$ is $-\frac{1}{a^2}$.

When $a = 1$, the slope of the tangent line at $(1, 1)$ is -1 , and the equation of the tangent line is $y = -x + 2$. The tangent line intersects the axes at $(0, 2)$ and $(2, 0)$. Thus, the area of the triangle is $\frac{1}{2}(2)(2) = 2$.

When $a = 2$, the slope of the tangent line at $\left(2, \frac{1}{2}\right)$ is $-\frac{1}{4}$, and the equation of the

tangent line is $y = -\frac{1}{4}x + 2$. The tangent line intersects the axes at $(0, 1)$ and $(4, 0)$.

Thus, the area of the triangle is $\frac{1}{2}(4)(1) = 2$.

In general, the equation of the tangent line is $y = -\left(\frac{1}{a^2}\right)x + \frac{2}{a}$. The tangent line inter-

sects the axes at $\left(0, \frac{2}{a}\right)$ and $(2a, 0)$. Thus, the area of the triangle is $\frac{1}{2}(2a)\left(\frac{2}{a}\right) = 2$.

42. For $y = \frac{1}{x^2} = x^{-2}$, we have

$$f'(x) = -2x^{-3} = -\frac{2}{x^3}$$

Thus, the slope of the tangent line at $x = a$ is $-\frac{2}{a^3}$.

When $a = 1$, the slope of the tangent line at $(1, 1)$ is -2 , and the equation of the tangent line is $y = -2x + 3$. The tangent line intersects the axes at $(0, 3)$ and $\left(\frac{3}{2}, 0\right)$. Thus

the area of the triangle is $\frac{1}{2}(3)\left(\frac{3}{2}\right) = \frac{9}{4}$.

When $a = 2$, the slope of the tangent line at $\left(2, \frac{1}{4}\right)$ is $-\frac{1}{4}$, and the equation of the

tangent line is

$y = -\frac{1}{4}x + \frac{3}{4}$. The tangent line intersects the axes at $\left(0, \frac{3}{4}\right)$ and $(3, 0)$. Thus the area of the triangle is $\frac{1}{2}\left(\frac{3}{4}\right)(3) = \frac{9}{8}$.

Since $\frac{9}{4} \neq \frac{9}{8}$, we see that the result for exercise 41 does not hold here.

$$\begin{aligned}
 43. \quad \text{(a)} \quad g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\max_{a \leq t \leq x+h} f(t) - \max_{a \leq t \leq x} f(t) \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} [f(x+h) - f(x)] \\
 &= f'(x)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\max_{a \leq t \leq x+h} f(t) - \max_{a \leq t \leq x} f(t) \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} [f(a) - f(a)] = 0
 \end{aligned}$$

$$\begin{aligned}
 44. \quad \text{(a)} \quad g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\min_{a \leq t \leq x+h} f(t) - \min_{a \leq t \leq x} f(t) \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} [f(a) - f(a)] = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\min_{a \leq t \leq x+h} f(t) - \min_{a \leq t \leq x} f(t) \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} [f(x+h) - f(x)] \\
 &= f'(x)
 \end{aligned}$$

45. Try $f(x) = cx^4$ for some constant c . Then $f'(x) = 4cx^3$ so c must be 1. One possible answer is $f(x) = x^4$.

46. Try $f(x) = cx^5$ for some constant c . Then $f'(x) = 5cx^4$ so c must be 1. One possible answer is $f(x) = x^5$.

47. $f'(x) = \sqrt{x} = x^{1/2}$
 $f(x) = \frac{2}{3}x^{3/2}$ is one possible function

48. If $f'(x) = x^{-2}$, then $f(x) = -x^{-1}$ is one possible function.

49. $w(b) = cb^{3/2}$
 $w'(b) = \frac{3c}{2}b^{1/2} = \frac{3c\sqrt{b}}{2}$

$w'(b) > 1$ when

$$\frac{3c\sqrt{b}}{2} > 1,$$

$$\sqrt{b} > \frac{2}{3c}$$

$$b > \frac{4}{9c^2}$$

Since c is constant, when b is large enough

b will be greater than $\frac{4}{9c^2}$. After this point,

when b increases by 1 *unit*, the leg width w is increasing by more than 1 *unit*, so that leg width is increasing faster than body length. This puts a limitation on the size of land animals since, eventually, the body will not be long enough to accomodate the width of the legs.

50. World Record Times Mens Track

<i>Dist.</i>	<i>Time</i>	<i>Ave.</i>	<i>f(d)</i>
400	43.18	9.26	9.25
800	101.11	7.91	8.17
1000	131.96	7.58	7.86
1500	206.00	7.28	9.25
2000	284.79	7.02	6.95

Here, distance is in meters, time is in seconds and hence average in meters per second.

The function $f(d)$ is quite close to predicting the average speed of world record pace.

$v'(d)$ represents the rate of change in average speed over d meters per meter. $v'(d)$ tells us how much $v(d)$ would change if d changed to $d + 1$.

51. We can approximate

$f'(2000) \approx \frac{9039.5 - 8690.7}{2001 - 1999} = 174.4$. This is the rate of change of the GDP in billions of dollars per year.

To approximate $f''(2000)$, we first estimate

$$f'(1999) \approx \frac{9016.8 - 8347.3}{2000 - 1998} = 334.75$$

$$\text{and } f'(1998) \approx \frac{8690.7 - 8004.5}{1999 - 1997} = 343.1$$

Since these values are decreasing, $f''(2000)$ is negative. We estimate

$$f''(2000) \approx \frac{174.4 - 334.75}{2000 - 1999} = -160.35$$

This represents the rate of change of the rate of change of the GDP over time. In 2000, the GDP is increasing by a rate of 174.4 billion dollars per year, but this increase is decreasing by a rate of 160.35 billion dollars-per-year per year.

52. $f'(2000)$ can be approximated by the average rate of change from 1995 to 2000.

$$f'(2000) \approx \frac{4619 - 4353}{2000 - 1995} = 53.2$$

This is the rate of change of weight of SUVs over time. In 2000 the weight of SUVs is increasing by 53.2 pounds per year.

Similarly approximate $f'(1995) \approx 32.8$ and $f'(1990) \approx 26.8$. The second derivative is definitely positive. We can approximate

$$f''(2000) \approx \frac{53.2 - 32.8}{2000 - 1995} = 4.08.$$

This is the rate of change in the rate of change of the weight of SUVs. Notonly are SUVs getting heavier at a rate of 53.2 pounds per year, this rate is itself increasing at a rate of about 4 pounds-per-year per year.

53. Newton's Law states that force equals mass times acceleration. That is, if $F(t)$ is the driving force at time t , then $m \cdot f''(t) = m \cdot a(t) = F(t)$ in which m is the mass, appropriately unitized. The third derivative of the distance function is then

$$f'''(t) = a'(t) = \frac{1}{m}F'(t).$$

It is both the derivative of the acceleration and directly proportional to the rate of change in force. Thus an abrupt change in acceleration or "jerk" is the direct consequence of an abrupt change in force.

2.4 The Product and Quotient Rules

1. $f(x) = (x^2 + 3)(x^3 - 3x + 1)$

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^2 + 3) \cdot (x^3 - 3x + 1) \\ &\quad + (x^2 + 3) \cdot \frac{d}{dx}(x^3 - 3x + 1) \\ &= (2x)(x^3 - 3x + 1) \\ &\quad + (x^2 + 3)(3x^2 - 3) \end{aligned}$$

2. $f(x) = (x^3 - 2x^2 + 5)(x^4 - 3x^2 + 2)$

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^3 - 2x^2 + 5)(x^4 - 3x^2 + 2) \\ &\quad + (x^3 - 2x^2 + 5) \frac{d}{dx}(x^4 - 3x^2 + 2) \\ &= (3x^2 - 4x)(x^4 - 3x^2 + 2) \\ &\quad + (x^3 - 2x^2 + 5)(4x^3 - 6x) \end{aligned}$$

3. $f(x) = (\sqrt{x} + 3x) \left(5x^2 - \frac{3}{x} \right)$

$$= (x^{1/2} + 3x)(5x^2 - 3x^{-1})$$

- $$f'(x) = \left(\frac{1}{2}x^{-1/2} + 3\right)(5x^2 - 3x^{-1})$$
- $$+ (x^{1/2} + 3x)(10x + 3x^{-2})$$
4. $f(x) = (x^{3/2} - 4x)(x^4 - 3x^{-2} + 2)$
- $$f'(x) = \frac{d}{dx}(x^{3/2} - 4x)(x^4 - 3x^{-2} + 2)$$
- $$+ (x^{3/2} - 4x)\frac{d}{dx}(x^4 - 3x^{-2} + 2)$$
- $$= \left(\frac{3}{2}x^{1/2} - 4\right)(x^4 - 3x^{-2} + 2)$$
- $$+ (x^{3/2} - 4x)(4x^3 + 6x^{-3})$$
5. $g(t) = \frac{3t - 2}{5t + 1}$
- $$g'(t) = \frac{\left((5t+1)\frac{d}{dt}(3t-2)\right) - \left((3t-2)\frac{d}{dt}(5t+1)\right)}{(5t+1)^2}$$
- $$= \frac{3(5t+1) - 5(3t-2)}{(5t+1)^2}$$
- $$= \frac{15t + 3 - 15t + 10}{(5t+1)^2} = \frac{13}{(5t+1)^2}$$
6. $g(t) = \frac{t^2 + 2t + 5}{t^2 - 5t + 1}$
- $$g'(t) = \frac{\left((t^2-5t+1)\frac{d}{dt}(t^2+2t+5)\right) - \left((t^2+2t+5)\frac{d}{dt}(t^2-5t+1)\right)}{(t^2-5t+1)^2}$$
- $$= \frac{(t^2 - 5t + 1)(2t + 2) - (t^2 + 2t + 5)(2t - 5)}{(t^2 - 5t + 1)^2}$$
7. $f(x) = \frac{3x - 6\sqrt{x}}{5x^2 - 2} = \frac{3(x - 2x^{1/2})}{5x^2 - 2}$
- $$f'(x) = \frac{3\left((5x^2-2)\frac{d}{dx}(x-2x^{1/2}) - (x-2x^{1/2})\frac{d}{dx}(5x^2-2)\right)}{(5x^2-2)^2}$$
- $$= 3 \frac{\left((5x^2-2)(1-x^{-1/2}) - (x-2x^{1/2})(10x)\right)}{(5x^2-2)^2}$$
8. $f(x) = \frac{6x - 2x^{-1}}{x^2 + x^{1/2}}$
- $$f'(x) = \frac{(x^2+x^{1/2})\frac{d}{dx}(6x-2x^{-1}) - (6x-2x^{-1})\frac{d}{dx}(x^2+x^{1/2})}{(x^2+x^{1/2})^2}$$
- $$= \frac{(x^2+x^{1/2})(6+2x^{-2}) - (6x-2x^{-1})(2x+\frac{1}{2}x^{-1/2})}{(x^2+x^{1/2})^2}$$
9. $f(u) = \frac{(u+1)(u-2)}{u^2-5u+1} = \frac{u^2-u-2}{u^2-5u+1}$
- $$f'(u) = \frac{\left((u^2-5u+1)\frac{d}{du}(u^2-u-2)\right) - \left((u^2-u-2)\frac{d}{du}(u^2-5u+1)\right)}{(u^2-5u+1)^2}$$
- $$= \frac{(u^2-5u+1)(2u-1) - (u^2-u-2)(2u-5)}{(u^2-5u+1)^2}$$
- $$= \frac{2u^3-10u^2+2u-u^3+5u-1-2u^3+2u^2+4u+5u^2-5u-10}{(u^2-5u+1)^2}$$
- $$= \frac{-4u^2+6u-11}{(u^2-5u+1)^2}$$
10. $f(u) = \frac{(2u)(u+3)}{u^2+1} = \frac{2u^2+6u}{u^2+1}$
- $$f'(u) =$$

$$\frac{\left((u^2+1)\frac{d}{du}(2u^2+6u)\right) - \left((2u^2+6u)\frac{d}{du}(u^2+1)\right)}{(u^2+1)^2}$$

$$= \frac{(u^2+1)(4u+6) - (2u^2+6u)(2u)}{(u^2+1)^2}$$

$$= \frac{4u^3+6u^2+4u+6-4u^3-12u^2}{(u^2+1)^2}$$

$$= \frac{-6u^2+4u+6}{(u^2+1)^2}$$

$$= \frac{2(-3u^2+2u+3)}{(u^2+1)^2}$$

11. We do not recommend treating this one as a quotient, but advise preliminary simplification.

$$f(x) = \frac{x^2 + 3x - 2}{\sqrt{x}}$$

$$= \frac{x^2}{\sqrt{x}} + \frac{3x}{\sqrt{x}} - \frac{2}{\sqrt{x}}$$

$$= x^{3/2} + 3x^{1/2} - 2x^{-1/2}$$

$$f'(x) = \frac{3}{2}x^{1/2} + \frac{3}{2}x^{-1/2} + x^{-3/2}$$

12. $f(x) = \frac{x^2 - 2x}{x^2 + 5x}$

$$f'(x) = \frac{(x^2+5x)\frac{d}{dx}(x^2-2x) - (x^2-2x)\frac{d}{dx}(x^2+5x)}{(x^2+5x)^2}$$

$$= \frac{(x^2+5x)(2x-2) - (x^2-2x)(2x+5)}{(x^2+5x)^2}$$

13. We simplify instead of using the product rule.

$$h(t) = t(\sqrt[3]{t} + 3) = t^{4/3} + 3t$$

$$h'(t) = \frac{4}{3}\sqrt[3]{t} + 3$$

14. $h(t) = \frac{t^2}{3} + \frac{5}{t^2} = \frac{1}{3}t^2 + 5t^{-2}$

$$h'(t) = \frac{2}{3}t - 10t^{-3}$$

15. $f(x) = (x^2 - 1)\frac{x^3 + 3x^2}{x^2 + 2}$

$$f'(x) = \frac{d}{dx}(x^2 - 1) \cdot \left(\frac{x^3 + 3x^2}{x^2 + 2}\right)$$

$$+ (x^2 - 1) \cdot \frac{d}{dx}\left(\frac{x^3 + 3x^2}{x^2 + 2}\right)$$

We have

$$\frac{d}{dx}\left(\frac{x^3 + 3x^2}{x^2 + 2}\right) = \frac{(x^2+2)\frac{d}{dx}(x^3+3x^2) - (x^3+3x^2)\frac{d}{dx}(x^2+2)}{(x^2+2)^2}$$

$$= \frac{(x^2+2) \cdot (3x^2+6x) - (x^3+3x^2) \cdot (2x)}{(x^2+2)^2}$$

$$= \frac{3x^4+6x^2+6x^3+12x - (2x^4+6x^3)}{(x^2+2)^2}$$

$$= \frac{x^4 + 6x^2 + 12x}{(x^2 + 2)^2}$$

So, $f'(x) =$
 $(2x) \cdot \left(\frac{x^3 + 3x^2}{x^2 + 2}\right) + (x^2 - 1) \cdot \frac{x^4 + 6x^2 + 12x}{(x^2 + 2)^2}$

$$\begin{aligned} 16. \quad f(x) &= \frac{(x+2)(x-1)(x+1)}{x(x+1)} \\ &= \frac{x^2 + x - 2}{x} \\ &= x + 1 - 2x^{-1} \\ f'(x) &= 1 + 2x^{-2} \end{aligned}$$

$$\begin{aligned} 17. \quad f(x) &= (x^2 + 2x)(x^4 + x^2 + 1) \\ f'(x) &= \left[\frac{d}{dx}(x^2 + 2x) \right] (x^4 + x^2 + 1) \\ &\quad + \left[\frac{d}{dx}(x^4 + x^2 + 1) \right] (x^2 + 2x) \\ &= (2x + 2)(x^4 + x^2 + 1) \\ &\quad + (4x^3 + 2x)(x^2 + 2x) \end{aligned}$$

At $x = a = 0$, we get:

$$f(0) = 0$$

$$f'(0) = 2$$

Therefore, the line with slope 2 and passing through the point $(0, 0)$ has equation $y = 2x$.

$$\begin{aligned} 18. \quad f(x) &= (x^3 + x + 1)(3x^2 + 2x - 1) \\ f'(x) &= \left[\frac{d}{dx}(x^3 + x + 1) \right] (3x^2 + 2x - 1) \\ &\quad + \left[\frac{d}{dx}(3x^2 + 2x - 1) \right] (x^3 + x + 1) \\ &= (3x^2 + 1)(3x^2 + 2x - 1) \\ &\quad + (6x + 2)(x^3 + x + 1) \end{aligned}$$

At $x = a = 1$, we get:

$$f(1) = 12$$

$$f'(1) = (3+1)(3+2-1) + (6+2)(1+1+1) = 40$$

Therefore, the line with slope 40 and passing through the point $(1, 12)$ has equation $y = 40(x - 1) + 12$.

$$\begin{aligned} 19. \quad f(x) &= \frac{x+1}{x+2} \\ \text{By The Quotient Rule, we have} \\ f'(x) &= \frac{((x+2)\frac{d}{dx}(x+1)) - ((x+1)\frac{d}{dx}(x+2))}{(x+2)^2} \\ &= \frac{(x+2) - (x+1)}{(x+2)^2} = \frac{1}{(x+2)^2}. \end{aligned}$$

$$\begin{aligned} \text{At } x = a = 0, \\ f(0) &= \frac{0+1}{0+2} = \frac{1}{2} \end{aligned}$$

$$f'(0) = \frac{1}{4}.$$

The line with slope $\frac{1}{4}$ and passing through the point $\left(0, \frac{1}{2}\right)$ has equation $y = \frac{1}{4}x + \frac{1}{2}$.

$$\begin{aligned} 20. \quad f(x) &= \frac{x+3}{x^2+1} \\ \text{By The Quotient Rule, we have} \\ f'(x) &= \frac{((x^2+1)\frac{d}{dx}(x+3)) - ((x+3)\frac{d}{dx}(x^2+1))}{(x^2+1)^2} \\ &= \frac{(x^2+1) - (x+3)(2x)}{(x^2+1)^2} \\ &= \frac{(x^2+1) - (2x^2+6x)}{(x^2+1)^2} \\ &= \frac{x^2+1-2x^2-6x}{(x^2+1)^2} \\ &= \frac{-x^2-6x+1}{(x^2+1)^2}. \end{aligned}$$

At $x = a = 1$,

$$f(1) = \frac{1+3}{1^2+1} = 2$$

$$f'(1) = \frac{-1-6+1}{(1+1)^2} = -\frac{6}{4} = -\frac{3}{2}.$$

The line with slope $-\frac{3}{2}$ and passing through the point $(1, 2)$ has equation $y = -\frac{3}{2}(x - 1) + 2$.

$$\begin{aligned} 21. \quad h(x) &= f(x)g(x) \\ h'(x) &= f'(x)g(x) + g'(x)f(x) \end{aligned}$$

(a) At $x = a = 0$,

$$h(0) = f(0)g(0) = (-1)(3) = -3$$

$$h'(0) = f'(0)g(0) + g'(0)f(0)$$

$$= (-1)(3) + (-1)(-1) = -2.$$

So, the equation of the tangent line is $y = -2x - 3$.

(b) At $x = a = 1$,

$$h(1) = f(1)g(1) = (-2)(1) = -2$$

$$h'(1) = f'(1)g(1) + g'(1)f(1)$$

$$= (3)(1) + (-2)(-2) = 7.$$

So, the equation of the tangent line is $y = 7(x - 1) - 2$ or $y = 7x - 9$.

$$\begin{aligned} 22. \quad h(x) &= \frac{f(x)}{g(x)} \\ h'(x) &= \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2} \end{aligned}$$

(a) At $x = a = 1$,

$$h(1) = \frac{f(1)}{g(1)} = -\frac{2}{1} = -2$$

$$\begin{aligned}
 h'(1) &= \frac{g(1)f'(1) - f(1)g'(1)}{(g(1))^2} \\
 &= \frac{(1)(3) - (-2)(-2)}{1^2} \\
 &= \frac{3 - 4}{1} = -1.
 \end{aligned}$$

So, the equation of the tangent line is $y = -(x - 1) - 2$.

(b) At $x = a = 0$,

$$\begin{aligned}
 h(0) &= \frac{f(0)}{g(0)} = -\frac{1}{3} \\
 h'(0) &= \frac{g(0)f'(0) - f(0)g'(0)}{(g(0))^2} \\
 &= \frac{(-1)(3) - (-1)(-1)}{(3)^2} \\
 &= \frac{-3 - 1}{9} \\
 &= -\frac{4}{9}.
 \end{aligned}$$

So, the equation of the tangent line is $y = -\frac{4}{9}x - \frac{1}{3}$.

23. $h(x) = x^2 f(x)$
 $h'(x) = 2xf(x) + x^2 f'(x)$

(a) At $x = a = 1$,

$$\begin{aligned}
 h(1) &= 1^2 f(1) = -2 \\
 h'(1) &= 2 \times 1 \times f(1) + 1^2 f'(1) \\
 &= (2)(-2) + (3) = -4 + 3 = -1.
 \end{aligned}$$

So, the equation of the tangent is $y = -1(x - 1) - 2$ or $y = -x - 1$.

(b) At $x = a = 0$,

$$\begin{aligned}
 h(0) &= 0^2 f(0) = 0 \\
 h'(0) &= 2 \times 0 \times f(0) + 0^2 f'(0) = 0.
 \end{aligned}$$

So, the equation of the tangent is $y = 0$.

24. $h(x) = \frac{x^2}{g(x)}$
 $h'(x) = \frac{2xg(x) - x^2 g'(x)}{(g(x))^2}$

(a) At $x = a = 1$,

$$\begin{aligned}
 h(1) &= \frac{1^2}{g(1)} = \frac{1}{1} = 1 \\
 h'(1) &= \frac{2 \times 1 \times g(1) - 1^2 g'(1)}{(g(1))^2} \\
 &= \frac{(2)(1)(1) - (1)(-2)}{1^2} \\
 &= \frac{2 + 2}{1} = 4.
 \end{aligned}$$

So, the equation of tangent line is $y = 4(x - 1) + 1$.

(b) At $x = a = 0$,

$$\begin{aligned}
 h(0) &= \frac{0^2}{g(0)} = \frac{0}{3} = 0 \\
 h'(0) &= \frac{2 \times 0 \times g(0) - 0^2 g'(0)}{(g(0))^2} = 0.
 \end{aligned}$$

So, the equation of the tangent line is $y = 0$.

25. The rate at which the quantity Q changes is Q' . Since the amount is said to be “decreasing at a rate of 4%” we have to ask “4% of what?” The answer in this type of context is usually 4% of itself. In other words, $Q' = -0.04Q$.

As for P , the 3% rate of increase would translate as $P' = 0.03P$. By the product rule with $R = PQ$, we have:

$$\begin{aligned}
 R' &= (PQ)' = P'Q + PQ' \\
 &= (0.03P)Q + P(-0.04Q) \\
 &= -(0.01)PQ = (-0.01)R.
 \end{aligned}$$

In other words, revenue is decreasing at a rate of 1%.

26. Revenue will be constant when the derivative is 0. Substituting, $Q' = -0.04Q$ and, $P' = aP$ into the expression for R' gives,

$$\begin{aligned}
 R' &= -0.04QP + aQP \\
 R' &= (-0.04 + a)QP.
 \end{aligned}$$

This is zero when $a = 0.04$, so price must increase by 4%.

27. $R' = Q'P + QP'$
 At a certain moment of time (call it t_0) we are given $P(t_0) = 20$ (\$/item), $Q(t_0) = 20,000$ (items)
 $P'(t_0) = 1.25$ (\$/item/year)
 $Q'(t_0) = 2,000$ (item/year)
 $R'(t_0) = 2,000(20) + (20,000)1.25$
 $R'(t_0) = 65,000$ (\$/year).
 So, revenue is increasing by \$65,000/year at the time t_0 .

28. We are given $P = \$14$, $Q = 12,000$ and $Q' = 1,200$. We want $R' = \$20,000$. Substituting these values in to the expression for R' (see exercise 25) yields:
 $20,000 = 1200 \cdot 14 + 12,000 \cdot P'$
 Solve to get $P' = 0.27$ dollars per year.

29. If $u(m) = \frac{82.5m - 6.75}{m + 0.15}$ then using the quotient rule,

$$\begin{aligned}
 \frac{du}{dm} &= \frac{(m + 0.15)(82.5) - (82.5m - 6.75)1}{(m + 0.15)^2} \\
 &= \frac{19.125}{(m + 0.15)^2}
 \end{aligned}$$

which is clearly positive. It seems to be saying that initial ball speed is an increasing function of the mass of the bat. Meanwhile,

$$u'(1) = \frac{19.125}{1.15^2} \approx 14.46$$

$$u'(1.2) = \frac{19.125}{1.35^2} \approx 10.49,$$

which suggests that the rate at which this speed is increasing is decreasing.

$$\begin{aligned} 30. \quad u'(M) &= \frac{(M + 1.05) \frac{d}{dM}(86.625 - 45M)}{(M + 1.05)^2} \\ &\quad - \frac{\frac{d}{dM}(M + 1.05)(86.625 - 45M)}{(M + 1.05)^2} \\ &= \frac{(-45M - 47.25) - (86.625 - 45M)}{(M + 1.05)^2} \\ &= \frac{-133.875}{(M + 1.05)^2} \end{aligned}$$

This quantity is negative. In baseball terms, as the mass of the baseball increases, the initial velocity decreases.

$$\begin{aligned} 31. \quad \text{If } u(m) &= \frac{14.11}{m + 0.05} = \frac{282.2}{20m + 1}, \text{ then} \\ \frac{du}{dm} &= \frac{(20m + 1) \cdot 0 - 282.2(20)}{(20m + 1)^2} \\ &= \frac{-5644}{(20m + 1)^2}. \end{aligned}$$

This is clearly negative, which means that impact speed of the ball is a decreasing function of the weight of the club. It appears that the explanation may have to do with the stated fact that the speed of the club is inversely proportional to its mass. Although the lesson of Example 4.6 was that a heavier club makes for greater ball velocity, that was assuming a fixed club speed, quite a different assumption from this problem.

$$32. \quad u'(v) = \frac{0.2822}{0.217} \approx 1.3. \text{ The initial speed of the ball increases 1.3 times more than the increase in club speed.}$$

$$\begin{aligned} 33. \quad \frac{d}{dx} [f(x)g(x)h(x)] &= \frac{d}{dx} [(f(x)g(x))h(x)] \\ &= (f(x)g(x))h'(x) + h(x)\frac{d}{dx}(f(x)g(x)) \\ &= (f(x)g(x))h'(x) \\ &\quad + h(x)(f(x)g'(x) + g(x)f'(x)) \\ &= f'(x)g(x)h(x) + f(x)g'(x)h(x) \\ &\quad + f(x)g(x)h'(x) \end{aligned}$$

In the general case of a product of n functions, the derivative will have n terms to be added, each term a product of all but one of

the functions multiplied by the derivative of the missing function.

$$\begin{aligned} 34. \quad \text{The derivative of } g(x)^{-1} &= \frac{1}{g(x)} \text{ is} \\ \frac{d}{dx} [g(x)^{-1}] &= \frac{g(x) \frac{d}{dx}(1) - (1) \frac{d}{dx}g(x)}{g(x)^2} \\ &= -\frac{g'(x)}{g(x)^2} = -g'(x)(g(x))^{-2} \end{aligned}$$

as claimed. The derivative of $f(x)(g(x))^{-1}$ is then $f'(x)(g(x))^{-1} + f(x)(-g'(x)(g(x))^{-2})$.

$$\begin{aligned} 35. \quad f'(x) &= \left[\frac{d}{dx}(x^{2/3}) \right] (x^2 - 2)(x^3 - x + 1) \\ &\quad + x^{2/3} \left[\frac{d}{dx}(x^2 - 2) \right] (x^3 - x + 1) \\ &\quad + x^{2/3}(x^2 - 2) \frac{d}{dx}(x^3 - x + 1) \\ &= \frac{2}{3}x^{-1/3}(x^2 - 2)(x^3 - x + 1) \\ &\quad + x^{2/3}(2x)(x^3 - x + 1) \\ &\quad + x^{2/3}(x^2 - 2)(3x^2 - 1) \end{aligned}$$

$$\begin{aligned} 36. \quad f'(x) &= 1(x^3 - 2x + 1)(3 - 2/x) \\ &\quad + (x + 4)(3x^2 - 2)(3 - 2/x) \\ &\quad + (x + 4)(x^3 - 2x + 1)(2/x^2) \end{aligned}$$

$$\begin{aligned} 37. \quad f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ f'(0) &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{hg(h) - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{hg(h)}{h} = \lim_{h \rightarrow 0} g(h) = g(0) \end{aligned}$$

Since, g is continuous at $x = 0$. When $g(x) = |x|$, $g(x)$ is continuous but not differentiable at $x = 0$. We have

$$f(x) = x|x| = \begin{cases} -x^2 & x < 0 \\ x^2 & x \geq 0 \end{cases}$$

This is differentiable at $x = 0$.

$$\begin{aligned} 38. \quad f(x) &= (x - a)g(x) \\ f'(x) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(a+h-a)g(a+h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{hg(a+h)}{h} \\ &= \lim_{h \rightarrow 0} g(a+h) \\ &= g(a) \end{aligned}$$

As g is continuous at $x = a$, hence $f(x)$ is differentiable.

$$\begin{aligned}
 39. \quad f(x) &= \frac{x}{x^2+1} \\
 f'(x) &= \frac{(x^2+1) - x(2x)}{(x^2+1)^2} \\
 &= \frac{x^2+1-2x^2}{(x^2+1)^2} = \frac{-x^2+1}{(x^2+1)^2} \\
 f''(x) &= \frac{(x^2+1)^2(-2x) - (-x^2+1)2(x^2+1)(2x)}{(x^2+1)^4} \\
 &= \frac{(x^2+1)(-2x) - (-x^2+1)(4x)}{(x^2+1)^3} \\
 &= \frac{-2x^3-2x+4x^3-4x}{(x^2+1)^3} = \frac{2x^3-6x}{(x^2+1)^3}
 \end{aligned}$$

At maxima or minima of f' , we have $f''(x) = 0$. So, $2x^3 - 6x = 0$

$$\begin{aligned}
 2x(x^2 - 3) &= 0 \\
 2x = 0, \quad x^2 - 3 &= 0 \\
 x = 0, \quad x &= \pm\sqrt{3}
 \end{aligned}$$

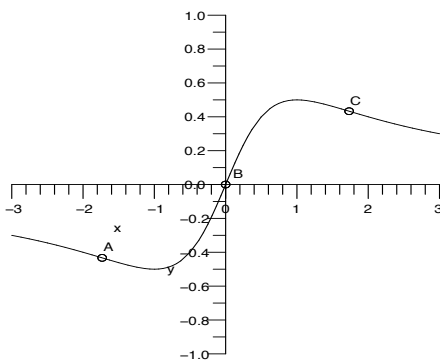
$$f'(0) = \frac{-0^2+1}{(0^2+1)^2} = 1$$

$$\begin{aligned}
 f'(\pm\sqrt{3}) &= \frac{-(\pm\sqrt{3})^2+1}{((\pm\sqrt{3})^2+1)^2} \\
 &= \frac{-3+1}{(3+1)^2} = -\frac{2}{16} = -\frac{1}{8}
 \end{aligned}$$

Therefore, $-\frac{1}{8} \leq m = f'(x) \leq 1$.

So, the function f has maximum slope $m = 1$ at $x = 0$ and minimum slope $m = -\frac{1}{8}$ at $x = \pm\sqrt{3}$.

In the graph of $f(x)$ in below, the point $B(0, 0)$ has maximum slope 1 and the points $A(-\sqrt{3}, -\frac{\sqrt{3}}{4})$, $C(\sqrt{3}, \frac{\sqrt{3}}{4})$ have minimum slope $-\frac{1}{8}$.



$$40. \quad f(x) = \frac{x}{\sqrt{x^2+1}}$$

$$\begin{aligned}
 f'(x) &= \frac{(\sqrt{x^2+1}) - x\left(\frac{1}{2\sqrt{x^2+1}} \times 2x\right)}{(x^2+1)} \\
 &= \frac{(\sqrt{x^2+1}) - \frac{x^2}{\sqrt{x^2+1}}}{(x^2+1)} \\
 &= \frac{x^2+1-x^2}{(x^2+1)^{\frac{3}{2}}} = (x^2+1)^{-\frac{3}{2}}
 \end{aligned}$$

Since $x^2+1 > 0$, $m > 0$.

$$f''(x) = -\frac{3}{2}(x^2+1)^{-\frac{5}{2}}(2x)$$

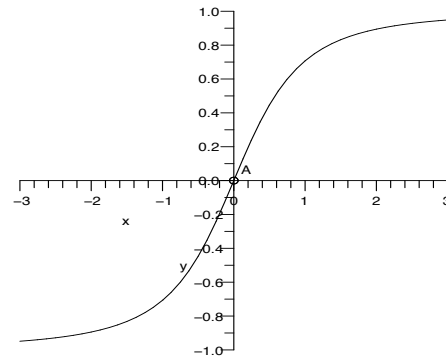
$$= -3x(x^2+1)^{-\frac{5}{2}} = -3x$$

For maxima or minima of $f''(x)$, we have $f''(x) = 0$. So, $x = 0$

$$f'(0) = (0^2+1)^{-\frac{3}{2}} = 1$$

Therefore $0 < m = f'(x) \leq 1$

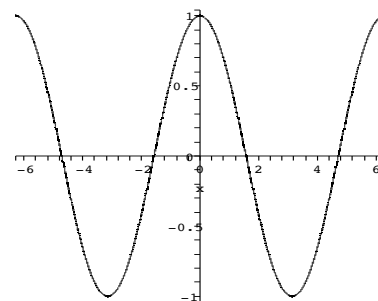
In the graph of $f(x)$ in below, the point $A(0, 0)$ has maximum slope 1.



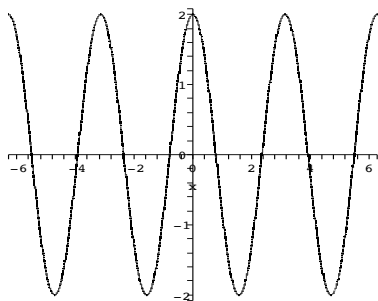
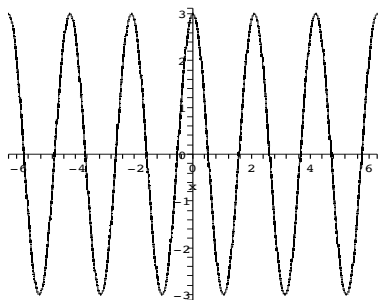
41. Answers depend on CAS.

42. For any constant k , the derivative of $\sin kx$ is $k \cos kx$.

Graph of $\frac{d}{dx} \sin x$:



Graph of $\frac{d}{dx} \sin 2x$

Graph of $\frac{d}{dx} \sin 3x$ 

43. CAS answers may vary.

44. The function $f(x)$ simplifies to $f(x) = 2x$, so $f'(x) = 2$. CAS answers vary, but should simplify to 2.

45. If $F(x) = f(x)g(x)$, then
 $F'(x) = f'(x)g(x) + f(x)g'(x)$ and
 $F''(x) = f''(x)g(x) + f'(x)g'(x)$
 $\quad + f'(x)g'(x) + f(x)g''(x)$
 $F'''(x) = f'''(x)g(x) + f''(x)g'(x)$
 $\quad + 2f''(x)g'(x) + 2f'(x)g''(x)$
 $\quad + f'(x)g''(x) + f(x)g'''(x)$
 $= f'''(x)g(x) + 3f''(x)g'(x)$
 $\quad + 3f'(x)g''(x) + f(x)g'''(x).$

One can see obvious parallels to the binomial coefficients as they come from Pascal's Triangle:

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3.$$

On this basis, one could correctly predict the pattern of the fourth or any higher derivative.

46. $F^{(4)}(x) = f^{(4)}g + 4f'''g' + 6f''g'' + 4f'g''' + f g^{(4)}$

47. If $g(x) = [f(x)]^2 = f(x)f(x)$, then
 $g'(x) = f'(x)f(x) + f(x)f'(x) = 2f(x)f'(x).$

48. $g(x) = f(x)[f(x)]^2$, so
 $g'(x) = f'(x)[f(x)]^2 + f(x)(2f(x)f'(x))$
 $= 3[f(x)]^2 f'(x)$

The derivative of $[f(x)]^n$ is $n[f(x)]^{n-1} f'(x)$.

49. $\lim_{x \rightarrow 0} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = 1$. Without any activator there is no enzyme. With unlimited amount of activator, the amount of enzyme approaches 1.

$$f(x) = \frac{x^{2.7}}{1 + x^{2.7}}$$

$$f'(x) = \frac{(1 + x^{2.7})(2.7)x^{1.7} - (2.7)x^{2.7}x^{1.7}}{(1 + x^{2.7})^2}$$

$$= \frac{2.7x^{1.7}}{(1 + x^{2.7})^2}$$

The fact that $0 < f(x) < 1$ when $x > 0$ suggest to us that f may be a kind of concentration ratio or percentage of presence of the allosteric enzymes in some systems. If so, the derivative would be interpreted as rate of change of concentration per unit activator.

50. $\lim_{x \rightarrow 0} f(x) = 1$ and $\lim_{x \rightarrow \infty} f(x) = 0$. Without any inhibitor the amount of enzyme approaches 1. With unlimited amount of inhibitor, the amount of enzyme approaches

$$0. \quad f'(x) = -\frac{2.7x^{1.7}}{(1 + x^{2.7})^2}$$

For positive x , f' is negative. Increase in the amount of inhibitor leads to a decrease in the amount of enzyme.

$$51. \text{ (a) } r = \frac{1}{\frac{0.55}{c} + \frac{0.45}{h}} = \left[\frac{0.55}{c} + \frac{0.45}{h} \right]^{-1}$$

$$\frac{d}{dc}(r) = \frac{d}{dc} \left[\frac{0.55}{c} + \frac{0.45}{h} \right]^{-1}$$

$$= \frac{-1}{\left[\frac{0.55}{c} + \frac{0.45}{h} \right]^2} \frac{d}{dc} \left[\frac{0.55}{c} + \frac{0.45}{h} \right]$$

$$= \frac{0.55}{c^2 \left[\frac{0.55}{c} + \frac{0.45}{h} \right]^2}$$

Therefore, from the above equation we can say that $\frac{dr}{dc} > 0$, for every c .

$$\text{(b) Similarly, } \frac{dr}{dh} = \frac{0.45}{h^2 \left[\frac{0.55}{c} + \frac{0.45}{h} \right]^2}.$$

Hence, from the above equation we can say that $\frac{dr}{dh} > 0$, for every h .

$$\text{(c) } r = \frac{1}{\frac{0.55}{c} + \frac{0.45}{h}}$$

When $c = h$, we get

$$r = \frac{1}{\frac{0.55}{h} + \frac{0.45}{h}} = h = c.$$

(d) If $c < h$

$$r = \frac{ch}{0.55h + 0.45c}$$

$$\frac{r}{c} = \frac{h}{0.55h + 0.45c} > \frac{h}{0.55h + 0.45h} = 1$$

So, $r > c$. And

$$r = \frac{ch}{0.55h + 0.45c}$$

$$\frac{r}{h} = \frac{c}{0.55h + 0.45c} < \frac{c}{0.55c + 0.45c} = 1$$

So, $r < h$ and hence $c < r < h$.

Now, r is an increasing function and $h = c$, we have $r = f(h) = c$. Hence for any value of h greater than c , we have the corresponding value of r greater than c .

$$\begin{aligned} \frac{dr}{dh} &= \frac{0.55}{c^2 \left[\frac{0.55}{c} + \frac{0.45}{h} \right]^2} \\ &= \frac{0.45c^2}{(0.55h + 0.45c)^2} < \frac{0.45c^2}{c^2} = 0.45 \end{aligned}$$

Also, from part (b),

$$\frac{dr}{dh} = \frac{0.45c^2}{(0.55h + 0.45c)^2}$$

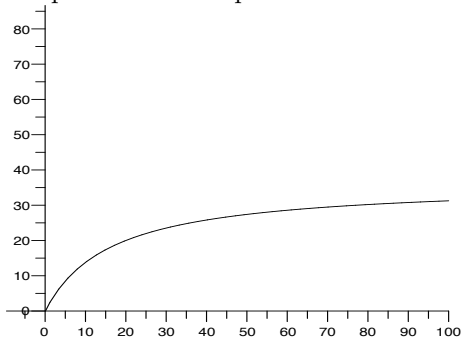
and from part (d),

$$\frac{r}{h} = \frac{c}{0.55h + 0.45c}$$

$$\Rightarrow 0.45 \left(\frac{r}{h} \right)^2 < 0.45$$

$$\Rightarrow r < h$$

Graph of r with respect to h when $c = 20$:



When c is constant, r remain stable for large h .

2.5 The Chain Rule

1. $f(x) = (x^3 - 1)^2$

Using the chain rule:

$$f'(x) = 2(x^3 - 1)(3x^2) = 6x^2(x^3 - 1)$$

Using the product rule:

$$f(x) = (x^3 - 1)(x^3 - 1)$$

$$f'(x) = (3x^2)(x^3 - 1) + (x^3 - 1)(3x^2)$$

$$= 2(3x^2)(x^3 - 1)$$

$$= 6x^2(x^3 - 1)$$

Using preliminary multiplication:

$$f(x) = x^6 - 2x^3 + 1$$

$$f'(x) = 6x^5 - 6x^2$$

$$= 6x^2(x^3 - 1)$$

2. $f(x) = (x^2 + 2x + 1)(x^2 + 2x + 1)$

Using the product rule:

$$f'(x) = (2x + 2)(x^2 + 2x + 1)$$

$$+ (x^2 + 2x + 1)(2x + 2)$$

Using the chain rule:

$$f'(x) = 2(x^2 + 2x + 1)(2x + 2)$$

3. $f(x) = (x^2 + 1)^3$

Using the chain rule:

$$f'(x) = 3(x^2 + 1)^2 \cdot 2x$$

Using preliminary multiplication:

$$f(x) = x^6 + 3x^4 + 3x^2 + 1$$

$$f'(x) = 6x^5 + 12x^3 + 6x$$

4. $f(x) = (2x + 1)^4$

Using preliminary multiplication:

$$f(x) = 16x^4 + 32x^3 + 24x^2 + 8x + 1$$

$$f'(x) = 64x^3 + 96x^2 + 48x + 8.$$

Using the chain rule:

$$f'(x) = 4(2x + 1)^3(2) = 8(2x + 1)^3$$

5. (a) By the chain rule:

$$f'(x) = 3(x^3 - x)^2 \frac{d}{dx} (x^3 - x)$$

$$= 3(x^3 - x)^2 (3x^2 - 1)$$

(b) By the chain rule:

$$f'(x) = \frac{1}{2\sqrt{x^2 + 4}} \frac{d}{dx} (x^2 + 4)$$

$$= \frac{1}{2\sqrt{x^2 + 4}} \cdot 2x = \frac{x}{\sqrt{x^2 + 4}}$$

6. (a) By the chain rule:

$$f'(x) = 4(x^3 + x - 1)^3 \frac{d}{dx} (x^3 + x - 1)$$

$$= 4(x^3 + x - 1)^3 (3x^2 + 1)$$

(b) By the chain rule:

$$\begin{aligned}
 f'(x) &= \frac{1}{2\sqrt{4x - \frac{1}{x}}} \frac{d}{dx} \left(4x - \frac{1}{x} \right) \\
 &= \frac{1}{2\sqrt{4x - \frac{1}{x}}} \left(4 + \frac{1}{x^2} \right) \\
 &= \frac{4 + \frac{1}{x^2}}{2\sqrt{4x - \frac{1}{x}}}
 \end{aligned}$$

7. (a) $f(t) = t^5 \sqrt{t^3 + 2}$
 By the product rule:

$$f'(t) = 5t^4 \sqrt{t^3 + 2} + t^5 \frac{d}{dt} (\sqrt{t^3 + 2})$$

 By the chain rule:

$$\begin{aligned}
 f'(t) &= 5t^4 \sqrt{t^3 + 2} + t^5 \frac{1}{2\sqrt{t^3 + 2}} \frac{d}{dt} (t^3 + 2) \\
 &= 5t^4 \sqrt{t^3 + 2} + t^5 \frac{1}{2\sqrt{t^3 + 2}} 3t^2 \\
 &= \frac{(5t^4 \sqrt{t^3 + 2})(2\sqrt{t^3 + 2}) + 3t^7}{2\sqrt{t^3 + 2}} \\
 &= \frac{10t^4 (t^3 + 2) + 3t^7}{2\sqrt{t^3 + 2}} \\
 &= \frac{10t^7 + 20t^4 + 3t^7}{2\sqrt{t^3 + 2}} \\
 &= \frac{13t^7 + 20t^4}{2\sqrt{t^3 + 2}}
 \end{aligned}$$

(b) $f(t) = (t^3 + 2) \sqrt{t}$
 By the product rule:

$$\begin{aligned}
 f'(t) &= 3t^2 \sqrt{t} + (t^3 + 2) \frac{1}{2\sqrt{t}} \\
 &= \frac{6t^3 + t^3 + 2}{2\sqrt{t}} = \frac{7t^3 + 2}{2\sqrt{t}}
 \end{aligned}$$

8. (a) $f(t) = (t^4 + 2) \sqrt{t^2 + 1}$
 By the product rule:

$$f'(t) = 4t^3 \sqrt{t^2 + 1} + (t^4 + 2) \frac{d}{dt} (\sqrt{t^2 + 1})$$

 By the chain rule:

$$\begin{aligned}
 f'(t) &= 4t^3 \sqrt{t^2 + 1} + (t^4 + 2) \frac{1}{2\sqrt{t^2 + 1}} (2t) \\
 &= 4t^3 \sqrt{t^2 + 1} + \frac{t(t^4 + 2)}{\sqrt{t^2 + 1}} \\
 &= \frac{4t^3 (t^2 + 1) + t(t^4 + 2)}{\sqrt{t^2 + 1}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{4t^5 + 4t^3 + t^5 + 2t}{\sqrt{t^2 + 1}} \\
 &= \frac{5t^5 + 4t^3 + 2t}{\sqrt{t^2 + 1}}
 \end{aligned}$$

(b) $f(t) = \sqrt{t} (t^{4/3} + 3)$
 By the product rule:

$$\begin{aligned}
 f'(t) &= \frac{1}{2\sqrt{t}} (t^{4/3} + 3) + \frac{4}{3} t^{1/3} \sqrt{t} \\
 &= \frac{1}{2\sqrt{t}} (t^{4/3} + 3) + \frac{4}{3} t^{1/3} t^{1/2} \\
 &= \frac{1}{2\sqrt{t}} (t^{4/3} + 3) + \frac{4}{3} t^{5/6}
 \end{aligned}$$

9. (a) $f(u) = \frac{u^2 + 1}{u + 4}$
 By the quotient rule:

$$\begin{aligned}
 f'(u) &= \frac{(u + 4)(2u) - (u^2 + 1)}{(u + 4)^2} \\
 &= \frac{2u^2 + 8u - u^2 - 1}{(u + 4)^2} \\
 &= \frac{u^2 + 8u - 1}{(u + 4)^2}
 \end{aligned}$$

(b) $f(u) = \frac{u^3}{(u^2 + 4)^2}$
 By the quotient rule:

$$f'(u) = \frac{(u^2 + 4)^2 (3u^2) - (u^3) \frac{d}{du} (u^2 + 4)^2}{(u^2 + 4)^4}$$

 By the chain rule:

$$\begin{aligned}
 f'(u) &= \frac{(u^2 + 4)^2 (3u^2) - 2u^3 (u^2 + 4) (2u)}{(u^2 + 4)^4} \\
 &= \frac{(u^2 + 4) [3u^2 (u^2 + 4) - 4u^4]}{(u^2 + 4)^4} \\
 &= \frac{3u^2 (u^2 + 4) - 4u^4}{(u^2 + 4)^3} \\
 &= \frac{3u^4 + 12u^2 - 4u^4}{(u^2 + 4)^3} \\
 &= \frac{12u^2 - u^4}{(u^2 + 4)^3} = \frac{u^2 (12 - u^2)}{(u^2 + 4)^3}
 \end{aligned}$$

10. (a) $f(x) = \frac{x^2 - 1}{x^2 + 1}$
 By the quotient rule:

$$\begin{aligned}
 f'(x) &= \frac{(x^2 + 1)(2x) - (x^2 - 1)(2x)}{(x^2 + 1)^2} \\
 &= \frac{(2x)(x^2 + 1 - x^2 + 1)}{(x^2 + 1)^2} \\
 &= \frac{4x}{(x^2 + 1)^2}
 \end{aligned}$$

$$(b) f(x) = \frac{x^2 + 4}{(x^3)^2}$$

By the quotient rule:

$$\begin{aligned} f'(x) &= \frac{x^6(2x) - (x^2 + 4)(6x^5)}{(x^6)^2} \\ &= \frac{2x^7 - 6x^7 - 24x^5}{x^{12}} \\ &= \frac{-4x^7 - 24x^5}{x^{12}} \\ &= -\frac{4x^5(x^2 + 6)}{x^{12}} \\ &= -\frac{4(x^2 + 6)}{x^7} \end{aligned}$$

$$11. (a) g(x) = \frac{x}{\sqrt{x^2 + 1}}$$

By the quotient rule:

$$g'(x) = \frac{\sqrt{x^2 + 1} - (x) \frac{d}{dx}(\sqrt{x^2 + 1})}{(x^2 + 1)}$$

By the chain rule:

$$\begin{aligned} g'(x) &= \frac{\sqrt{x^2 + 1} - (x) \left(\frac{1}{2\sqrt{x^2 + 1}} \right) (2x)}{(x^2 + 1)} \\ &= \frac{\sqrt{x^2 + 1} - \frac{x^2}{\sqrt{x^2 + 1}}}{(x^2 + 1)} \\ &= \frac{x^2 + 1 - x^2}{\sqrt{x^2 + 1}(x^2 + 1)} \\ &= \frac{1}{\sqrt{x^2 + 1}(x^2 + 1)} \\ &= \frac{1}{(x^2 + 1)^{3/2}} \end{aligned}$$

$$(b) g(x) = \sqrt{\frac{x}{x^2 + 1}}$$

By the chain rule:

$$g'(x) = \frac{1}{2\sqrt{\frac{x}{x^2 + 1}}} \frac{d}{dx} \left(\frac{x}{x^2 + 1} \right)$$

By the quotient rule:

$$\begin{aligned} g'(x) &= \frac{1}{2\sqrt{\frac{x}{x^2 + 1}}} \left(\frac{(x^2 + 1) - x(2x)}{(x^2 + 1)^2} \right) \\ &= \frac{1}{2\sqrt{\frac{x}{x^2 + 1}}} \left(\frac{x^2 + 1 - 2x^2}{(x^2 + 1)^2} \right) \\ &= \frac{1}{2\sqrt{x}} \left(\frac{1 - x^2}{(x^2 + 1)^{3/2}} \right) \\ &= \frac{1 - x^2}{2\sqrt{x}(x^2 + 1)^{3/2}} \end{aligned}$$

$$12. (a) g(x) = x^2\sqrt{x+1}$$

By the product rule:

$$g'(x) = 2x\sqrt{x+1} + (x^2) \frac{d}{dx}(\sqrt{x+1})$$

By the chain rule:

$$\begin{aligned} g'(x) &= 2x\sqrt{x+1} + (x^2) \frac{1}{2\sqrt{x+1}} \\ &= 2x\sqrt{x+1} + \frac{x^2}{2\sqrt{x+1}} \\ &= \frac{4x(x+1) + x^2}{2\sqrt{x+1}} \\ &= \frac{4x^2 + 4x + x^2}{2\sqrt{x+1}} \\ &= \frac{5x^2 + 4x}{2\sqrt{x+1}} \end{aligned}$$

$$(b) g(x) = \sqrt{(x^2 + 1)(\sqrt{x+1})^3}$$

By the chain rule:

$$g'(x) = \frac{\frac{d}{dx}[(x^2 + 1)(\sqrt{x+1})^3]}{2\sqrt{(x^2 + 1)(\sqrt{x+1})^3}}$$

By the product rule:

$$g'(x) = \frac{(2x(\sqrt{x+1})^3) + (x^2 + 1) \frac{d}{dx}(\sqrt{x+1})^3}{2\sqrt{(x^2 + 1)(\sqrt{x+1})^3}}$$

By the chain rule:

$$\begin{aligned} g'(x) &= \frac{1}{2\sqrt{(x^2 + 1)(\sqrt{x+1})^3}} \left(2x(\sqrt{x+1})^3 \right. \\ &\quad \left. + 3(x^2 + 1)(\sqrt{x+1})^2 \frac{1}{2\sqrt{x}} \right) \end{aligned}$$

$$13. (a) h(x) = 6(x^2 + 4)^{-1/2}$$

By the chain rule:

$$\begin{aligned} h'(x) &= 6 \times \left(\frac{-1}{2} \right) (x^2 + 4)^{-3/2} (2x) \\ &= \frac{-6x}{(x^2 + 4)^{3/2}} \end{aligned}$$

$$(b) h(x) = \frac{\sqrt{x^2 + 4}}{6}$$

By the chain rule:

$$\begin{aligned} h'(x) &= \frac{1}{6} \cdot \frac{1}{2\sqrt{x^2 + 4}} \frac{d}{dx}(x^2 + 4) \\ &= \frac{1}{6} \cdot \frac{1}{2\sqrt{x^2 + 4}} (2x) \\ &= \frac{x}{6\sqrt{x^2 + 4}} \end{aligned}$$

$$14. (a) h(t) = \frac{(t^3 + 4)^5}{8}$$

By the chain rule:

$$\begin{aligned}
 h'(t) &= \frac{5}{8}(t^3 + 4)^4 \frac{d}{dt}(t^3 + 4) \\
 &= \frac{5}{8}(t^3 + 4)^4 (3t^2) \\
 &= \frac{15t^2}{8}(t^3 + 4)^4
 \end{aligned}$$

(b) $h(t) = 8(t^3 + 4)^{-5}$

By the chain rule:

$$\begin{aligned}
 h'(t) &= 8 \times (-5)(t^3 + 4)^{-6} \frac{d}{dt}(t^3 + 4) \\
 &= -40(t^3 + 4)^{-6} (3t^2) \\
 &= -120t^2(t^3 + 4)^{-6}
 \end{aligned}$$

15. (a) $f(x) = (\sqrt{x^3 + 2} + 2x)^{-2}$

By the chain rule:

$$\begin{aligned}
 f'(x) &= -2(\sqrt{x^3 + 2} + 2x)^{-3} \frac{d}{dx}(\sqrt{x^3 + 2} + 2x) \\
 &= -2(\sqrt{x^3 + 2} + 2x)^{-3} \left(\frac{3x^2}{2\sqrt{x^3 + 2}} + 2 \right) \\
 &= \frac{-2}{(\sqrt{x^3 + 2} + 2x)^3} \left(\frac{3x^2 + 4\sqrt{x^3 + 2}}{2\sqrt{x^3 + 2}} \right) \\
 &= -\frac{3x^2 + 4(\sqrt{x^3 + 2})}{(\sqrt{x^3 + 2} + 2x)^3 \sqrt{x^3 + 2}}
 \end{aligned}$$

(b) $f(x) = \sqrt{x^3 + 2} + 2x^{-2}$

By the chain rule:

$$\begin{aligned}
 f'(x) &= \frac{1}{2\sqrt{x^3 + 2} + 2x^{-2}} \frac{d}{dx}(x^3 + 2 + 2x^{-2}) \\
 &= \frac{1}{2\sqrt{x^3 + 2} + 2x^{-2}} (3x^2 - 4x^{-3}) \\
 &= \frac{3x^2 - 4x^{-3}}{2\sqrt{x^3 + 2} + 2x^{-2}}
 \end{aligned}$$

16. (a) $f(x) = \sqrt{4x^2 + (8 - x^2)^2}$

By the chain rule:

$$\begin{aligned}
 f'(x) &= \frac{8x - 4x(8 - x^2)}{2\sqrt{4x^2 + (8 - x^2)^2}} \\
 &= \frac{8x - 32x + 4x^3}{2\sqrt{4x^2 + (8 - x^2)^2}} \\
 &= \frac{-24x + 4x^3}{2\sqrt{4x^2 + (8 - x^2)^2}} \\
 &= \frac{2x^3 - 12x}{\sqrt{4x^2 + (8 - x^2)^2}}
 \end{aligned}$$

(b) $f(x) = (\sqrt{4x^2 + 8} - x^2)^2$

By the chain rule:

$$\begin{aligned}
 f'(x) &= 2(\sqrt{4x^2 + 8} - x^2) \frac{d}{dx}(\sqrt{4x^2 + 8} - x^2) \\
 &= 2(\sqrt{4x^2 + 8} - x^2) \left(\frac{4x}{\sqrt{4x^2 + 8}} - 2x \right)
 \end{aligned}$$

$$\begin{aligned}
 &= 2(\sqrt{4x^2 + 8} - x^2) \left(\frac{4x - 2x\sqrt{4x^2 + 8}}{\sqrt{4x^2 + 8}} \right) \\
 &= 4(\sqrt{4x^2 + 8} - x^2) \left(\frac{2x - x\sqrt{4x^2 + 8}}{\sqrt{4x^2 + 8}} \right)
 \end{aligned}$$

17. $f(x) = x^3 + 4x - 1$ is a one-to-one function with $f(0) = -1$ and $f'(0) = 4$. Therefore $g(-1) = 0$ and

$$g'(-1) = \frac{1}{f'(g(-1))} = \frac{1}{f'(0)} = \frac{1}{4}.$$

18. $f(x) = x^5 + 4x - 2$ is a one-to-one function with $f(0) = -2$ and $f'(0) = 4$. Therefore $g(-2) = 0$ and

$$g'(-2) = \frac{1}{f'(g(-2))} = \frac{1}{f'(0)} = \frac{1}{4}.$$

19. $f(x) = x^5 + 3x^3 + x$ is a one-to-one function with $f(1) = 5$ and $f'(1) = 5 + 9 + 1 = 15$. Therefore $g(5) = 1$ and

$$g'(5) = \frac{1}{f'(g(5))} = \frac{1}{f'(1)} = \frac{1}{15}.$$

20. $f(x) = x^3 + 2x + 1$ is a one-to-one function with $f(-1) = -2$ and $f'(-1) = 5$. Therefore $g(-2) = -1$ and

$$g'(-2) = \frac{1}{f'(g(-2))} = \frac{1}{f'(-1)} = \frac{1}{5}.$$

21. $f(x) = \sqrt{x^3 + 2x + 4}$ is a one-to-one function and $f(0) = 2$ so $g(2) = 0$. Meanwhile,

$$\begin{aligned}
 f'(x) &= \frac{1}{2\sqrt{x^3 + 2x + 4}} (3x^2 + 2) \\
 f'(0) &= 1/2 \\
 g'(2) &= \frac{1}{f'(g(2))} = \frac{1}{f'(0)} = 2.
 \end{aligned}$$

22. $f(x) = \sqrt{x^5 + 4x^3 + 3x + 1}$ is a one-to-one function and $f(1) = 3$ so $g(3) = 1$. Meanwhile,

$$\begin{aligned}
 f'(x) &= \frac{5x^4 + 12x^2 + 3}{2\sqrt{x^5 + 4x^3 + 3x + 1}} \\
 f'(1) &= \frac{20}{6} = \frac{10}{3} \\
 g'(3) &= \frac{1}{f'(g(3))} = \frac{1}{f'(1)} = \frac{3}{10}.
 \end{aligned}$$

23. $f(x) = \sqrt[3]{x\sqrt{x^4 + 2x}\sqrt[4]{\frac{8}{x+2}}}$

Use Chain rule to find the derivative of the function. We can also use Product rule.

$$24. f(x) = \frac{3x^2 + 2\sqrt{x^3 + \frac{4}{x^4}}}{(x^3 - 4)\sqrt{x+2}}$$

Use Quotient rule to find the derivative of the function. We can also use Chain rule and Product rule.

$$25. f(t) = \sqrt{t^2 + \frac{4}{t^3}} \left(\frac{8t+5}{2t-1} \right)^3$$

Use product rule to find the derivative of the function. We can also use chain rule and Quotient rule.

$$26. f(t) = \left(3t + \frac{4\sqrt{t^2+1}}{t-5} \right)^3$$

Use Chain rule to find the derivative of the function. We can also use Quotient rule.

$$27. f(x) = \sqrt{x^2 + 16}, \quad a = 3, \quad f(3) = 5$$

$$f'(x) = \frac{1}{2\sqrt{x^2+16}}(2x) = \frac{x}{\sqrt{x^2+16}}$$

$$f'(3) = \frac{3}{\sqrt{3^2+16}} = \frac{3}{5}$$

So, the tangent line is $y = \frac{3}{5}(x-3) + 5$ or

$$y = \frac{3}{5}x + \frac{16}{5}.$$

$$28. f(-2) = \frac{3}{4}$$

$$f'(x) = \frac{-12x}{(x^2+4)^2}$$

$$f'(-2) = \frac{24}{64} = \frac{3}{8}$$

The equation of the tangent line is

$$y = \frac{3}{8}(x+2) + \frac{3}{4}.$$

$$29. s(t) = \sqrt{t^2 + 8}$$

$$v(t) = s'(t) = \frac{2t}{2\sqrt{t^2+8}} = \frac{t}{\sqrt{t^2+8}} \text{ m/s}$$

$$v(2) = \frac{2}{\sqrt{12}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \text{ m/s}$$

$$30. s(t) = \frac{60t}{\sqrt{t^2+1}}$$

$$v(t) = \frac{\sqrt{t^2+1}(60) - 60t \cdot \frac{1}{2\sqrt{t^2+1}} \cdot 2t}{t^2+1} \text{ m/s}$$

$$v(2) = \frac{60\sqrt{5} - \frac{240}{\sqrt{5}}}{5} = \frac{12\sqrt{5}}{5} \text{ m/s}$$

$$31. h'(x) = f'(g(x))g'(x)$$

$$h'(1) = f'(g(1))g'(1) = f'(2) \cdot (-2) = -6$$

$$32. h'(x) = f'(g(x))g'(x)$$

$$h'(2) = f'(g(2))g'(2) = f'(3) \cdot (4) = -12$$

33. As a temporary device given any f , set $g(x) = f(-x)$. Then by the chain rule,

$$g'(x) = f'(-x)(-1) = -f'(-x).$$

In the even case ($g = f$) this reads $f'(-x) = -f'(x)$ and shows f' is odd.

In the odd case ($g = -f$ and therefore $g' = -f'$), this reads $-f'(x) = -f'(-x)$ or $f'(x) = f'(-x)$ and shows f' is even.

34. To say that $f(x)$ is symmetric about the line $x = a$ is the same as saying that $f(a+x) = f(a-x)$. Taking derivatives (using the chain rule), we have

$$\frac{d}{dx}f(a+x) = f'(a+x)$$

$$\frac{d}{dx}f(a-x) = f'(a-x)(-1) = -f'(a-x).$$

Thus, $f'(a+x) = -f'(a-x)$ and the graph of $f'(x)$ is symmetric through the point $(a, 0)$.

35. (a) Chain rule gives,

$$\begin{aligned} \frac{d}{dx}f(x^2) &= f'(x^2) \frac{d}{dx}(x^2) \\ &= f'(x^2)(2x) \\ &= 2xf'(x^2). \end{aligned}$$

(b) Chain rule gives,

$$\begin{aligned} \frac{d}{dx}[f(x)]^2 &= 2f(x) \frac{d}{dx}f(x) \\ &= 2f(x)f'(x). \end{aligned}$$

(c) Chain rule gives,

$$\begin{aligned} \frac{d}{dx}f(f(x)) &= f'(f(x)) \frac{d}{dx}f(x) \\ &= f'(f(x))f'(x). \end{aligned}$$

36. (a) Chain rule gives,

$$\begin{aligned} \frac{d}{dx}f(\sqrt{x}) &= f'(\sqrt{x}) \frac{d}{dx}(\sqrt{x}) \\ &= f'(\sqrt{x}) \frac{1}{2\sqrt{x}}. \end{aligned}$$

(b) Chain rule gives,

$$\begin{aligned} \frac{d}{dx}(\sqrt{f(x)}) &= \frac{1}{2\sqrt{f(x)}} \frac{d}{dx}f(x) \\ &= \frac{1}{2\sqrt{f(x)}} f'(x). \end{aligned}$$

(c) Chain rule gives,

$$\begin{aligned} \frac{d}{dx}[f(xf(x))] &= f'(xf(x)) \frac{d}{dx}(xf(x)) \\ &\text{and product rule gives} \\ &= f'(xf(x)) \left(f(x) + x \frac{d}{dx}f(x) \right) \\ &= f'(xf(x)) (f(x) + xf'(x)). \end{aligned}$$

37. (a) Chain rule gives,

$$\begin{aligned}\frac{d}{dx} \left[f \left(\frac{1}{x} \right) \right] &= f' \left(\frac{1}{x} \right) \cdot \frac{d}{dx} \left(\frac{1}{x} \right) \\ &= f' \left(\frac{1}{x} \right) \cdot \left(-\frac{1}{x^2} \right) \\ &= - \left(\frac{1}{x^2} \right) \cdot f' \left(\frac{1}{x} \right).\end{aligned}$$

(b) Chain rule gives,

$$\begin{aligned}\frac{d}{dx} \left(\frac{1}{f(x)} \right) &= \left(-\frac{1}{f(x)^2} \right) \cdot \frac{d}{dx} f(x) \\ &= \left(-\frac{1}{f(x)^2} \right) \cdot f'(x).\end{aligned}$$

(c) Chain rule gives,

$$\begin{aligned}\frac{d}{dx} \left[f \left(\frac{x}{f(x)} \right) \right] &= f' \left(\frac{x}{f(x)} \right) \frac{d}{dx} \left(\frac{x}{f(x)} \right) \\ \text{and quotient rule gives,} \\ &= f' \left(\frac{x}{f(x)} \right) \left(\frac{f(x) - x f'(x)}{[f(x)]^2} \right).\end{aligned}$$

38. (a) Chain rule gives,

$$\begin{aligned}\frac{d}{dx} (1 + f(x^2)) &= f'(x^2) \frac{d}{dx} (x^2) \\ &= f'(x^2) (2x) = 2x f'(x^2).\end{aligned}$$

(b) Chain rule gives,

$$\begin{aligned}\frac{d}{dx} [1 + f(x)]^2 &= 2 [1 + f(x)] \frac{d}{dx} (1 + f(x)) \\ &= 2 [1 + f(x)] f'(x) = 2 f'(x) [1 + f(x)].\end{aligned}$$

(c) Chain rule gives,

$$\begin{aligned}\frac{d}{dx} [f(1 + f(x))] &= f'(1 + f(x)) \cdot \frac{d}{dx} (1 + f(x)) \\ &= f'(1 + f(x)) f'(x) \\ &= f'(x) f'(1 + f(x)).\end{aligned}$$

39. $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$

(a) At $x = 0$: $g'(0) = 1$, $g(0) = 1$,

$$\begin{aligned}\frac{d}{dx} f(g(0)) &= f'(g(0)) g'(0) \\ &= f'(1) \cdot g'(0) = 3 \times 1 = 3\end{aligned}$$

(b) At $x = 1$: $g'(1)$ does not exist.

So $\frac{d}{dx} f(g(1))$ does not exist.

(c) At $x = 3$:

$$\begin{aligned}g'(3) &= 3, g(3) = 1 \\ \frac{d}{dx} f(g(3)) &= f'(g(3)) g'(3) \\ &= f'(1) \cdot g'(3) = 3 \times 3 = 9\end{aligned}$$

40. $\frac{d}{dx} g(f(x)) = g'(f(x)) f'(x)$

(a) At $x = 0$:

$f'(0)$ does not exist. So $\frac{d}{dx} g(f(0))$ does not exist.

(b) At $x = 1$:

$$\begin{aligned}f'(1) &= 3, f(1) = 0, \\ \frac{d}{dx} g(f(1)) &= g'(f(1)) f'(1) \\ &= g'(0) \cdot f'(1) = 1 \times 3 = 3\end{aligned}$$

(c) At $x = 3$:

$$\begin{aligned}f'(3) &= 0, f(3) = 3, \\ \frac{d}{dx} g(f(3)) &= g'(f(3)) f'(3) \\ &= g'(3) \cdot f'(3) = 3 \times 0 = 0\end{aligned}$$

41. (a) $f(x) = \sqrt{x^2 + 4}$

By the chain rule:

$$\begin{aligned}f'(x) &= \frac{1}{2\sqrt{x^2 + 4}} \frac{d}{dx} (x^2 + 4) \quad \text{By the} \\ &= \frac{2x}{2\sqrt{x^2 + 4}} \\ &= \frac{x}{\sqrt{x^2 + 4}}\end{aligned}$$

quotient rule:

$$\begin{aligned}f''(x) &= \frac{\sqrt{x^2 + 4} - x \frac{d}{dx} (\sqrt{x^2 + 4})}{(x^2 + 4)} \\ &= \frac{\sqrt{x^2 + 4} - x \frac{2x}{2\sqrt{x^2 + 4}}}{(x^2 + 4)} \\ &= \frac{\sqrt{x^2 + 4} - \frac{x^2}{\sqrt{x^2 + 4}}}{(x^2 + 4)} \\ &= \frac{x^2 + 4 - x^2}{\sqrt{x^2 + 4} (x^2 + 4)} \\ &= \frac{4}{(x^2 + 4)^{3/2}}\end{aligned}$$

(b) $f(t) = 2(t^2 + 4)^{-1/2}$

By the chain rule:

$$\begin{aligned}f'(t) &= 2 \cdot \frac{-1}{2} (t^2 + 4)^{-3/2} \frac{d}{dt} (t^2 + 4) \\ &= \frac{-1}{(t^2 + 4)^{3/2}} (2t) \\ &= \frac{-2t}{(t^2 + 4)^{3/2}}\end{aligned}$$

By the quotient rule:

$$\begin{aligned}
 f''(t) &= -2 \left[\frac{(t^2+4)^{3/2} - t \frac{d}{dt}(t^2+4)^{3/2}}{(t^2+4)^3} \right] \\
 &= -2 \left[\frac{(t^2+4)^{3/2} - t(\frac{3}{2})(t^2+4)^{1/2} \frac{d}{dt}(t^2+4)}{(t^2+4)^3} \right] \\
 &= -2 \left[\frac{(t^2+4)^{3/2} - t(\frac{3}{2})(t^2+4)^{1/2} 2t}{(t^2+4)^3} \right] \\
 &= -2 \left[\frac{(t^2+4)^{3/2} - 3t^2(t^2+4)^{1/2}}{(t^2+4)^3} \right] \\
 &= \left[\frac{-2(t^2+4)^{3/2} + 6t^2(t^2+4)^{1/2}}{(t^2+4)^3} \right] \\
 &= \left[\frac{(t^2+4)^{1/2} [-2(t^2+4) + 6t^2]}{(t^2+4)^3} \right] \\
 &= \left[\frac{-2t^2 - 8 + 6t^2}{(t^2+4)^{5/2}} \right] \\
 &= \frac{4t^2 - 8}{(t^2+4)^{5/2}} = \frac{4(t^2-2)}{(t^2+4)^{5/2}}
 \end{aligned}$$

42. (a) By the chain rule:

$$\begin{aligned}
 h'(t) &= 2(t^3+3) \frac{d}{dt}(t^3+3) \\
 &= 2(t^3+3)(3t^2) = 6t^5 + 18t^2 \\
 h''(t) &= 30t^4 + 36t
 \end{aligned}$$

(b) $g(s) = 3(s^2+1)^{-2}$

By the chain rule:

$$\begin{aligned}
 g'(s) &= 3(-2)(s^2+1)^{-3} \frac{d}{ds}(s^2+1) \\
 &= -6(s^2+1)^{-3}(2s) \\
 &= \frac{-12s}{(s^2+1)^3}
 \end{aligned}$$

By the product and chain rule:

$$\begin{aligned}
 g''(s) &= \frac{d}{ds}(-12s(s^2+1)^{-3}) \\
 &= -12 \left((s^2+1)^{-3} - 6s^2(s^2+1)^{-4} \right) \\
 &= -12(s^2+1)^{-4}(s^2+1-6s^2) \\
 &= -\frac{12(1-5s^2)}{(s^2+1)^4}
 \end{aligned}$$

43. (a) $f(x) = (x^3 - 3x^2 + 2x)^{1/3}$

$$\begin{aligned}
 f'(x) &= \frac{\frac{d}{dx}(x^3 - 3x^2 + 2x)}{3(x^3 - 3x^2 + 2x)^{2/3}} \\
 &= \frac{3x^2 - 6x + 2}{3(x^3 - 3x^2 + 2x)^{2/3}}
 \end{aligned}$$

The derivative of f does not exist at values of x for which

$$x^3 - 3x^2 + 2x = 0$$

$$x(x^2 - 3x + 2) = 0$$

$$x(x-1)(x-2) = 0.$$

Thus, the derivative of f does not exist for $x = 0, 1$, and 2 . The derivative fails to exist at these points because the tangent lines at these points are vertical.

$$\begin{aligned}
 \text{(b)} \quad f(x) &= \sqrt{x^4 - 3x^3 + 3x^2 - x} \\
 f'(x) &= \frac{\frac{d}{dx}(x^4 - 3x^3 + 3x^2 - x)}{2\sqrt{x^4 - 3x^3 + 3x^2 - x}} \\
 &= \frac{4x^3 - 9x^2 + 6x - 1}{2\sqrt{x^4 - 3x^3 + 3x^2 - x}}
 \end{aligned}$$

The derivative of f does not exist at values of x for which

$$x^4 - 3x^3 + 3x^2 - x = 0$$

$$x(x^3 - 3x^2 + 3x - 1) = 0$$

$$x(x-1)^3 = 0.$$

Thus, the derivative of x does not exist for $x = 0$ and 1 . The derivative fails to exist at these points because the tangent lines at these points are vertical.

44. Multiply numerator and denominator by $g(x+h) - g(x)$.

$$\lim_{h \rightarrow 0} \left(\frac{f(g(x+h)) - f(g(x))}{h} \right) \left(\frac{g(x+h) - g(x)}{g(x+h) - g(x)} \right)$$

The above step is not well documented and in this step we use the assumption that $g'(x) \neq 0$. Since $g'(x) \neq 0$ implies that $g(x+h) - g(x) \neq 0$ for $h \neq 0$.

45. $f(x) = (x^2 + 3)^2 \cdot 2x$

Recognizing the “ $2x$ ” as the derivative of $x^2 + 3$, we guess $g(x) = c(x^2 + 3)^3$ where c is some constant.

$$g'(x) = 3c(x^2 + 3)^2 \cdot 2x$$

which will be $f(x)$ only if $3c = 1$, so $c = 1/3$, and

$$g(x) = \frac{(x^2 + 3)^3}{3}.$$

46. A good initial guess is $(x^3 + 4)^{5/3}$, then adjust the constant to get

$$g(x) = \frac{1}{5}(x^3 + 4)^{5/3}.$$

47. $f(x) = \frac{x}{\sqrt{x^2 + 1}}$.

Recognizing the “ x ” as half the derivative of $x^2 + 1$, and knowing that differentiation throws the square root into the denominator, we guess $g(x) = c\sqrt{x^2 + 1}$ where c is some constant and find that

$$g'(x) = \frac{c}{2\sqrt{x^2 + 1}}(2x)$$

will match $f(x)$ if $c = 1$, so

$$g(x) = \sqrt{x^2 + 1}.$$

48. A good initial guess is $(x^2 + 1)^{-1}$, then adjust the constant to get

$$g(x) = -\frac{1}{2}(x^2 + 1)^{-1}.$$

2.6 Derivatives of Trigonometric Functions

$$\begin{aligned} 1. \quad f(x) &= 4 \sin 3x - x \\ f'(x) &= 4(\cos 3x)(3) - 1 \\ &= 12 \cos 3x - 1 \end{aligned}$$

$$\begin{aligned} 2. \quad f(x) &= 4x^2 - 3 \tan 2x \\ f'(x) &= 4(2x) - 3 \sec^2(2x)(2) \\ &= 8x - 6 \sec^2(2x) \end{aligned}$$

$$\begin{aligned} 3. \quad f(t) &= \tan^3 2t - \csc^4 3t \\ f'(t) &= 3 \tan^2(2t) \sec^2(2t)(2) \\ &\quad - 4 \csc^3(3t)[- \csc(3t) \cot(3t)](3) \\ &= 6 \tan^2(2t) \sec^2(2t) \\ &\quad + 12 \csc^4(3t) \cot(3t) \end{aligned}$$

$$\begin{aligned} 4. \quad f(t) &= t^2 + 2 \cos^2 4t \\ f'(t) &= 2t + 4 \cos(4t)[- \sin(4t)](4) \\ &= 2t - 16 \sin(4t) \cos(4t) \end{aligned}$$

$$\begin{aligned} 5. \quad f(x) &= x \cos 5x^2 \\ f'(x) &= (1) \cos 5x^2 + x(-\sin 5x^2) \cdot 10x \\ &= \cos 5x^2 - 10x^2 \sin 5x^2 \end{aligned}$$

$$\begin{aligned} 6. \quad f(x) &= x^2 \sec 4x \\ f'(x) &= x^2(\sec 4x \tan 4x)4 + (\sec 4x)2x \\ &= 4x^2(\sec 4x \tan 4x) + 2x \sec(4x) \end{aligned}$$

$$\begin{aligned} 7. \quad f(x) &= \frac{\sin(x^2)}{x^2} \\ f'(x) &= \frac{x^2 \cos(x^2) \cdot 2x - \sin(x^2) \cdot 2x}{x^4} \\ &= \frac{2x[x^2 \cos(x^2) - \sin(x^2)]}{x^4} \\ &= \frac{2[x^2 \cos(x^2) - \sin(x^2)]}{x^3} \end{aligned}$$

$$\begin{aligned} 8. \quad f(x) &= \frac{x^2}{\csc^4(2x)} \\ f'(x) &= \end{aligned}$$

$$\begin{aligned} &\frac{2x[\csc^4(2x)] - 4x^2[\csc^3(2x)][-\csc(2x) \cot(2x)](2)}{[\csc^4(2x)]^2} \\ &= \frac{2x}{\csc^4(2x)} + \frac{8x^2[\csc^4(2x) \cot(2x)]}{[\csc^4(2x)]^2} \\ &= \frac{2x}{\csc^4(2x)} + \frac{8x^2 \cot(2x)}{\csc^4(2x)} \\ &= \frac{2x + 8x^2 \cot(2x)}{\csc^4(2x)} \end{aligned}$$

$$\begin{aligned} 9. \quad f(t) &= \sin 3t \sec 3t = \tan 3t \\ f'(t) &= \frac{d}{dt} [\tan(3t)] = \sec^2(3t)(3) \\ &= 3 \sec^2(3t) \end{aligned}$$

$$\begin{aligned} 10. \quad f(t) &= \sqrt{\cos 5t \sec 5t} \\ &= \sqrt{\cos 5t \cdot \frac{1}{(\cos 5t)}} = 1 \\ f'(t) &= \frac{d}{dt}(1) = 0 \end{aligned}$$

$$\begin{aligned} 11. \quad f(w) &= \frac{1}{\sin 4w} \\ f'(w) &= \frac{-1}{(\sin 4w)^2} \cos 4w(4) \\ &= \frac{-4 \cos 4w}{\sin^2 4w} \end{aligned}$$

$$\begin{aligned} 12. \quad f(w) &= w^2 \sec^2 3w \\ f'(w) &= w^2(2 \sec 3w)(\sec 3w \tan 3w)(3) \\ &\quad + \sec^2(3w)(2w) \\ &= 6w^2 \sec^2 3w \tan 3w + 2w \sec^2 3w \end{aligned}$$

$$\begin{aligned} 13. \quad f(x) &= 2 \sin(2x) \cos(2x) \\ f'(x) &= 2\{\sin(2x)[- \sin(2x)](2) \\ &\quad + \cos(2x)[\cos(2x)](2)\} \\ &= -4 \sin^2(2x) + 4 \cos^2(2x) \\ &= 4 \cos^2(2x) - 4 \sin^2(2x) \end{aligned}$$

$$\begin{aligned} 14. \quad f(x) &= 4 \sin^2(3x) + 4 \cos^2(3x) \\ &= 4[\sin^2(3x) + \cos^2(3x)] = 4 \\ f'(x) &= \frac{d}{dx}(4) = 0 \end{aligned}$$

$$\begin{aligned} 15. \quad f(x) &= \tan \sqrt{x^2 + 1} \\ f'(x) &= (\sec^2 \sqrt{x^2 + 1}) \\ &\quad \cdot \left(\frac{1}{2}\right)(x^2 + 1)^{-1/2}(2x) \\ &= \frac{x}{\sqrt{x^2 + 1}} \sec^2 \sqrt{x^2 + 1} \end{aligned}$$

$$\begin{aligned} 16. \quad f(x) &= 4x^2 \sin x \sec 3x \\ f'(x) &= 8x \sin x \sec 3x + 4x^2[\cos x \sec 3x \\ &\quad + \sin x \sec 3x \tan 3x(3)] \end{aligned}$$

17. $f(x) = \sin^3(\cos \sqrt{x^3 + 2x^2})$
 $f'(x) = 3\sin^2(\cos \sqrt{x^3 + 2x^2})$
 $\cdot \cos(\cos \sqrt{x^3 + 2x^2})$
 $\cdot (-\sin \sqrt{x^3 + 2x^2})$
 $\cdot \frac{1}{2}(x^3 + 2x^2)^{-1/2}(3x^2 + 4x)$
 $= \frac{3}{2}(3x^2 + 4x)(x^3 + 2x^2)^{-1/2}$
 $\cdot \sin^2(\cos \sqrt{x^3 + 2x^2})$
 $\cdot \cos(\cos \sqrt{x^3 + 2x^2})$
 $\cdot (-\sin \sqrt{x^3 + 2x^2})$
18. $f(x) = \tan^4[\sin^2(x^3 + 2x)]$
 $f'(x) = 4[\tan^3(\sin^2(x^3 + 2x))]$
 $\cdot [\sec^2(\sin^2(x^3 + 2x))]$
 $\cdot [2\sin(x^3 + 2x)]$
 $\cdot [\cos(x^3 + 2x)] \cdot (3x^2 + 2)$
19. (a) $f(x) = \sin x^2$
 $f'(x) = \cos(x^2) \cdot (2x) = 2x \cos(x^2)$
 (b) $f(x) = \sin^2 x$
 $f'(x) = 2 \sin x \cos x$
 (c) $f(x) = \sin 2x$
 $f'(x) = \cos 2x (2) = 2 \cos 2x$
20. (a) $f(x) = \cos \sqrt{x}$
 $f'(x) = (-\sin \sqrt{x}) \cdot \frac{1}{2}(x)^{-1/2}$
 $= -\frac{1}{2}(x)^{-1/2} \sin \sqrt{x}$
 (b) $f(x) = \sqrt{\cos x}$
 $f'(x) = \frac{1}{2}(\cos x)^{-1/2} \cdot (-\sin x)$
 $= -\frac{1}{2} \sin x (\cos x)^{-1/2}$
 (c) $f(x) = \cos\left(\frac{1}{2}x\right)$
 $f'(x) = -\sin\left(\frac{1}{2}x\right) \cdot \left(\frac{1}{2}\right)$
 $= -\frac{1}{2} \sin\left(\frac{1}{2}x\right)$
21. (a) $f(x) = \sin x^2 \tan x$
 $f'(x) = \sin x^2 (\sec^2 x) + 2x \cos x^2 \tan x$
 (b) $f(x) = \sin^2(\tan x)$
 $f'(x) = 2 \sin(\tan x) \cdot \cos(\tan x) \cdot \sec^2 x$
 (c) $f(x) = \sin(\tan^2 x)$

- $f'(x) = [\cos(\tan^2 x)](2 \tan x)(\sec^2 x)$
 $= (2 \tan x)(\sec^2 x)[\cos(\tan^2 x)]$
22. (a) $f(x) = \sec x^2 \tan x^2$
 $f'(x) = \sec^3(x^2)(2x)$
 $+ \tan^2(x^2) \sec(x^2)(2x)$
 $= 2x \sec x^2 [\sec^2 x^2 + \tan^2 x^2]$
 (b) $f(x) = \sec^2(\tan x)$
 $f'(x) = 2 \sec(\tan x)[\sec(\tan x)$
 $\cdot \tan(\tan x)](\sec^2 x)$
 (c) $f(x) = \sec(\tan^2 x)$
 $f'(x) = [\sec(\tan^2 x) \tan(\tan^2 x)]$
 $\cdot (2 \tan x)(\sec^2 x)$
 $= (2 \tan x \sec^2 x)$
 $\cdot [\sec(\tan^2 x) \tan(\tan^2 x)]$
23. $f\left(\frac{\pi}{8}\right) = \sin \frac{\pi}{2} = 1$
 $f'(x) = 4 \cos 4x$
 $f'\left(\frac{\pi}{8}\right) = 4 \cos \frac{\pi}{2} = 0$
 So, the equation of the tangent line is
 $y = 0\left(x - \frac{\pi}{8}\right) + 1$ i.e. $y = 1$.
24. $f(0) = 0$
 $f'(x) = 3 \sec^2 3x,$
 $f'(0) = 3.$
 So, the equation of tangent line is $y = 3x$.
25. $f\left(\frac{\pi}{2}\right) = \left(\frac{\pi}{2}\right)^2 \cos\left(\frac{\pi}{2}\right) = 0$
 $f'(x) = x^2(-\sin x) + \cos x(2x)$
 $= -x^2(\sin x) + (2x) \cos x$
 $f'\left(\frac{\pi}{2}\right) = -\frac{\pi^2}{4} \sin \frac{\pi}{2} + 2 \cdot \frac{\pi}{2} \cos \frac{\pi}{2} = -\frac{\pi^2}{4}$
 So, the equation of the tangent line is
 $y = -\frac{\pi^2}{4}\left(x - \frac{\pi}{2}\right).$
26. $f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$
 $f'(x) = \sin x + x \cos x$, so $f'\left(\frac{\pi}{2}\right) = 1.$
 So, the equation of the tangent line is $y = x$.
27. $s(t) = t^2 - \sin(2t), t_0 = 0$
 $v(t) = s'(t) = 2t - 2 \cos(2t)$
 $v(0) = 0 - 2 \cos(0) = 0 - 2 = -2 \text{ ft/s}$
28. $s(t) = 4 + 3 \sin t, t_0 = \pi$
 $v(t) = s'(t) = 3 \cos t$
 $v(\pi) = -3 \text{ ft/s}$

$$\begin{aligned}
 29. \quad s(t) &= \frac{\cos t}{t}, \quad t_0 = \pi \\
 v(t) &= s'(t) = \frac{-1}{t^2} \cos t + \frac{1}{t}(-\sin t) \\
 v(\pi) &= -\frac{\cos \pi}{\pi^2} - \frac{\sin \pi}{\pi} \\
 &= \frac{1}{\pi^2} - \frac{1}{\pi}(0) = \frac{1}{\pi^2} \text{ ft/s}
 \end{aligned}$$

$$\begin{aligned}
 30. \quad s(t) &= t \cos(t^2 + \pi), \quad t_0 = 0 \\
 v(t) &= s'(t) = \cos(t^2 + \pi) - 2t^2 \sin(t^2 + \pi) \\
 v(0) &= \cos \pi - 0 = -1 \text{ ft/s}.
 \end{aligned}$$

31. (a) $f(t) = 4 \sin 3t$. The velocity at time t is $f'(t) = 12 \cos 3t$.
 (b) The maximum speed is 12.
 (c) The maximum speed of 12 occurs when the vertical position is zero.

32. (a) The velocity is $f'(t) = 12 \cos 3t$ which is 0 when $3t = \frac{k\pi}{2}$ or $t = \frac{k\pi}{6}$ for any odd integer k .
 (b) The location of the spring at these times is given (for any odd integer k) by $f(\frac{k\pi}{6}) = 4 \sin(3 \cdot \frac{k\pi}{6}) = 4 \sin(\frac{k\pi}{2}) = \pm 4$.
 (c) The spring changes directions at the top and bottom.

$$\begin{aligned}
 33. \quad (a) \quad \lim_{x \rightarrow 0} \frac{\sin 3x}{x} &= \lim_{x \rightarrow 0} \frac{3 \sin 3x}{3x} \\
 &= 3 \cdot \lim_{x \rightarrow 0} \frac{\sin(3x)}{(3x)} \\
 &= 3 \cdot 1 = 3
 \end{aligned}$$

$$(b) \quad \lim_{t \rightarrow 0} \frac{\sin t}{4t} = \frac{1}{4} \lim_{t \rightarrow 0} \frac{\sin t}{t} = \frac{1}{4} \cdot 1 = \frac{1}{4}$$

$$(c) \quad \lim_{x \rightarrow 0} \frac{\cos x - 1}{5x} = \frac{1}{5} \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$

$$(d) \quad \text{Let } u = x^2: \text{ then } u \rightarrow 0 \text{ as } x \rightarrow 0, \text{ and}$$

$$\lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = \lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$$

$$34. \quad (a) \quad \lim_{t \rightarrow 0} \frac{2t}{\sin t} = \lim_{t \rightarrow 0} \frac{2}{\frac{\sin t}{t}} = 2$$

$$(b) \quad \text{Let } u = x^2: \text{ then } u \rightarrow 0 \text{ as } x \rightarrow 0, \text{ and}$$

$$\lim_{x \rightarrow 0} \frac{\cos x^2 - 1}{x^2} = \lim_{u \rightarrow 0} \frac{\cos u - 1}{u} = 0$$

$$(c) \quad \lim_{x \rightarrow 0} \frac{\sin 6x}{\sin 5x} = \lim_{x \rightarrow 0} \frac{\frac{6 \sin 6x}{6x}}{\frac{5 \sin 5x}{5x}} = \frac{6}{5}$$

$$\begin{aligned}
 (d) \quad \lim_{x \rightarrow 0} \frac{\tan 2x}{x} &= \lim_{x \rightarrow 0} \frac{\frac{\sin 2x}{\cos 2x}}{x} \\
 &= \lim_{x \rightarrow 0} \frac{2 \sin 2x}{2x} \cdot \frac{1}{\cos 2x} = 2
 \end{aligned}$$

$$\begin{aligned}
 35. \quad f(x) &= \sin(2x) = 2^0 \sin(2x) \\
 f'(x) &= 2 \cos 2x = 2^1 \cos(2x)
 \end{aligned}$$

$$\begin{aligned}
 f''(x) &= -4 \sin 2x = -2^2 \sin(2x) \\
 f'''(x) &= -8 \cos 2x = -2^3 \cos(2x) \\
 f^{(4)}(x) &= 16 \sin 2x = 2^4 \sin(2x)
 \end{aligned}$$

$$\begin{aligned}
 f^{(75)}(x) &= \left(f^{(72)}\right)^{(3)}(x) \\
 &= \left(f^{(18 \cdot 4)}\right)^{(3)}(x) \\
 &= 2^{72} f'''(x) \\
 &= 2^{72} [-2^3 \cos(2x)] \\
 &= -2^{75} \cos(2x)
 \end{aligned}$$

$$\begin{aligned}
 f^{(150)}(x) &= \left(f^{(148)}\right)^{(2)}(x) \\
 &= \left(f^{(37 \cdot 4)}\right)^{(2)}(x) \\
 &= 2^{148} f''(x) \\
 &= 2^{148} [-2^2 \sin(2x)] \\
 &= -2^{150} \sin(2x)
 \end{aligned}$$

$$\begin{aligned}
 36. \quad f(x) &= \cos(3x) = 3^0 \cos(3x) \\
 f'(x) &= -3 \sin 3x = -3^1 \sin(3x) \\
 f''(x) &= -9 \cos 3x = -3^2 \cos(3x) \\
 f'''(x) &= 27 \sin 3x = 3^3 \sin(3x) \\
 f^{(4)}(x) &= 81 \cos 3x = 3^4 \cos(3x)
 \end{aligned}$$

$$\begin{aligned}
 f^{(77)}(x) &= \left(f^{(76)}\right)^{(1)}(x) \\
 &= \left(f^{(19 \cdot 4)}\right)^{(1)}(x) \\
 &= 3^{76} f'(x) \\
 &= 3^{76} [-3^1 \sin(3x)] \\
 &= -3^{77} \sin(3x)
 \end{aligned}$$

$$\begin{aligned}
 f^{(120)}(x) &= \left(f^{(120)}\right)(x) \\
 &= \left(f^{(30 \cdot 4)}\right)(x) \\
 &= 3^{120} \cos(3x)
 \end{aligned}$$

37. Since, $0 \leq \sin \theta \leq \theta$, we have $-\theta \leq -\sin \theta \leq 0$ which implies $-\theta \leq \sin(-\theta) \leq 0$, so for $-\frac{\pi}{2} \leq \theta \leq 0$ we have $\theta \leq \sin \theta \leq 0$. We also know that $\lim_{\theta \rightarrow 0^-} \theta = 0 = \lim_{\theta \rightarrow 0^-} \sin \theta$, so the Squeeze Theorem implies that $\lim_{\theta \rightarrow 0^-} \sin \theta = 0$.

38. Since $\cos^2 \theta + \sin^2 \theta = 1$, we have $\cos \theta = \sqrt{1 - \sin^2 \theta}$. Then $\lim_{\theta \rightarrow 0} \cos \theta = \lim_{\theta \rightarrow 0} \sqrt{1 - \sin^2 \theta} = \pm 1$. Since $\cos \theta$ is a continuous function and $\cos 0 = 1$, we conclude that $\lim_{\theta \rightarrow 0} \cos \theta = 1$.

39. If $f(x) = \cos(x)$, then

$$\begin{aligned} & \frac{f(x+h) - f(x)}{h} \\ &= \frac{\cos(x+h) - \cos(x)}{h} \\ &= \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \\ &= (\cos x) \frac{(\cos h - 1)}{h} - (\sin x) \left(\frac{\sin h}{h} \right). \end{aligned}$$

Taking the limit according to lemma 6.1

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= (\cos x) \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} \\ &\quad - (\sin x) \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= \cos x \cdot 0 - \sin x \cdot 1 = -\sin x \end{aligned}$$

$$\begin{aligned} 40. \quad \frac{d}{dx} \cot x &= \frac{d}{dx} \left(\frac{\cos x}{\sin x} \right) \\ &= \frac{\sin x(-\sin x) - \cos x \cos x}{\sin^2 x} \\ &= -\frac{1}{\sin^2 x} = -\csc^2 x \\ \frac{d}{dx} \sec x &= \frac{d}{dx} \left(\frac{1}{\cos x} \right) \\ &= \frac{\cos x \cdot 0 - 1(-\sin x)}{\cos^2 x} \\ &= \frac{\sin x}{\cos x} \left(\frac{1}{\cos x} \right) = \sec x \tan x. \\ \frac{d}{dx} \csc x &= \frac{d}{dx} \left(\frac{1}{\sin x} \right) = \frac{\sin x \cdot 0 - 1 \cos x}{\sin^2 x} \\ &= -\frac{1}{\sin x} \left(\frac{\cos x}{\sin x} \right) = -\csc x \cot x. \end{aligned}$$

41. Answers depend on CAS.

42. Answers depend on CAS.

43. Answers depend on CAS.

44. Answers depend on CAS.

45. (a) If $x \neq 0$, then f is continuous by Theorem 4.2 in Section 1.4, and f is differentiable by the Quotient rule (Theorem 4.2 in Section 2.4) Thus, we only need to check $x = 0$. To see that f is continuous at $x = 0$.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

(By Lemma 6.3)

Since $\lim_{x \rightarrow 0} f(x) = f(0)$, f is continuous at $x = 0$.

To see that f is differentiable at $x = 0$.

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{\frac{\sin x}{x} - 1}{x}$$

In the proof of Lemma 6.3, equation 6.8 was derived:

$$1 > \frac{\sin x}{x} > \cos x.$$

Thus, $0 > \frac{\sin x}{x} - 1 > \cos x - 1$ and therefore if $x > 0$,

$$0 > \frac{\frac{\sin x}{x} - 1}{x} > \frac{\cos x - 1}{x}$$

and if $x < 0$,

$$0 < \frac{\frac{\sin x}{x} - 1}{x} < \frac{\cos x - 1}{x}$$

By lemma 6.4, $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 1$.

By applying squeeze theorem to previous two inequalities, we obtain

$$\lim_{x \rightarrow 0} \frac{\frac{\sin x}{x} - 1}{x} = 1 \text{ so, } f'(0) = 0.$$

(b) From part(a) and quotient rule we have,

$$f'(x) = \begin{cases} 0 & x = 0 \\ \frac{x \cos x - \sin x}{x^2} & x \neq 0 \end{cases}$$

Thus to show that $f'(x)$ is continuous, we need only to show that

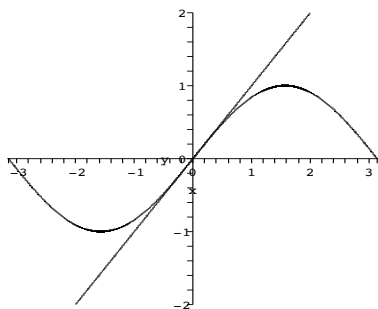
$$\lim_{x \rightarrow 0} f'(x) = f'(0) = 0.$$

$$\begin{aligned} \lim_{x \rightarrow 0} f'(x) &= \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{x \left(\cos x - \frac{\sin x}{x} \right)}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{\left(\cos x - \frac{\sin x}{x} \right)}{x} = 0 \end{aligned}$$

$$\text{Since, } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

46. We use the assumption that x is in radians in Lemma 6.3. The derivative of $\sin x^\circ = \sin\left(\frac{\pi}{180^\circ}x\right)$ is $\frac{\pi}{180^\circ} \cos(x^\circ)$. The factor of $\frac{\pi}{180^\circ}$ comes from applying the chain rule.

47. The Sketch: $y = x$ and $y = \sin x$



It is not possible visually to either detect or rule out intersections near $x = 0$ (other than zero itself).

We have that $f'(x) = \cos x$, which is less than 1 for $0 < x < 1$. If $\sin x \geq x$ for some x in the interval $(0, 1)$, then there would be a point on the graph of $y = \sin x$ which lies above the line $y = x$, but then (since $\sin x$ is continuous) the slope of the tangent line of $\sin x$ would have to be greater than 1 or equal to at some point in that interval, contradicting $f'(x) < 1$. Since $\sin x < x$ for $0 < x < 1$, we have for $-\sin x > -x$ for $0 < x < 1$. Then $-\sin x = \sin(-x)$ so $\sin(-x) > -x$ for $0 < x < 1$, which is the same as saying $\sin x > x$ for $-1 < x < 0$.

Since $-1 \leq \sin x \leq 1$, the only interval on which $y = \sin x$ might intersect $y = x$ is $[-1, 1]$. We know they intersect at $x = 0$ and we just showed that they do not intersect on the intervals $(-1, 0)$ and $(0, 1)$. So the only other points they might intersect are $x = \pm 1$, but we know that $\sin(\pm 1) \neq \pm 1$, so these graphs intersect only at $x = 0$.

48. $0 < k \leq 1$ produces one intersection. For $1 < k < 7.8$ (roughly) there are exactly three intersections. For $k \approx 7.8$ there are 5 intersections. For $k > 7.8$ there are 7 or more intersections.

2.7 Derivatives of Exponential and Logarithmic Functions

1. $f'(x) = 3x^2 \cdot e^x + x^3 \cdot e^x = x^2 e^x (x + 3)$
2. $f'(x) = 2e^{2x} \cos 4x + e^{2x} (-\sin 4x) 4$

3. $f(t) = t + 2^t$
 $f'(t) = 1 + 2^t \log 2$
4. $f(t) = t4^{3t}$
 $f'(t) = 4^{3t} + t4^{3t} (\ln 4) 3 = 4^{3t} (1 + 3t \ln 4)$
5. $f'(x) = 2e^{4x+1} (4) = 8e^{4x+1}$
6. $f'(x) = e^{-x}$, so $f'(x) = -e^{-x}$.
7. $h(x) = (\frac{1}{3})^{x^2}$
 $h'(x) = \ln(\frac{1}{3}) \cdot 2x \cdot (\frac{1}{3})^{x^2}$
 $= 2x \cdot \ln(\frac{1}{3}) \cdot (\frac{1}{3})^{x^2}$
 $= -2x \cdot \ln(3) \cdot (\frac{1}{3})^{x^2}$
8. $h(x) = 4^{-x^2}$
 $h'(x) = 4^{-x^2} \cdot \ln(4) \cdot (-2x)$
 $= -2x \cdot 4^{-x^2} \cdot \ln(4)$
9. $f(u) = e^{u^2+4u}$
 $f'(u) = e^{u^2+4u} \cdot \frac{d}{du}(u^2 + 4u)$
 $= e^{u^2+4u} \cdot (2u + 4)$
10. $f(x) = 3e^{\tan x}$
 $f'(x) = 3e^{\tan x} \cdot \frac{d}{dx}(\tan x)$
 $= 3e^{\tan x} \sec^2 x$
11. $f(w) = \frac{e^{4w}}{w}$
 $f'(w) = \frac{w \cdot 4e^{4w} - e^{4w} \cdot 1}{w^2}$
 $= \frac{e^{4w}(4w - 1)}{w^2}$
12. $f(w) = \frac{w}{e^{6w}}$
 $f'(w) = \frac{1 \cdot e^{6w} - w \cdot e^{6w} \cdot 6}{(e^{6w})^2}$
 $= \frac{e^{6w} - 6we^{6w}}{(e^{6w})^2} = \frac{(1 - 6w)}{e^{6w}}$
13. $f'(x) = \frac{1}{2x} \cdot (2) = \frac{1}{x}$
14. $f(x) = \frac{1}{2} \ln 8 + \frac{1}{2} \ln x$
 $f'(x) = \frac{1}{2x}$
15. $f(t) = \ln(t^3 + 3t)$
 $f'(t) = \frac{1}{t^3 + 3t} \cdot (3t + 3)$
 $= \frac{3t^2 + 3}{t^3 + 3t} = \frac{3(t^2 + 1)}{t(t^2 + 3)}$

16. $f(t) = t^3 \ln(t)$

$$f'(t) = 3t^2 \cdot \ln(t) + t^3 \cdot \frac{1}{t}$$

$$= 3t^2 \ln(t) + t^2$$

17. $g(x) = \ln(\cos x)$

$$g'(x) = \frac{1}{\cos x}(-\sin x) = -\tan x$$

18. $g(x) = \cos x \ln(x^2 + 1)$

$$g'(x) = \ln(x^2 + 1) \cdot (-\sin x) + \frac{2x \cos x}{x^2 + 1}$$

$$= \frac{2x \cos x}{x^2 + 1} - \sin x \cdot \ln(x^2 + 1)$$

19. (a) $f(x) = \sin(\ln x^2)$

$$f'(x) = \cos(\ln x^2) \cdot \frac{2x}{x^2}$$

$$= \frac{2 \cos(\ln x^2)}{x}$$

(b) $g(t) = \ln(\sin t^2)$

$$g'(t) = \frac{1}{\sin t^2} \cdot \cos t^2 \cdot 2t$$

$$= \frac{\cos t^2 \cdot 2t}{\sin t^2} = 2t \cot(t^2)$$

20. (a) $f(x) = \frac{\sqrt{\ln x}}{x}$

$$f'(x) = \frac{x \cdot \frac{1}{2}(\ln x)^{-\frac{1}{2}} \cdot \frac{1}{x} - (\ln x)^{\frac{1}{2}} \cdot 1}{x^2}$$

$$= \frac{\frac{1}{2\sqrt{\ln x}} - \sqrt{\ln x}}{x^2}$$

$$= \frac{1 - 2 \ln x}{2x^2 \sqrt{\ln x}}$$

(b) $g(t) = \frac{\ln \sqrt{t}}{t}$

$$g'(t) = \frac{t \cdot \frac{1}{2\sqrt{t}} \cdot t^{-\frac{1}{2}} - \ln \sqrt{t}}{t^2}$$

$$= \frac{\frac{1}{2} - \ln \sqrt{t}}{t^2}$$

$$= \frac{1 - 2 \cdot \ln \sqrt{t}}{2t^2}$$

21. (a) $h(x) = e^x \cdot \ln x$

$$h'(x) = e^x \cdot \frac{1}{x} + \ln x \cdot e^x$$

(b) $f(x) = e^{\ln x}$

$$f'(x) = e^{\ln x} \cdot \frac{1}{x}$$

22. (a) $h(x) = 2^{e^x}$

$$h'(x) = 2^{e^x} \cdot e^x \cdot \ln 2$$

(b) $f(x) = \frac{e^x}{2^x}$

$$f'(x) = \frac{2^x \cdot e^x - e^x \cdot 2^x \cdot \ln 2}{(2^x)^2}$$

$$= \frac{e^x(1 - \ln 2)}{2^x}$$

23. (a) $f(x) = \ln(\sin x)$

$$f'(x) = \frac{1}{\sin x} \cdot \cos x = \cot x$$

(b) $f(t) = \ln(\sec t + \tan t)$

$$f'(t) = \frac{\sec t \tan t + \sec^2 t}{\sec t + \tan t} = \sec t$$

24. (a) $f(x) = \sqrt[3]{e^{2x} \cdot x^3}$

$$f'(x) = \frac{1}{3}(e^{2x} \cdot x^3)^{-\frac{2}{3}} \cdot (3x^2 e^{2x} + 2x^3 e^{2x})$$

$$= \frac{x^2 \cdot e^{2x} \cdot (3 + 2x)}{3(e^{2x} \cdot x^3)^{\frac{2}{3}}}$$

(b) $f(w) = \sqrt[3]{e^{2w} + w^3}$

$$f'(w) = \frac{1}{3}(e^{2w} + w^3)^{-\frac{2}{3}} \cdot (2e^{2w} + 3w^2)$$

25. $f(x) = 3e^{x^2}$

$$f(1) = 3e^{1^2} = 3e$$

$$f'(x) = 3e^{x^2} 2x$$

$$f'(1) = 3e^{1^2} 2(1) = 6e$$

So, the equation of the tangent line is,

$$y = 6e(x - 1) + 3e.$$

26. $f(x) = 3^{x^e}$

$$f(1) = 3^{1^e} = 3$$

$$f'(x) = 3^{x^e} \ln 3 \cdot e x^{(e-1)}$$

$$f'(1) = 3 \ln 3 \cdot e$$

So, the equation of the tangent line is,

$$y = 3 \ln 3 \cdot e(x - 1) + 3.$$

27. $f(1) = 0$

$$f'(x) = 2x \ln x + x^2 \cdot \frac{1}{x} = 2x \ln x + x$$

$$f'(1) = 2 \cdot 1 \ln 1 + 1 = 2 \cdot 0 + 1 = 1$$

So the equation of tangent line is

$$y = 1(x - 1) + 0 \text{ or } y = x - 1.$$

28. $f(x) = 2 \ln x^3$

$$f'(x) = \frac{2}{x^3} \cdot 3x^2 = \frac{6}{x}$$

Slope = $f'(x)$ at $x = 1$.

$$\text{Slope } m = \frac{6}{1} = 6.$$

Equation of the line passing through (x_1, y_1) with slope m is $y - y_1 = m(x - x_1)$.

At $x_1 = 1$, $y_1 = f(1) = 2 \cdot \ln 1^3 = 0$.

Therefore equation is $y - 0 = 6 \cdot (x - 1)$ or $y = 6x - 6$.

29. (a) $f(x) = xe^{-2x}$

Given that, the tangent line to $f(x)$ is horizontal. Therefore slope is zero.

Hence,

$$f'(x) = e^{-2x} - 2xe^{-2x} = 0$$

$$e^{-2x}(1 - 2x) = 0$$

$$x = \frac{1}{2}.$$

(b) $f(x) = x \cdot e^{-3x}$

Given that, the tangent line to $f(x)$ is horizontal. Therefore slope is zero.

Hence,

$$f'(x) = x \cdot (-3e^{-3x}) + e^{-3x} = 0.$$

$$\Rightarrow e^{-3x}(-3x + 1) = 0$$

$$\Rightarrow 3x - 1 = 0$$

$$\Rightarrow x = \frac{1}{3}$$

30. (a) $f(x) = x^2 \cdot e^{-2x}$

Given that, the tangent line to $f(x)$ is horizontal. Therefore slope is zero.

Hence,

$$f'(x) = x^2 \cdot (-2e^{-2x}) + 2x \cdot e^{-2x} = 0$$

$$\Rightarrow -x + 1 = 0$$

$$\Rightarrow x = 1$$

(b) $f(x) = x^2 \cdot e^{-3x}$

Given that, the tangent line to $f(x)$ is horizontal. Therefore slope is zero.

Hence,

$$f'(x) = x^2 \cdot (-3e^{-3x}) + 2x \cdot e^{-3x} = 0$$

$$\Rightarrow -3x + 2 = 0$$

$$\Rightarrow x = \frac{2}{3}.$$

31. $v'(t) = 100.3^t \ln 3$

$$\frac{v'(t)}{v(t)} = \frac{100.3^t \ln 3}{100.3^t} = \ln 3 \approx 1.10$$

So, the percentage change is about 110%

32. $v'(t) = 1004^t (\ln 4)$

$$\frac{v'(t)}{v(t)} = \ln 4 \approx 1.3863$$

The instantaneous percentage rate of change is 138.6%

33. $v(t) = 40e^{0.4t}$

$$v'(t) = 40e^{0.4t} (0.4) = 16e^{0.4t}$$

$$\frac{v'(t)}{v(t)} = \frac{16e^{0.4t}}{40e^{0.4t}} = 0.4$$

The instantaneous percentage rate of change is 40%.

34. $v(t) = 60e^{-0.2t}$

$$v'(t) = 60e^{-0.2t} (-0.2) = -12e^{-0.2t}$$

$$\frac{v'(t)}{v(t)} = \frac{-12e^{-0.2t}}{60e^{-0.2t}} = -0.2$$

The instantaneous percentage rate of change is -20%.

35. $p(t) = 200.3^t$

$$\ln(p(t)) = \ln(200) + t \ln(3)$$

$$\frac{p'(t)}{p(t)} = \frac{d}{dt} [\ln(p(t))] = \ln 3 \approx 1.099,$$

so the rate of change of population is about 110% per unit of time.

36. The population after t days will be $p(t) = 500.2^{t/4}$. The rate of change is $p'(t) = 500.2^{t/4} (\ln 2) (1/4)$. So the relative rate of change is $\frac{\ln 2}{4} \approx 0.1733$. Therefore the percentage rate of change is about 17.3%.

37. $c(t) = \frac{6}{2e^{-8t} + 1} = 6(2e^{-8t} + 1)^{-1}$
 $c'(t) = -6(2e^{-8t} + 1)^{-2} \cdot (-16e^{-8t})$
 $= \frac{96e^{-8t}}{(2e^{-8t} + 1)^2}$

Since $e^{-8t} > 0$ for any t both numerator and denominator are positive, so that $c'(t) > 0$. Then, since $c(t)$ is an increasing function with a limiting value of 6 (as t goes to infinity) the concentration never exceeds (indeed, never reaches) the value of 6.

38. $c'(t) = -10(9e^{-10t} + 2)^{-2} (-90e^{-10t})$
 $= \frac{900e^{-10t}}{(9e^{-10t} + 2)^2}$

Since $e^{-10t} > 0$ for all t , $c'(t) > 0$ for all t , and $c(t)$ is increasing for all t . This forces, $c(t) < \lim_{t \rightarrow \infty} c(t) = 5$

39. $f(x) = x^{\sin x}$

$$\ln f(x) = \sin x \cdot \ln x$$

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} (\sin x \cdot \ln x)$$

$$= \cos x \cdot \ln x + \frac{\sin x}{x}$$

$$f'(x) = x^{\sin x} \left(\frac{x \cos x \cdot \ln x + \sin x}{x} \right)$$

40. $f(x) = x^{4-x^2}$

$$\ln f(x) = (4 - x^2) \ln x$$

$$\frac{f'(x)}{f(x)} = -2x \ln x + (4 - x^2) \frac{1}{x}$$

$$f'(x) = x^{4-x^2} \left(-2x \ln x + (4 - x^2) \frac{1}{x} \right)$$

41. $f(x) = (\sin x)^x$

$$\ln f(x) = x \cdot \ln(\sin x)$$

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} (x \cdot \ln(\sin x))$$

$$= \frac{x \cos x}{\sin x} + \ln(\sin x)$$

$$= x \cot x + \ln(\sin x)$$

$$f'(x) = (\sin x)^x \cdot (x \cot x + \ln(\sin x))$$

42. $f(x) = (x^2)^{4x}$

$$\ln f(x) = 8x \ln x$$

$$\frac{f'(x)}{f(x)} = 8 \ln x + 8x \frac{1}{x}$$

$$f'(x) = (x^2)^{4x} (8 \ln x + 8)$$

43. $f(x) = x^{\ln x}$

$$\ln f(x) = \ln x \cdot \ln x = \ln^2 x$$

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} (\ln^2 x) = \frac{2 \ln x}{x}$$

$$f'(x) = x^{\ln x} \left[\frac{2 \ln x}{x} \right] = 2x^{[(\ln x) - 1] \ln x}$$

44. $f(x) = x^{\sqrt{x}}$

$$\ln f(x) = \sqrt{x} \ln x$$

$$\frac{f'(x)}{f(x)} = \frac{1}{2\sqrt{x}} \ln x + \sqrt{x} \frac{1}{x}$$

$$f'(x) = x^{\sqrt{x}} \left(\frac{1}{2\sqrt{x}} \ln x + \frac{1}{\sqrt{x}} \right)$$

45. Let $(a, \ln a)$ be the point of intersection of the tangent line and the graph of $y = f(x)$.

$$f(x) = \ln x$$

$$f'(x) = \frac{1}{x}$$

$$m = f'(a) = \frac{1}{a}$$

Since the tangent line passes through the origin, the equation of the tangent line is

$$y = mx = \frac{1}{a}x.$$

Since $(a, \ln a)$ is a point on the tangent line

$$\ln a = \frac{1}{a}a = 1 \text{ so, } a = e.$$

Second part: Let (a, e^a) be the point of intersection of the tangent line and the graph of $y = f(x)$.

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$m = f'(a) = e^a$$

Since the tangent passes through the origin, the equation of the tangent line is

$$y = mx = e^a x.$$

Since (a, e^a) is a point on the tangent line,

$$e^a = e^a a$$

so, $a = 1$. The slope of the tangent line in $y = \ln x$ is $1/e$ while the slope of the tangent line in $y = e^x$ is e .

46. We approximate $\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$ for $a = 3$.

h	$\frac{a^h - 1}{h}$
0.01	1.10466919
0.001	1.09921598
0.0001	1.09867264
0.00001	1.09861832
-0.01	1.09259958
-0.001	1.09800903
-0.0001	1.09855194

The limit seems to be approaching approximately 1.0986, which is very close to $\ln 3 \approx 1.09861$

Second part: We approximate $\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$ for $a = \frac{1}{3}$.

h	$\frac{a^h - 1}{h}$
0.01	-1.09259958
0.001	-1.09800904
0.0001	-1.09855194
0.00001	-1.09860625
-0.01	-1.10466919
-0.001	-1.09921598
-0.0001	-1.09867264

The limit seems to be approaching approximately, -1.0986 , which is very close to $\ln \frac{1}{3} \approx 1.09861228867$

47. Answers depend on CAS.

48. Answers depend on CAS.

49. $f(x) = \frac{a + bx}{1 + cx}$

$$f(0) = a$$

$$f'(x) = \frac{b(1 + cx) - (a + bx)c}{(1 + cx)^2} = \frac{b - ac}{(1 + cx)^2}$$

$$f'(0) = b - ac$$

$$f''(x) = \frac{-2c(b - ac)}{(1 + cx)^3}$$

$$f''(0) = -2c(b - ac)$$

Now,

$$f(0) = 1 \Rightarrow a = 1.$$

$$f'(0) = 1 \Rightarrow b - ac = 1 \Rightarrow b - c = 1$$

$$f''(0) = 1 \Rightarrow -2c(b - ac) = 1$$

$$\Rightarrow 2c(b - c) = -1$$

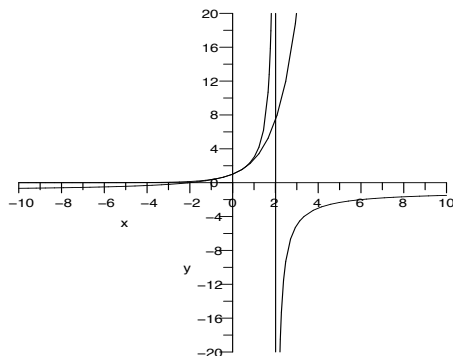
$$\Rightarrow 2c = -1$$

$$\Rightarrow c = -\frac{1}{2}$$

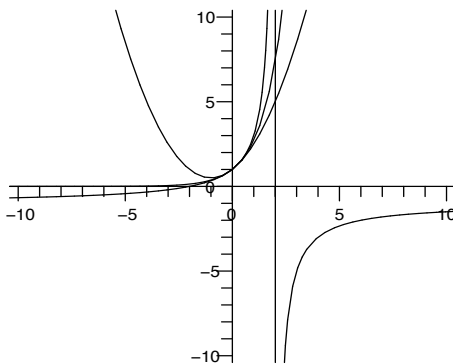
So, $a = 1$, $b = 1 + c = \frac{1}{2}$, $c = -\frac{1}{2}$ and

$$f(x) = \frac{2+x}{2-x}.$$

The graphs of e^x and $\frac{2+x}{2-x}$ are as follows:



50. If $g(x) = e^x$, then
 $g'(x) = e^x$ and $g''(x) = e^x$ so
 $g(0) = g'(0) = g''(0) = 1$
 If $f(x) = a + bx + cx^2$, then $f(0) = a$,
 $f'(x) = b + 2cx$
 $f'(0) = b$
 $f''(x) = 2c$
 $f''(0) = 2c$
 $1 = g(0) = f(0) = a$ so, $a = 1$
 $1 = g'(0) = f'(0) = b$ so, $b = 1$
 $1 = g''(0) = f''(0) = 2c$ so, $c = \frac{1}{2}$
 In summary, $a = 1$, $b = 1$, $c = \frac{1}{2}$ and
 $g(x) = 1 + x + \frac{1}{2}x^2$. The graphs of the
 functions e^x , $1 + x + \frac{1}{2}x^2$ and the Pade ap-
 proximation of e^x , which is $\frac{2+x}{2-x}$ are as
 follows:



51. $f(x) = e^{-x^2/2}$

$$f'(x) = e^{-x^2/2} \cdot (-2x/2)$$

$$= -xe^{-x^2/2}$$

$$f''(x) = -\left[x \left(-xe^{-x^2/2}\right) + 1 \cdot e^{-x^2/2}\right]$$

$$= xe^{-x^2/2} (x^2 - 1)$$

This will be zero only when $x = \pm 1$

52. $f(x) = e^{-x^2/8}$, $f'(x) = (-x/4)e^{-x^2/8}$

and

$$f''(x) = (-1/4)e^{-x^2/8} + (x^2/16)e^{-x^2/8}$$

$$= e^{-x^2/8} \left((-1/4) + x^2/16\right).$$

This is zero when $x = \pm 2$. The graph is flatter in the middle, but the tails are thicker.

53. It helps immensely to leave the name f as it was in #51 and give a new name g to the new function here, so that

$$g(x) = e^{-(x-m)^2/2c^2} = f(u)$$

in which $u = \frac{x-m}{c}$. Then

$$g'(x) = f'(u) \frac{du}{dx} = \frac{f'(u)}{c} = \frac{-uf(u)}{c}$$

$$= \frac{-(x-m)e^{-(x-m)^2/2c^2}}{c^2}$$

$$g''(x) = \frac{d}{dx} \left(\frac{f'(u)}{c} \right) = \frac{f''(u) \frac{du}{dx}}{c}$$

$$= \frac{f''(u)}{c^2} = \frac{(u^2 - 1)f(u)}{c^2}$$

$$= \frac{\left((x-m)^2 - c^2\right)e^{-(x-m)^2/2c^2}}{c^4}$$

This will be zero only when, $x = m \pm c$.

54. $f(x) = e^{-(x-m)^2/2c^2}$

$$f'(x) = \frac{-(x-m)}{c^2} e^{-(x-m)^2/2c^2},$$

and this is equal to zero when $x = m$.

55. $f(t) = e^{-t} \cos t$

$$v(t) = f'(t) = -e^{-t} \cos t + e^{-t} (-\sin t)$$

$$= -e^{-t} (\cos t + \sin t)$$

If the velocity is zero, it is because $\cos t = -\sin t$, so

$$t = \frac{3\pi}{4}, \frac{7\pi}{4}, \dots, \frac{(3+4n)\pi}{4}, \dots$$

Position when velocity is zero:

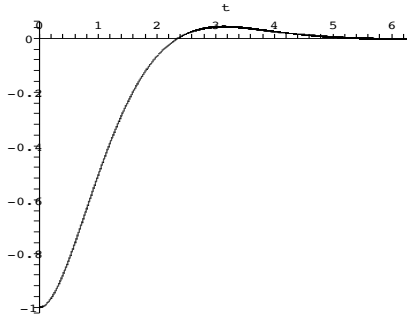
$$f(3\pi/4) = e^{-3\pi/4} \cos(3\pi/4)$$

$$= e^{-3\pi/4} \left(-1/\sqrt{2}\right) \approx -.067020$$

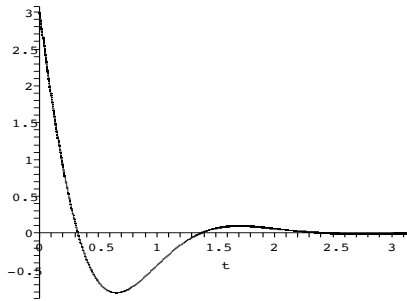
$$f(7\pi/4) = e^{-7\pi/4} \cos(7\pi/4)$$

$$= e^{-7\pi/4} \left(-1/\sqrt{2}\right) \approx .002896$$

Graph of the velocity function:



$$\begin{aligned} 56. \quad f'(t) &= -2e^{-2t} \sin 3t + 3e^{-2t} \cos 3t \\ &= e^{-2t} (-2 \sin 3t + 3 \cos 3t) \end{aligned}$$



The velocity of the spring is zero when it is changing direction at the top and bottom of the motion. This occurs when $3 \cos 3t = 2 \sin 3t$ or $\tan 3t = 3/2$. The *i.e.*, at $t = \frac{1}{3} \tan^{-1}(3/2) \approx 0.3276$ position of the spring at this time is approximate.

57. Graphically the maximum velocity seems to occur at $t = \pi$.
58. Graphically, the maximum velocity seems to occur at $t = 0$; the maximum velocity is not reached on $t > 0$.
59. Consider $f(x) = \frac{Ax^n}{\left(\frac{\theta}{x}\right)^n + 1}$ for $A, n, \theta > 0$

$$f(x) = \frac{A}{\left(\frac{\theta}{x}\right)^n + 1}$$

$$\ln f(x) = \ln A - \ln \left[\left(\frac{\theta}{x} \right)^n + 1 \right]$$

On differentiating with respect to x

$$\frac{1}{f(x)} f'(x) = - \frac{1}{\left[\left(\frac{\theta}{x} \right)^n + 1 \right]} \cdot n \left(\frac{\theta}{x} \right)^{n-1} \cdot \left(-\frac{\theta}{x^2} \right)$$

$$f'(x) = \frac{An}{\left[\left(\frac{\theta}{x} \right)^n + 1 \right]^2} \frac{\left(\frac{\theta}{x} \right)^n}{x}$$

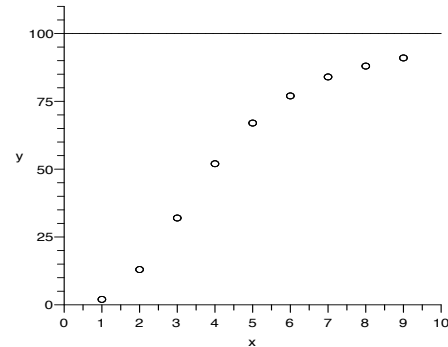
$f'(x) > 0$ if and only if $x > 0$ ($A, n, \theta > 0$)

$$\text{Also, } \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left[\frac{A}{\left(\frac{\theta}{x} \right)^n + 1} \right] = A$$

$$\begin{aligned} u &= \ln \left(\frac{f(x)/A}{1 - f(x)/A} \right) \\ &= \ln \left(\frac{\frac{1}{\left(\frac{\theta}{x} \right)^n + 1}}{1 - \frac{1}{\left(\frac{\theta}{x} \right)^n + 1}} \right) \\ &= \ln \left(\frac{1}{\left(\frac{\theta}{x} \right)^n} \right) \\ &= -n \ln \left(\frac{\theta}{x} \right) \\ &= -n (\ln \theta - \ln x) \\ &= -n \ln \theta + n \ln x \\ &= nv - n \ln \theta \end{aligned}$$

Therefore, u is a linear function of v .

Graph of (x, y) in below:

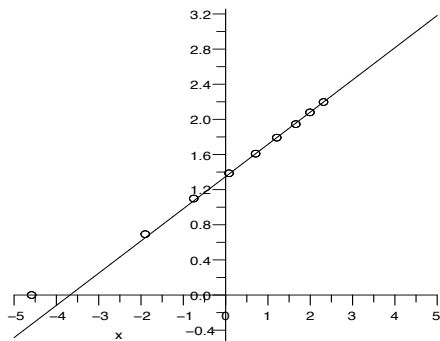


From the graph, we can see that $y = f(x) \rightarrow 100$ as $x \rightarrow \infty$.

The table gives (u, v) values as follows:

x	y	$u = \ln \frac{y}{100 - y}$	$v = \ln x$
1	2	-3.8918	0
2	13	-1.9009	0.6931
3	32	-.75377	1.098
4	52	.80012	1.3863
5	67	.70818	1.6094
6	77	1.2083	1.7918
7	84	1.6582	1.9459
8	88	1.9924	2.0794
9	91	2.3136	2.1972

The graph of (u, v) points are as below which are almost linear.



Comparing the line passing through the points (u, v) with $v = \frac{1}{n}u + \ln \theta$, we get $\frac{1}{n} = 0.3679$, $\ln \theta = 1.3458$ and hence $n = 2.7174$, $\theta = 3.8413$.

- 60.** Answers very depending on source. Linear growth corresponds to constant slope. In other words the population changes by the same fixed amount per year. In exponential growth, the size of the change depends on the size of the population. The percentage change is the same though from year to year.

2.8 Implicit Differentiation and Inverse Trigonometric Function

- 1.** Explicitly:

$$4y^2 = 8 - x^2$$

$$y^2 = \frac{8 - x^2}{4}$$

$$y = \pm \frac{\sqrt{8 - x^2}}{2} \text{ (choose plus to fit (2,1))}$$

$$\text{For } y = \frac{\sqrt{8 - x^2}}{2},$$

$$y' = \frac{1}{2} \frac{(-2x)}{2\sqrt{8 - x^2}} = \frac{-x}{2\sqrt{8 - x^2}},$$

$$y'(2) = \frac{-1}{2}.$$

Implicitly:

$$\frac{d}{dx}(x^2 + 4y^2) = \frac{d}{dx}(8)$$

$$2x + 8y \cdot y' = 0$$

$$y' = -\frac{2x}{8y} = -\frac{x}{4y}$$

$$\text{At } (2, 1) : y' = -\frac{2}{4} = -\frac{1}{2}$$

- 2.** Explicitly:

$$y = \frac{4\sqrt{x}}{x^3 - x^2}$$

$$y' = \frac{(x^3 - x^2) \frac{2}{\sqrt{x}} - 4\sqrt{x}(3x^2 - 2x)}{(x^3 - x^2)^2}$$

Implicitly differentiating:

$$3x^2y + x^3y' - \frac{2}{\sqrt{x}} = 2xy + x^2y',$$

And we solve for y' to get

$$y' = \frac{2xy + \frac{2}{\sqrt{x}} - 3x^2y}{x^3 - x^2}.$$

Substitute $x = 2$ into the first expression, and $(x, y) = (2, \sqrt{2})$, into the second to

$$\text{get } y' = -\frac{7\sqrt{2}}{4}.$$

- 3.** Explicitly:

$$y(1 - 3x^2) = \cos x$$

$$y = \frac{\cos x}{1 - 3x^2}$$

$$\begin{aligned} y'(x) &= \frac{(1 - 3x^2)(-\sin x) - \cos x(-6x)}{(1 - 3x^2)^2} \\ &= \frac{-\sin x + 3x^2 \sin x + 6x \cos x}{(1 - 3x^2)^2} \end{aligned}$$

$$y'(0) = 0.$$

Implicitly:

$$\frac{d}{dx}(y - 3x^2y) = \frac{d}{dx}(\cos x)$$

$$y' - 3x^2y' - 6xy = -\sin x$$

$$y'(1 - 3x^2) = 6xy - \sin x$$

$$y' = \frac{6xy - \sin x}{1 - 3x^2}$$

$$\text{At } (0, 1) : y' = 0(\text{again})$$

- 4.** Explicitly:

$$y = -x \pm \sqrt{x^2 - 4}$$

At the point $(-2, 2)$, the sign is irrelevant, so we choose

$$y = -x + \sqrt{x^2 - 4}$$

$$y' = -1 + \frac{2x}{2\sqrt{x^2 - 4}} = -1 + \frac{x}{\sqrt{x^2 - 4}}$$

Implicitly differentiating:

$$y' + 2y + 2xy' = 0,$$

and we solve for y' :

$$y' = \frac{-2y}{2x + 2y}$$

Substitute $x = -2$ in the first expression and $(x, y) = (-2, 2)$ in to the second expression to see that y' is undefined. There is a vertical tangent at this point.

- 5.** $\frac{d}{dx}(x^2y^2 + 3y) = \frac{d}{dx}(4x)$

$$2xy^2 + x^22yy' + 3y' = 4$$

$$y'(2x^2y + 3) = 4 - 2xy^2$$

$$y' = \frac{4 - 2xy^2}{2x^2y + 3}$$

- 6.** $3y^3 + 3x(3y^2)y' - 4 = 20yy'$

- $(9xy^2 - 20y)y' = 4 - 3y^3$
 $y' = \frac{3y^3 - 4}{20y - 9xy^2}$
- 7.** $\frac{d}{dx}(\sqrt{xy} - 4y^2) = \frac{d}{dx}(12)$
 $\frac{1}{2\sqrt{xy}} \cdot \frac{d}{dx}(xy) - 8y \cdot y' = 0$
 $\frac{1}{2\sqrt{xy}}(xy' + y) - 8y \cdot y' = 0$
 $(xy' + y) - 16y \cdot y' \sqrt{xy} = 0$
 $y'(x - 16y\sqrt{xy}) = -y$
 $y' = \frac{-y}{(x - 16y\sqrt{xy})} = \frac{y}{16y\sqrt{xy} - x}$
- 8.** $\cos(xy)(y + xy') = 2x$
 $y' = \frac{2x - y \cos(xy)}{x \cos(xy)}$
- 9.** $x + 3 = 4xy + y^3$
 $1 = \frac{d}{dx}(4xy + y^3) = 4(xy' + y) + 3y^2y'$
 $1 - 4y = (4x + 3y^2)y'$
 $y' = \frac{1 - 4y}{3y^2 + 4x}$
- 10.** $3x + y^3 - \frac{4y}{x+2} = 10x^2$
 Differentiating with respect to x ,
 $\frac{d}{dx}\left(3x + y^3 - \frac{4y}{x+2}\right) = \frac{d}{dx}(10x^2)$
 By the Chain rule and Product rule,
 $3 + 3y^2y' - \left[\frac{(x+2)4y' - 4y}{(x+2)^2}\right] = 20x$
 $3(x+2)^2 + 3y^2y'(x+2)^2 - 4y'(x+2) + 4y = 20x(x+2)^2$
 $3y^2y'(x+2)^2 - 4y'(x+2) = 20x(x+2)^2 - 3(x+2)^2 - 4y$
 $y'(x+2)[3y^2(x+2) - 4] = (x+2)^2(20x - 3) - 4y$
 $y' = \frac{(x+2)^2(20x - 3) - 4y}{(x+2)[3y^2(x+2) - 4]}$
- 11.** $\frac{d}{dx}(e^{x^2y} - e^y) = \frac{d}{dx}(x)$
 $e^{x^2y} \frac{d}{dx}(e^{x^2y}) - e^y y' = 1$
 $e^{x^2y}(2xy + x^2y') - e^y y' = 1$
 $y'(x^2e^{x^2y} - e^y) = 1 - 2xye^{x^2y}$
 $y' = \frac{1 - 2xye^{x^2y}}{(x^2e^{x^2y} - e^y)}$
- 12.** $e^y + xe^y y' - 3y' \sin x - 3y \cos x = 0$
 $y' = \frac{3y \cos x - e^y}{xe^y - 3 \sin x}$
- 13.** $y^2\sqrt{x+y} - 4x^2 = y$
 Differentiating with respect to x ,
 $\frac{d}{dx}(y^2\sqrt{x+y} - 4x^2) = \frac{d}{dx}(y)$
 By the Chain rule and Product rule,
 $\frac{d}{dx}(y^2\sqrt{x+y}) - 4\frac{d}{dx}(x^2) = \frac{d}{dx}(y)$
 $\left[y^2\left(\frac{1}{2\sqrt{x+y}}\right)(1+y')\right] + 2yy'\sqrt{x+y} - 8x = y'$
 $y^2 + y^2y' + 4yy'(x+y) - 16x\sqrt{x+y} = y'$
 $= 2y'\sqrt{x+y}$
 $y^2y' + 4yy'(x+y) - 2y'\sqrt{x+y} = 16x\sqrt{x+y} - y^2$
 $y'[y^2 + 4y(x+y) - 2\sqrt{x+y}] = 16x\sqrt{x+y} - y^2$
 $y' = \frac{16x\sqrt{x+y} - y^2}{y^2 + 4y(x+y) - 2\sqrt{x+y}}$
- 14.** $x \cos(x+y) - y^2 = 8$
 Differentiating with respect to x ,
 $\frac{d}{dx}(x \cos(x+y) - y^2) = \frac{d}{dx}(8)$
 By the Chain rule and Product rule,
 $\frac{d}{dx}(x \cos(x+y)) - \frac{d}{dx}(y^2) = \frac{d}{dx}(8)$
 $\cos(x+y) - x \sin(x+y)(1+y') - 2yy' = 0$
 $\cos(x+y) - x \sin(x+y) - x \sin(x+y)y' - 2yy' = 0$
 $y'(-x \sin(x+y) - 2y) = x \sin(x+y) - \cos(x+y)$
 $y' = \frac{x \sin(x+y) - \cos(x+y)}{-x \sin(x+y) - 2y}$
 $y' = \frac{\cos(x+y) - x \sin(x+y)}{x \sin(x+y) + 2y}$
- 15.** $e^{4y} - \ln(y^2 + 3) = 2x$
 Differentiating with respect to x ,
 $\frac{d}{dx}(e^{4y} - \ln(y^2 + 3)) = \frac{d}{dx}(2x)$
 By the Chain rule and Product rule,
 $\frac{d}{dx}(e^{4y}) - \frac{d}{dx}(\ln(y^2 + 3)) = \frac{d}{dx}(2x)$
 $e^{4y}(4y') - \frac{2yy'}{y^2 + 3} = 2$
 $4e^{4y}(y^2 + 3)y' - 2yy' = 2(y^2 + 3)$
 $y'(4e^{4y}(y^2 + 3) - 2y) = 2(y^2 + 3)$
 $y' = \frac{2(y^2 + 3)}{4e^{4y}(y^2 + 3) - 2y}$
- 16.** $e^{x^2}y - 3\sqrt{y^2 + 2} = x^2 + 1$
 Differentiating with respect to x ,

$$\frac{d}{dx} (e^{x^2} y - 3\sqrt{y^2 + 2}) = \frac{d}{dx} (x^2 + 1)$$

By the Chain rule and Product rule,

$$\frac{d}{dx} (e^{x^2} y) - 3 \frac{d}{dx} (\sqrt{y^2 + 2}) = 2x$$

$$e^{x^2} (2x) y + e^{x^2} y' - 3 \cdot \frac{2yy'}{2\sqrt{y^2 + 2}} = 2x$$

$$2xye^{x^2} + e^{x^2} y' - \frac{3yy'}{\sqrt{y^2 + 2}} = 2x$$

$$2xye^{x^2} \sqrt{y^2 + 2} + e^{x^2} y' \sqrt{y^2 + 2} - 3yy'$$

$$= 2x\sqrt{y^2 + 2}$$

$$y' (e^{x^2} \sqrt{y^2 + 2} - 3y) = 2x\sqrt{y^2 + 2}$$

$$-2xye^{x^2} \sqrt{y^2 + 2}$$

$$y' = \frac{2x\sqrt{y^2 + 2} (1 - ye^{x^2})}{e^{x^2} \sqrt{y^2 + 2} - 3y}$$

17. Rewrite: $x^2 = 4y^3$

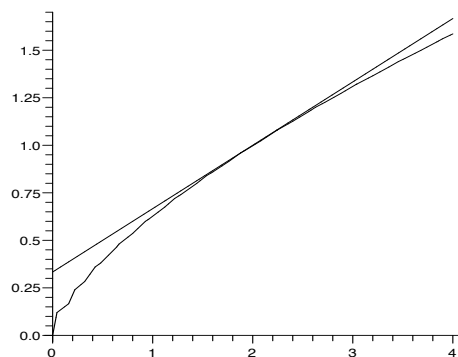
Differentiate by x : $2x = 12y^2 y'$

$$y' = \frac{2x}{12y^2}$$

$$\text{At } (2, 1): y' = \frac{2}{6 \cdot 1^2} = \frac{1}{3}$$

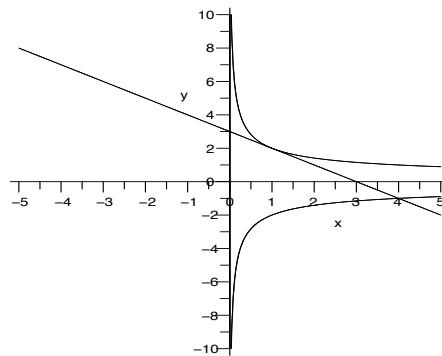
The equation of the tangent line is

$$y - 1 = \frac{1}{3}(x - 2) \text{ or } y = \frac{1}{3}(x + 1).$$



18. $2xy^2 + x^2 2y \cdot y' = 4$, so $y' = \frac{4 - 2xy^2}{2x^2 y}$.

y' at $(1, 2)$ is -1 , and the equation of the line is $y = -1(x - 1) + 2$.



19. $x^2 y^2 = 3y + 1$

Differentiating with respect to x ,

$$\frac{d}{dx} (x^2 y^2) = \frac{d}{dx} (3y + 1)$$

By using the Product Rule we have,

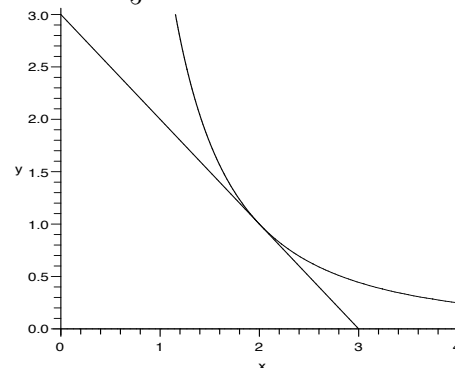
$$2xy^2 + 2yy'x^2 = 3y'$$

$$y' = \frac{2xy^2}{3 - 2yx^2}$$

$$\text{At } (2, 1), y' = -\frac{4}{5}.$$

The equation of the tangent line is given by

$$y - 1 = -\frac{4}{5}(x - 2).$$



20. $x^3 y^2 = -2xy - 3$

Differentiating with respect to x ,

$$\frac{d}{dx} (x^3 y^2) = \frac{d}{dx} (-2xy - 3)$$

By using Product Rule,

$$3x^2 y^2 + 2yy'x^3 = -2y - 2y'$$

$$y' (2x^3 y + 2x) = -2y - 3x^2 y^2$$

$$y' = -\frac{2y + 3x^2 y^2}{2x^3 y + 2x}$$

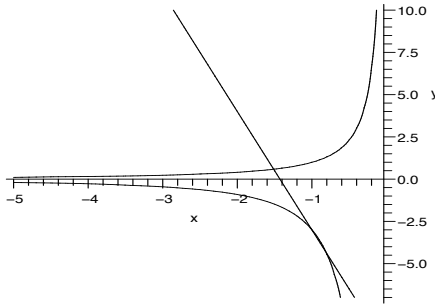
Substituting $x = -1$ and $y = -3$,

$$y'(-1) = -\frac{2(-3) + 3(-1)^2(-3)^2}{2(-1)^3(-3) + 2(-1)}$$

$$= -\frac{-6 + 27}{-6 - 2} = -\frac{21}{4}$$

The equation of the tangent line is

$$(y + 3) = -\frac{21}{4}(x + 1).$$



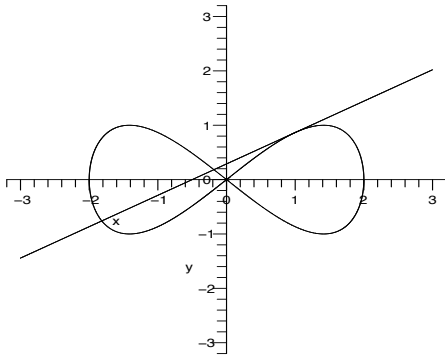
$$\begin{aligned} 21. \quad & 4y^2 = 4x^2 - x^4 \\ & 8yy' = 8x - 4x^3 \\ & y' = \frac{x(2 - x^2)}{2y}. \end{aligned}$$

The slope of the tangent line at $(1, \frac{\sqrt{3}}{2})$ is

$$m = \frac{1(2 - 1^2)}{2 \cdot (\frac{\sqrt{3}}{2})} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

The equation of the tangent line is

$$\begin{aligned} y - \frac{\sqrt{3}}{2} &= \frac{\sqrt{3}}{3}(x - 1) \\ y &= \frac{\sqrt{3}}{3}x + \frac{\sqrt{3}}{6}. \end{aligned}$$

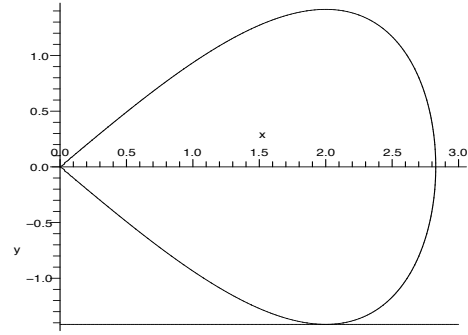


$$\begin{aligned} 22. \quad & x^4 - 8x^2 = -8y^2 \\ & 4x^3 - 16x = -16yy' \\ & y' = \frac{-(4x^3 - 16x)}{16y} = \frac{4x(4 - x^2)}{16y} \end{aligned}$$

The slope of the tangent line at $(2, -\sqrt{2})$ is

$$m = \frac{2(4 - 2^2)}{4(-\sqrt{2})} = 0.$$

The equation of the tangent line is $y = -\sqrt{2}$.



$$\begin{aligned} 23. \quad & \frac{d}{dx}(x^2y^2 + 3x - 4y) = \frac{d}{dx}(5) \\ & x^2 2yy' + 2xy^2 + 3 - 4y' = 0 \end{aligned}$$

Differentiate both sides of this with respect to x :

$$\begin{aligned} & \frac{d}{dx}(x^2 2yy' + 2xy^2 + 3 - 4y') = \frac{d}{dx}(0) \\ & 2(2xyy' + x^2(y')^2 + x^2yy'') + 2(2xyy' + y^2) - 4y'' = 0. \end{aligned}$$

$$\begin{aligned} & y' + x^2(y')^2 + x^2yy'' + 2xyy' + y^2 - 2y'' = 0. \\ & y' + x^2(y')^2 + y^2 = y''(2 - x^2y) \end{aligned}$$

$$y'' = \frac{4xyy' + x^2(y')^2 + y^2}{2 - x^2y}$$

$$\begin{aligned} 24. \quad & \frac{d}{dx}(x^{2/3} + y^{2/3}) = \frac{d}{dx}(4) \\ & \frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}y' = 0 \end{aligned}$$

Multiply by $\frac{3}{2}$ and implicitly differentiate again:

$$-\frac{1}{3}x^{-4/3} - \frac{1}{3}y^{-4/3}y'y' + y^{-1/3}y'' = 0$$

$$\text{so } y'' = \frac{x^{-4/3} + y^{-4/3}(y')^2}{3y^{-1/3}}$$

$$\begin{aligned} 25. \quad & \frac{d}{dx}(y^2) = \frac{d}{dx}(x^3 - 6x + 4 \cos y) \\ & 2yy' = 3x^2 - 6 - 4 \sin y \cdot y'. \\ & \text{Differentiating again with respect to } x : \\ & 2yy'' + 2(y')^2 = 6x - 4 [\sin y \cdot y'' + \cos y \cdot (y')^2] \\ & yy'' + (y')^2 = 3x - 2 \sin y \cdot y'' - 2 \cos y \cdot (y')^2 \\ & y''(y + 2 \sin y) = 3x - (2 \cos y + 1)(y')^2 \\ & y'' = \frac{3x - (2 \cos y + 1)(y')^2}{y + 2 \sin y} \end{aligned}$$

$$\begin{aligned} 26. \quad & \frac{d}{dx}(e^{xy} + 2y - 3x) = \frac{d}{dx}(\sin y) \\ & e^{xy}(y + xy') + 2y' - 3 = \cos y \cdot y' \\ & \text{Differentiating again with respect to } x : \\ & e^{xy}(y + xy')^2 + e^{xy}(y' + y' + xy'') + 2y'' \\ & = -\sin y (y')^2 + \cos y \cdot y'' \\ & \text{and} \end{aligned}$$

$$y'' = \frac{e^{xy}(y + xy')^2 + 2e^{xy}y' + \sin y(y')^2}{\cos y - xe^{xy} - 2}$$

27. $(y - 1)^2 = 3xy + e^{4y}$

Differentiating with respect to x ,

$$\frac{d}{dx}(y - 1)^2 = \frac{d}{dx}(3xy + e^{4y})$$

By the Chain and Product rule,

$$2(y - 1)y' = 3y + 3xy' + 4e^{4y}y'$$

Differentiating with respect to x ,

$$\frac{d}{dx}[2(y - 1)y'] = \frac{d}{dx}[3y + 3xy' + 4e^{4y}y']$$

By the Chain and Product rule,

$$2(y - 1)y'' + 2(y')^2 = 3y' + 3xy'' + 3y' + 4e^{4y}y'' + 16e^{4y}(y')^2$$

$$2(y - 1)y'' - 3xy'' - 4e^{4y}y''$$

$$= 6y'(x) + 16e^{4y}(y')^2 - 2(y')^2$$

$$y''[2(y - 1) - 3x - 4e^{4y}]$$

$$= 2y'(3 + 8e^{4y}y' - y')$$

$$y'' = \frac{2y'(3 + 8e^{4y}y' - y')}{2(y - 1) - 3x - 4e^{4y}}$$

28. $(x + y)^2 - e^{y+1} = 3x$

Differentiating with respect to x ,

$$\frac{d}{dx}[(x + y)^2 - e^{y+1}] = \frac{d}{dx}(3x)$$

By the Chain rule,

$$2(x + y)(1 + y') - e^{y+1}y' = 3$$

Differentiating with respect to x ,

$$\frac{d}{dx}[2(x + y)(1 + y') - e^{y+1}y'] = 0$$

By the Chain and Product rule,

$$2(x + y)y'' + 2(1 + y')^2 - e^{y+1}y'' - e^{y+1}(y')^2 = 0$$

$$y''[2(x + y) - e^{y+1}]$$

$$= e^{y+1}(y')^2 - 2(1 + y')^2$$

$$y'' = \frac{e^{y+1}(y')^2 - 2(1 + y')^2}{2(x + y) - e^{y+1}}$$

29. (a) $f(x) = \sin^{-1}(x^3 + 1)$

Differentiating with respect to x ,

$$f'(x) = \frac{d}{dx}[\sin^{-1}(x^3 + 1)]$$

By the Chain rule we get,

$$f'(x) = \frac{1}{\sqrt{1 - (x^3 + 1)^2}} \frac{d}{dx}(x^3 + 1)$$

$$= \frac{1}{\sqrt{1 - (x^3 + 1)^2}} (3x^2)$$

$$= \frac{3x^2}{\sqrt{1 - (x^3 + 1)^2}}$$

(b) $f(x) = \sin^{-1}(\sqrt{x})$

Differentiating with respect to x ,

$$f'(x) = \frac{d}{dx}[\sin^{-1}(\sqrt{x})]$$

By the Chain rule, we get

$$f'(x) = \frac{1}{\sqrt{1 - (\sqrt{x})^2}} \frac{d}{dx}(\sqrt{x})$$

$$= \frac{1}{\sqrt{1 - x}} \left(\frac{1}{2\sqrt{x}} \right)$$

$$= \frac{1}{2\sqrt{x(1 - x)}}$$

30. (a) $f(x) = \cos^{-1}(x^2 + x)$

Differentiating with respect to x ,

$$f'(x) = \frac{d}{dx}[\cos^{-1}(x^2 + x)]$$

By using Chain rule,

$$f'(x) = \frac{-1}{\sqrt{1 - (x^2 + x)^2}} \frac{d}{dx}(x^2 + x)$$

$$= \frac{-(2x + 1)}{\sqrt{1 - (x^2 + x)^2}}$$

(b) $f(x) = \cos^{-1}\left(\frac{2}{x}\right)$

Differentiating with respect to x ,

$$f'(x) = \frac{d}{dx}\left[\cos^{-1}\left(\frac{2}{x}\right)\right]$$

By using Chain rule,

$$f'(x) = \frac{-1}{\sqrt{1 - \left(\frac{2}{x}\right)^2}} \frac{d}{dx}\left(\frac{2}{x}\right)$$

$$= \frac{-1}{\sqrt{1 - \left(\frac{4}{x^2}\right)}} \left(\frac{-2}{x^2}\right)$$

$$= \frac{2}{x\sqrt{x^2 - 4}}$$

31. (a) $f(x) = \tan^{-1}(\sqrt{x})$

Differentiating with respect to x ,

$$f'(x) = \frac{d}{dx}[\tan^{-1}(\sqrt{x})]$$

By the Chain rule,

$$f'(x) = \frac{1}{1 + (\sqrt{x})^2} \frac{d}{dx}(\sqrt{x})$$

$$= \frac{1}{(1 + x)} \left(\frac{1}{2\sqrt{x}} \right)$$

$$= \frac{1}{2\sqrt{x}(1 + x)}$$

(b) $f(x) = \tan^{-1}\left(\frac{1}{x}\right)$

Differentiating with respect to x ,

$$f'(x) = \frac{d}{dx}\left[\tan^{-1}\left(\frac{1}{x}\right)\right]$$

By the Chain rule,

$$\begin{aligned}
 f'(x) &= \frac{1}{1 + \left(\frac{1}{x}\right)^2} \frac{d}{dx} \left(\frac{1}{x} \right) \\
 &= \frac{1}{\left(1 + \frac{1}{x^2}\right)} \left(\frac{-1}{x^2} \right) \\
 &= \frac{-1}{(x^2 + 1)}
 \end{aligned}$$

32. (a) $f(x) = \sqrt{2 + \tan^{-1}x}$
 Differentiating with respect to x ,
 $f'(x) = \frac{d}{dx} \left(\sqrt{2 + \tan^{-1}x} \right)$.
 By the Chain rule,

$$f'(x) = \frac{1}{2\sqrt{2 + \tan^{-1}x}} \frac{d}{dx} (2 + \tan^{-1}x)$$

$$= \frac{1}{2\sqrt{2 + \tan^{-1}x}} \left(\frac{1}{1 + x^2} \right)$$

$$= \frac{1}{2(1 + x^2)\sqrt{2 + \tan^{-1}x}}$$

(b) $f(x) = e^{\tan^{-1}x}$
 Differentiating with respect to x ,
 $f'(x) = \frac{d}{dx} \left(e^{\tan^{-1}x} \right)$.
 By the Chain rule,

$$f'(x) = \left(e^{\tan^{-1}x} \right) \left(\frac{1}{1 + x^2} \right)$$

$$= \frac{e^{\tan^{-1}x}}{1 + x^2}$$

33. (a) $f(x) = 4 \sec(x^4)$
 Differentiating with respect to x ,
 $f'(x) = \frac{d}{dx} (4 \sec(x^4))$
 By Chain rule,

$$f'(x) = 4 \sec(x^4) \tan(x^4) \frac{d}{dx} (x^4)$$

$$= 4 \sec(x^4) \tan(x^4) (4x^3)$$

$$= 16x^3 \sec(x^4) \tan(x^4)$$

(b) $f(x) = 4 \sec^{-1}(x^4)$
 Differentiating with respect to x ,
 $f'(x) = \frac{d}{dx} (4 \sec^{-1}(x^4))$.
 By Chain rule,

$$f'(x) = 4 \frac{1}{x^4 \sqrt{(x^4)^2 - 1}} \frac{d}{dx} (x^4)$$

$$= 4 \frac{1}{x^4 \sqrt{x^8 - 1}} (4x^3)$$

$$= \frac{16}{x \sqrt{x^8 - 1}}$$

34. (a) $f(x) = \sin^{-1} \left(\frac{1}{x} \right)$
 Differentiating with respect to x ,

$$f'(x) = \frac{d}{dx} \left(\sin^{-1} \left(\frac{1}{x} \right) \right).$$

By the Chain rule,

$$\begin{aligned}
 f'(x) &= \frac{1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \frac{d}{dx} \left(\frac{1}{x} \right) \\
 &= \frac{x}{\sqrt{x^2 - 1}} \left(\frac{-1}{x^2} \right) \\
 &= -\frac{1}{x\sqrt{x^2 - 1}}
 \end{aligned}$$

(b) $f(x) = c \sec^{-1}(x)$
 Differentiating with respect to x ,
 $f'(x) = \frac{d}{dx} c \sec^{-1}(x)$.
 By the Chain rule,

$$f'(x) = -\frac{1}{x\sqrt{x^2 - 1}}.$$

35. In example 8.6, we are given

$$\theta'(d) = \frac{2(-130)}{4 + d^2}$$

Setting this equal to -3 and solving for d gives $d^2 = 82 \Rightarrow d = 9$ feet. The better can track the ball after they would have to start swinging (when the ball is 30 feet away), but not all the way to home plate.

36. From example 8.6, the rate of angle is

$$\theta'(t) = \frac{1}{1 + \left[\frac{d(t)}{2}\right]^2} \left(\frac{d'(t)}{2} \right)$$

Given a maximum rotational rate of $\theta'(t) = -3$ (radians/second), the distance from the plate at which a player can track the ball can be obtained by solving the equation

$$-3 = \frac{2d'(t)}{4 + [d(t)]^2}$$

for $d(t)$ in terms of $d'(t)$. This leads to

$$d(t) = \frac{\sqrt{-6 \cdot d'(t) - 36}}{3}$$

if $d'(t) \leq -6$ which may be reasonable since the distance is decreasing as the ball approaches the plate. We get $d(t) = 4$ for $d'(t) = -30$ ft/sec and $d(t) = 9.45$ for $d'(t) = -140$ ft/sec. This would mean a player can track the ball to within 4 feet from the plate in slowpitch, but only to within 9.45 feet from the plate in the major leagues.

37. Suppose that d is the distance from ball to home plate and θ is the angle of gaze. Since distance is changing with time, therefore $d = d(t)$. The velocity 130 ft/sec means that $d'(t) = -130$

$$\theta(t) = \tan^{-1} \left[\frac{d(t)}{3} \right]$$

The rate of change of angle is then

$$\begin{aligned}\theta'(t) &= \frac{1}{1 + \left(\frac{d(t)}{3}\right)^2} \frac{d'(t)}{3} \\ &= \frac{3d'(t)}{9 + [d(t)]^2} \text{ radians/second}\end{aligned}$$

when $d''(t) = 0$.

The rate of the change is then

$$\theta'(t) = \frac{3(-130)}{9} = -43.33 \text{ radians/sec.}$$

- 38.** Let d is the distance from ball to home plate and θ is the angle of gaze, Since distance is changing with time therefore $d = d(t)$. The velocity 130 ft/sec means that $d'(t) = -130$,

$$\theta(t) = \tan^{-1} \left[\frac{d(t)}{x} \right]$$

The rate of change of angle is then

$$\begin{aligned}\theta'(t) &= \frac{1}{1 + \left(\frac{d(t)}{x}\right)^2} \frac{d'(t)}{x} \\ &= \frac{xd'(t)}{x^2 + [d(t)]^2} \text{ radians/second}\end{aligned}$$

when $d(t) = 0$,

The rate of the change is then

$$\begin{aligned}\theta'(t) &= \frac{x(-130)}{x^2} \text{ radians/second} \\ &= \frac{-130}{x} = -3 \text{ radians/second}\end{aligned}$$

Therefore, $x = \frac{-130}{-3} = 43.33$

- 39.** $\frac{d}{dx}(x^2 + y^2 - 3y) = \frac{d}{dx}(0)$
 $2x + 2y \cdot y' - 3y' = 0$
 $y'(2y - 3) = -2x$
 $y' = \frac{2x}{3 - 2y}$

Horizontal tangents:

From the formula, $y' = 0$ only when $x = 0$. When $x = 0$ we have $0 + y^2 - 3y = 0$. Therefore $y = 0$ and $y = 3$ are the horizontal tangents.

Vertical tangents:

The denominator in y' must be zero.

$$3 - 2y = 0$$

$$y = 1.5$$

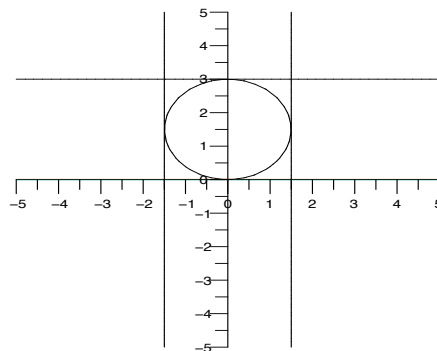
When $y = 1.5$,

$$x^2 + (1.5)^2 - 3(1.5) = 0$$

$$x^2 = 2.25$$

$$x = \pm 1.5$$

$x = \pm 1.5$ are the vertical tangents.



- 40.** $\frac{d}{dx}(x^2 + y^2 - 2y) = \frac{d}{dx}(3)$
 $2x + 2yy' - 2y' = 0$
 $x + y'(y - 1) = 0$
 $y'(y - 1) = -x$
 $y' = \frac{x}{1 - y}$

Horizontal tangents:

The curve has horizontal tangents when $y' = 0$ i.e. when $x = 0$.

$$\text{At } x = 0, y = \frac{2 \pm \sqrt{4 - 4(-3)}}{2} = \frac{2 \pm 4}{2}$$

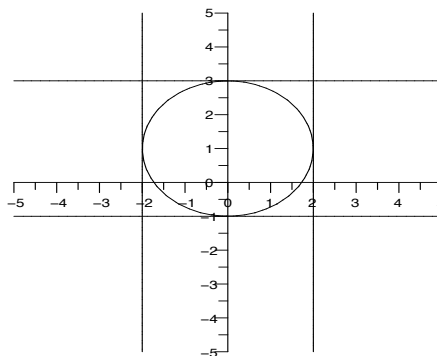
which gives $y = 3$ or $y = -1$. Therefore $y = 3$ and $y = -1$ are the horizontal tangents to the curve.

Vertical tangents:

The curve has vertical tangents when the denominator in y' is 0 which gives $y = 1$.

$$\text{At } y = 1, x = \pm 2$$

Therefore, $x = \pm 2$ are the vertical tangents to curve.



- 41. (a)** $x^2y^2 + 3y = 4x$

To find the derivative of y , we use Implicit differentiation.

- (b)** $x^2y + 3y = 4x$

The derivative of y can be found directly and implicitly.

- (c)** $3xy + 6x^2 \cos x = y \sin x$

The derivative of y can be found directly and implicitly.

- (d)** $3xy + 6x^2 \cos y = y \sin x$

By using Implicit differentiation we can find the derivative of y .

42. $f(x) = \sin^{-1}(\sin x)$

$$= \sin^{-1}(\sin[2n\pi - (2n\pi - x)])$$

$$= \sin^{-1}(-\sin(2n\pi - x))$$

$$= -\sin^{-1}[\sin(2n\pi - x)]$$

In the interval $\left(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2}\right)$,

$$-\frac{\pi}{2} \leq 2n\pi - x \leq \frac{\pi}{2}.$$

So, $f(x) = -(2n\pi - x) = x - 2n\pi$.

Again,

$$f(x) = \sin^{-1}(\sin x)$$

$$= \sin^{-1}(\sin[(2n\pi + \pi) - (2n\pi + \pi - x)])$$

$$= \sin^{-1}(\sin(2n\pi + \pi - x))$$

$$= \sin^{-1}[\sin(2n\pi + \pi - x)]$$

In the interval $\left[2n\pi + \pi - \frac{\pi}{2}, 2n\pi + \pi + \frac{\pi}{2}\right]$,

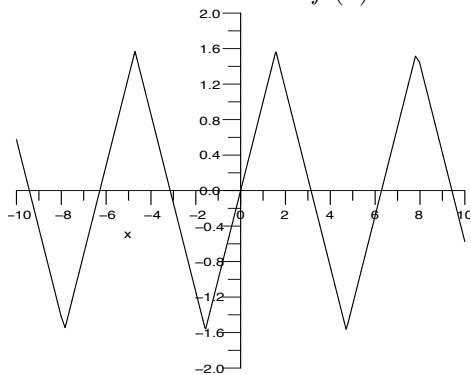
$$-\frac{\pi}{2} \leq 2n\pi + \pi - x \leq \frac{\pi}{2}.$$

So, $f(x) = (2n + 1)\pi - x$.

Therefore $f'(x) = 1$ for all $x \in (2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2})$ and $f'(x) = -1$ for all $x \in (2n\pi + \pi - \frac{\pi}{2}, 2n\pi + \pi + \frac{\pi}{2})$. At the points $x = n\pi \pm \frac{\pi}{2}$, $f'(x)$ is not defined.

Here n is any integer.

From the graph of $f(x)$ in below, we can check the above values of $f'(x)$.



43. Let $y = \sin^{-1}x + \cos^{-1}x$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$$

Therefore, $y = c$, where c is a constant. To determine c , substitute any convenient value of x , such as $x = 0$

$$\sin^{-1}x + \cos^{-1}x = c$$

$$\sin^{-1}0 + \cos^{-1}0 = c, \text{ so } c = \frac{\pi}{2}$$

$$\text{Thus } \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$$

44. Let $y = \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{1 - \left(\frac{x}{\sqrt{x^2+1}}\right)^2}} \cdot \frac{d}{dx} \left(\frac{x}{\sqrt{x^2+1}}\right) \\ &= \left(\frac{1}{\sqrt{1 - \frac{x^2}{x^2+1}}}\right) \cdot \left(\frac{\sqrt{x^2+1} - x(1/2)(x^2+1)^{-1/2}(2x)}{x^2+1}\right) \\ &= \frac{1 - \frac{x^2}{x^2+1}}{\sqrt{1 - \frac{x^2}{x^2+1}}} \cdot \frac{\sqrt{x^2+1}}{x^2+1} \\ &= \frac{\sqrt{1 - \frac{x^2}{x^2+1}}}{\sqrt{x^2+1}} \cdot \left(\frac{\sqrt{x^2+1}}{\sqrt{x^2+1}}\right) \\ &= \frac{1}{1+x^2} \end{aligned}$$

Thus $\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = y = \int \frac{1}{1+x^2} dx$

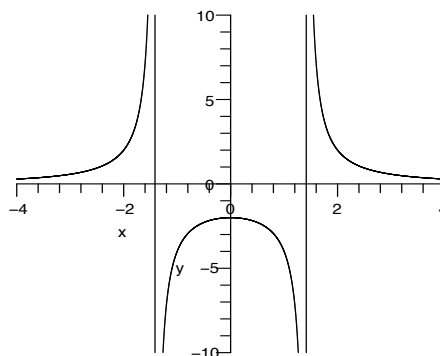
$$= \tan^{-1}(x) + c \text{ for some constant } c.$$

Substitute $x = 0$ in to the above expression to find $c = 0$ and so

$$\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \tan^{-1}x$$

45. $\frac{d}{dx}(x^2y - 2y) = \frac{d}{dx}(4)$
 $2xy + x^2y' - 2y' = 0$
 $y'(x^2 - 2) = -2xy$
 $y' = \frac{2xy}{(x^2 - 2)}$

The derivative is undefined at $x = \pm\sqrt{2}$, suggesting that there might be vertical tangent lines at these points. Similarly, $y' = 0$ at $y = 0$ suggesting that there might be a horizontal tangent line at this point. However, plugging $x = \pm\sqrt{2}$ into the original equation gives $0 = 4$, a contradiction which shows that there are no points on the curve with x value $\pm\sqrt{2}$. Likewise, plugging $y = 0$ in the original equation gives $0 = 4$. Again, this is a contradiction which shows that there are no points on the graph with y value of 4. Sketching the graph, we see that there is a horizontal asymptote at $y = 0$ and vertical asymptote at $x = \pm\sqrt{2}$



46. For the first type of curve, $y + xy' = 0$ and $y' = \frac{-y}{x}$.

For the second type of curve, $2x - 2yy' = 0$ and $y' = \frac{x}{y}$.

At any point of intersection, the tangent line to the first curve is perpendicular to the tangent line to the second curve.

47. If $y_1 = \frac{c}{x}$ then $y_1' = -\frac{c}{x^2} = -\frac{y_1}{x}$.
If $y_2^2 = x^2 + k$ then $2y_2y_2' = 2x$ and $y_2' = \frac{x}{y_2}$. If we are at a particular point (x_0, y_0) on both graphs, this means $y_1(x_0) = y_0 = y_2(x_0)$ and $y_1' \cdot y_2' = \left(\frac{-y_0}{x_0}\right) \cdot \left(\frac{x_0}{y_0}\right) = -1$.

This means that the slopes are negative reciprocals and the curves are orthogonal.

48. For the first type of curve, $2x + 2yy' = c$ and $y' = \frac{c - 2x}{2y}$.

For the second type of curve, $2x + 2yy' = ky'$ and $y' = \frac{2x}{k - 2y}$.

Multiply the first x/x and the second by y/y .

This gives $y' = \frac{cx - 2x^2}{2xy} = \frac{y^2 - x^2}{2xy}$, and

$$y' = \frac{2xy}{ky - y^2} = \frac{2xy}{x^2 - y^2}.$$

These are negative reciprocals of each other, so the families of the curve are orthogonal.

49. For the first type of curve, $y' = 3cx^2$.
For the second type of curve, $2x + 6yy' = 0$,
 $y' = -\frac{2x}{6y} = -\frac{x}{3y} = -\frac{x}{3cx^3} = -\frac{1}{3cx^2}$.
These are negative reciprocals of each other, so the families of the curve are orthogonal.

50. For the first type of curve, $y' = 4cx^3$.
For the second type of curve, $2x + 8yy' = 0$.
 $y' = \frac{-2x}{8y} = \frac{-x}{4y} = \frac{-x}{4cx^4} = \frac{-1}{4cx^3}$.
These are negative reciprocals of each other, so the families of the curve are orthogonal.

51. Conjecture: The family of functions $\{y_1 = cx^n\}$ is orthogonal to the family of functions $\{x^2 + ny^2 = k\}$ wherever $n \neq 0$.

If $y_1 = cx^n$, then $y_1' = ncx^{n-1} = \frac{ny_1}{x}$.

If $ny_2^2 = -x^2 + k$, then $2ny_2 \cdot (y_2') = -2x$ and $y_2' = \frac{-x}{ny_2}$.

If we are at a particular point (x_0, y_0) on both graphs, this means $y_1(x_0) = y_0 = y_2(x_0)$ and

$$y_1' \cdot y_2' = \left(\frac{ny_0}{x_0}\right) \cdot \left(\frac{-x_0}{ny_0}\right) = -1.$$

This means that the slopes are negative reciprocals and the curves are orthogonal.

52. The domain of the function $\sin^{-1}x$ is $[-1, 1]$ and the domain of the function $\sec^{-1}x$ is $(-\infty, -1) \cup (1, \infty)$. Therefore, the function $\sin^{-1}x + \sec^{-1}x$ is not defined.

53. (a) Both of the points $(-3, 0)$ and $(0, 3)$ are on the curve:

$$0^2 = (-3)^3 - 6(-3) + 9$$

$$3^2 = 0^3 - 6(0) + 9$$

The equation of the line through these points has slope $= \frac{0-3}{-3-0} = 1$ and y-intercept 3, so $y = x + 3$.

This line intersects the curve at:

$$y^2 = x^3 - 6x + 9$$

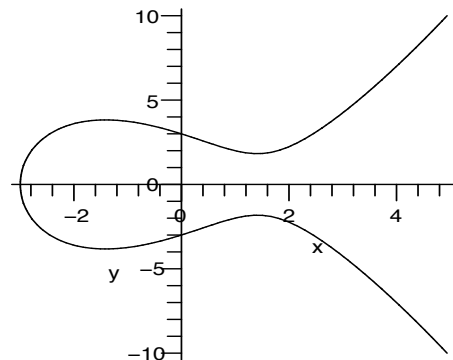
$$(x+3)^2 = x^3 - 6x + 9$$

$$x^2 + 6x + 9 = x^3 - 6x + 9$$

$$x^3 - 12x - x^2 = 0$$

$$x(x^2 - x - 12) = 0$$

Therefore $x = 0, -3$ or 4 and so third point is $(4, 7)$.



- (b) $3^2 = (-1)^3 - 6(-1) + 4$ is true.

$$2yy' = 3x^2 - 6, \text{ so } y' = \frac{3x^2 - 6}{2y} \text{ and}$$

at $(-1, 3)$ the slope is $-\frac{1}{2}$. The line is $y = -\frac{1}{2}(x+1) + 3$.

To find the other point of intersection, substitute the equation of the line in to the equation for the elliptic curve and simplify:

$$\left(-\frac{1}{2}x + \frac{5}{2}\right)^2 = x^3 - 6x + 4$$

$$x^2 - 10x + 25 = 4x^3 - 24x + 16$$

$$4x^3 - x^2 - 14x - 9 = 0.$$

We know already that $x = -1$ is a solution (actually a double solution) so we can factor out $(x+1)$. Long division

yields $(x+1)^2(4x^2-9)$.

The second point has x -coordinate $\frac{9}{4}$, which can be substituted into the equation for the line to get $y = \frac{11}{8}$.

54. The equation of the circle is $x^2 + (y-c)^2 = r^2$. Differentiating implicitly gives $2x + 2(y-c)y' = 0$ so $y' = \frac{x}{c-y}$.

At the point of tangency, the derivatives must be the same. Since the derivative of $y = x^2$ is $2x$, we must solve the equation $2x = \frac{x}{c-y}$. This gives $y = c - \frac{1}{2}$, as desired. Since $y = x^2$, plugging, $y = c - \frac{1}{2}$ into the equation of the circle gives $\left(c - \frac{1}{2}\right)^2 + \left(c - \frac{1}{2} - c\right)^2 = r^2$
 $c - \frac{1}{2} + \frac{1}{4} = r^2$
 $c = r^2 + \frac{1}{4}$

55. The viewing angle is given by the formula $\theta(x) = \tan^{-1}\left(\frac{3}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right)$.

This will be maximum where the derivative is zero.

$$\begin{aligned}\theta'(x) &= \frac{1}{1 + \left(\frac{3}{x}\right)^2} \cdot \frac{-3}{x^2} - \frac{1}{1 + x^2} \cdot \frac{-1}{x^2} \\ &= \frac{1}{1 + x^2} - \frac{3}{9 + x^2}.\end{aligned}$$

$$\begin{aligned}\text{This is zero when } \frac{1}{1 + x^2} &= \frac{3}{9 + x^2} \Rightarrow x^2 = 3 \\ 3 \Rightarrow x &= \sqrt{3}\end{aligned}$$

56. If A is the viewing angle formed between the rays from the person's eye to the top of the frame and to the bottom of the frame, and if x is the distance between the person and the wall, then since the frame extends from 6 to 8 feet, we have $\tan A = \frac{2}{x}$, or $A = \arctan\left(\frac{2}{x}\right)$.

$$\begin{aligned}\text{Then} \\ \frac{dA}{dx} &= \frac{1}{1 + \left(\frac{2}{x}\right)^2} \cdot \left(\frac{-2}{x^2}\right) = \frac{-2}{x^2 + 4}\end{aligned}$$

Since the derivative is negative, the angle is decreasing function of x . Strictly speaking $\arctan\left(\frac{2}{x}\right)$ is undefined at $x = 0$ but

$\arctan\left(\frac{2}{x}\right) \rightarrow \frac{\pi}{2}$ as $x \rightarrow 0$. The angle a continues to enlarge (upto a right angle) as x decreases to zero. In this case, the maximal viewing angle is not a feasible one.

57. $x^2 + y^2 = 9$

Differentiating the above equation implicitly, we get $2x + 2yy' = 0$
 $x + yy' = 0 \Rightarrow y' = -\frac{x}{y}$

At $(2.9, 0.77)$, y' gives slope of the tangent.

$$y'|_{(2.9, 0.77)} = \frac{-2.9}{0.77} = -3.77$$

Therefore the equation of the tangent line is $y - 0.77 = -3.77(x - 2.9) \Rightarrow y = -3.77x + 11.7$

Let (x_1, y_1) be any point on the line such that the distance is 300 feet. Therefore $(x_1 - 2.9)^2 + (y_1 - 0.77)^2 = 300^2$. Substitute the value of x as x_1 and y_1 , as $y_1 = -3.77x_1 + 11.7$ into the above equation we get,
 $(x_1 - 2.9)^2 + (-3.77x_1 + 11.7 - 0.77)^2 = 90000$

$$\begin{aligned}(x_1 - 2.9)^2 + (-3.77x_1 + 10.93)^2 &= 90000 \\ 15.21x_1^2 - 88.41x_1 - 89872.13 &= 0\end{aligned}$$

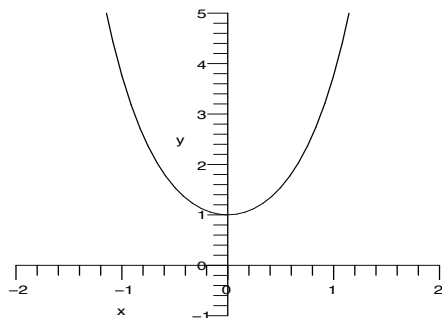
Solving the above quadratic equation, we get $x_1 = 79.83$, $x_1 = -74.02$

Since the sling shot is rotating in the counter clockwise direction, we have to consider the negative value of x_1 . Therefore substituting the negative value of x_1 into the equation, $y_1 = -3.77x_1 + 11.70$

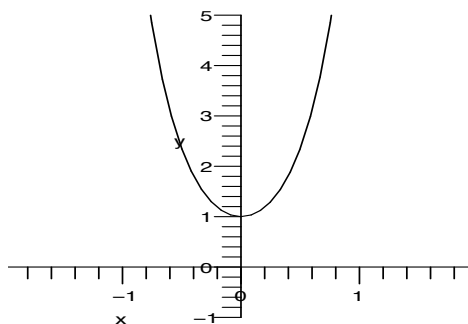
we get $y_1 = -3.77(-74.02) + 11.7 = 290.75$
 Therefore $(-74.02, 290.75)$ is the required point.

2.9 The Hyperbolic Functions

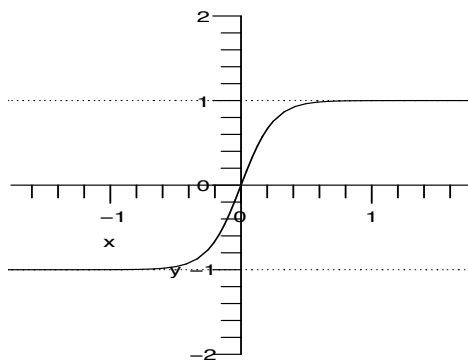
1. Graph of $f(x) = \cosh(2x)$ is:



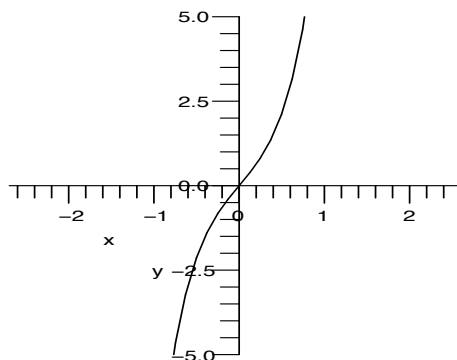
2. Graph of $f(x) = \cosh(3x)$ is:



3. Graph of $f(x) = \tanh(4x)$ is:



4. Graph of $f(x) = \sinh(3x)$ is:



$$\begin{aligned} 5. \quad (a) \quad f'(x) &= \frac{d}{dx} (\cosh 4x) \\ &= \sinh 4x \frac{d}{dx} (4x) \\ &= 4 \sinh 4x \end{aligned}$$

$$\begin{aligned} (b) \quad f'(x) &= \frac{d}{dx} \cosh^4 x \\ &= \frac{d}{dx} (\cosh x)^4 \\ &= 4(\cosh x)^3 (\sinh x) \\ &= 4 \sinh x \cdot \cosh^3 x \end{aligned}$$

$$\begin{aligned} 6. \quad (a) \quad f'(x) &= \frac{d}{dx} (\sinh(\sqrt{x})) \\ &= \cosh(\sqrt{x}) \frac{d}{dx} (\sqrt{x}) \\ &= \cosh(\sqrt{x}) \left(\frac{1}{2\sqrt{x}} \right) \\ &= \frac{\cosh(\sqrt{x})}{2\sqrt{x}} \end{aligned}$$

$$\begin{aligned} (b) \quad f'(x) &= \frac{d}{dx} (\sqrt{\sinh x}) \\ &= \frac{1}{2\sqrt{\sinh x}} \frac{d}{dx} (\sinh x) \\ &= \frac{1}{2\sqrt{\sinh x}} (\cosh x) \\ &= \frac{\cosh x}{2\sqrt{\sinh x}} \end{aligned}$$

$$\begin{aligned} 7. \quad (a) \quad f'(x) &= \frac{d}{dx} (\tanh x^2) \\ &= \operatorname{sech}^2 x^2 \cdot \frac{d}{dx} (x^2) \\ &= (\operatorname{sech}^2 x^2) \cdot (2x) \\ &= 2x \operatorname{sech}^2 x^2 \end{aligned}$$

$$\begin{aligned} (b) \quad f'(x) &= \frac{d}{dx} (\tanh x)^2 \\ &= 2 \tanh x \operatorname{sech}^2 x \end{aligned}$$

$$\begin{aligned}
 8. \quad (a) \quad f'(x) &= \frac{d}{dx} (\operatorname{sech} 3x) \\
 &= -\operatorname{sech} 3x \tanh 3x \frac{d}{dx} (3x) \\
 &= -3 \operatorname{sech} 3x \tanh 3x
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad f'(x) &= \frac{d}{dx} (\operatorname{csch} x)^3 \\
 &= 3(\operatorname{csch}^2 x) \frac{d}{dx} (\operatorname{csch} x) \\
 &= 3(\operatorname{csch}^2 x) (-\operatorname{csch} x \coth x) \\
 &= -3 \operatorname{csch}^3 x \coth x
 \end{aligned}$$

$$\begin{aligned}
 9. \quad (a) \quad f'(x) &= \frac{d}{dx} (x^2 \sinh 5x) \\
 &= x^2 \frac{d}{dx} (\sinh 5x) + \sinh 5x \frac{d}{dx} (x^2) \\
 &= x^2 \cosh 5x \frac{d}{dx} (5x) + \sinh 5x (2x) \\
 &= 5x^2 \cosh 5x + 2x \sinh 5x
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad f(x) &= \frac{x^2 + 1}{\operatorname{csch}^3 x} = (x^2 + 1) \sinh^3 x \\
 f'(x) &= 2x \sinh^3 x + (x^2 + 1) \frac{d}{dx} (\sinh^3 x) \\
 &= 2x \sinh^3 x + (x^2 + 1) 3 \sinh^2 x \cosh x \\
 &= 2x \sinh^3 x + 3(x^2 + 1) \sinh^2 x \cosh x
 \end{aligned}$$

$$\begin{aligned}
 10. \quad (a) \quad f'(x) &= \frac{d}{dx} \left(\frac{\cosh 4x}{x+2} \right) \\
 &= \frac{(x+2) \frac{d}{dx} \cosh 4x - \cosh 4x \frac{d}{dx} (x+2)}{(x+2)^2} \\
 &= \frac{(x+2) \sinh 4x (4) - \cosh 4x (1)}{(x+2)^2} \\
 &= \frac{4(x+2) \sinh 4x - \cosh 4x}{(x+2)^2}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad f'(x) &= \frac{d}{dx} (x^2 \tanh (x^3 + 4)) \\
 &= x^2 \frac{d}{dx} \tanh (x^3 + 4) \\
 &\quad + \tanh (x^3 + 4) \frac{d}{dx} (x^2) \\
 &= x^2 \operatorname{sech}^2 (x^3 + 4) (3x^2) \\
 &\quad + \tanh (x^3 + 4) (2x) \\
 &= 3x^4 \operatorname{sech}^2 (x^3 + 4) + 2x \tanh (x^3 + 4)
 \end{aligned}$$

$$\begin{aligned}
 11. \quad (a) \quad f'(x) &= \frac{d}{dx} (\cosh^{-1} 2x) \\
 &= \frac{1}{\sqrt{(2x)^2 - 1}} \frac{d}{dx} (2x) \\
 &= \frac{2}{\sqrt{4x^2 - 1}}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad f'(x) &= \frac{d}{dx} (\sinh^{-1} x^2) \\
 &= \frac{1}{\sqrt{1+x^4}} \frac{d}{dx} (x^2) \\
 &= \frac{2x}{\sqrt{1+x^4}}
 \end{aligned}$$

$$\begin{aligned}
 12. \quad (a) \quad f'(x) &= \frac{d}{dx} (\tanh^{-1} 3x) \\
 &= \frac{1}{1-(3x)^2} \frac{d}{dx} (3x) \\
 &= \frac{3}{1-9x^2}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad f'(x) &= \frac{d}{dx} (x^2 \cosh^{-1} 4x) \\
 &= x^2 \frac{d}{dx} (\cosh^{-1} 4x) + \cosh^{-1} 4x \frac{d}{dx} (x^2) \\
 &= x^2 \frac{1}{\sqrt{(4x)^2 - 1}} (4) + \cosh^{-1} 4x (2x) \\
 &= \frac{4x^2}{\sqrt{16x^2 - 1}} + 2x \cosh^{-1} 4x
 \end{aligned}$$

$$\begin{aligned}
 13. \quad \frac{d}{dx} (\cosh x) &= \frac{d}{dx} \left(\frac{e^x + e^{-x}}{2} \right) \\
 &= \frac{e^x - e^{-x}}{2} = \sinh x
 \end{aligned}$$

$$\begin{aligned}
 &\frac{d}{dx} (\tanh x) \\
 &= \frac{d}{dx} \left(\frac{\sinh x}{\cosh x} \right) \\
 &= \frac{\cosh x \frac{d}{dx} (\sinh x) - \sinh x \frac{d}{dx} (\cosh x)}{\cosh^2 x} \\
 &= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x
 \end{aligned}$$

$$\begin{aligned}
 14. \quad \frac{d}{dx} [\coth x] &= \frac{d}{dx} \left[\frac{\cosh x}{\sinh x} \right] \\
 &= \frac{\sinh x \cdot \sinh x - \cosh x \cdot \cosh x}{(\sinh x)^2} \\
 &= \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} \\
 &= \frac{-1}{\sinh^2 x} \\
 &= -\operatorname{csch}^2 x
 \end{aligned}$$

$$\begin{aligned}
\frac{d}{dx}[\operatorname{sech} x] &= \frac{d}{dx} \left[\frac{1}{\cosh x} \right] \\
&= -\frac{1}{\cosh^2 x} \sinh x \\
&= -\frac{1}{\cosh x} \cdot \frac{\sinh x}{\cosh x} \\
&= -\operatorname{sech} x \tanh x
\end{aligned}$$

$$\begin{aligned}
\frac{d}{dx}[\operatorname{csch} x] &= \frac{d}{dx} \left[\frac{1}{\sinh x} \right] \\
&= -\frac{1}{\sinh^2 x} \cosh x \\
&= -\frac{1}{\sinh x} \cdot \frac{\cosh x}{\sinh x} \\
&= -\operatorname{csch} x \coth x
\end{aligned}$$

- 15.** First, $e^x > e^{-x}$ if $x > 0$ and $e^x < e^{-x}$ if $x < 0$. Since $\sinh x = \frac{e^x - e^{-x}}{2}$, we have that $e^x - e^{-x} > 0$ if $x > 0$ and $e^x - e^{-x} < 0$ if $x < 0$. Therefore $\sinh x > 0$ if $x > 0$ and $\sinh x < 0$ if $x < 0$.

$$\begin{aligned}
\text{16. } \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\
&= \frac{1}{4} [(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})] \\
&= \frac{1}{4}(4) = 1
\end{aligned}$$

- 17.** If $y = \cosh^{-1} x$ then $x = \cosh y$ and $x = \frac{e^y + e^{-y}}{2}$.

Also $\sinh y = \frac{e^y - e^{-y}}{2}$. Then

$$\begin{aligned}
e^y &= \cosh y + \sinh y \\
&= \cosh y + \sqrt{\sinh^2 y} \\
&= \cosh y + \sqrt{\cosh^2 y - 1} \\
&= x + \sqrt{x^2 - 1}
\end{aligned}$$

So, $y = \cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$

- 18.** If $y = \tanh^{-1} x$ then $x = \tanh y$ and $x = \frac{e^y - e^{-y}}{e^y + e^{-y}}$
Applying Componendo and Dividendo Rule,

$$\frac{1+x}{1-x} = \frac{2e^y}{2e^{-y}}$$

$$\frac{1+x}{1-x} = e^{2y}$$

$$e^{2y} = \frac{1+x}{1-x}$$

$$y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

19. $\cosh x + \sinh x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} = e^x$

20. $\cosh(-x) = \frac{e^{-x} + e^x}{2} = \cosh x$

$$\sinh(-x) = \frac{e^{-x} - e^x}{2} = -\sinh x$$

- 21.** Since e^{-x} term tend to 0 as x tend to ∞ .

$$\lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = 1,$$

$$\lim_{x \rightarrow -\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \lim_{x \rightarrow -\infty} \frac{e^{-x} - e^x}{e^{-x} + e^x} = -1$$

22. $\tanh x = \frac{\frac{e^x + e^{-x}}{2}}{\frac{e^x + e^{-x}}{2}} \cdot \frac{2e^x}{2e^x} = \frac{e^{2x} - 1}{e^{2x} + 1}$

- 23.** Given, $y = a \cosh \left(\frac{x}{b} \right)$. The hanging cable is as shown in the figure: From figure, $a = 10$ and $y = 10 \cosh \left(\frac{x}{b} \right)$. The point $B(20, 20)$ is on the catenary.

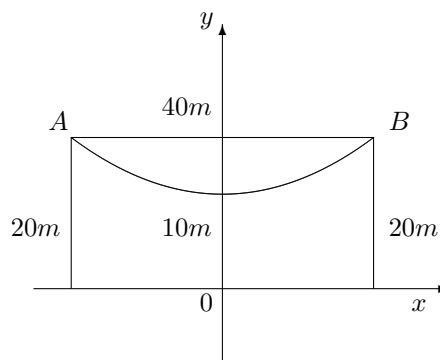
$$\Rightarrow 20 = 10 \cosh \left(\frac{20}{b} \right)$$

$$\Rightarrow 2 = \cosh \left(\frac{20}{b} \right)$$

$$\Rightarrow \frac{20}{b} = \cosh^{-1}(2) = \ln(2 + \sqrt{3})$$

$$\left[\cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1}) \right]$$

$$\Rightarrow b = \frac{20}{\ln(2 + \sqrt{3})}$$



- 24.** Given, $y = a \cosh \left(\frac{x}{b} \right)$. The hanging cable is as shown in the figure: From figure, $a = 10$ and $y = 10 \cosh \left(\frac{x}{b} \right)$. Let

$A \equiv (-x_1, 30)$ and $B \equiv (x_2, 20)$ such that
 $AB = 40$.

$$d(A, B) = \sqrt{(x_1 + x_2)^2 + 100}$$

$$1600 = (x_1 + x_2)^2 + 100$$

$$x_1 + x_2 = \sqrt{1500} \quad \dots(1)$$

The point $A(-x_1, 10)$ is on the catenary.

$$30 = 10 \cosh\left(\frac{-x_1}{b}\right)$$

$$3 = \cosh\left(\frac{x_1}{b}\right)$$

$$\Rightarrow x_1 = b \cosh^{-1}(3)$$

$$\Rightarrow x_1 = b \ln(3 + \sqrt{8})$$

The point $A(x_2, 20)$ is on the catenary.

$$20 = 10 \cosh\left(\frac{x_2}{b}\right)$$

$$2 = \cosh\left(\frac{x_2}{b}\right)$$

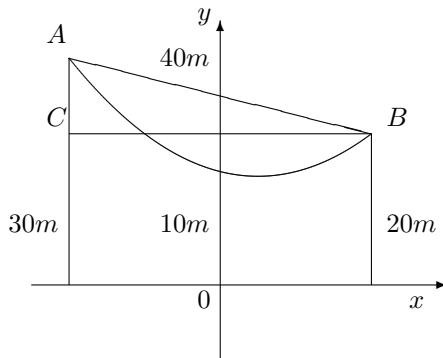
$$\Rightarrow x_2 = b \cosh^{-1}(2)$$

$$\Rightarrow x_2 = b \ln(2 + \sqrt{3})$$

By using (1),

$$b \ln[(3 + \sqrt{8})(2 + \sqrt{3})] = \sqrt{1500}$$

$$b = \frac{\sqrt{1500}}{\ln[(3 + \sqrt{8})(2 + \sqrt{3})]}$$



25. (a) Given that

$$v(t) = -\sqrt{\frac{mg}{k}} \tanh\left\{\sqrt{\frac{kg}{m}} t\right\}$$

Now, find terminal velocity(V)

$$V = \lim_{t \rightarrow \infty} v(t)$$

$$= -\sqrt{\frac{mg}{k}} \lim_{t \rightarrow \infty} \tanh\left\{\sqrt{\frac{kg}{m}} t\right\}$$

$$= -\sqrt{\frac{mg}{k}} \lim_{t \rightarrow \infty} \tanh\{ct\}$$

By putting $\sqrt{\frac{kg}{m}} = c$,

$$V = -\sqrt{\frac{mg}{k}} \lim_{t \rightarrow \infty} \frac{\sinh ct}{\cosh ct}$$

$$= -\sqrt{\frac{mg}{k}} \lim_{t \rightarrow \infty} \left(\frac{\frac{e^{ct} - e^{-ct}}{2}}{\frac{e^{ct} + e^{-ct}}{2}}\right)$$

$$= -\sqrt{\frac{mg}{k}} \lim_{t \rightarrow \infty} \frac{e^{2ct} - 1}{e^{2ct} + 1}$$

$$= -\sqrt{\frac{mg}{k}} \lim_{t \rightarrow \infty} \frac{\frac{d}{dt}\{e^{2ct} - 1\}}{\frac{d}{dt}\{e^{2ct} + 1\}}$$

By L'Hospital's rule,

$$= -\sqrt{\frac{mg}{k}} \lim_{t \rightarrow \infty} \frac{2ce^{2ct}}{2e^{2ct}}$$

$$= -\sqrt{\frac{mg}{k}} (1)$$

$$\lim_{t \rightarrow \infty} v(t) = -\sqrt{\frac{mg}{k}}$$

(b) From (a), we get

$$V = -\sqrt{\frac{mg}{k}}$$

$$V^2 = \frac{mg}{k}$$

$$mg = kV^2$$

26. For the first skydiver:

Terminal velocity is -80m/s.

Distance in 2 seconds is 19.41m.

Distance in 4 second is 75.45m.

For the second skydiver:

Terminal velocity is -40m/s.

Distance in 2 seconds is 18.86m.

Distance in 4 seconds is 68.35m.

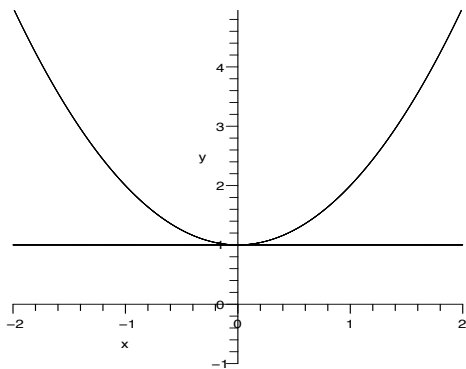
27. For an initial velocity $v_0 = 2000$, we set the derivative of the velocity equal to 0 and solve the resulting equation in a CAS. The maximum acceleration of -9.797 occurs at about 206 seconds.

2.10 The Mean Value Theorem

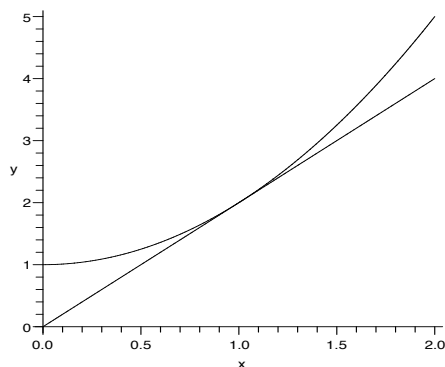
1. $f(x) = x^2 + 1, [-2, 2]$

$$f(-2) = 5 = f(2).$$

As a polynomial $f(x)$ is continuous on $[-2, 2]$, differentiable on $(-2, 2)$, and the condition's of Roll's Theorem hold. There exists $c \in (-2, 2)$ such that $f'(c) = 0$. But $f'(c) = 2c \Rightarrow c = 0$



2. $f(x) = x^2 + 1, [0, 2]$
 $f(x)$ is continuous on $[0, 2]$ and differentiable on $(0, 2)$, so the conditions of the Mean Value Theorem hold. We need to find c so that
- $$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{5 - 1}{2 - 0} = 2.$$
- $f'(x) = 2x = 2$ when $x = 1$, so $c = 1$.



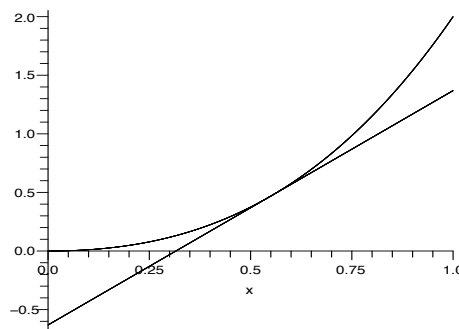
3. $f(x) = x^3 + x^2$ on $[0, 1]$ with $f(0) = 0$, $f(1) = 2$. As a polynomial, $f(x)$ is continuous on $[0, 1]$ and differentiable on $(0, 1)$. Since the conditions of the Mean Value Theorem hold, there exists a number $c \in (0, 1)$ such that
- $$f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{2 - 0}{1 - 0} = 2.$$
- But $f'(c) = 3c^2 + 2c$.
 $\Rightarrow 3c^2 + 2c = 2 \Rightarrow 3c^2 + 2c - 2 = 0$.
 By the quadratic formula,

$$c = \frac{-2 \pm \sqrt{2^2 - 4(3)(-2)}}{2(3)}$$

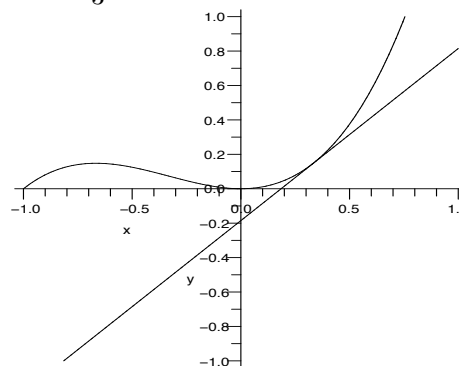
$$= \frac{-2 \pm \sqrt{28}}{6}$$

$$= \frac{-2 \pm 2\sqrt{7}}{6} = \frac{-1 \pm \sqrt{7}}{3}$$

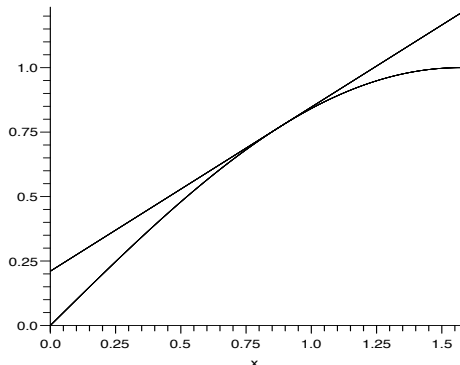
$$\Rightarrow c \approx -1.22 \text{ or } c \approx 0.55$$
 But since $-1.22 \notin (0, 1)$, we accept only the other alternatives: $c = \frac{-1 \pm \sqrt{7}}{3} \approx 0.55$



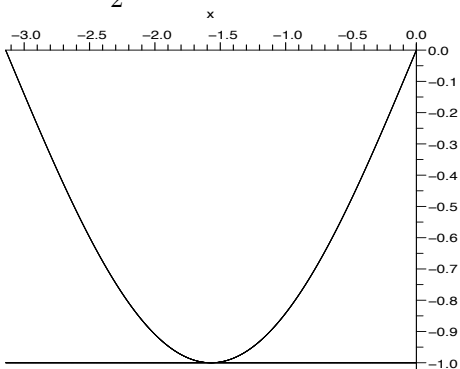
4. $f(x) = x^3 + x^2$ on $[-1, 1]$
 $f(x)$ is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$. So the conditions of the Mean Value Theorem hold. We need to find c so that
- $$f'(c) = \frac{f(1) - f(-1)}{1 - (-1)} = \frac{2 - 0}{2} = 1.$$
- $f'(x) = 3x^2 + 2x = 1$ when $x = -1$ or $x = \frac{1}{3}$,
 so $c = \frac{1}{3}$



5. $f(x) = \sin x, [0, \pi/2]$,
 $f(0) = 0, f(\pi/2) = 1$.
 As a trig function, $f(x)$ is continuous on $[0, \pi/2]$ and differentiable on $(0, \pi/2)$. The conditions of the Mean Value Theorem hold, and there exists $c \in (0, \pi/2)$ such that
- $$f'(c) = \frac{f(\frac{\pi}{2}) - f(0)}{\frac{\pi}{2} - 0}$$
- $$= \frac{1 - 0}{\frac{\pi}{2} - 0} = \frac{2}{\pi}$$
- But $f'(c) = \cos(c)$, and c is to be in the first quadrant, therefore $c = \cos^{-1}(\frac{2}{\pi}) \approx .88$



6. $f(x) = \sin x$, $[-\pi, 0]$
 $f(x)$ is continuous on $[-\pi, 0]$, and differentiable on $(-\pi, 0)$. Also, $\sin(-\pi) = 0 = \sin(0)$. So the conditions of Roll's Theorem hold. We need to find c so that $f'(c) = 0$.
 $f'(x) = \cos x = 0$, on $(-\pi, 0)$ when $x = -\frac{\pi}{2}$,
 so $c = -\frac{\pi}{2}$.



7. Let $f(x) = x^3 + 5x + 1$. As a polynomial, $f(x)$ is continuous and differentiable, for all x , with $f'(x) = 3x^2 + 5$, which is positive for all x . So $f(x)$ is strictly increasing for all x . Therefore the equation can have at most one solution. Since $f(x)$ is negative at $x = -1$ and positive at $x = 1$ and $f(x)$ is continuous, there must be a solution to $f(x) = 0$.
8. The derivative is $3x^2 + 4 > 0$ for all x . Therefore the function is strictly increasing, and so the equation can have at most one solution. Because the function is negative at $x = 0$ and positive at $x = 1$, and continuous, we know the equation has exactly one solution.
9. Let $f(x) = x^4 + 3x^2 - 2$. The derivative is $f'(x) = 4x^3 + 6x$. This is negative for negative x , and positive for positive x so $f(x)$ strictly decreasing on $(-\infty, 0)$ and strictly increasing on $(0, \infty)$. Since $f(0) = -2 \neq 0$, $f(x)$ can have at most one zero for $x < 0$

and one zero for $x > 0$. The function is continuous everywhere and $f(-1) = 2 = f(1)$, $f(0) < 0$. Therefore $f(x) = 0$ has exactly one solution between $x = -1$ and $x = 0$, and $f(x) = 0$ has exactly one solution between $x = 0$ and $x = 1$, and no other solutions.

10. Let $f(x) = x^4 + 6x^2 - 1$. The derivative is $f'(x) = 4x^3 + 12x$. This is negative for negative x , and positive for positive x so $f(x)$ is strictly decreasing on $(-\infty, 0)$ and strictly increasing on $(0, \infty)$. Since $f(0) = -1 \neq 0$, $f(x)$ can have at most one zero for $x < 0$ and one zero for $x > 0$. The function is continuous everywhere and $f(-1) = 6 = f(1)$, $f(0) < 0$. Therefore $f(x) = 0$ has exactly one solution between $x = -1$ and $x = 0$, exactly one solution between $x = 0$ and $x = 1$, and no other solutions.
11. $f(x) = x^3 + ax + b$, $a > 0$. Any cubic (actually any odd degree) polynomial heads in opposite directions ($\pm\infty$) as x goes to the oppositely signed infinities, and therefore by the Intermediate Value Theorem $f(x)$ has at least one root. For the uniqueness, we look at the derivative, in this case $3x^2 + a$. Because $a > 0$ by assumption, this expression is strictly positive. The function is strictly increasing and can have at most one root. Hence $f(x)$ has exactly one root.
12. The derivative is $f'(x) = 4x^3 + 2ax$. This is negative for negative x , and positive for positive x so $f(x)$ is strictly decreasing on $(-\infty, 0)$ and strictly increasing on $(0, \infty)$. So can have at most one zero for $x < 0$ and one zero for $x > 0$. The function is continuous everywhere and $f(0) = -b < 0$ and $\lim_{x \rightarrow \pm\infty} f(x) = \infty$, therefore $f(x)$ has exactly one solution for $x < 0$, and similarly exactly one solution for $x > 0$, and no other solutions.
13. $f(x) = x^5 + ax^3 + bx + c$, $a > 0$, $b > 0$. Here is another odd degree polynomial (see #11) with at least one root. $f'(x) = 5x^4 + 3ax^2 + b$ is evidently strictly positive because of our assumption about a , b . Exactly as in #11, $f(x)$ has exactly one root.
14. A third degree polynomial $p(x)$ has at least one zero because

$$\lim_{x \rightarrow \pm\infty} p(x) = \pm \lim_{x \rightarrow \infty} p(x) = \pm\infty,$$

and it is continuous. Say this zero is at $x = c$. Then we know $p(x)$ factors into

$p(x) = (x-c)q(x)$, where $q(x)$ is a quadratic polynomial. Quadratic polynomials have at most two zeros so $p(x)$ has at most three zeros.

15. $f(x) = x^2$.

One candidate: $g_0(x) = kx^3$.

Because we require $x^2 = g'_0(x) = 3kx^2$, we must have $3k = 1$, $k = 1/3$.

Most general solution: $g(x) = g_0(x) + c = x^3/3 + c$, where c is an arbitrary constant.

16. If $g'(x) = 9x^4$, then $g(x) = \frac{9}{5}x^5 + c$ for any constant c .

17. Although the obvious first candidate is $g_0(x) = -1/x$, due to disconnection of the domain by the discontinuity at $x = 0$, we could add different constants, one for negative x , another for positive x . Thus the most general solution is:

$$g(x) = \begin{cases} -1/x + a & \text{when } x > 0 \\ -1/x + b & \text{when } x < 0 \end{cases}$$

18. If $g'(x) = \sqrt{x}$, then $g(x) = \frac{2}{3}x^{3/2} + c$ for any constant c .

19. If $g'(x) = \sin x$, then $g(x) = -\cos x + c$ for any constant c .

20. If $g'(x) = \cos x$, then $g(x) = \sin x + c$ for any constant c .

21. If $g'(x) = \frac{4}{1+x^2}$ then $g(x) = 4\tan^{-1}(x) + c$.

22. If $g'(x) = \frac{2}{\sqrt{1-x^2}}$ then $g(x) = 2\sin^{-1}(x) + c$.

23. If derivative $g'(x)$ is positive at a single point $x = b$, then $g(x)$ is an increasing function for x sufficiently near b , i.e., $g(x) > g(b)$ for $x > b$ but sufficiently near b . In this problem, we will apply that remark to f' at $x = 0$, and conclude from $f''(0) > 0$ that $f'(x) > f'(0) = 0$ for $x > 0$ but sufficiently small. This being true about the derivative f' , it tells us that f itself is increasing on some interval $(0, a)$ and in particular that $f(x) > f(0) = 0$ for $0 < x < a$. On the other side (the negative side) f' is negative, f is decreasing (to zero) and therefore likewise positive. In summary, $x = 0$ is a genuine relative minimum.

24. The function $\cos x$ is continuous and differentiable everywhere, so for any u and v we can apply the Mean Value Theorem to get

$\frac{\cos u - \cos v}{u - v} = \sin c$ for some c between u and v . We know $-1 \leq \sin x \leq 1$, so taking absolute values, we get $\left| \frac{\cos u - \cos v}{u - v} \right| \leq 1$, or $|\cos u - \cos v| \leq |u - v|$.

25. Consider the function $g(x) = x - \sin x$, obviously with $g(0) = 0$ and $g'(x) = 1 - \cos x$. If there was ever point $a > 0$ with $\sin(a) \geq a$, ($g(a) \leq 0$), then by the MVT applied to g on the interval $[0, a]$, there would be a point c ($0 < c < a$) with $g'(c) = \frac{g(a) - g(0)}{a - 0}$

$$= \frac{g(a)}{a} \leq 0.$$

This would read $1 - \cos c = g'(c) \leq 0$ or $\cos c \geq 1$. The latter condition is possible only if $\cos(c) = 1$ and $\sin(c) = 0$, in which case c (being positive) would be at minimum π . But even this unlikely case we still would have $\sin(a) \leq 1 < \pi \leq c < a$.

Since $\sin a < a$ for all $a > 0$, we have $-\sin a > -a$ for all $a > 0$, but $-\sin a = \sin(-a)$ so we have $\sin(-a) > -a$ for all $a > 0$. This is the same as saying $\sin a > a$ for all $a < 0$ so in absolute value we have $|\sin a| < |a|$ for all $a \neq 0$.

Thus the only possible solution to the equation $\sin x = x$ is $x = 0$, which we know to be true.

26. The function $\tan^{-1}x$ is continuous and differentiable everywhere, so for any $a \neq 0$ we can apply the Mean Value Theorem to get $\frac{\tan^{-1}a - \tan^{-1}0}{a - 0} = \frac{1}{1+c^2}$ for some c between 0 and a . Taking absolute values, we get $\left| \frac{\tan^{-1}a}{a} \right| = \frac{1}{1+c^2} < 1$, so $|\tan^{-1}a| < |a|$ for $a \neq 0$. This means that the only solution to $\tan^{-1}x = x$ is $x = 0$.

27. Since the inverse sine function is increasing on the interval $[0, 1)$ (it has positive derivative) we start from the previously proven inequality $\sin x < x$ for $0 < x$. If indeed $0 < x < 1$, we can apply the inverse sine and conclude $x = \sin^{-1}(\sin x) < \sin^{-1}(x)$.

28. The function $\tan x$ is continuous and differentiable for $|x| < \pi/2$, so for any $a \neq 0$ in $(-\pi/2, \pi/2)$, we can apply the Mean Value Theorem to get $\frac{\tan a - \tan 0}{a - 0} = \sec^2 c$ for some c between 0 and a . Taking absolute values, we get $\left| \frac{\tan a}{a} \right| = |\sec^2 c| > 1$, so

$|\tan a| > |a|$ for $a \neq 0$. Of course $\tan 0 = 0$, so $|\tan a| \geq |a|$ for all $|a| < \pi/2$.

29. If $f'(x) > 0$ for all x then for each (a, b) with $a < b$ we know there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c) > 0.$$

$a < b$ makes the denominator positive, and so we must have the numerator also positive, which implies $f(a) < f(b)$.

30. Let $a < b$. f is differentiable on (a, b) and continuous on $[a, b]$, since it is differentiable for all x . This means that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

for some $c \in (a, b)$. Therefore $f(b) - f(a) = f'(c)(b - a)$ is negative, and $f(a) > f(b)$.

31. $f'(x) = 3x^2 + 5$. This is positive for all x , so $f(x)$ is increasing.
32. $f'(x) = 5x^4 + 9x^2 \geq 0$ for all x . $f' = 0$ only at $x = 0$, so $f(x)$ is increasing.
33. $f'(x) = -3x^2 - 3$. This is negative for all x , so $f(x)$ is decreasing.
34. $f'(x) = 4x^3 + 4x$ is negative for negative x , and positive for positive x , so $f(x)$ is neither an increasing function nor a decreasing function.
35. $f'(x) = e^x$. This is positive for all x , so $f(x)$ is increasing.
36. $f'(x) = -e^{-x} < 0$ for all x , so $f(x)$ is a decreasing function.
37. $f'(x) = \frac{1}{x}$
 $f'(x) > 0$ for $x > 0$, that is, for all x in the domain of f . So $f(x)$ is increasing.
38. $f'(x) = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$ is negative for negative x , and positive for positive x , so $f(x)$ is neither an increasing function nor a decreasing function.

39. The average velocity on $[a, b]$ is $\frac{s(b) - s(a)}{b - a}$. By the Mean Value Theorem, there exists a $c \in (a, b)$ such that $s'(c) = \frac{s(b) - s(a)}{b - a}$. Thus, the instantaneous velocity at $t = c$ is equal to the average velocity between times $t = a$ and $t = b$.

40. Let $f(t)$ be the distance the first runner has gone after time t and let $g(t)$ be the distance the second runner has gone after time t . The functions $f(t)$ and $g(t)$ will be continuous and differentiable. Let $h(t) = f(t) - g(t)$. At $t = 0$, $f(0) = 0$ and $g(0) = 0$ so $h(0) = 0$. At $t = a$, $f(a) > g(a)$ so $h(a) > 0$. Similarly, at $t = b$, $f(b) < g(b)$ so $h(b) < 0$. Thus, by the Intermediate Value Theorem, there is time $t = t_0$ for $t_0 \in (a, b)$ where $h(t_0) = 0$. Rolle's Theorem then says that there is time $t = c$ where $c \in (0, t_0)$ such that $h'(c) = 0$. But $h'(t) = f'(t) - g'(t)$, so $h'(c) = f'(c) - g'(c) = 0$ implies that $f'(c) = g'(c)$, i.e., at time $t = c$ the runners are going exactly the same speed.

41. Define $h(x) = f(x) - g(x)$. Then h is differentiable because f and g are, and $h(a) = h(b) = 0$. Apply Rolle's Theorem to h on $[a, b]$ to conclude that there exists $c \in (a, b)$ such that $h'(c) = 0$. Thus, $f'(c) = g'(c)$, and so f and g have parallel tangent lines at $x = c$.

42. As in #41, let $h(x) = f(x) - g(x)$. Again, h is continuous and differentiable on the appropriate intervals because f and g are. Since $f(a) - f(b) = g(a) - g(b)$ (by assumption), we have $f(a) = g(a) - g(b) + f(b)$. Then,

$$\begin{aligned} h(a) &= f(a) - g(a) \\ &= g(a) - g(b) + f(b) - g(a) \\ &= f(b) - g(b) = h(b). \end{aligned}$$

Rolle's Theorem then tells us that there exists $c \in (a, b)$ such that $h'(c) = 0$ or $f'(c) = g'(c)$ so that f and g have parallel tangent lines at $x = c$.

43. $f(x) = 1/x$ on $[-1, 1]$. We easily see that $f(1) = 1$, $f(-1) = -1$, and $f'(x) = -1/x^2$. If we try to find the c in the interval $(-1, 1)$ for which

$$f'(c) = \frac{f(1) - f(-1)}{1 - (-1)} = \frac{1 - (-1)}{1 - (-1)} = 1,$$

the equation would be $-1/c^2 = 1$ or $c^2 = -1$. There is of course no such c , and the explanation is that the function is not defined for $x = 0 \in (-1, 1)$ and so the function is not continuous.

The hypotheses for the Mean Value Theorem are not fulfilled.

44. $f(x)$ is not continuous on $[-1, 2]$, and not differentiable on $(-1, 2)$. Can we find

$$f'(c) = \frac{f(2) - f(-1)}{2 - (-1)} = \frac{\frac{1}{4} - 1}{3} = -\frac{1}{4}?$$

$f'(x) = -\frac{2}{x^3} = -\frac{1}{4}$ when $x = 2$. This is not in $(-1, 2)$, so no c makes the conclusion of Mean Value Theorem true.

45. $f(x) = \tan x$ on $[0, \pi]$, $f'(x) = \sec^2 x$. We know the tangent has a massive discontinuity at $x = \pi/2$, so as in #44, we should not be surprised if the Mean Value Theorem does not apply. As applied to the interval $[0, \pi]$ it would say

$$\begin{aligned}\sec^2 c = f'(c) &= \frac{f(\pi) - f(0)}{\pi - 0} \\ &= \frac{\tan \pi - \tan 0}{\pi - 0} = 0.\end{aligned}$$

But secant $= 1/\cosine$ is never 0 in the interval $(-1, 1)$, so no such c exists.

46. $f(x)$ is not differentiable on $(-1, 1)$. Can we find c with

$$f'(c) = \frac{f(1) - f(-1)}{1 - (-1)} = \frac{1 - (-1)}{2} = 1?$$

$f'(x) = \frac{1}{3}x^{-2/3} = 1$ when $x = \pm(\frac{1}{3})^{3/2}$. These are both in $(-1, 1)$, so we can use either of these as c to make the conclusion of Mean Value Theorem true.

47. $f(x) = \begin{cases} 2x & \text{when } x \leq 0 \\ 2x - 4 & \text{when } x > 0 \end{cases}$
 $f(x) = 2x - 4$ is continuous and differentiable on $(0, 2)$. Also, $f(0) = 0 = f(2)$. But $f'(x) = 2$ on $(0, 2)$, so there is no c such that $f'(c) = 0$. Rolle's Theorem requires that $f(x)$ be continuous on the closed interval, but we have a jump discontinuity at $x = 0$, which is enough to preclude the applicability of Rolle's.

48. $f(x) = x^2$ is counter-example. The flaw in the proof is that we do not have $f'(c) = 0$.

Ch. 2 Review Exercises

1. $\frac{3.4 - 2.6}{1.5 - 0.5} = \frac{0.8}{1} = 0.8$

2. C (large negative), B (small negative), A (small positive), and D (large positive)

3. $f'(2) = \frac{f(2+h) - f(2)}{h}$
 $= \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2(2+h) - (0)}{h}$

$$\begin{aligned}&= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4 - 2h}{h} \\ &= \lim_{h \rightarrow 0} \frac{2h + h^2}{h} \\ &= \lim_{h \rightarrow 0} 2 + h = 2\end{aligned}$$

4. $f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$
 $= \lim_{x \rightarrow 1} \frac{1 + \frac{1}{x} - 2}{x - 1}$
 $= \lim_{x \rightarrow 1} \frac{\frac{x - 1}{x}}{x - 1}$
 $= \lim_{x \rightarrow 1} \frac{-1}{x} = -1$

5. $f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$
 $= \lim_{h \rightarrow 0} \frac{\sqrt{1+h} - 1}{h}$
 $= \lim_{h \rightarrow 0} \frac{\sqrt{1+h} - 1}{h} \cdot \frac{\sqrt{1+h} + 1}{\sqrt{1+h} + 1}$
 $= \lim_{h \rightarrow 0} \frac{1 + h - 1}{h(\sqrt{1+h} + 1)}$
 $= \lim_{h \rightarrow 0} \frac{1}{\sqrt{1+h} + 1} = \frac{1}{2}$

6. $f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$
 $= \lim_{x \rightarrow 0} \frac{x^3 - 2x}{x}$
 $= \lim_{x \rightarrow 0} x^2 - 2 = -2$

7. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{(x+h)^3 + (x+h) - (x^3 + x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 + h}{h}$
 $= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 + 1$
 $= 3x^2 + 1$

8. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{\frac{3}{x+h} - \frac{3}{x}}{h}$
 $= \lim_{h \rightarrow 0} \frac{\frac{3x - 3(x+h)}{x(x+h)}}{h}$
 $= \lim_{h \rightarrow 0} \frac{\frac{-3h}{x(x+h)}}{h}$
 $= \lim_{h \rightarrow 0} \frac{-3}{x(x+h)} = \frac{-3}{x^2}$

9. The point is $(1, 0)$. $y' = 4x^3 - 2$ so the slope at $x = 1$ is 2, and the equation of the tangent line is $y - 0 = 2(x - 1)$ or $y = 2x - 2$.

10. The point is $(0, 0)$. $y' = 2 \cos 2x$, so the slope at $x = 0$ is 2, and the equation of the tangent line is $y = 2x$.

11. The point is $(0, 3)$. $y' = 6e^{2x}$, so the slope at $x = 0$ is 6, and the equation of the tangent line is $y - 3 = 6(x - 0)$ or $y = 6x + 3$.

12. The point is $(0, 1)$. $y' = \frac{2x}{2\sqrt{x^2 + 1}}$, so the slope at $x = 0$ is 0, and the equation of the tangent line is $y = 1$.

13. Find the slope to $y - x^2y^2 = x - 1$ at $(1, 1)$.

$$\frac{d}{dx}(y - x^2y^2) = \frac{d}{dx}(x - 1)$$

$$y' - 2xy^2 - x^2 \cdot 2y \cdot y' = 1$$

$$y'(1 - 2x^2y) = 1 + 2xy^2$$

$$y' = \frac{1 + 2xy^2}{1 - 2x^2y}$$

At $(1, 1)$:

$$y' = \frac{1 + 2(1)(1)^2}{1 - 2(1)^2(1)} = \frac{3}{-1} = -3$$

The equation of the tangent line is

$$y - 1 = -3(x - 1) \text{ or } y = -3x + 4.$$

14. Implicitly differentiating:

$$2yy' + e^y + xe^y y' = -1, \text{ and}$$

$$y' = \frac{-1 - e^y}{2y + xe^y}.$$

At $(2, 0)$ the slope is -1 , and the equation of the tangent line is $y = -(x - 2)$.

15. $s(t) = -16t^2 + 40t + 10$

$$v(t) = s'(t) = -32t + 40$$

$$a(t) = v'(t) = -32$$

16. $s(t) = -9.8t^2 - 22t + 6$

$$v(t) = s'(t) = -19.6t - 22$$

$$a(t) = s''(t) = -19.6$$

17. $s(t) = 10e^{-2t} \sin 4t$

$$v(t) = s'(t)$$

$$= 10(-2e^{-2t} \sin 4t + 4e^{-2t} \cos 4t)$$

$$a(t) = v'(t)$$

$$= 10 \cdot (-2) [-2e^{-2t} \sin 4t + e^{-2t} 4 \cos 4t]$$

$$+ 10(4) \cdot [-2e^{-2t} \cos 4t - e^{-2t} 4 \sin 4t]$$

$$= 160e^{-2t} \cos 4t - 120e^{-2t} \sin 4t$$

18. $s(t) = \sqrt{4t + 16} - 4$

$$v(t) = s'(t) = \frac{4}{2\sqrt{4t + 16}}$$

$$= \frac{2}{\sqrt{4t + 16}}$$

$$a(t) = s''(t)$$

$$= \frac{-2 \cdot 4}{2(4t + 16)^{3/2}} = \frac{-4}{(4t + 16)^{3/2}}$$

19. $v(t) = s'(t) = -32t + 40$

$$v(1) = -32(1) + 40 = 8$$

The ball is rising.

$$v(2) = -32(2) + 40 = -24$$

The ball is falling.

20. $v(t) = s'(t) = 20e^{-2t}(2 \cos 4t - \sin 4t)$

$v(0) = 40$ and $v(\pi) = 40e^{-2\pi} \approx 0.075$. The mass attached to the spring is moving in the same direction, much faster at $t = 0$.

$$\begin{aligned} 21. \text{ (a) } m_{\text{sec}} &= \frac{f(2) - f(1)}{2 - 1} \\ &= \frac{\sqrt{3} - \sqrt{2}}{1} \approx .318 \end{aligned}$$

$$\begin{aligned} \text{ (b) } m_{\text{sec}} &= \frac{f(1.5) - f(1)}{1.5 - 1} \\ &= \frac{\sqrt{2.5} - \sqrt{2}}{.5} \approx .334 \end{aligned}$$

$$\begin{aligned} \text{ (c) } m_{\text{sec}} &= \frac{f(1.1) - f(1)}{1.1 - 1} \\ &= \frac{\sqrt{2.1} - \sqrt{2}}{.1} \approx .349 \end{aligned}$$

Best estimate for the slope of the tangent line: (c) (approximately .349).

22. Point at $x = 1$ is $(1, 7.3891)$.

$$\begin{aligned} \text{ (a) } m_{\text{sec}} &= \frac{f(2) - f(1)}{2 - 1} \\ &= \frac{e^4 - e^2}{1} \approx 47.2091 \end{aligned}$$

$$\begin{aligned} \text{ (b) } m_{\text{sec}} &= \frac{f(1.5) - f(1)}{1.5 - 1} \\ &= \frac{e^3 - e^2}{.5} \approx 25.3928 \end{aligned}$$

$$\begin{aligned} \text{ (c) } m_{\text{sec}} &= \frac{f(1.1) - f(1)}{1.1 - 1} \\ &= \frac{e^{2.2} - e^2}{.1} \approx 16.3590 \end{aligned}$$

Best estimate for the slope of the tangent line: (c) (approximately 16.3590).

23. $f'(x) = 4x^3 - 9x^2 + 2$

$$24. f'(x) = \frac{2}{3}x^{-1/3} - 8x$$

$$\begin{aligned} 25. f'(x) &= -\frac{3}{2}x^{-3/2} - 10x^{-3} \\ &= \frac{-3}{2x\sqrt{x}} - \frac{10}{x^3} \end{aligned}$$

$$\begin{aligned} 26. f'(x) &= \frac{\sqrt{x}(-3 + 2x)}{(2 - 3x + x^2)^{\frac{1}{2\sqrt{x}}}} \\ &\quad - \frac{x}{(2 - 3x + x^2)^{\frac{1}{2\sqrt{x}}}} \end{aligned}$$

$$\begin{aligned} 27. f'(t) &= 2t(t+2)^3 + t^2 \cdot 3(t+2)^2 \cdot 1 \\ &= 2t(t+2)^3 + 3t^2(t+2)^2 \\ &= t(t+2)^2(5t+4) \end{aligned}$$

$$28. f'(t) = 2t(t^3 - 3t + 2) + (t^2 + 1)(3t^2 - 3)$$

$$\begin{aligned}
 29. \quad g'(x) &= \frac{(3x^2 - 1) \cdot 1 - x(6x)}{(3x^2 - 1)^2} \\
 &= \frac{3x^2 - 1 - 6x^2}{(3x^2 - 1)^2} \\
 &= -\frac{3x^2 + 1}{(3x^2 - 1)^2}
 \end{aligned}$$

$$\begin{aligned}
 30. \quad g(x) &= 3x - \frac{1}{x} \\
 g'(x) &= 3 + \frac{1}{x^2}
 \end{aligned}$$

$$31. \quad f'(x) = 2x \sin x + x^2 \cos x$$

$$32. \quad f'(x) = 2x \cos x^2$$

$$33. \quad f'(x) = \sec^2 \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$$

$$34. \quad f'(x) = \frac{1}{2\sqrt{\tan x}} \sec^2 x$$

$$\begin{aligned}
 35. \quad f'(t) &= \csc t \cdot 1 + t \cdot (-\csc t \cdot \cot t) \\
 &= \csc t - t \csc t \cot t
 \end{aligned}$$

$$36. \quad f'(t) = 3 \cos 3t \cos 4t - 4 \sin 3t \sin 4t$$

$$37. \quad u'(x) = 2e^{-x^2}(-2x) = -4xe^{-x^2}$$

$$38. \quad u'(x) = 2(2e^{-x})(-2e^{-x}) = -8e^{-2x}$$

$$\begin{aligned}
 39. \quad f'(x) &= 1 \cdot \ln x^2 + x \cdot \frac{1}{x^2} \cdot 2x \\
 &= \ln x^2 + 2
 \end{aligned}$$

$$40. \quad f'(x) = \frac{1}{2\sqrt{\ln x + 1}} \cdot \frac{1}{x}$$

$$41. \quad f'(x) = \frac{1}{2} \cdot \frac{1}{\sin 4x} \cdot \cos 4x \cdot 4 = 2 \cot 4x$$

$$\begin{aligned}
 42. \quad f'(x) &= e^{\tan(x^2+1)} \cdot \sec^2(x^2+1) \cdot 2 \cdot x \\
 &= 2xe^{\tan(x^2+1)} \sec^2(x^2+1)
 \end{aligned}$$

$$\begin{aligned}
 43. \quad f'(x) &= 2 \left(\frac{x+1}{x-1} \right) \frac{d}{dx} \left(\frac{x+1}{x-1} \right) \\
 &= 2 \left(\frac{x+1}{x-1} \right) \frac{(x-1) - (x+1)}{(x-1)^2} \\
 &= 2 \left(\frac{x+1}{x-1} \right) \frac{-2}{(x-1)^2} \\
 &= \frac{-4(x+1)}{(x-1)^3}
 \end{aligned}$$

$$44. \quad f'(x) = \frac{3}{2\sqrt{3x}} e^{\sqrt{3x}}$$

$$45. \quad f'(t) = e^{4t} \cdot 1 + te^{4t} \cdot 4 = (1+4t)e^{4t}$$

$$46. \quad f'(x) = \frac{(x-1)^2 6 - 6x \cdot 2(x-1)}{(x-1)^4}$$

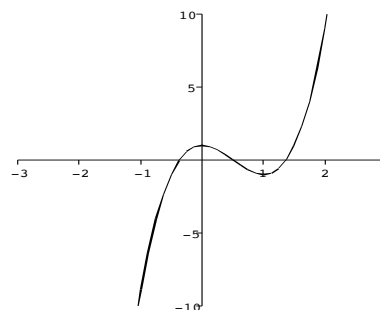
47. The given function is well defined only for $x = 0$. Hence it is not differentiable.

$$\begin{aligned}
 48. \quad f'(x) &= \cos(\cos^{-1}(x^2)) \cdot \left(\frac{-2x}{\sqrt{1-(x^2)^2}} \right) \\
 &= \frac{-2x^3}{\sqrt{1-x^4}}
 \end{aligned}$$

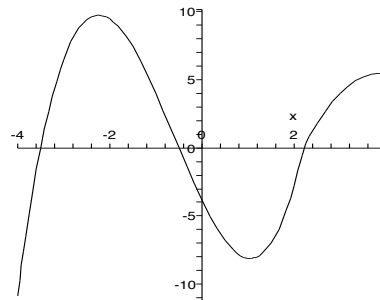
$$49. \quad \frac{1}{1 + (\cos 2x)^2} \cdot (-2 \sin 2x)$$

$$50. \quad \frac{1}{3x^2 \sqrt{(3x^2)^2 - 1}} \cdot 6x$$

51. The derivative should look roughly like:



52. The derivative should look roughly like:



$$\begin{aligned}
 53. \quad f(x) &= x^4 - 3x^3 + 2x^2 - x - 1 \\
 f'(x) &= 4x^3 - 9x^2 + 4x - 1 \\
 f''(x) &= 12x^2 - 18x + 4
 \end{aligned}$$

$$\begin{aligned}
 54. \quad f(x) &= (x+1)^{1/2} \\
 f'(x) &= \frac{1}{2}(x+1)^{-1/2} \\
 f''(x) &= \frac{-1}{4}(x+1)^{-3/2} \\
 f'''(x) &= \frac{3}{8}(x+1)^{-5/2}
 \end{aligned}$$

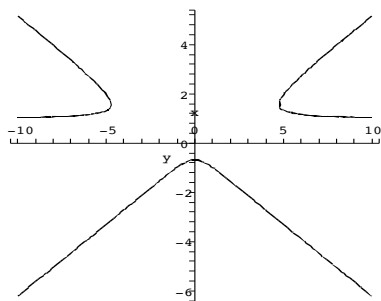
$$\begin{aligned}
 55. \quad f(x) &= xe^{2x} \\
 f'(x) &= 1 \cdot e^{2x} + xe^{2x} \cdot 2 = e^{2x} + 2xe^{2x} \\
 f''(x) &= e^{2x} \cdot 2 + 2 \cdot (e^{2x} + 2xe^{2x}) \\
 &= 4e^{2x} + 4xe^{2x} \\
 f'''(x) &= 4e^{2x} \cdot 2 + 4(e^{2x} + 2xe^{2x}) \\
 &= 12e^{2x} + 8xe^{2x}
 \end{aligned}$$

56. $f(x) = 4(x+1)^{-1}$
 $f'(x) = -4(x+1)^{-2}$
 $f''(x) = 8(x+1)^{-3}$
57. $f''(x) = 2 \cdot 2 \cdot \sec(2x) \cdot \sec(2x) \tan(2x) \cdot 2x$
 $= 8\sec^2(2x) \tan(2x)$
58. Let $f(x) = [p(x)]^2$, where
 $p(x) = x^6 - 3x^4 + 2x^3 - 7x + 1$
 $p'(x) = 6x^5 - 12x^3 + 6x^2 - 7$
 $p''(x) = 30x^4 - 36x^2 + 12x$
 $p'''(x) = 120x^3 - 72x + 12$
 $p^{(4)}(x) = 360x^2 - 72$
Then
 $f^{(4)}(x) = 6[p''(x)]^2 + 8[p'(x)][p'''(x)] + 2[p(x)][p^{(4)}(x)]$
59. $f(x) = \sin 3x$
 $f'(x) = \cos 3x \cdot 3 = 3 \cos 3x$
 $f''(x) = 3(-\sin 3x \cdot 3) = -9 \sin 3x$
 $f'''(x) = -9 \cos 3x \cdot 3 = -27 \cos 3x$
 $f^{(26)}(x) = -3^{26} \sin 3x$
60. For $f(x) = e^{-2x}$, each derivative multiplies by a factor of -2 , so
 $f^{(31)}(x) = (-2)^{31} e^{-2x}$.
61. $R(t) = P(t)Q(t)$
 $R'(t) = Q'(t) \cdot P(t) + Q(t) \cdot P'(t)$
 $P(0) = 2.4(\text{\$})$
 $Q(0) = 12$ (thousands)
 $Q'(t) = -1.5$ (thousands per year)
 $P'(t) = 0.1$ (\\$ per year)
 $R'(0) = (-1.5) \cdot (2.4) + 12 \cdot (0.1)$
 $= -2.4$ (thousand \\$ per year)
Revenue is decreasing at a rate of \\$2400 per year.
62. The relative rate of change is $\frac{v'(t)}{v(t)}$. $v'(t) = 200(\frac{3}{2})^t \ln \frac{3}{2}$, so the relative rate of change is $\ln \frac{3}{2} \approx 0.4055$, giving an instantaneous percentage rate of change of 40.55%.
63. $f(t) = 4 \cos 2t$
 $v(t) = f'(t) = 4(-\sin 2t) \cdot 2$
 $= -8 \sin 2t$
- (a) The velocity is zero when
 $v(t) = -8 \sin 2t = 0$, i.e., when
 $2t = 0, \pi, 2\pi, \dots$ so when
 $t = 0, \pi/2, \pi, 3\pi/2, \dots$
 $f(t) = 4$ for $t = 0, \pi, 2\pi, \dots$
 $f(t) = 4 \cos 2t = -4$ for
 $t = \pi/2, 3\pi/2, \dots$
The position of the spring when the velocity is zero is 4 or -4 .
- (b) The velocity is a maximum when
 $v(t) = -8 \sin 2t = 8$, i.e., when
- $2t = 3\pi/2, 7\pi/2, \dots$ so
 $t = 3\pi/4, 7\pi/4, \dots$
 $f(t) = 4 \cos 2t = 0$ for
 $t = 3\pi/4, 7\pi/4, \dots$
The position of the spring when the velocity is at a maximum is zero.
- (c) Velocity is at a minimum when
 $v(t) = -8 \sin 2t = -8$, i.e., when
 $2t = \pi/2, 5\pi/2, \dots$ so
 $t = \pi/4, 5\pi/4, \dots$
 $f(t) = 4 \cos 2t = 0$ for
 $t = \pi/4, 5\pi/4, \dots$
The position of the spring when the velocity is at a minimum is also zero.
64. The velocity is given by
 $f'(t) = -2e^{-2t} \sin 3t + 3e^{-2t} \cos 3t$.
65. $\frac{d}{dx}(x^2y - 3y^3) = \frac{d}{dx}(x^2 + 1)$
 $2xy + x^2y' - 3 \cdot 3y^2 \cdot y' = 2x$
 $y'(x^2 - 9y^2) = 2x - 2xy$
 $y' = \frac{2x(1-y)}{x^2 - 9y^2}$
66. $\frac{d}{dx}(\sin(xy) + x^2) = \frac{d}{dx}(x - y)$
 $\cos(xy)(y + xy') + 2x = 1 - y'$
 $y' = \frac{1 - 2x - y \cos(xy)}{x \cos(xy) + 1}$.
67. $\frac{d}{dx}\left(\frac{y}{x+1} - 3y\right) = \frac{d}{dx} \tan x$
 $\frac{(x+1)y' - y \cdot (1)}{(x+1)^2} - 3y' = \sec^2 x$
 $y'(x+1) - y = (x+1)^2(3y' + \sec^2 x)$
 $y' = \frac{\sec^2 x(x+1)^2 + y}{(x+1)[1 - 3(x+1)]}$
68. $\frac{d}{dx}(x - 2y^2) = \frac{d}{dx}(3e^{x/y})$
 $1 - 2yy' = 3e^{x/y} \cdot \frac{y - xy'}{y^2}$
 $1 - 2yy' = \frac{3e^{x/y}}{y} - \frac{3e^{x/y}xy'}{y^2}$
 $y' = \frac{\frac{3e^{x/y}}{y} - 1}{\frac{3xe^{x/y}}{y^2} - 2y}$
69. When $x = 0$, $-3y^3 = 1$, $y = \frac{-1}{\sqrt[3]{3}}$ (call this a).
From our formula (#65), we find $y' = 0$ at this point. To find y'' , implicitly differentiate the first derivative (second line in #65):
 $2(xy' + y) + (2xy' + x^2y'')$
 $- 9[2y(y')^2 + y^2y''] = 2$
At $(0, a)$ with $y' = 0$, we find

$$2a - 9a^2y'' = 2,$$

$$y'' = \frac{-2\sqrt[3]{3}}{9} \left(\sqrt[3]{3} + 1 \right)$$

Below is a sketch of the graph of $x^2y - 3y^3 = x^2 + 1$.



- 70.** Plugging in $x = 0$ gives $-2y = 0$ so $y = 0$. Plugging $(0, 0)$ into the formula for y' gives a slope of $-1/2$. Implicitly differentiating the third line of the solution to #37 gives

$$y''(x+1) + y' - y' = 2(x+1)(3y' + \sec^2 x) + (x+1)^2(3y'' + 2\sec x \cdot \sec x \tan x)$$

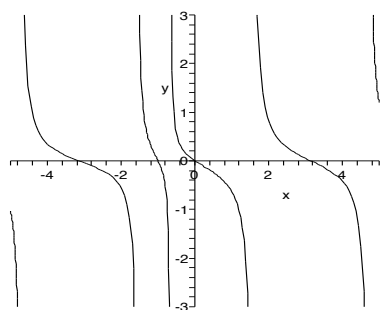
Plugging in $x = 0$, $y = 0$ and $y' = -1/2$ gives

$$y'' = 2(-3/2 + \sec^2(0)) + (1)^2(3y'' + 2\sec^2(0)\tan(0))$$

$$y'' = 1 + 3y''.$$

So at $x = 0$, $y'' = -1/2$.

The graph is:



71. $y' = 3x^2 - 12x = 3x(x - 4)$

- (a) $y' = 0$ for $x = 0$ ($y = 1$), and $x = 4$ ($y = -31$) so there are horizontal tangent lines at $(0, 1)$ and $(4, -31)$.

- (b) y' is defined for all x , so there are no vertical tangent lines.

72. $y' = \frac{2}{3}x^{-1/3}$

- (a) The derivative is never 0, so the tangent line is never horizontal.
- (b) The derivative is undefined at $x = 0$ and the tangent is vertical there.

73. $\frac{d}{dx}(x^2y - 4y) = \frac{d}{dx}x^2$

$$2xy + x^2y' - 4y' = 2x$$

$$y'(x^2 - 4) = 2x - 2xy$$

$$y' = \frac{2x - 2xy}{x^2 - 4} = \frac{2x(1 - y)}{x^2 - 4}$$

- (a) $y' = 0$ when $x = 0$ or $y = 1$.

At $y = 1$, $x^2 \cdot 1 - 4 \cdot 1 = x^2$

$$x^2 - 4 = x^2$$

This is impossible, so there is no x for which $y = 1$.

At $x = 0$, $0^2 \cdot y - 4y = 0^2$, so $y = 0$.

Therefore, there is a horizontal tangent line at $(0, 0)$.

- (b) y' is not defined when $x^2 - 4 = 0$, or $x = \pm 2$. At $x = \pm 2$, $4y - 4y = 4$ so the function is not defined at $x = \pm 2$. There are no vertical tangent lines.

74. $y' = 4x^3 - 2x = 2x(2x^2 - 1)$.

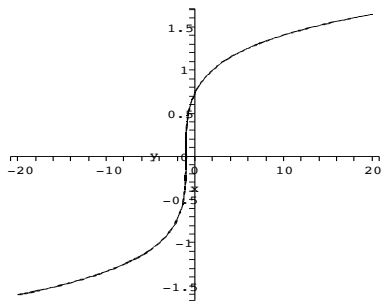
- (a) The derivative is 0 at $x = 0$ and $x = \pm\sqrt{1/2}$, and the tangent line is horizontal at those points.

- (b) The tangent line is never vertical.

- 75.** $f(x)$ is continuous and differentiable for all x , and $f'(x) = 3x^2 + 7$, which is positive for all x . By Theorem 9.2, if the equation $f(x) = 0$ has two solutions, then $f'(x) = 0$ would have at least one solution, but it has none. We discussed at length (Section 2.9) why every odd degree polynomial has at least one root, so in this case there is exactly one root.

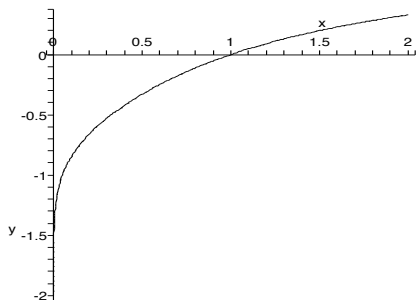
- 76.** The derivative is $5x^4 + 9x^2$. This is non-negative for all x . $f(x)$ is increasing function so can have at most one zero. Since $f(0) = -2$, $f(1) = 2$, $f(x)$ has exactly one solution.

- 77.** $f(x) = x^5 + 2x^3 - 1$ is a one-to-one function with $f(1) = 2$, $f'(1) = 11$. If g is the name of the inverse, then $g(2) = 1$ and
- $$g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(1)} = \frac{1}{11}.$$



78. Since $e^{0^3+2\cdot 0} = 1$, the derivative of the inverse at $x = 1$ will be one over the derivative of e^{x^3+2x} at $x = 0$. The derivative of e^{x^3+2x} is $(3x^2 + 2)e^{x^3+2x}$ and this is 2 when $x = 0$. Therefore the derivative of the inverse to e^{x^3+2x} at $x = 1$ is $1/2$.

The graph is the graph of e^{x^3+2x} reflected across the line $y = x$.



79. Let $a > 0$. We know that $f(x) = \cos x - 1$ is continuous and differentiable on the interval $(0, a)$. Also $f'(x) = \sin x \leq 1$ for all x . The Mean Value Theorem implies that there exists some c in the interval $(0, a)$ such that $f'(c) = \sin c$. But

$$\begin{aligned} f'(c) &= \frac{\cos a - 1 - (\cos 0 - 1)}{a - 0} \\ &= \frac{\cos a - 1}{a}. \end{aligned}$$

Since this is equal to $\sin c$ and $\sin c \leq 1$ for any c , we get that

$$\cos a - 1 \leq a$$

as desired. This works for all positive a , but since $\cos x - 1$ is symmetric about the y axis, we get

$$|\cos x - 1| \leq |x|.$$

They are actually equal at $x = 0$.

80. This is an example of a Taylor polynomial. Later, Taylor's theorem will be used to prove such inequalities. For now, one can use multiple derivatives and argue that the rate of the rate of change (etc.) increases as one moves left to right through the inequalities.

81. To show that $g(x)$ is continuous at $x = a$, we need to show that the limit as x approaches a of $g(x)$ exists and is equal to $g(a)$. But

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

which is the definition of the derivative of $f(x)$ at $x = a$. Since $f(x)$ is differentiable at $x = a$, we know this limit exists and is equal to $f'(a)$, which, in turn, is equal to $g(a)$. Thus $g(x)$ is continuous at $x = a$.

82. We have

$$\begin{aligned} f(x) - T(x) &= f(x) - f(a) - f'(a)(x - a) \\ &= \left(\frac{f(x) - f(a)}{x - a} - f'(a) \right) (x - a) \end{aligned}$$

Letting $e(x) = \frac{f(x) - f(a)}{x - a} - f'(a)$, we obtain the desired form. Since $f(x)$ is differentiable at $x = a$, we know that

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a)$$

so

$$\begin{aligned} \lim_{x \rightarrow a} e(x) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} - f'(a) \\ &= 0. \end{aligned}$$

83. $f(x) = x^2 - 2x$ on $[0, 2]$

$$f(2) = 0 = f(0)$$

$$\text{If } f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{0 - 0}{2} = 0$$

then $2c - 2 = f'(c) = 0$ so $c = 1$.

84. $f(x)$ is continuous on $[0, 2]$ and differentiable on $(0, 2)$, so the Mean Value Theorem applies. We need to find c so that

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{6 - 0}{2 - 0} = 3.$$

$$f'(x) = 3x^2 - 1 = 3 \text{ when } x = \sqrt{4/3}, \text{ so } c = 2\sqrt{3}/3.$$

85. $f(x) = 3x^2 - \cos x$

One trial: $g_o(x) = kx^3 - \sin x$

$$g'_o(x) = 3kx^2 - \cos x$$

Need $3k = 3$, $k = 1$, and the general solution is

$$g(x) = g_o(x) + c = x^3 - \sin x + c$$

for c an arbitrary constant.

86. If $g'(x) = x^3 - e^{2x}$, then $g(x)$ must be $\frac{1}{4}x^4 - \frac{1}{2}e^{2x} + c$, for any constant c .

87. $x = 1$ is to be double root of

$$\begin{aligned} f(x) &= (x^3 + 1) - [m(x - 1) + 2] \\ &= (x^3 + 1 - 2) - m(x - 1) \\ &= (x^3 - 1) - m(x - 1) \\ &= (x - 1)[x^2 + x + 1 - m] \end{aligned}$$

Let $g(x) = x^2 + x + 1 - m$. Then $x = 1$ is a *double* root of f only if $(x - 1)$ is a *factor* of g , in which case $g(1) = 0$. Therefore we require $0 = g(1) = 3 - m$ or $m = 3$. Now $g(x) = x^2 + x - 2 = (x - 1)(x + 2)$, $f(x) = (x - 1)g(x) = (x - 1)^2(x + 2)$ and $x = 1$ is a double root.

The line tangent to the curve $y = x^3 + 1$ at the point $(1, 2)$ has slope $y' = 3x^2 = 3(1) = 3 (= m)$. The equation of the tangent line is $y - 2 = 3(x - 1)$ or $y = 3x - 1 (= m(x - 1) + 2)$.

88. We are asked to find m so that

$$\begin{aligned} &x^3 + 2x - [m(x - 2) + 12] \\ &= x^3 + (2 - m)x + (2m - 12) \end{aligned}$$

has a double root. A cubic with a double root factors as

$$\begin{aligned} &(x - a)^2(x - b) \\ &= x^3 - (2a + b)x^2 + (2ab + a^2)x - a^2b. \end{aligned}$$

Equating like coefficients gives a system of equations

$$2a + b = 0,$$

$$2ab + a^2 = 2 - m, \text{ and}$$

$$-a^2b = 2m - 12.$$

The first equation gives $b = -2a$. Substituting this into the second equation gives $m = 2 + 3a^2$. Substituting these results into the third equation gives a cubic polynomial in a with zeros $a = -1$ and $a = 2$. This gives two solutions: $m = 5$ and $m = 14$.

$f'(x) = 3x^2 + 2$, so $f'(2) = 14$. The tangent line at $(2, 12)$ is $y = 14(x - 2) + 12$.

The second solution corresponds to the tangent line to $f(x)$ at $x = -1$, which happens to pass through the point $(2, 12)$.

89. Given, $f = \frac{1}{2L}\sqrt{\frac{T}{P}} \Rightarrow \frac{df}{dT} = \frac{1}{4L\sqrt{pT}}$ as T is an independent variable and p, L are constants. Tightening the string means increasing the tension, resulting in decrease in $\frac{df}{dT}$, which means there is a decrease in the rate of change of frequency with respect to the tension in the string. On the other end, loosening the string means decreasing the tension, resulting in increase in $\frac{df}{dT}$, which means there is a increase in the rate of change of frequency with respect to the tension in the string. Also, $f = \frac{1}{2L}\sqrt{\frac{T}{P}} \Rightarrow \frac{df}{dL} = -\frac{1}{2L^2}\sqrt{\frac{T}{p}}$. When the guitarist plays the notes by pressing the string against a fret; he is increasing the length and hence decreasing the rate of change of frequency of vibration with respect to the length of the string.

Chapter 3

Applications of Differentiation

3.1 Linear Approximations and Newtons Method

$$\begin{aligned} 1. \quad (a) \quad & f(x) = \sqrt{x}, x_0 = 1 \\ & f(x_0) = f(1) = \sqrt{1} = 1 \\ & f'(x) = \frac{1}{2}x^{-1/2} \end{aligned}$$

$$f'(x_0) = f'(1) = \frac{1}{2}$$

So,

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$= 1 + \frac{1}{2}(x - 1)$$

$$= \frac{1}{2} + \frac{1}{2}x$$

$$(b) \text{ Using the approximation } L(x) \text{ to estimate } \sqrt{1.2}, \text{ we get } \sqrt{1.2} = f(1.2) \approx L(1.2) = \frac{1}{2} + \frac{1}{2}(1.2) = 1.1$$

$$\begin{aligned} 2. \quad (a) \quad & f(x_0) = f(0) = 1 \text{ and} \\ & f'(x) = \frac{1}{3}(x+1)^{-2/3} \\ \text{So, } & f'(0) = \frac{1}{3} \\ \text{The Linear approximation is,} \\ & L(x) = 1 + \frac{1}{3}(x - 0) = 1 + \frac{1}{3}x \end{aligned}$$

$$(b) \text{ Using the approximation } L(x) \text{ to estimate } \sqrt[3]{1.2}, \text{ we get } \sqrt[3]{1.2} = f(0.2) \approx L(0.2) = 1 + \frac{1}{3}(0.2) = 1.066$$

$$\begin{aligned} 3. \quad (a) \quad & f(x) = \sqrt{2x+9}, x_0 = 0 \\ & f(x_0) = f(0) = \sqrt{2 \cdot 0 + 9} = 3 \\ & f'(x) = \frac{1}{2}(2x+9)^{-1/2} \cdot 2 \\ & = (2x+9)^{-1/2} \end{aligned}$$

$$f'(x_0) = f'(0) = (2 \cdot 0 + 9)^{-1/2} = \frac{1}{3}$$

So,

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$= 3 + \frac{1}{3}(x - 0)$$

$$= 3 + \frac{1}{3}x$$

$$(b) \text{ Using the approximation } L(x) \text{ to estimate } \sqrt{8.8}, \text{ we get } \sqrt{8.8} = f(-0.1) \approx L(-0.1) = 3 + \frac{1}{3}(-0.1) = 3 - 0.033 = 2.967$$

$$\begin{aligned} 4. \quad (a) \quad & f(x) = \frac{2}{x}, x_0 = 1 \\ & f(x_0) = f(1) = 2 \\ & f'(x) = -\frac{2}{x^2} \text{ and so } f'(1) = -2 \\ \text{The linear approximation is} \\ & L(x) = 2 + (-2)(x - 1) \end{aligned}$$

$$(b) \text{ Using the approximation } L(x) \text{ to estimate } \frac{2}{0.99}, \text{ we get } \frac{2}{0.99} = f(0.99) \approx L(0.99) = 2 + (-2)(0.99 - 1) = 2.02$$

$$\begin{aligned} 5. \quad (a) \quad & f(x) = \sin 3x, x_0 = 0 \\ & f(x_0) = \sin(3 \cdot 0) = 0 \\ & f'(x) = 3 \cos 3x \\ & f'(x_0) = f'(0) = 3 \cos(3 \cdot 0) = 3 \\ \text{So,} \\ & L(x) = f(x_0) + f'(x_0)(x - x_0) \\ & = 0 + 3(x - 0) \\ & = 3x \end{aligned}$$

$$(b) \text{ Using the approximation } L(x) \text{ to estimate } \sin(0.3), \text{ we get } \sin(0.3) = f(0.1) \approx L(0.1) = 3(0.1) = 0.3$$

$$\begin{aligned} 6. \quad (a) \quad & f(x) = \sin x, x_0 = \pi \\ & f(x_0) = \sin \pi = 0 \\ & f'(x) = \cos x \\ & f'(x_0) = f'(\pi) = \cos \pi = -1 \\ \text{The linear approximation is,} \\ & L(x) = f(x_0) + f'(x_0)(x - x_0) \\ & = 0 + (-1)(x - \pi) = \pi - x \end{aligned}$$

$$(b) \text{ Using the approximation } L(x) \text{ to estimate } \sin(3.0), \text{ we get } \sin(3.0) = f(3.0) \approx L(3.0) = \pi - 3.0$$

$$\begin{aligned} 7. \quad (a) \quad & f(x) = \sqrt[4]{16+x}, x_0 = 0 \\ & f(0) = \sqrt[4]{16+0} = 2 \\ & f'(x) = \frac{1}{4}(16+x)^{-3/4} \\ & f'(0) = \frac{1}{4}(16+0)^{-3/4} = \frac{1}{32} \end{aligned}$$

$$L(x) = f(0) + f'(0)(x - 0)$$

$$= 2 + \frac{1}{32}x$$

$$= 2 + \frac{1}{32}(0.04) = 2.00125$$

$$(b) \quad L(0.08) = 2 + \frac{1}{32}(0.08) = 2.0025$$

$$(c) \quad L(0.16) = 2 + \frac{1}{32}(0.16) = 2.005$$

$$8. \quad (a) \quad f(x) = \sin x, \quad x_0 = 0$$

$$f(0) = 0$$

$$f'(x) = \cos x$$

$$f'(0) = \cos 0 = 1$$

$$L(x) = f(0) + f'(0)(x - 0)$$

$$= 0 + 1 \cdot x$$

$$L(0.1) = 0.1$$

$$(b) \quad f(x) = \sin x, \quad x_0 = \frac{\pi}{3}$$

$$f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$f'\left(\frac{\pi}{3}\right) = \cos \frac{\pi}{3} = \frac{1}{2}$$

$$L(x) = f\left(\frac{\pi}{3}\right) + f'\left(\frac{\pi}{3}\right)\left(x - \frac{\pi}{3}\right)$$

$$L(1) = \frac{\sqrt{3}}{2} + \frac{1}{2}\left(1 - \frac{\pi}{3}\right) \approx 0.842$$

$$(c) \quad f(x) = \sin x, \quad x_0 = \frac{2\pi}{3}$$

$$f\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$f'\left(\frac{2\pi}{3}\right) = \cos \frac{2\pi}{3} = -\frac{1}{2}$$

$$L(x) = f\left(\frac{2\pi}{3}\right) + f'\left(\frac{2\pi}{3}\right)\left(x - \frac{2\pi}{3}\right)$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{2}\left(x - \frac{2\pi}{3}\right)$$

$$L\left(\frac{9}{4}\right) = \frac{\sqrt{3}}{2} - \frac{1}{2}\left(\frac{9}{4} - \frac{2\pi}{3}\right) \approx 0.788$$

$$9. \quad (a) \quad L(x) = f(20) + \frac{18 - 14}{20 - 30}(x - 20)$$

$$L(24) \approx 18 - \frac{4}{10}(24 - 20)$$

$$= 18 - 0.4(4)$$

$$= 16.4 \text{ games}$$

$$(b) \quad L(x) = f(40) + \frac{14 - 12}{30 - 40}(x - 40)$$

$$f(36) \approx 12 - \frac{2}{10}(36 - 40)$$

$$= 12 - 0.2(-4)$$

$$= 12.8 \text{ games}$$

$$10. \quad (a) \quad L(x) = f(80) + \frac{120 - 84}{80 - 60}(x - 80)$$

$$L(72) = 120 + \frac{36}{20}(72 - 80)$$

$$= 120 + 1.8(-8)$$

$$= 105.6 \text{ cans}$$

$$(b) \quad L(x) = f(100) + \frac{168 - 120}{100 - 80}(x - 100)$$

$$L(94) = 168 - \frac{48}{20}(94 - 100)$$

$$= 168 - 2.4(-6)$$

$$= 182.4 \text{ cans}$$

$$11. \quad (a) \quad L(x) = f(200) + \frac{142 - 128}{220 - 200}(x - 200)$$

$$L(208) = 128 + \frac{14}{20}(208 - 200)$$

$$= 128 + 0.7(8) = 133.6$$

$$(b) \quad L(x) = f(240) + \frac{142 - 136}{220 - 240}(x - 240)$$

$$L(232) = 136 - \frac{6}{20}(232 - 240)$$

$$= 136 - 0.3(-8) = 138.4$$

$$12. \quad (a) \quad L(x) = f(10) + \frac{14 - 8}{10 - 5}(x - 10)$$

$$L(8) = 14 + \frac{6}{5}(-2) = 11.6$$

$$(b) \quad L(x) = f(10) + \frac{14 - 8}{10 - 5}(x - 10)$$

$$L(12) = 14 + \frac{6}{5}(2) = 16.4$$

$$13. \quad f(x) = x^3 + 3x^2 - 1 = 0, \quad x_0 = 1$$

$$f'(x) = 3x^2 + 6x$$

$$(a) \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1 - \frac{1^3 + 3 \cdot 1^2 - 1}{3 \cdot 1^2 + 6 \cdot 1}$$

$$= 1 - \frac{3}{9} = \frac{2}{3}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= \frac{2}{3} - \frac{\left(\frac{2}{3}\right)^3 + 3\left(\frac{2}{3}\right)^2 - 1}{3\left(\frac{2}{3}\right)^2 + 6\left(\frac{2}{3}\right)}$$

$$= \frac{79}{144} \approx 0.5486$$

$$(b) \quad 0.53209$$

$$14. \quad f(x) = x^3 + 4x^2 - x - 1, \quad x_0 = -1$$

$$f'(x) = 3x^2 + 8x - 1$$

$$\begin{aligned}
 \text{(a)} \quad x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
 &= -1 - \frac{3}{-6} = -\frac{1}{2} \\
 x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= -\frac{1}{2} - \frac{0.375}{-4.25} = -0.4117647
 \end{aligned}$$

(b) The root is $x \approx -0.4064206546$.

15. $f(x) = x^4 - 3x^2 + 1 = 0$, $x_0 = 1$
 $f'(x) = 4x^3 - 6x$

$$\begin{aligned}
 \text{(a)} \quad x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
 &= 1 - \left(\frac{1^4 - 3 \cdot 1^2 + 1}{4 \cdot 1^3 - 6 \cdot 1} \right) = \frac{1}{2} \\
 x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= \frac{1}{2} - \left(\frac{\left(\frac{1}{2}\right)^4 - 3\left(\frac{1}{2}\right)^2 + 1}{4\left(\frac{1}{2}\right)^3 - 6\left(\frac{1}{2}\right)} \right) \\
 &= \frac{5}{8}
 \end{aligned}$$

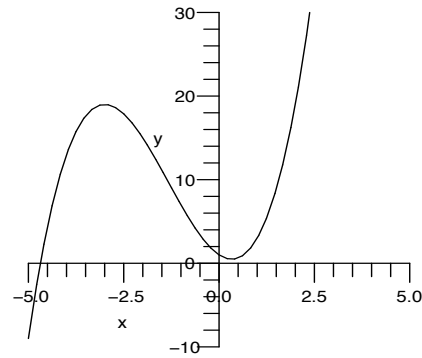
(b) 0.61803

16. $f(x) = x^4 - 3x^2 + 1$, $x_0 = -1$
 $f'(x) = 4x^3 - 6x$

$$\begin{aligned}
 \text{(a)} \quad x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
 &= -1 - \frac{-1}{2} = -\frac{1}{2} \\
 x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= -\frac{1}{2} - \frac{0.3125}{2.5} = -0.625
 \end{aligned}$$

(b) The root is $x \approx -0.6180339887$.

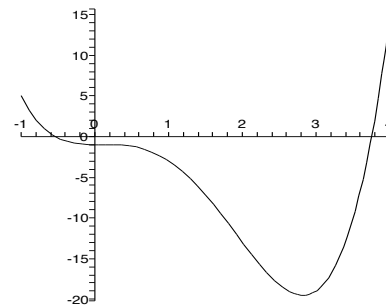
17. Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with
 $f(x) = x^3 + 4x^2 - 3x + 1$, and
 $f'(x) = 3x^2 + 8x - 3$



Start with $x_0 = -5$ to find the root near -5 :

$$\begin{aligned}
 x_1 &= -4.718750, \quad x_2 = -4.686202, \\
 x_3 &= -4.6857796, \quad x_4 = -4.6857795
 \end{aligned}$$

18. From the graph, we see two roots:



Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with

$$\begin{aligned}
 f(x) &= x^4 - 4x^3 + x^2 - 1, \text{ and} \\
 f'(x) &= 4x^3 - 12x^2 + 2x
 \end{aligned}$$

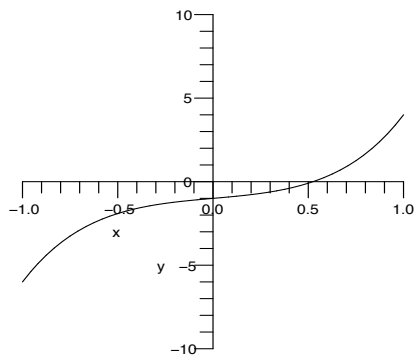
Start with $x_0 = 4$ to find the root below 4:

$$\begin{aligned}
 x_1 &= 3.791666667, \quad x_2 = 3.753630030, \quad x_3 = \\
 &= 3.7524339, \quad x_4 = 3.752432297
 \end{aligned}$$

Start with $x = -1$ to find the root just above -1 :

$$\begin{aligned}
 x_1 &= -0.7222222222, \\
 x_2 &= -0.5810217936, \\
 x_3 &= -0.5416512863, \\
 x_4 &= -0.5387668233, \\
 x_5 &= -0.5387519962
 \end{aligned}$$

19. Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with
 $f(x) = x^5 + 3x^3 + x - 1$, and
 $f'(x) = 5x^4 + 9x^2 + 1$

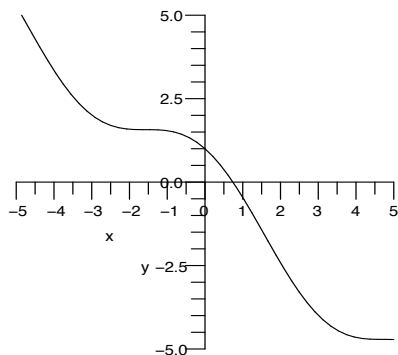


Start with $x_0 = 0.5$ to find the root near 0.5:

$$x_1 = 0.526316, x_2 = 0.525262,$$

$$x_3 = 0.525261, x_4 = 0.525261$$

20. Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with
 $f(x) = \cos x - x$, and
 $f'(x) = -\sin x - 1$

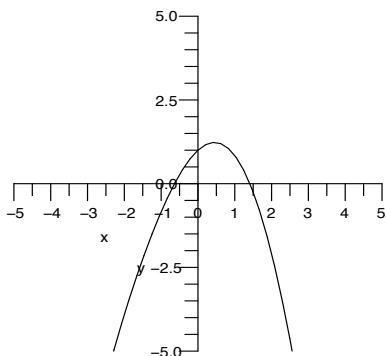


Start with $x_0 = 1$ to find the root near 1:

$$x_1 = 0.750364, x_2 = 0.739113,$$

$$x_3 = 0.739085, x_4 = 0.739085$$

21. Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with
 $f(x) = \sin x - x^2 + 1$, and
 $f'(x) = \cos x - 2x$



Start with $x_0 = -0.5$ to find the root near -0.5:

$$x_1 = -0.644108, x_2 = -0.636751$$

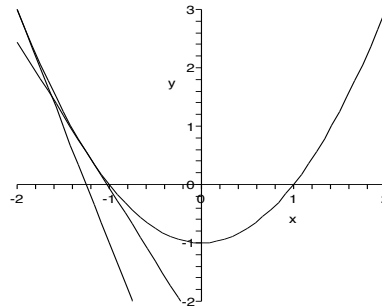
$$x_3 = -0.636733, x_4 = -0.636733$$

Start with $x_0 = 1.5$ to find the root near 1.5:

$$x_1 = 1.413799, x_2 = 1.409634$$

$$x_3 = 1.409624, x_4 = 1.409624$$

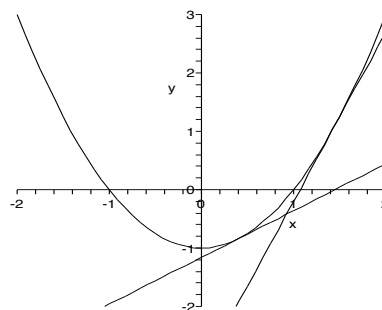
22. Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with
 $f(x) = \cos x^2 - x$, and
 $f'(x) = 2x \sin x^2 - 1$



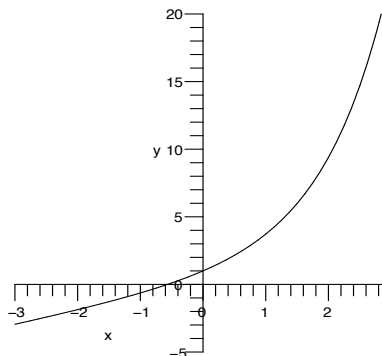
Start with $x_0 = 1$ to find the root between 0 and 1:

$$x_1 = 0.8286590991, x_2 = 0.8016918647,$$

$$x_3 = 0.8010710854, x_4 = 0.8010707652$$



23. Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with
 $f(x) = e^x + x$, and
 $f'(x) = e^x + 1$



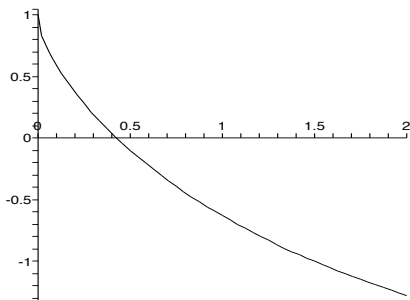
Start with $x_0 = -0.5$ to find the root between

0 and -1:

$$x_1 = -0.566311, x_2 = -0.567143$$

$$x_3 = -0.567143, x_4 = -0.567143$$

24. Use $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$ with
 $f(x) = e^{-x} - \sqrt{x}$, and
 $f'(x) = -e^{-x} - \frac{1}{2\sqrt{x}}$



Start with $x_0 = 0.5$ to find the root close to 0.5:

$$x_1 = 0.4234369253, x_2 = 0.4262982542,$$

$$x_3 = 0.4263027510$$

25. $f(x) = x^2 - 11$; $x_0 = 3$; $\sqrt{11} \approx 3.316625$
26. Newton's method for $\sqrt{x}$ near $x = 23$ is $x_{n+1} = \frac{1}{2}(x_n + 23/x_n)$. Start with $x_0 = 5$ to get:
 $x_1 = 4.8$, $x_2 = 4.7958333$, and $x_3 = 4.7958315$.
27. $f(x) = x^3 - 11$; $x_0 = 2$; $\sqrt[3]{11} \approx 2.22398$
28. Newton's method for $\sqrt[3]{x}$ near $x = 23$ is $x_{n+1} = \frac{1}{3}(2x_n + 23/x_n^2)$. Start with $x_0 = 3$ to get:
 $x_1 = 2.851851851$, $x_2 = 2.843889316$, and
 $x_3 = 2.884386698$
29. $f(x) = x^{4.4} - 24$; $x_0 = 2$; $\sqrt[4.4]{24} \approx 2.059133$
30. Newton's method for $\sqrt[4.6]{x}$ near $x = 24$ is $x_{n+1} = \frac{1}{4.6}(3.6x_n + 24/x_n^{3.6})$. Start with $x_0 = 2$ to get:
 $x_1 = 1.995417100$, $x_2 = 1.995473305$, and
 $x_3 = 1.995473304$
31. $f(x) = 4x^3 - 7x^2 + 1 = 0$, $x_0 = 0$
 $f'(x) = 12x^2 - 14x$
 $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{1}{0}$
 The method fails because $f'(x_0) = 0$. Roots are 0.3454, 0.4362, 1.659.

32. Newton's method fails because $f' = 0$. As long as the sequence avoids $x_n = 0$ and $x_n = \frac{7}{6}$ (the

zeros of f'), Newton's method will succeed. Which root is found depends on the starting place.

33. $f(x) = x^2 + 1$, $x_0 = 0$
 $f'(x) = 2x$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{1}{0}$$

The method fails because $f'(x_0) = 0$. There are no roots.

34. Newton's method fails because the function does not have a root!

35. $f(x) = \frac{4x^2 - 8x + 1}{4x^2 - 3x - 7} = 0$, $x_0 = -1$

Note: $f(x_0) = f(-1)$ is undefined, so Newton's Method fails because x_0 is not in the domain of f . Notice that $f(x) = 0$ only when $4x^2 - 8x + 1 = 0$. So using Newton's Method on $g(x) = 4x^2 - 8x + 1$ with $x_0 = -1$ leads to $x \approx .1339$. The other root is $x \approx 1.8660$.

36. The slope tends to infinity at the zero. For every starting point, the sequence does not converge.

37. (a) With $x_0 = 1.2$,
 $x_1 = 0.800000000$,
 $x_2 = 0.950000000$,
 $x_3 = 0.995652174$,
 $x_4 = 0.999962680$,
 $x_5 = 0.999999997$,
 $x_6 = 1.000000000$,
 $x_7 = 1.000000000$

- (b) With $x_0 = 2.2$,
 $x_0 = 2.200000$, $x_1 = 2.107692$,
 $x_2 = 2.056342$, $x_3 = 2.028903$,
 $x_4 = 2.014652$, $x_5 = 2.007378$,
 $x_6 = 2.003703$, $x_7 = 2.001855$,
 $x_8 = 2.000928$, $x_9 = 2.000464$,
 $x_{10} = 2.000232$, $x_{11} = 2.000116$,
 $x_{12} = 2.000058$, $x_{13} = 2.000029$,
 $x_{14} = 2.000015$, $x_{15} = 2.000007$,
 $x_{16} = 2.000004$, $x_{17} = 2.000002$,
 $x_{18} = 2.000001$, $x_{19} = 2.000000$,
 $x_{20} = 2.000000$
 The convergence is much faster with $x_0 = 1.2$.

38. Starting with $x_0 = 0.2$ we get a sequence that converges to 0 very slowly. (The 20th iteration is still not accurate past 7 decimal places). Starting with $x_0 = 3$ we get a sequence that quickly converges to π . (The third iteration is already accurate to 10 decimal places!)

39. (a) With
- $x_0 = -1.1$

$x_1 = -1.0507937,$
 $x_2 = -1.0256065,$
 $x_3 = -1.0128572,$
 $x_4 = -1.0064423,$
 $x_5 = -1.0032246,$
 $x_6 = -1.0016132,$
 $x_7 = -1.0008068,$
 $x_8 = -1.0004035,$
 $x_9 = -1.0002017,$
 $x_{10} = -1.0001009,$
 $x_{11} = -1.0000504,$
 $x_{12} = -1.0000252,$
 $x_{13} = -1.0000126,$
 $x_{14} = -1.0000063,$
 $x_{15} = -1.0000032,$
 $x_{16} = -1.0000016,$
 $x_{17} = -1.0000008,$
 $x_{18} = -1.0000004,$
 $x_{19} = -1.0000002,$
 $x_{20} = -1.0000001,$
 $x_{21} = -1.0000000,$
 $x_{22} = -1.0000000$

- (b) With
- $x_0 = 2.1$

$x_0 = 2.100000000,$
 $x_1 = 2.006060606,$
 $x_2 = 2.000024340,$
 $x_3 = 2.000000000,$
 $x_4 = 2.000000000$

The rate of convergence in (a) is slower than the rate of convergence in (b).

40. From exercise 37, $f(x) = (x-1)(x-2)^2$. Newton's method converges slowly near the double root. From exercise 39, $f(x) = (x-2)(x+1)^2$. Newton's method again converges slowly near the double root. In exercise 38, Newton's method converges slowly near 0, which is a zero of both x and $\sin x$ but converges quickly near π , which is a zero only of $\sin x$.

41. $f(x) = \tan x$, $f(0) = \tan 0 = 0$
 $f'(x) = \sec^2 x$, $f'(0) = \sec^2 0 = 1$
 $L(x) = f(0) + f'(0)(x-0)$ $L(0.01) = 0.01$
 $= 0 + 1(x-0) = x$
 $f(0.01) = \tan 0.01 \approx 0.0100003$
 $L(0.1) = 0.1$
 $f(0.1) = \tan(0.1) \approx 0.1003$
 $L(1) = 1$
 $f(1) = \tan 1 \approx 1.557$

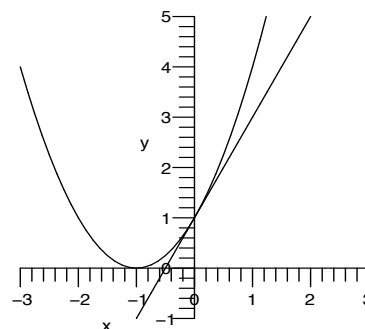
42. The linear approximation for $\sqrt{1+x}$ at $x = 0$ is $L(x) = 1 + \frac{1}{2}x$. The error when $x = 0.01$ is 0.0000124, when $x = 0.1$ is 0.00119, and when $x = 1$ is 0.0858.

43. $f(x) = \sqrt{4+x}$
 $f(0) = \sqrt{4+0} = 2$
 $f'(x) = \frac{1}{2}(4+x)^{-1/2}$
 $f'(0) = \frac{1}{2}(4+0)^{-1/2} = \frac{1}{4}$
 $L(x) = f(0) + f'(0)(x-0) = 2 + \frac{1}{4}x$
 $L(0.01) = 2 + \frac{1}{4}(0.01) = 2.0025$
 $f(0.01) = \sqrt{4+0.01} \approx 2.002498$
 $L(0.1) = 2 + \frac{1}{4}(0.1) = 2.025$
 $f(0.1) = \sqrt{4+0.1} \approx 2.0248$
 $L(1) = 2 + \frac{1}{4}(1) = 2.25$
 $f(1) = \sqrt{4+1} \approx 2.2361$

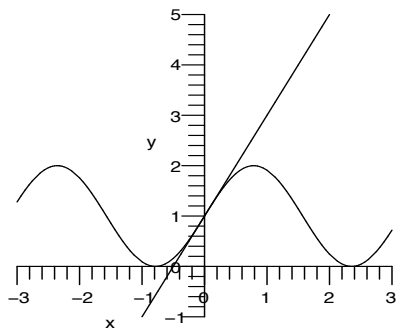
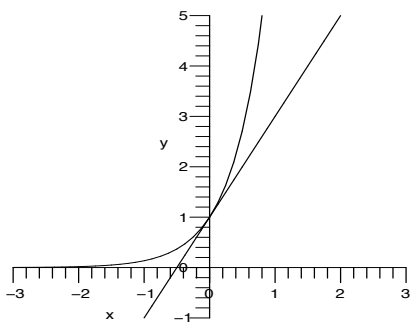
44. The linear approximation for e^x at $x = 0$ is $L(x) = 1 + x$. The error when $x = 0.01$ is 0.0000502, when $x = 0.1$ is 0.00517, and when $x = 1$ is 0.718.

45. (a) $f(0) = g(0) = h(0) = 1$, so all pass through the point $(0, 1)$.
 $f'(0) = 2(0+1) = 2$,
 $g'(0) = 2 \cos(2 \cdot 0) = 2$, and
 $h'(0) = 2e^{2 \cdot 0} = 2$,
 so all have slope 2 at $x = 0$.
 The linear approximation at $x = 0$ for all three functions is $L(x) = 1 + 2x$.

- (b) Graph of
- $f(x) = (x+1)^2$
- :

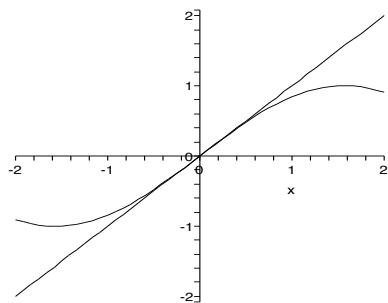
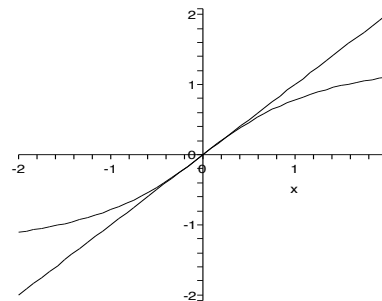
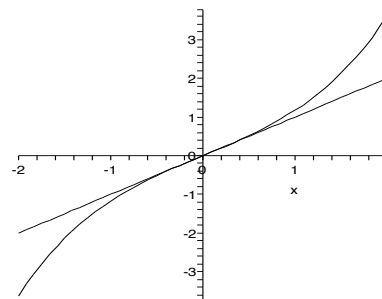


Graph of $f(x) = 1 + \sin(2x)$:

Graph of $f(x) = e^{2x}$:

46. (a) $f(0) = g(0) = h(0) = 0$, so all pass through the point $(0, 0)$.
 $f'(0) = \cos 0 = 1$,
 $g'(0) = \frac{1}{1+0^2} = 1$, and
 $h'(0) = \cosh 0 = 1$,
 so all have slope 1 at $x = 0$.
 The linear approximation at $x = 0$ for all three functions is $L(x) = x$.

(b) Graph of $f(x) = \sin x$:

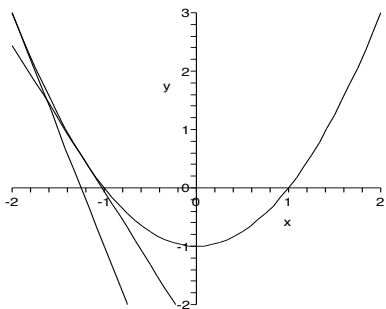
Graph of $g(x) = \tan^{-1} x$:Graph of $h(x) = \sinh x$:

$\sin x$ is the closest fit, but $\sinh x$ is close.

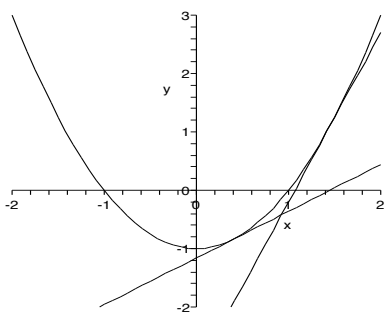
47. (a) $\sqrt[4]{16.04} = 2.0012488$
 $L(0.04) = 2.00125$
 $|2.0012488 - 2.00125| = .00000117$
 (b) $\sqrt[4]{16.08} = 2.0024953$
 $L(.08) = 2.0025$
 $|2.0024953 - 2.0025| = .00000467$
 (c) $\sqrt[4]{16.16} = 2.0049814$
 $L(.16) = 2.005$
 $|2.0049814 - 2.005| = .0000186$

48. If you graph $|\tan x - x|$, you see that the difference is less than .01 on the interval $-.306 < x < .306$ (In fact, a slightly larger interval would work as well).

49. The first tangent line intersects the x -axis at a point a little to the right of 1. So x_1 is about 1.25 (very roughly). The second tangent line intersects the x -axis at a point between 1 and x_1 , so x_2 is about 1.1 (very roughly). Newton's Method will converge to the zero at $x = 1$. Starting with $x_0 = -2$, Newton's method converges to $x = -1$.



Starting with $x_0 = 0.4$, Newton's method converges to $x = 1$.



50. It wouldn't work because $f'(0) = 0$. $x_0 = 0.2$ works better as an initial guess. After jumping to $x_1 = 2.55$, the sequence rapidly decreases toward $x = 1$. Starting with $x_0 = 10$, it takes several steps to get to 2.5, on the way to $x = 1$.

$$\begin{aligned}
 51. \quad x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\
 &= x_n - \left(\frac{x_n^2 - c}{2x_n} \right) \\
 &= x_n - \frac{x_n^2}{2x_n} + \frac{c}{2x_n} \\
 &= \frac{x_n}{2} + \frac{c}{2x_n} \\
 &= \frac{1}{2} \left(x_n + \frac{c}{x_n} \right)
 \end{aligned}$$

If $x_0 < \sqrt{a}$, then $a/x_0 > \sqrt{a}$, so $x_0 < \sqrt{a} < a/x_0$.

52. The root of $x^n - c$ is $\sqrt[n]{c}$, so Newton's method approximates this number.

Newton's method gives

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} = x_i - \frac{x_i^n - c}{nx_i^{n-1}}$$

$$= \frac{1}{n}(nx_i - x_i + cx_i^{1-n}),$$

as desired.

53. (a) $f(x) = x^2 - x - 1$

$$f'(x) = 2x - 1$$

$$\text{At } x_0 = \frac{3}{2}$$

$$f(x_0) = \left(\frac{3}{2}\right)^2 - \frac{3}{2} - 1 = -\frac{1}{4}$$

and

$$f'(x_0) = 2\left(\frac{3}{2}\right) - 1 = 2$$

By Newton's formula,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{3}{2} - \frac{-\frac{1}{4}}{2} = \frac{13}{8}$$

- (b) $f(x) = x^2 - x - 1$

$$f'(x) = 2x - 1$$

$$\text{At } x_0 = \frac{5}{3}$$

$$f(x_0) = \left(\frac{5}{3}\right)^2 - \frac{5}{3} - 1 = \frac{1}{9}$$

and

$$f'(x_0) = 2\left(\frac{5}{3}\right) - 1 = \frac{7}{3}$$

By Newton's formula,

$$\begin{aligned}
 x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
 &= \frac{5}{3} - \frac{\frac{1}{9}}{\frac{7}{3}} = \frac{5}{3} - \frac{1}{21} = \frac{34}{21}
 \end{aligned}$$

- (c) $f(x) = x^2 - x - 1$

$$f'(x) = 2x - 1$$

$$\text{At } x_0 = \frac{8}{5}$$

$$f(x_0) = \left(\frac{8}{5}\right)^2 - \frac{8}{5} - 1 = -\frac{1}{25}$$

and

$$f'(x_0) = 2\left(\frac{8}{5}\right) - 1 = \frac{11}{5}$$

By Newton's formula,

$$\begin{aligned}
 x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\
 &= \frac{8}{5} - \frac{-\frac{1}{25}}{\frac{11}{5}} = \frac{8}{5} + \frac{1}{55} = \frac{89}{55}
 \end{aligned}$$

- (d) From part (a),

$$\text{since } x_0 = \frac{F_4}{F_3}, \text{ hence } x_1 = \frac{F_7}{F_6}.$$

From part (b),

$$\text{since } x_0 = \frac{F_5}{F_4} \text{ hence } x_1 = \frac{F_9}{F_8}.$$

From part (c),

$$\text{since } x_0 = \frac{F_6}{F_5} \text{ hence } x_1 = \frac{F_{11}}{F_{10}}.$$

Thus in general if $x_0 = \frac{F_{n+1}}{F_n}$, then $x_1 =$

$$\frac{F_{2n+1}}{F_{2n}} \text{ implies } m = 2n + 1 \text{ and } k = 2n$$

- (e) Given $x_0 = \frac{3}{2}$, then $\lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n}$ will be

the zero of the function $f(x) = x^2 - x - 1$ which is 1.618034. Therefore,

$$\lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n} = 1.618034$$

54. The general form of function $f(x)$ is,

$$f_n(x) = \frac{1}{5} (2^{n+2}x - 3) \quad \text{for } \frac{1}{2^n} < x < \frac{1}{2^{n-1}}.$$

Hence

$$f'(x) = f'_n(x) = \frac{2^{n+2}}{5} \quad \text{for } \frac{1}{2^n} < x < \frac{1}{2^{n-1}}.$$

By Newton's method,

$$\begin{aligned} x_1 &= \frac{3}{4} - \frac{f\left(\frac{3}{4}\right)}{f'\left(\frac{3}{4}\right)} = \frac{3}{4} - \frac{f_1\left(\frac{3}{4}\right)}{f'_1\left(\frac{3}{4}\right)} \\ &= \frac{3}{4} - \frac{(3/5)}{(8/5)} = \frac{3}{8} = \frac{x_0}{2} \end{aligned}$$

$$\text{Similarly, } x_2 = \frac{x_1}{2} = \frac{x_0}{2^2} \text{ and } x_3 = \frac{x_0}{2^3}$$

Continuing this, we get, $x_{n-1} = \frac{x_0}{2^{n-1}}$. It may

also be observed that, for each $f_n(x)$

$$x_0 = \frac{(1/2^n) + (1/2^{n+1})}{2} = \frac{3}{2^{n+1}},$$

$x_n = \frac{x_0}{2^n} = \frac{3}{2^{2n+1}} \Rightarrow x_{n+1} = \frac{3}{2^{2n+2}}$ which is the zero of F . Therefore Newton's method converges to zero of F .

55. For small x we approximate e^x by $x + 1$ (see exercise 44)

$$\begin{aligned} &\frac{Le^{2\pi d/L} - e^{-2\pi d/L}}{e^{2\pi d/L} + e^{-2\pi d/L}} \\ &\approx \frac{L \left[\left(1 + \frac{2\pi d}{L}\right) - \left(1 - \frac{2\pi d}{L}\right) \right]}{\left(1 + \frac{2\pi d}{L}\right) + \left(1 - \frac{2\pi d}{L}\right)} \\ &\approx \frac{L \left(\frac{4\pi d}{L} \right)}{2} = 2\pi d \\ f(d) &\approx \frac{4.9}{\pi} \cdot 2\pi d = 9.8d \end{aligned}$$

56. If $f(x) = \frac{8\pi hcx^{-5}}{e^{hc/(kTx)} - 1}$, then using the linear approximation we see that
- $$f(x) \approx \frac{8\pi hcx^{-5}}{\left(1 + \frac{hc}{kTx}\right) - 1} = 8\pi kTx^{-4}$$
- as desired.

57. $W(x) = \frac{PR^2}{(R+x)^2}$, $x_0 = 0$

$$W'(x) = \frac{-2PR^2}{(R+x)^3}$$

$$\begin{aligned} L(x) &= W(x_0) + W'(x_0)(x - x_0) \\ &= \frac{PR^2}{(R+0)^2} + \left(\frac{-2PR^2}{(R+0)^3} \right) (x - 0) \\ &= P - \frac{2Px}{R} \end{aligned}$$

$$L(x) = 120 - .01(120) = P - \frac{2Px}{R}$$

$$= 120 - \frac{2 \cdot 120x}{R}$$

$$.01 = \frac{2x}{R}$$

$$x = .005R = .005(20,900,000)$$

$$= 104,500 \text{ ft}$$

58. If $m = m_0(1 - v^2/c^2)^{1/2}$, then $m' = (m_0/2)(1 - v^2/c^2)^{-1/2}(-2v/c^2)$, and $m' = 0$ when $v = 0$. The linear approximation is the constant function $m = m_0$ for small values v .

59. The only positive solution is 0.6407.

60. The smallest positive solution of the first equation is 0.132782, and for the second equation the smallest positive solution is 1, so the species modeled by the second equation is certain to go extinct. This is consistent with the models, since the expected number of offspring for the population modeled by the first equation is 2.2, while for the second equation it is only 1.3

61. The linear approximation for the inverse tangent function at $x = 0$ is

$$f(x) \approx f(0) + f'(0)(x - 0)$$

$$\tan^{-1}(x) \approx \tan^{-1}(0) + \frac{1}{1+0^2}(x - 0)$$

$$\tan^{-1}(x) \approx x$$

Using this approximation,

$$\phi = \tan^{-1} \left(\frac{3[1 - d/D] - w/2}{D - d} \right)$$

$$\phi \approx \frac{3[1 - d/D] - w/2}{D - d}$$

If $d = 0$, then $\phi \approx \frac{3-w/2}{D}$. Thus, if w or D increase, then ϕ decreases.

62. $d'(\theta) = D(w/6 \sin \theta)$
 $d(0) = D(1 - w/6)$ so
 $d(\theta) \approx d(0) + d'(\theta)(\theta - 0)$
 $= D(1 - w/6) + 0(\theta) = D(1 - w/6)$,
as desired.

3.2 Indeterminate Forms and L'Hôpital's Rule

$$\begin{aligned} 1. \quad &\lim_{x \rightarrow -2} \frac{x+2}{x^2-4} \\ &= \lim_{x \rightarrow -2} \frac{x+2}{(x+2)(x-2)} \\ &= \lim_{x \rightarrow -2} \frac{1}{x-2} = -\frac{1}{4} \end{aligned}$$

$$\begin{aligned} 2. \quad \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 3x + 2} &= \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)(x-1)} \\ &= \lim_{x \rightarrow 2} \frac{x+2}{x-1} = 4 \end{aligned}$$

$$\begin{aligned} 3. \quad \lim_{x \rightarrow \infty} \frac{3x^2 + 2}{x^2 - 4} &= \lim_{x \rightarrow \infty} \frac{3 + \frac{2}{x^2}}{1 - \frac{4}{x^2}} \\ &= \frac{3}{1} = 3 \end{aligned}$$

$$\begin{aligned} 4. \quad \lim_{x \rightarrow -\infty} \frac{x+1}{x^2 + 4x + 3} &\text{ is type } \frac{\infty}{\infty}; \\ &\text{we apply L'Hôpital's Rule to get} \\ &\lim_{x \rightarrow -\infty} \frac{1}{2x+4} = 0. \end{aligned}$$

$$\begin{aligned} 5. \quad \lim_{t \rightarrow 0} \frac{e^{2t} - 1}{t} &\text{ is type } \frac{0}{0}; \\ &\text{we apply L'Hôpital's Rule to get} \\ &\lim_{t \rightarrow 0} \frac{\frac{d}{dt}(e^{2t} - 1)}{\frac{d}{dt}t} \\ &\lim_{t \rightarrow 0} \frac{2e^{2t}}{1} = \frac{2}{1} = 2 \end{aligned}$$

$$\begin{aligned} 6. \quad \lim_{t \rightarrow 0} \frac{\sin t}{e^{3t} - 1} &\text{ is type } \frac{0}{0}; \\ &\text{we apply L'Hôpital's Rule to get} \\ &\lim_{t \rightarrow 0} \frac{\frac{d}{dt}(\sin t)}{\frac{d}{dt}(e^{3t} - 1)} = \lim_{t \rightarrow 0} \frac{\cos t}{3e^{3t}} = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} 7. \quad \lim_{t \rightarrow 0} \frac{\tan^{-1}t}{\sin t} &\text{ is type } \frac{0}{0}; \\ &\text{we apply L'Hôpital's Rule to get} \\ &\lim_{t \rightarrow 0} \frac{\frac{d}{dt}(\tan^{-1}t)}{\frac{d}{dt}(\sin t)} = \lim_{t \rightarrow 0} \frac{1/(1+t^2)}{\cos t} = 1 \end{aligned}$$

$$\begin{aligned} 8. \quad \lim_{t \rightarrow 0} \frac{\sin t}{\sin^{-1}t} &\text{ is type } \frac{0}{0}; \\ &\text{we apply L'Hôpital's Rule to get} \\ &\lim_{t \rightarrow 0} \frac{\frac{d}{dt}(\sin t)}{\frac{d}{dt}(\sin^{-1}t)} = \lim_{t \rightarrow 0} \frac{\cos t}{1/(\sqrt{1-t^2})} = 1 \end{aligned}$$

$$\begin{aligned} 9. \quad \lim_{x \rightarrow \pi} \frac{\sin 2x}{\sin x} &\text{ is type } \frac{0}{0}; \\ &\text{we apply L'Hôpital's Rule to get} \end{aligned}$$

$$\lim_{x \rightarrow \pi} \frac{2 \cos 2x}{\cos x} = \frac{2(1)}{-1} = -2.$$

$$10. \quad \lim_{x \rightarrow -1} \frac{\cos^{-1}x}{x^2 - 1} \text{ is undefined (numerator goes to } \pi, \text{ denominator goes to 0).}$$

$$\begin{aligned} 11. \quad \lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} &\text{ is type } \frac{0}{0}; \\ &\text{we apply L'Hôpital's Rule thrice to get} \\ &= \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{-\sin x}{6x} \\ &= \lim_{x \rightarrow 0} \frac{-\cos x}{6} = -\frac{1}{6}. \end{aligned}$$

$$\begin{aligned} 12. \quad \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} &\text{ is type } \frac{0}{0}; \\ &\text{we apply L'Hôpital's Rule to get} \\ &\lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2}. \end{aligned}$$

Apply L'Hôpital's Rule twice more to get

$$\begin{aligned} &\lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x}{6x} \\ &= \lim_{x \rightarrow 0} \frac{4 \sec^2 x \tan^2 x + 2 \sec^4 x}{6} = \frac{1}{3}. \end{aligned}$$

$$\begin{aligned} 13. \quad \lim_{t \rightarrow 1} \frac{\sqrt{t} - 1}{t - 1} &= \lim_{t \rightarrow 1} \frac{\sqrt{t} - 1}{t - 1} \cdot \frac{\sqrt{t} + 1}{\sqrt{t} + 1} \\ &= \lim_{t \rightarrow 1} \frac{(t - 1)}{(t - 1)\sqrt{t} + 1} \\ &= \lim_{t \rightarrow 1} \frac{1}{\sqrt{t} + 1} = \frac{1}{2} \end{aligned}$$

$$14. \quad \lim_{t \rightarrow 1} \frac{\ln t}{t - 1} \text{ is type } \frac{0}{0};$$

we apply L'Hôpital's Rule to get

$$\lim_{t \rightarrow 1} \frac{\frac{d}{dt}(\ln t)}{\frac{d}{dt}(t - 1)} = \lim_{t \rightarrow 1} \frac{\frac{1}{t}}{1} = 1$$

$$\begin{aligned} 15. \quad \lim_{x \rightarrow \infty} \frac{x^3}{e^x} &\text{ is type } \frac{\infty}{\infty}; \\ &\text{we apply L'Hôpital's Rule thrice to get} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{3x^2}{e^x} &= \lim_{x \rightarrow \infty} \frac{6x}{e^x} \\ &= \lim_{x \rightarrow \infty} \frac{6}{e^x} = 0. \end{aligned}$$

$$16. \quad \lim_{x \rightarrow \infty} \frac{e^x}{x^4} \text{ is type } \frac{\infty}{\infty};$$

we apply L'Hôpital's Rule four times to get

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{e^x}{4x^3} &= \lim_{x \rightarrow \infty} \frac{e^x}{12x^2} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{24x} = \lim_{x \rightarrow \infty} \frac{e^x}{24} = \infty. \end{aligned}$$

$$\begin{aligned} 17. \quad \lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x \sin^2 x} &\text{ is type } \frac{\infty}{\infty}; \\ &\text{we apply L'Hôpital's Rule twice to get} \\ &\lim_{x \rightarrow 0} \frac{\cos x - x \sin x - \cos x}{\sin^2 x + 2x \sin x \cos x} \\ &= \lim_{x \rightarrow 0} \frac{-x \sin x}{\sin x (\sin x + 2x \cos x)} \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{-x}{\sin x + 2x \cos x} \\
&= \lim_{x \rightarrow 0} \frac{-1}{\cos x + 2 \cos x - 2x \sin x} \\
&= -\frac{1}{3}.
\end{aligned}$$

18. Rewrite as one fraction, we have

$$\lim_{x \rightarrow 0} \left(\cot x - \frac{1}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{x \cos x - \sin x}{x \sin x} \right)$$

which is of type $\frac{0}{0}$

we apply L'Hôpital's Rule to get

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left(\frac{\cos x - x \sin x - \cos x}{\sin x + x \cos x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{\frac{d}{dx}(-x \sin x)}{\frac{d}{dx}(\sin x + x \cos x)} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{-\sin x - x \cos x}{\cos x + \cos x - x \sin x} \right) = 0
\end{aligned}$$

19. Rewrite as one fraction, we have

$$\begin{aligned}
&\lim_{x \rightarrow 0} \left(\frac{x+1}{x} - \frac{2}{\sin 2x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{(x+1) \sin 2x - 2x}{x \sin 2x} \right) \text{ is type } \frac{0}{0}; \\
&\text{we apply L'Hôpital's Rule four times to get} \\
&\lim_{x \rightarrow 0} \left(\frac{\frac{d}{dx}(x+1) \sin 2x - 2x}{\frac{d}{dx}(x \sin 2x)} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{\sin 2x + 2(x+1) \cos 2x - 2}{\sin 2x + 2x \cos 2x} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{\frac{d}{dx}(\sin 2x + 2(x+1) \cos 2x - 2)}{\frac{d}{dx}(\sin 2x + 2x \cos 2x)} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{2 \cos 2x + 2 \cos 2x - 4(x+1) \sin 2x}{2 \cos 2x + 2 \cos 2x - 4x \sin 2x} \right) \\
&= \frac{4}{4} = 1
\end{aligned}$$

20. $\lim_{x \rightarrow \frac{\pi}{2}} \left(\tan x + \frac{1}{x - \frac{\pi}{2}} \right)$

In this case the limit has the form $(\infty - \infty)$.

Rewrite $\tan x$ as $\frac{\sin x}{\cos x}$ and then as one fraction, we get

$$\begin{aligned}
&\lim_{x \rightarrow \frac{\pi}{2}} \left(\tan x + \frac{1}{x - \frac{\pi}{2}} \right) \\
&= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\sin x}{\cos x} + \frac{1}{x - \frac{\pi}{2}} \right) \\
&= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{(x - \frac{\pi}{2}) \sin x + \cos x}{(x - \frac{\pi}{2}) \cos x} \right) \text{ is type } \frac{0}{0}
\end{aligned}$$

we apply L'Hôpital's Rule to get

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\sin x + (x - \frac{\pi}{2}) \cos x - \sin x}{\cos x - (x - \frac{\pi}{2}) \sin x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{(x - \frac{\pi}{2}) \cos x}{\cos x - (x - \frac{\pi}{2}) \sin x} \right) = 0$$

21. $\lim_{x \rightarrow \infty} \frac{\ln x}{x^2}$ is type $\frac{\infty}{\infty}$
we apply L'Hôpital's Rule to get

$$\lim_{x \rightarrow \infty} \frac{1/x}{2x} = \lim_{x \rightarrow \infty} \frac{1}{2x^2} = 0.$$

22. $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}}$ is type $\frac{\infty}{\infty}$;
we apply L'Hôpital's Rule to get

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0.$$

23. $\lim_{t \rightarrow \infty} \frac{t}{e^t}$ is type $\frac{\infty}{\infty}$
we apply L'Hôpital's Rule to get

$$\lim_{t \rightarrow \infty} \frac{\frac{d}{dt}(t)}{\frac{d}{dt}(e^t)} = \lim_{t \rightarrow \infty} \frac{1}{e^t} = 0.$$

24. $\lim_{t \rightarrow \infty} \frac{\sin \frac{1}{t}}{\frac{1}{t}}$ is type $\frac{0}{0}$

we apply L'Hôpital's Rule to get

$$= \lim_{t \rightarrow \infty} \frac{-\frac{1}{t^2} \cos \frac{1}{t}}{-\frac{1}{t^2}} = \lim_{t \rightarrow \infty} \cos \frac{1}{t} = 1.$$

25. $\lim_{t \rightarrow 1} \frac{\ln(\ln t)}{\ln t}$

As t approaches 1 from below, $\ln t$ is a small negative number. Hence $\ln(\ln t)$ is undefined, so the limit is undefined.

26. $\lim_{t \rightarrow 0} \left(\frac{\sin(\sin t)}{\sin t} \right)$ is type $\frac{0}{0}$

we apply L'Hôpital's Rule to get

$$\lim_{t \rightarrow 0} \left(\frac{\cos(\sin t) \cos t}{\cos t} \right) = 1.$$

27. $\lim_{x \rightarrow 0} \left(\frac{\sin(\sinh x)}{\sinh(\sin x)} \right)$ is type $\frac{0}{0}$

we apply L'Hôpital's Rule to get

$$\lim_{x \rightarrow 0} \left(\frac{\cos(\sinh x) \cosh x}{\cosh(\sin x) \cos x} \right) = 1$$

28. $\lim_{x \rightarrow 0} \left(\frac{\sin x - \sinh x}{\cos x - \cosh x} \right)$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \left(\frac{2 \sin x - e^x + e^{-x}}{2 \cos x - e^x - e^{-x}} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{2e^x \sin x - e^{2x} + 1}{2e^x \cos x - e^{2x} - 1} \right) \text{ is type } \frac{0}{0}
\end{aligned}$$

we apply L'Hôpital's Rule twice to get

$$\begin{aligned}
&\lim_{x \rightarrow 0} \left(\frac{2e^x \cos x + 2e^x \sin x - 2e^{2x}}{-2e^x \sin x + 2e^x \cos x - 2e^{2x}} \right) \\
&= \lim_{x \rightarrow 0} \left(\frac{\cos x + \sin x - 1}{\cos x - \sin x - 1} \right) \text{ is type } \frac{0}{0} \\
&= \lim_{x \rightarrow 0} \left(\frac{-\sin x + \cos x}{-\sin x - \cos x} \right) = -1
\end{aligned}$$

29. $\lim_{x \rightarrow 0^+} \frac{\ln x}{\cot x}$ is type $\frac{\infty}{\infty}$
 we apply L'Hôpital's Rule to get

$$\lim_{x \rightarrow 0^+} \frac{1/x}{-\csc^2 x} = \lim_{x \rightarrow 0^+} \left(-\sin x \cdot \frac{\sin x}{x} \right) = (0)(1) = 0.$$

30. $\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\ln x} = 0$ (numerator goes to 0 and denominator goes to $-\infty$).

31. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x)$

$$\begin{aligned} &= \lim_{x \rightarrow \infty} \left((\sqrt{x^2 + 1} - x) \frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1} + x} \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{x^2 + 1 - x^2}{\sqrt{x^2 + 1} + x} \right) \\ &= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2 + 1} + x} = 0 \end{aligned}$$

32. $\lim_{x \rightarrow \infty} \ln x - x = \lim_{x \rightarrow \infty} \frac{\ln x - 1}{\frac{1}{x}} = -\infty$ since the numerator goes to -1 and the denominator goes to 0^+ . (Recall Example 2.8 which shows $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$.)

33. Let $y = \left(1 + \frac{1}{x}\right)^x$
 $\Rightarrow \ln y = x \ln \left(1 + \frac{1}{x}\right)$. Then

$$\begin{aligned} \lim_{x \rightarrow \infty} \ln y &= \lim_{x \rightarrow \infty} x \ln \left(1 + \frac{1}{x}\right) \\ &= \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{x}\right)}{1/x} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{1}{1+\frac{1}{x}} \left(-\frac{1}{x^2}\right)}{-1/x^2} \\ &= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} = 1. \end{aligned}$$

Hence $\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln y} = e$.

34. Notice that the limit in question has the indeterminate form 1^∞ . Also, note that as x gets large, $\left|\frac{x+1}{x-2}\right| = \frac{x+1}{x-2}$.

Define $y = \left(\frac{x+1}{x-2}\right)^{\sqrt{x^2-4}}$. Then

$$\ln y = \sqrt{x^2-4} \ln \left(\frac{x+1}{x-2}\right) \text{ and}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \left(\sqrt{x^2-4} \ln \left(\frac{x+1}{x-2}\right) \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{\ln \left(\frac{x+1}{x-2}\right)}{\frac{1}{\sqrt{x^2-4}}} \right)$$

This last limit has indeterminate form $\frac{0}{0}$, so we can apply L'Hôpital's Rule and simplify to find that the above is equal to

$$\lim_{x \rightarrow \infty} \frac{-3(x^2-4)^{3/2}}{-x^3+x^2+2x} \text{ and this is equal to 3. So}$$

$$\lim_{x \rightarrow \infty} \ln y = 3.$$

Thus $\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln y} = e^3 \approx 20.086$.

35. $\lim_{x \rightarrow 0^+} \left(\frac{1}{\sqrt{x}} - \frac{\sqrt{x}}{\sqrt{x+1}} \right)$

$$\begin{aligned} &= \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x+1} - (\sqrt{x})^2}{\sqrt{x}\sqrt{x+1}} \right) \\ &= \lim_{x \rightarrow 0^+} \left(\frac{\sqrt{x+1} - x}{\sqrt{x}\sqrt{x+1}} \right) \\ &= \infty. \end{aligned}$$

36. $\lim_{x \rightarrow 1} \frac{\sqrt{5-x}-2}{\sqrt{10-x}-3}$ is type $\frac{0}{0}$
 we apply L'Hôpital's Rule to get

$$\begin{aligned} &\lim_{x \rightarrow 1} \frac{\frac{1}{2}(5-x)^{-1/2}(-1)}{\frac{1}{2}(10-x)^{-1/2}(-1)} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{10-x}}{\sqrt{5-x}} = \frac{3}{2}. \end{aligned}$$

37. Let $y = (1/x)^x$. Then $\ln y = x \ln(1/x)$. Then

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln(1/x) = 0, \text{ by Exercise 27. Thus } \lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\ln y} = 1.$$

38. Let $y = \lim_{x \rightarrow 0^+} (\cos x)^{1/x}$. Then

$$\begin{aligned} \ln y &= \lim_{x \rightarrow 0^+} \frac{1}{x} \ln \cos x \\ &= \lim_{x \rightarrow 0^+} \frac{\ln(\cos x)}{x} \text{ is type } \frac{0}{0} \end{aligned}$$

so apply L'Hôpital's Rule to get

$$\lim_{x \rightarrow 0^+} \frac{-\tan x}{1} = 0.$$

Therefore the limit is $y = e^0 = 1$.

39. $\lim_{t \rightarrow \infty} \left(\frac{t-3}{t+2} \right)^t = \lim_{t \rightarrow \infty} \frac{(t-3)^t}{(t+2)^t}$

$$\begin{aligned} &= \lim_{t \rightarrow \infty} \frac{\left(1 - \frac{3}{t}\right)^t}{\left(1 + \frac{2}{t}\right)^t} = \frac{\lim_{t \rightarrow \infty} \left(1 - \frac{3}{t}\right)^t}{\lim_{t \rightarrow \infty} \left(1 + \frac{2}{t}\right)^t} \\ &= \frac{\lim_{t \rightarrow \infty} \left(1 + \frac{-3}{t}\right)^t}{\lim_{t \rightarrow \infty} \left(1 + \frac{2}{t}\right)^t} = \frac{e^{-3}}{e^2} = e^{-5} \end{aligned}$$

40. $\lim_{t \rightarrow \infty} \left(\frac{t-3}{2t+1} \right)^t = \lim_{t \rightarrow \infty} \left(\frac{1 - \frac{3}{t}}{2 + \frac{1}{t}} \right)^t$

$$= \lim_{t \rightarrow \infty} \frac{(1 - \frac{3}{t})^t}{2^t (1 + \frac{1}{t})^t} = \lim_{t \rightarrow \infty} \frac{e^{-3}}{2^t e^{1/2}} = 0$$

41. L'Hôpital's rule does not apply. As $x \rightarrow 0$, the numerator gets close to 1 and the denominator is small and positive. Hence the limit is ∞ .

42. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x^2}$ is type $\frac{0}{0}$, but $\lim_{x \rightarrow 0} \frac{e^x}{2x}$ is not, so L'Hôpital's Rule does not apply to this limit.

43. L'Hôpital's rule does not apply. As $x \rightarrow 0$, the numerator is small and positive while the denominator goes to $-\infty$. Hence the limit is 0. Also $\lim_{x \rightarrow 0} \frac{2x}{2/x}$, which equals $\lim_{x \rightarrow 0} x^2$, is not of the form $\frac{0}{0}$ so L'Hôpital's rule doesn't apply here either.

44. $\lim_{x \rightarrow 0} \frac{\sin x}{x^2}$ is type $\frac{0}{0}$, but $\lim_{x \rightarrow 0} \frac{\cos x}{2x}$ is not, so L'Hôpital's rule does not apply. This limit is undefined because the numerator goes to 1 and the denominator goes to 0.

45. $\lim_{x \rightarrow 0^+} \frac{\csc x}{\sqrt{x}}$
In this case limit has the form $\frac{\infty}{0}$, L'Hôpital's Rule should not be used.

46. $\lim_{x \rightarrow 0^+} \frac{x^{-3/2}}{\ln x}$ is type $\frac{\infty}{-\infty}$. In this case L'Hôpital's Rule should be used.

47. $\lim_{x \rightarrow \infty} \frac{x^2 - 3x + 1}{\tan^{-1} x} = \infty$. In this case limit has the form ∞ . So L'Hôpital's Rule should not be used.

48. $\lim_{x \rightarrow \infty} \frac{\ln(x^2)}{e^{x/3}}$ is type $\frac{\infty}{\infty}$. So L'Hôpital's Rule should be used.

49. (a) Starting with $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 2x}$, we cannot "cancel sin" to get $\lim_{x \rightarrow 0} \frac{3x}{2x}$. We can cancel the x 's in the last limit to get the final answer of $3/2$. The first step is likely to give a correct answer because the linear approximation of $\sin 3x$ is $3x$, and the linear approximation of $\sin 2x$ is $2x$. The linear approximations are better the closer x is to zero, so the limits are likely to be the same.

- (b) $\lim_{x \rightarrow 0} \frac{\sin nx}{\sin mx}$ is type $\frac{0}{0}$;

we apply L'Hôpital's Rule to get

$$\lim_{x \rightarrow 0} \frac{n \cos nx}{m \cos mx} = \frac{n}{m}.$$

50. (a) $\lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = \lim_{x \rightarrow 0} \frac{2x \cos x^2}{2x}$

$$= \lim_{x \rightarrow 0} \cos x^2 = 1,$$

 which is the same as $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

- (b) $\lim_{x \rightarrow 0} \frac{1 - \cos x^2}{x^4}$

$$= \lim_{x \rightarrow 0} \frac{2x \sin x^2}{4x^3} = \lim_{x \rightarrow 0} \frac{\sin x^2}{2x^2}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = \frac{1}{2} \text{ (by part (a))},$$

while

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2}(1) = \frac{1}{2}$$

so both of these limits are the same.

- (c) Based on the patterns found in exercise 45, we should guess

$$\lim_{x \rightarrow 0} \frac{\sin x^3}{x^3} = 1 \text{ and } \lim_{x \rightarrow 0} \frac{1 - \cos x^3}{x^6} = \frac{1}{2}.$$

51. (a) $\frac{(x+1)(2+\sin x)}{x(2+\cos x)}$

(b) $\frac{x}{e^x}$

(c) $\frac{3x+1}{x-7}$

(d) $\frac{3-8x}{1+2x}$

52. (a) $\lim_{x \rightarrow \infty} x - \ln x = \infty$ (see exercise 32).

- (b) $\lim_{x \rightarrow \infty} \sqrt{x^2 + 1} - x = 0$ (see exercise 31).

(c) $\lim_{x \rightarrow \infty} \sqrt{x^2 + 4x} - x$

$$= \lim_{x \rightarrow \infty} (\sqrt{x^2 + 4x} - x)$$

$$= \lim_{x \rightarrow \infty} \frac{4x}{\sqrt{x^2 + 4x} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{4x \frac{1}{x}}{(\sqrt{x^2 + 4x} + x) \frac{1}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{4}{\sqrt{1 + \frac{4}{x}} + 1} = 2,$$

where to get from the second to the third line, we have multiplied by $\frac{(\sqrt{x^2 + 4x} + x)}{(\sqrt{x^2 + 4x} + x)}$.

53. $\lim_{x \rightarrow \infty} e^x = \lim_{x \rightarrow \infty} x^n = \infty$
 $\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty$. Since n applications of
 L'Hôpital's rule yields
 $\lim_{x \rightarrow \infty} \frac{e^x}{n!} = \infty$.
 Hence e^x dominates x^n .

54. $\lim_{x \rightarrow \infty} \ln x = \lim_{x \rightarrow \infty} x^p = \infty$.
 $\lim_{x \rightarrow \infty} \frac{\ln x}{x^p}$ is of type $\frac{\infty}{\infty}$
 we use L'Hôpital's Rule to get
 $\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{px^{p-1}} = \lim_{x \rightarrow \infty} \frac{1}{px^p} = 0$ (since $p > 0$).
 Therefore, x^p dominates $\ln x$.

55. $\lim_{t \rightarrow \infty} (e^{\frac{t}{2}} - t^3)$ Since $e^{\frac{t}{2}}$ dominates t^3 . So
 $\lim_{t \rightarrow \infty} (e^{\frac{t}{2}} - t^3) = \infty$

56. $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x} - \ln x}{\sqrt{x}} \right)$ is type $\frac{\infty}{\infty}$.
 we apply L'Hôpital's Rule to get
 $\lim_{x \rightarrow \infty} \left(\frac{\frac{1}{2\sqrt{x}} - \frac{1}{x}}{\frac{1}{2\sqrt{x}}} \right) = \lim_{x \rightarrow \infty} \left(\frac{x - 2\sqrt{x}}{x} \right)$
 $= \lim_{x \rightarrow \infty} \left(1 - \frac{2}{\sqrt{x}} \right) = 1$.

57. $\lim_{x \rightarrow \infty} \frac{\ln(x^3 + 2x + 1)}{\ln(x^2 + x + 2)}$
 we apply L'Hôpital's Rule
 $\lim_{x \rightarrow \infty} \left(\frac{\frac{d}{dx}(\ln(x^3 + 2x + 1))}{\frac{d}{dx}(\ln(x^2 + x + 2))} \right)$
 $= \lim_{x \rightarrow \infty} \left(\frac{\frac{3x^2 + 2}{x^3 + 2x + 1}}{\frac{2x + 1}{x^2 + x + 2}} \right)$
 $= \lim_{x \rightarrow \infty} \left(\frac{3x^4 + 3x^3 + 8x^2 + 2x + 4}{2x^4 + x^3 + 4x^2 + 4x + 1} \right) = \frac{3}{2}$
 In general, for numerator and denominator the
 highest degree of polynomials p and q , such that
 $p(x) > 0$ and $q(x) > 0$ for $x > 0$,
 should be the $\lim_{x \rightarrow \infty} \frac{\ln(p(x))}{\ln(q(x))}$.

58. $\lim_{x \rightarrow \infty} \frac{\ln(e^{3x} + x)}{\ln(e^{2x} + 4)}$ is $\frac{\infty}{\infty}$;
 we apply L'Hôpital's Rule
 $\lim_{x \rightarrow \infty} \left(\frac{\frac{d}{dx}(\ln(e^{3x} + x))}{\frac{d}{dx}(\ln(e^{2x} + 4))} \right)$
 $= \lim_{x \rightarrow \infty} \left(\frac{\frac{3e^{3x} + 1}{e^{3x} + x}}{\frac{2e^{2x}}{e^{2x} + 4}} \right)$
 $= \lim_{x \rightarrow \infty} \left(\frac{3e^{5x} + 12e^{3x} + e^{2x} + 4}{2e^{5x} + 2xe^{2x}} \right) = \frac{3}{2}$

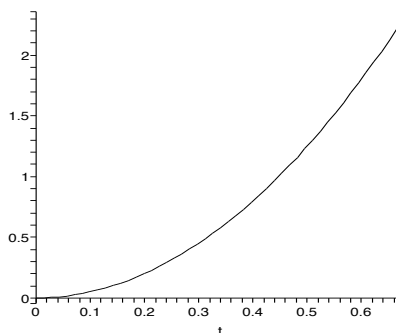
In general, when the degree of exponential term in the numerator and denominator are different, then the $\lim_{x \rightarrow \infty} \frac{\ln(e^{kx} + p(x))}{\ln(e^{cx} + q(x))}$ for polynomials p and q and positive numbers. k and c will be the fraction of degrees that is $\frac{k}{c}$.

59. If $x \rightarrow 0$, then $x^2 \rightarrow 0$, so if $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = L$,
 then $\lim_{x \rightarrow 0} \frac{f(x^2)}{g(x^2)} = L$ (but not conversely). If
 $a \neq 0$ or 1 , then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ involves the be-
 havior of the quotient near a , while $\lim_{x \rightarrow a} \frac{f(x^2)}{g(x^2)}$
 involves the behavior of the quotient near the
 different point a^2 .

60. Functions $f(x) = |x|$ and $g(x) = x$ work.
 $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ does not exist as it approaches -1
 from the left and it approaches 1 from the
 right, but $\lim_{x \rightarrow 0} \frac{f(x^2)}{g(x^2)} = 1$.

61. $\lim_{\omega \rightarrow 0} \frac{2.5(4\omega t - \sin 4\omega t)}{4\omega^2}$
 $= \lim_{\omega \rightarrow 0} \frac{2.5(4t - 4t \cos 4\omega t)}{8\omega}$
 $= \lim_{\omega \rightarrow 0} \frac{2.5(16t^2 \sin 4\omega t)}{8} = 0$

62. $\lim_{\omega \rightarrow 0} \frac{2.5 - 2.5 \sin(4\omega t + \frac{\pi}{2})}{4\omega^2}$ is type $\frac{0}{0}$;
 we apply L'Hôpital's Rule to get
 $\lim_{\omega \rightarrow 0} \frac{-10t \cos(4\omega t + \frac{\pi}{2})}{8\omega}$
 $= \lim_{\omega \rightarrow 0} \frac{40t^2 \sin(4\omega t + \frac{\pi}{2})}{8} = 5t^2$.



63. The area of triangular region 1 is
 $(1/2)(\text{base})(\text{height})$

$$= (1/2)(1 - \cos \theta)(\sin \theta).$$

Let P be the center of the circle. The area of region 2 equals the area of sector APC minus the area of triangle APB . The area of the sector is $\theta/2$, while the area of triangle APB is

$$(1/2)(\text{base})(\text{height})$$

$$= (1/2)(\cos \theta)(\sin \theta).$$

Hence the area of region 1 divided by the area of region 2 is

$$\frac{(1/2)(1 - \cos \theta)(\sin \theta)}{\theta/2 - (1/2)(\cos \theta)(\sin \theta)}$$

$$= \frac{(1 - \cos \theta)(\sin \theta)}{\theta - (\cos \theta)(\sin \theta)}$$

$$= \frac{\theta - \cos \theta \sin \theta}{\sin \theta - \cos \theta \sin \theta}$$

$$= \frac{\theta - \cos \theta \sin \theta}{\sin \theta - (1/2) \sin 2\theta}$$

$$= \frac{\theta - (1/2) \sin 2\theta}{\theta - (1/2) \sin 2\theta}$$

$$\text{Then } \lim_{\theta \rightarrow 0} \frac{\sin \theta - (1/2) \sin 2\theta}{\theta - (1/2) \sin 2\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{\cos \theta - \cos 2\theta}{1 - \cos 2\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{-\sin \theta + 2 \sin 2\theta}{2 \sin 2\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{2 \sin 2\theta}{-\cos \theta + 4 \cos 2\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{4 \cos 2\theta}{-1 + 4(1)} = \frac{3}{4}$$

$$= \frac{4(1)}{4(1)} = 1$$

64. $\lim_{x \rightarrow 0^+} \frac{160x^{-0.4} + 90}{8x^{-0.4} + 10}$
 $= \lim_{x \rightarrow 0^+} \frac{160 + 90x^{0.4}}{8 + 10x^{0.4}} = \frac{160}{8} = 20$. If there is no light, the pupils will expand to this size. This is the largest the pupils can get.
 $\lim_{x \rightarrow \infty} \frac{160x^{-0.4} + 90}{8x^{-0.4} + 10} = \frac{90}{10} = 9$. As the amount of light grows, the pupils shrink, and the size approaches 6mm in the limit. This is the smallest possible size of the pupils.

65. (a) $V = \sqrt{40mg} \tanh(\sqrt{\frac{g}{40m}} t)$, therefore
 $\lim_{t \rightarrow \infty} V$
 $= \lim_{t \rightarrow \infty} \sqrt{40mg} \left(\frac{e^{\sqrt{\frac{g}{40m}} t} - e^{-\sqrt{\frac{g}{40m}} t}}{e^{\sqrt{\frac{g}{40m}} t} + e^{-\sqrt{\frac{g}{40m}} t}} \right)$
 $= \sqrt{40mg} \lim_{t \rightarrow \infty} \left(\frac{1 - e^{-2\sqrt{\frac{g}{40m}} t}}{1 + e^{-2\sqrt{\frac{g}{40m}} t}} \right)$
 $= \sqrt{40mg}$
as $t \rightarrow \infty$; $2\sqrt{\frac{g}{40m}} t \rightarrow \infty$ and
 $e^{-2\sqrt{\frac{g}{40m}} t} \rightarrow 0$ This means, when the time increases indefinitely, its velocity reaches $\sqrt{40mg}$.

- (b) $\lim_{m \rightarrow 0} V$
 $= \lim_{m \rightarrow 0} \sqrt{40mg} \left(\frac{e^{\sqrt{\frac{g}{40m}} t} - e^{-\sqrt{\frac{g}{40m}} t}}{e^{\sqrt{\frac{g}{40m}} t} + e^{-\sqrt{\frac{g}{40m}} t}} \right)$
 $= \lim_{m \rightarrow 0} \sqrt{40mg} \left(\frac{1 - e^{-2\sqrt{\frac{g}{40m}} t}}{1 + e^{-2\sqrt{\frac{g}{40m}} t}} \right)$
 $= \lim_{m \rightarrow 0} \sqrt{40mg} \lim_{m \rightarrow 0} \left(\frac{1 - e^{-2\sqrt{\frac{g}{40m}} t}}{1 + e^{-2\sqrt{\frac{g}{40m}} t}} \right)$
 $= 0$
as $m \rightarrow 0$; $2\sqrt{\frac{g}{40m}} t \rightarrow \infty$ and
 $e^{-2\sqrt{\frac{g}{40m}} t} \rightarrow 0$. This means, when the mass is negligible, its velocity is 0.

- (c) $\lim_{m \rightarrow \infty} V$
 $= \lim_{m \rightarrow \infty} \sqrt{40mg} \left(\frac{e^{\sqrt{\frac{g}{40m}} t} - e^{-\sqrt{\frac{g}{40m}} t}}{e^{\sqrt{\frac{g}{40m}} t} + e^{-\sqrt{\frac{g}{40m}} t}} \right)$
 $= \lim_{m \rightarrow \infty} \sqrt{40mg} \left(\frac{e^{2\sqrt{\frac{g}{40m}} t} - 1}{e^{2\sqrt{\frac{g}{40m}} t} + 1} \right)$
as $m \rightarrow \infty$; $2\sqrt{\frac{g}{40m}} t \rightarrow 0$ and
 $e^{2\sqrt{\frac{g}{40m}} t} \rightarrow 1$
 $= \lim_{m \rightarrow \infty} \left(\frac{1}{e^{2\sqrt{\frac{g}{40m}} t} + 1} \right) \lim_{m \rightarrow \infty} \left(\frac{e^{2\sqrt{\frac{g}{40m}} t} - 1}{(1/\sqrt{40mg})} \right)$
 $= (1/2) \lim_{(2\sqrt{\frac{g}{40m}} t) \rightarrow 0} \left(\frac{e^{2\sqrt{\frac{g}{40m}} t} - 1}{2\sqrt{\frac{g}{40m}} t} \right) (2\sqrt{g} t)$
 $= \sqrt{g} t$
This means, when the increases indefinitely, its velocity reaches $\sqrt{g} t$.

66. $\lim_{c \rightarrow \infty} S = \lim_{c \rightarrow \infty} \left\{ \frac{8\pi}{3} c^2 \left[\left(\frac{d^2}{16c^2} + 1 \right)^{3/2} - 1 \right] \right\}$
 $= \frac{8\pi}{3} \lim_{c \rightarrow \infty} \left(\frac{c^2 [(d^2 + 16c^2)^{3/2} - 64c^3]}{64c^3} \right)$
 $= \frac{\pi}{24} \lim_{c \rightarrow \infty} \left(\frac{[(d^2 + 16c^2)^{3/2} - 64c^3]}{c} \right)$ is type $\frac{\infty}{\infty}$;
we apply L'Hôpital's Rule to get
 $\frac{\pi}{24} \lim_{c \rightarrow \infty} \left[\frac{3}{2} (d^2 + 16c^2)^{1/2} (32c) - 192c^2 \right]$
 $= 2\pi \lim_{c \rightarrow \infty} c [(d^2 + 16c^2)^{1/2} - 4c^2]$
which on rationalising gives
 $2\pi \lim_{c \rightarrow \infty} \frac{c [(d^2 + 16c^2) - 16c^2]}{[(d^2 + 16c^2)^{1/2} + 4c^2]} = \frac{\pi d^2}{4}$

3.3 Maximum and Minimum Values

1. (a) $f(x) = \frac{1}{x^2 - 1}$ on $(0, 1) \cup (1, \infty)$
 $f'(x) = \frac{-2x}{(x^2 - 1)^2}$
 $x = 0$ is critical point.
 $f(0) = -1$ is absolute maximum value but 0 is not included. Hence f has no absolute extrema on interval $(0, 1) \cup (1, \infty)$.
- (b) $f(x) = \frac{1}{x^2 - 1}$ on $(-1, 1)$
 $f'(x) = \frac{-2x}{(x^2 - 1)^2}$
 $x = 0$ is the only critical point.
 $f(0) = -1$ is absolute maximum value of $f(x)$. Hence f has no absolute minimum on interval $(-1, 1)$.
- (c) No absolute extrema. (They would be at the endpoints which are not included in the interval.)
- (d) $f(x) = \frac{1}{x^2 - 1}$ on $\left[-\frac{1}{2}, \frac{1}{2}\right]$
 $f'(x) = \frac{-2x}{(x^2 - 1)^2}$
 $x = 0$ is critical point.
 f has an absolute maximum value of $f(0) = -1$. f assumes its minimum at two points $x = \pm \frac{1}{2}$ and minimum value is $f\left(-\frac{1}{2}\right) = f\left(\frac{1}{2}\right) = -\frac{4}{3}$.
2. (a) $f(x) = \frac{x^2}{(x-1)^2}$ on $(-\infty, 1) \cup (1, \infty)$
 $f'(x) = \frac{2x(x-1)^2 - 2x^2(x-1)}{(x-1)^4} = 0$
 $x = 0$ is critical point.
 f has an absolute minimum value of $f(0) = 0$ at $x = 0$ and no absolute maximum occurs.
- (b) $f(x) = \frac{x^2}{(x-1)^2}$ on $(-1, 1)$
 $f'(x) = \frac{2x(x-1)^2 - 2x^2(x-1)}{(x-1)^4} = 0$
 $x = 0$ is critical point.
 f has an absolute minimum value $f(0) = 0$ at $x = 0$ and there is no absolute maximum.
- (c) The function does not have a maximum or minimum. The minimum would be at $x = 0$ (not included in this interval) while

the asymptote at $x = 1$ precludes an absolute maximum.

- (d) $f(x) = \frac{x^2}{(x-1)^2}$ on $[-2, -1]$
 $f'(x) = \frac{2x(x-1)^2 - 2x^2(x-1)}{(x-1)^4}$
 $= \frac{-2x(x-1)}{(x-1)^4} < 0$ on $[-2, -1]$
 $f(x)$ is decreasing function on $[-2, -1]$.
 $f(x)$ is maximum at $x = -2$ and minimum at $x = -1$.
3. (a) $f(x) = x^2 + 5x - 1$
 $f'(x) = 2x + 5$
 $2x + 5 = 0$
 $x = -5/2$ is a critical number.
This is a parabola opening upward, so we have a minimum at $x = -5/2$.
- (b) $f(x) = -x^2 + 4x + 2$
 $f'(x) = -2x + 4 = 0$ when $x = 2$.
This is a parabola opening downward, so we have a maximum at $x = 2$.
4. (a) $f(x) = x^3 - 3x + 1$
 $f'(x) = 3x^2 - 3$
 $= 3(x^2 - 1)$
 $= 3(x+1)(x-1) = 0$
 $x = \pm 1$ are critical numbers and $f(1) = -1$, $f(-1) = 3$.
This is a cubic with a positive leading coefficient so $x = -1$ is a local max, $x = 1$ is a local min.
- (b) $f(x) = -x^3 + 6x^2 + 2$
 $f'(x) = -3x^2 + 12x = -3x(x+4) = 0$ when $x = 0$ and $x = -4$.
 $f(0) = 2$, $f(-4) = 162$.
This is a cubic with a negative leading coefficient so $x = 0$ is a local min and $x = -4$ is a local max.
5. (a) $f(x) = x^3 - 3x^2 + 6x$
 $f'(x) = 3x^2 - 6x + 6$
 $3x^2 - 6x + 6 = 3(x^2 - 2x + 2) = 0$
We can use the quadratic formula to find the roots, which are $x = 1 \pm \sqrt{-1}$. These are imaginary so there are no real critical points.
- (b) $f(x) = -x^3 + 3x^2 - 3x$
 $f'(x) = -3x^2 + 6x - 3$
 $= 3(-x^2 + 2x - 1)$
 $= -3(x^2 - 2x + 1)$
 $= -3(x-1)^2$

$$f'(x) = 3(x-1)^2 = 0 \text{ when } x = 1.$$

Since $f(x)$ is a cubic with only one critical number it is neither local min nor max.

6. (a) $f(x) = x^4 - 2x^2 + 1$

$$f'(x) = 4x^3 - 4x$$

$$= 4x(x^2 - 1)$$

$$= 4x(x-1)(x+1)$$

$$f'(x) = 0 \text{ when } x = 0, \pm 1.$$

$x = 0, \pm 1$ are critical numbers. $x = 0$ is local maximum and $x = \pm 1$ are local minimum.

(b) $f(x) = x^4 - 3x^3 + 2$

$$f'(x) = 4x^3 - 9x^2$$

$$= x^2(4x - 9)$$

$$f'(x) = 0 \text{ when } x = 0, \frac{9}{4}.$$

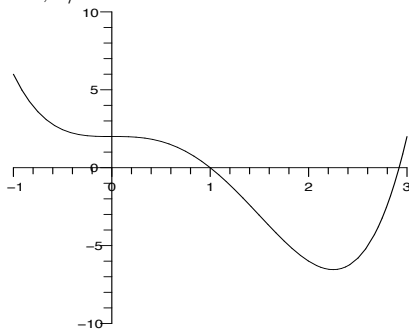
$x = 0, \frac{9}{4}$ are critical points. $x = \frac{9}{4}$ is local minimum and $x = 0$ is neither max nor min.

7. $f(x) = x^4 - 3x^3 + 2$

$$f'(x) = 4x^3 - 9x^2$$

$$4x^3 - 9x^2 = x^2(4x - 9) = 0$$

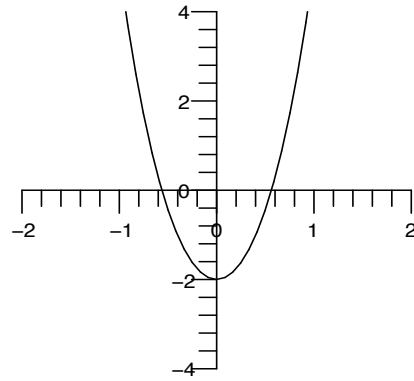
$x = 0, 9/4$ are critical numbers



$x = 9/4$ is a local min; $x = 0$ is neither a local max nor min.

8. $f(x) = x^4 + 6x^2 - 2$

$$f'(x) = 4x^3 + 12x = 0 \text{ when } x = 0 \text{ (minimum).}$$

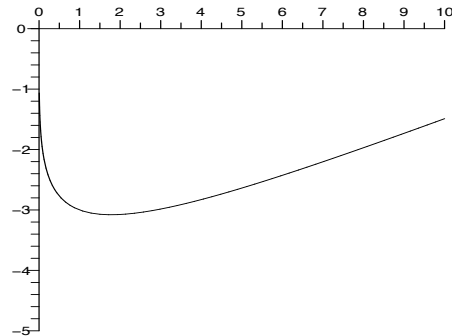


9. $f(x) = x^{3/4} - 4x^{1/4}$

$$f'(x) = \frac{3}{4x^{1/4}} - \frac{1}{x^{3/4}}$$

If $x \neq 0$, $f'(x) = 0$ when $3x^{3/4} = 4x^{1/4}$
 $x = 0, 16/9$ are critical numbers.

$x = 16/9$ is a local min, $x = 0$ is a local maximum.



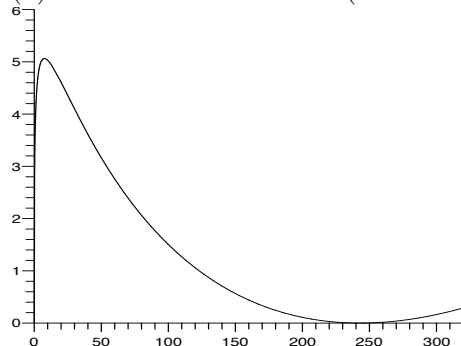
10. $f(x) = (x^{2/5} - 3x^{1/5})^2$

$$f'(x) = 2(x^{2/5} - 3x^{1/5}) \left(\frac{2}{5x^{3/5}} - \frac{3}{5x^{4/5}} \right)$$

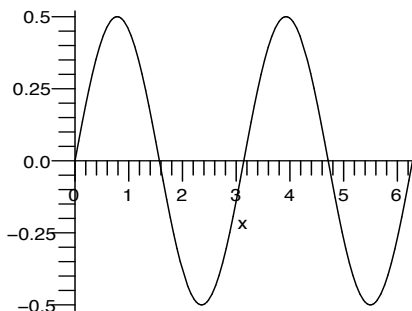
$$f'(x) = 0 \text{ when } x = 3^5 \text{ (minimum) and}$$

$$x = \left(\frac{3}{2}\right)^5 \text{ (maximum).}$$

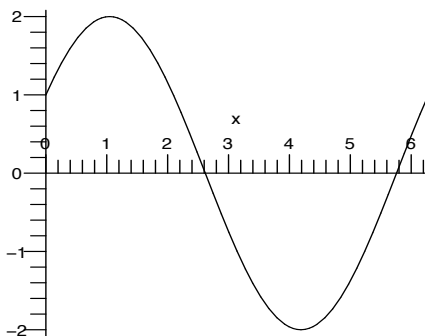
$f'(x)$ is undefined when $x = 0$ (minimum).



11. $f(x) = \sin x \cos x$ on $[0, 2\pi]$
 $f'(x) = \cos x \cos x + \sin x(-\sin x)$
 $= \cos^2 x - \sin^2 x$
 $\cos^2 x - \sin^2 x = 0$
 $\cos^2 x = \sin^2 x$
 $\cos x = \pm \sin x$
 $x = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$
are critical numbers.
 $x = \pi/4, 5\pi/4$ are local max, $x = 3\pi/4, 7\pi/4$
are local min.
Also $x = 0$ is local minimum and $x = 2\pi$ is
local maximum.

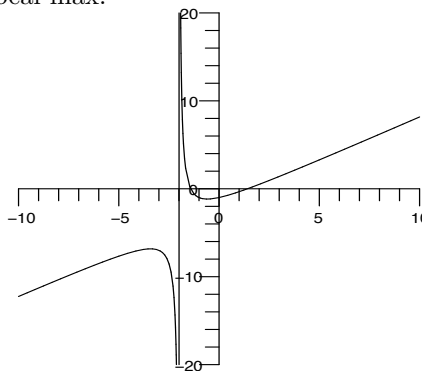


12. $f(x) = \sqrt{3} \sin x + \cos x$
 $f'(x) = \sqrt{3} \cos x - \sin x = 0$ when $\tan(x) = \sqrt{3}$
or $x = \pi/3 + k\pi$ for any integer k (maxima for
even k and minima for odd k).



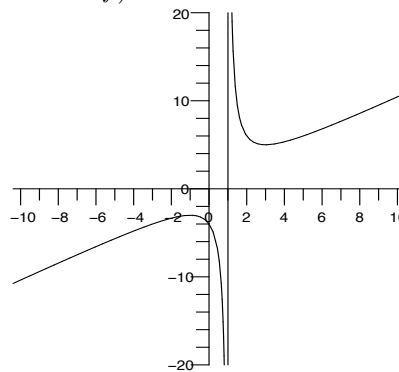
13. $f(x) = \frac{x^2 - 2}{x + 2}$
Note that $x = -2$ is not in the domain of f .
 $f'(x) = \frac{(2x)(x+2) - (x^2 - 2)(1)}{(x+2)^2}$
 $= \frac{2x^2 + 4x - x^2 + 2}{(x+2)^2}$
 $= \frac{x^2 + 4x + 2}{(x+2)}$
 $f'(x) = 0$ when $x^2 + 4x + 2 = 0$, so the critical
numbers are $x = -2 \pm \sqrt{2}$.

$x = -2 + \sqrt{2}$ is a local min; $x = -2 + \sqrt{2}$ is a
local max.

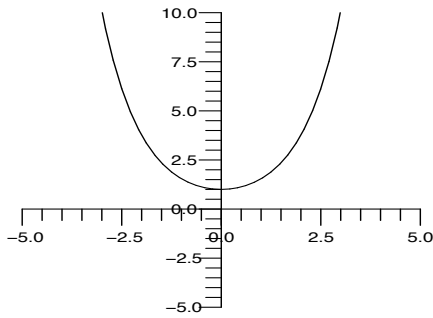


14. $f(x) = \frac{x^2 - x + 4}{x - 1}$
 $f'(x) = \frac{(x-1)(2x-1) - (x^2 - x + 4)}{(x-1)^2}$
 $= \frac{(x-3)(x+1)}{(x-1)^2} = 0$

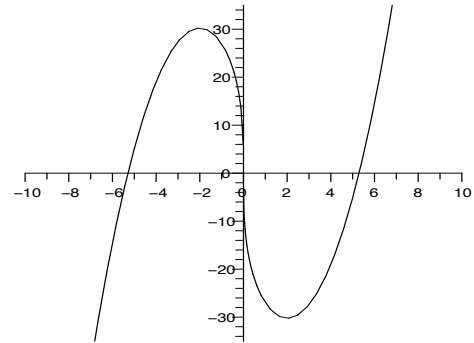
when $x = -1$ (maximum) and $x = 3$ (mini-
mum). $f'(x)$ is undefined when $x = 1$ (not in
domain of f).



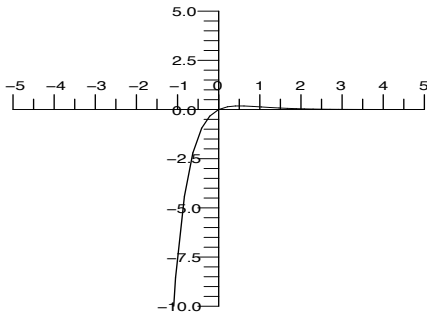
15. $f(x) = \frac{e^x + e^{-x}}{2}$
 $f'(x) = \frac{e^x - e^{-x}}{2}$
 $f'(x) = 0$ when $e^x = e^{-x}$, that is, $x = 0$.
 $f'(x)$ is defined for all x , so $x = 0$ is a critical
number. $x = 0$ is a local min.



$f'(x)$ is undefined at $x = 0$ (neither)



16. $f(x) = xe^{-2x}$
 $f'(x) = e^{-2x} - 2xe^{-2x} = 0$ when $x = \frac{1}{2}$ (maximum).

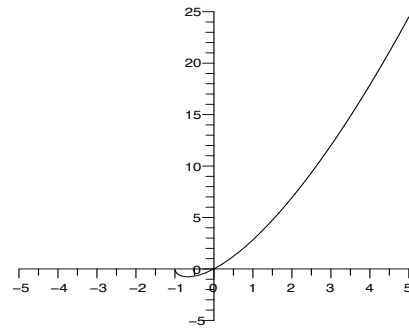


19. $f(x) = 2x\sqrt{x+1} = 2x(x+1)^{1/2}$
 Domain of f is all $x \geq -1$.
 $f'(x) = 2(x+1)^{1/2} + 2x(\frac{1}{2}(x+1)^{-1/2})$

$$= \frac{2(x+1) + x}{\sqrt{x+1}}$$

$$= \frac{3x+2}{\sqrt{x+1}}$$

 $f'(x) = 0$ for $3x+2 = 0$, $x = -2/3$.
 $x = -2/3$ is critical numbers.
 $f'(x)$ is undefined for $x = -1$.

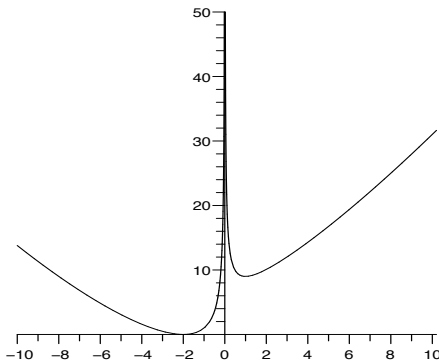


17. $f(x) = x^{4/3} + 4x^{1/3} + 4x^{-2/3}$
 f is not defined at $x = 0$.
 $f'(x) = \frac{4}{3}x^{1/3} + \frac{4}{3}x^{-2/3} - \frac{8}{3}x^{-5/3}$

$$= \frac{4}{3}x^{-5/3}(x^2 + x - 2)$$

$$= \frac{4}{3}x^{-5/3}(x-1)(x+2)$$

 $x = -2, 1$ are critical numbers.
 $x = -2$ and $x = 1$ are local minima.



$x = -2/3$ is a local min. $x = -1$ is an endpoint and local maximum.

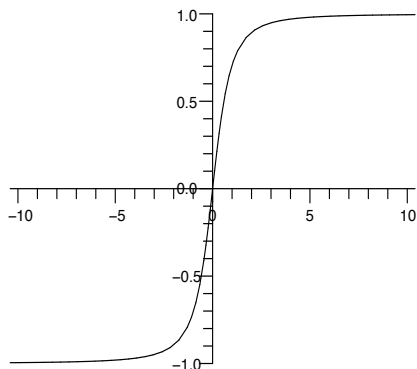
18. $f(x) = x^{7/3} - 28x^{1/3}$
 $f'(x) = \frac{7}{3}x^{4/3} - \frac{28}{3}x^{-2/3} = 0$ when $x = -2$ (local maximum) and $x = 2$ (local minimum).

20. $f(x) = \frac{x}{\sqrt{x^2+1}}$

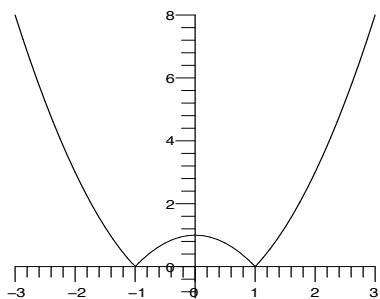
$$f'(x) = \frac{\sqrt{x^2+1} - \frac{x^2}{\sqrt{x^2+1}}}{x^2+1}$$

$$= \frac{1}{(x^2+1)^{3/2}} \neq 0 \text{ for any } x, \text{ and } f(x)$$

 has no critical points.

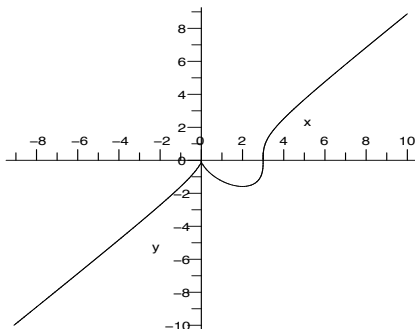


21. Because of the absolute value sign, there may be critical numbers where the function $x^2 - 1$ changes sign; that is, at $x = \pm 1$. For $x > 1$ and for $x < -1$, $f(x) = x^2 - 1$ and $f'(x) = 2x$, so there are no critical numbers on these intervals. For $-1 < x < 1$, $f(x) = 1 - x^2$ and $f'(x) = -2x$, so 0 is a critical number.



The graph confirms this analysis and shows there is a local max at $x = 0$ and local min at $x = \pm 1$.

22. $f(x) = \sqrt[3]{(x^3 - 3x^2)} = (x^3 - 3x^2)^{\frac{1}{3}}$
 $f'(x) = \frac{1}{3} \cdot \frac{3x^2 - 6x}{(x^3 - 3x^2)^{\frac{2}{3}}} = \frac{1}{3} \cdot \frac{3x^2 - 6x}{(x^3 - 3x^2)^{\frac{2}{3}}} = 0$
 when $x = 2$.
 $x = 2$ is critical number. $x = 2$ is local minimum. $x = 0$ is local maximum.



23. First, let's find the critical numbers for $x < 0$. In this case,

$$f(x) = x^2 + 2x - 1$$

$$f'(x) = 2x + 2 = 2(x + 1)$$

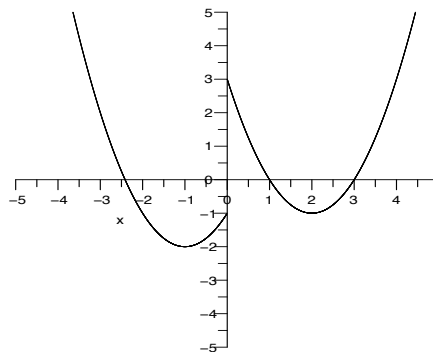
so the only critical number in this interval is $x = -1$ and it is a local minimum.

Now for $x > 0$,

$$f(x) = x^2 - 4x + 3$$

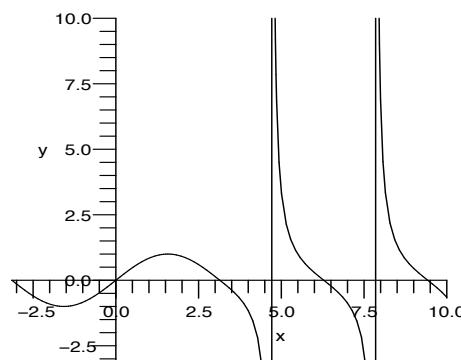
$$f'(x) = 2x - 4 = 2(x - 2)$$

so the only critical number is $x = 2$ and it is a local minimum.



Finally, since f is not continuous and hence not differentiable at $x = 0$. Indeed, $x = 0$ is a local maximum.

24. $f'(x) = \cos x$ for $-\pi < x < \pi$, and $f'(x) = -\sec^2 x$ for $|x| \geq \pi$.
 $f'(x) = 0$ for $x = -\pi/2$ (minimum) and $x = \pi/2$ (maximum).



$f'(x)$ is undefined for $x = (2k+1)\frac{\pi}{2}$ for integers $k \neq -1$ or 0 (not in domain of f).

25. $f(x) = x^3 - 3x + 1$
 $f'(x) = 3x^2 - 3 = 3(x^2 - 1)$
 $f'(x) = 0$ for $x = \pm 1$.

(a) On $[0, 2]$, 1 is the only critical number.

We calculate:

$$f(0) = 1$$

$$f(1) = -1 \text{ is the abs min.}$$

$$f(2) = 3 \text{ is the abs max.}$$

- (b) On the interval $[-3, 2]$, we have both 1 and -1 as critical numbers.

We calculate:

$$f(-3) = -17 \text{ is the abs min.}$$

$$f(-1) = 3 \text{ is the abs max.}$$

$$f(1) = -1$$

$$f(2) = 3 \text{ is also the abs max.}$$

26. $f(x) = x^4 - 8x^2 + 2$

$$f'(x) = 4x^3 - 16x = 0 \text{ when } x = 0 \text{ and } x = \pm 2.$$

- (a) On $[-3, 1]$:

$$f(-3) = 11, f(-2) = -14, f(0) = 2, \text{ and}$$

$$f(1) = -5.$$

The abs min on this interval is $f(-2) = -14$ and the abs max is $f(-3) = 11$.

- (b) On $[-1, 3]$:

$$f(-1) = -5, f(2) = -14, \text{ and } f(3) = 11.$$

The abs min on this interval is $f(2) = -14$ and the abs max is $f(3) = 11$.

27. $f(x) = x^{2/3}$

$$f'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3\sqrt[3]{x}}$$

$f'(x) \neq 0$ for any x , but $f'(x)$ undefined for $x = 0$, so $x = 0$ is critical number.

- (a) On $[-4, -2]$:

$0 \notin [-4, -2]$ so we only look at endpoints.

$$f(-4) = \sqrt[3]{16} \approx 2.52$$

$$f(-2) = \sqrt[3]{4} \approx 1.59$$

So $f(-4) = \sqrt[3]{16}$ is the abs max and

$$f(-2) = \sqrt[3]{4} \text{ is the abs min.}$$

- (b) On $[-1, 3]$, we have 0 as a critical number.

$$f(-1) = 1$$

$$f(0) = 0 \text{ is the abs min.}$$

$$f(3) = 3^{2/3} \text{ is the abs max.}$$

28. $f(x) = \sin x + \cos x$

$$f'(x) = \cos x - \sin x = 0 \text{ when } x = \frac{\pi}{4} + k\pi \text{ for integers } k.$$

- (a) On $[0, 2\pi]$:

$$f(0) = 1, f(\pi/4) = \sqrt{2}, f(5\pi/4) = -\sqrt{2}, \text{ and } f(2\pi) = 1.$$

The abs min on this interval is $f(5\pi/4) = -\sqrt{2}$ and the abs max is $f(\pi/4) = \sqrt{2}$.

- (b) On $[\pi/2, \pi]$:

$$f(\pi/2) = 1, f(\pi) = -1.$$

The abs min on this interval is $f(\pi) = -1$ and the abs max is $f(\pi/2) = 1$.

29. $f(x) = e^{-x^2}$

$$f'(x) = -2xe^{-x^2}$$

Hence $x = 0$ is the only critical number.

- (a) On $[0, 2]$:

$$f(0) = 1 \text{ is the abs max.}$$

$$f(2) = e^{-4} \text{ is the abs min.}$$

- (b) On $[-3, 2]$:

$$f(-3) = e^{-9} \text{ is the abs min.}$$

$$f(0) = 1 \text{ is the abs max.}$$

$$f(2) = e^{-4}$$

30. $f(x) = x^2e^{-4x}$

$$f'(x) = 2xe^{-4x} - 4x^2e^{-4x} = 0 \text{ when } x = 0 \text{ and } x = 1/2.$$

- (a) On $[-2, 0]$:

$$f(-2) = 4e^8, f(0) = 0.$$

The abs min is $f(0) = 0$ and the abs max is $f(-2) = 4e^8$.

- (b) On $[0, 4]$:

$$f(1/2) = e^{-2}/4, f(4) = 16e^{-16}.$$

The abs min is $f(0) = 0$ and the abs max is $f(1/2) = e^{-2}/4$.

31. $f(x) = \frac{3x^2}{x-3}$

Note that $x = 3$ is not in the domain of f .

$$\begin{aligned} f'(x) &= \frac{6x(x-3) - 3x^2(1)}{(x-3)^2} \\ &= \frac{6x^2 - 18x - 3x^2}{(x-3)^2} \\ &= \frac{3x^2 - 18x}{(x-3)^2} \\ &= \frac{3x(x-6)}{(x-3)^2} \end{aligned}$$

The critical points are $x = 0, x = 6$.

- (a) On $[-2, 2]$:

$$f(-2) = -12/5$$

$$f(2) = -12$$

$$f(0) = 0$$

Hence abs max is $f(0) = 0$ and abs min is $f(2) = -12$.

- (b) On $[2, 8]$, the function is not continuous and in fact has no absolute max or min.

32. $f(x) = \tan^{-1}(x^2)$

$$f'(x) = \frac{2x}{1+x^4} = 0 \text{ when } x = 0.$$

- (a) On $[0, 1]$:

$$f(0) = 0 \text{ and } f(1) = \pi/4.$$

The abs min is $f(0) = 0$ and the abs max is $f(1) = \pi/4$.

- (b) On $[-3, 4]$:

$$f(-3) \approx 1.46, f(0) = 0, \text{ and } f(4) \approx 1.51.$$

The abs min is $f(0) = 0$ and the abs max is $f(4) = \tan^{-1} 16$.

$$\begin{aligned}
 33. \quad f(x) &= \frac{x}{x^2 + 1} \\
 f'(x) &= \frac{(x^2 + 1) \cdot 1 - x \cdot (2x)}{(x^2 + 1)^2} \\
 &= \frac{(x^2 + 1) \cdot 1 - x \cdot (2x)}{(x^2 + 1)^2} = \frac{-x^2 + 1}{(x^2 + 1)^2} = 0
 \end{aligned}$$

when $x = \pm 1$.

$x = \pm 1$ are critical numbers.

$$\begin{aligned}
 (a) \quad &\text{On } [0, 2]: \\
 f(0) &= \frac{0}{0^2 + 1} = 0 \text{ is the abs minimum.} \\
 f(2) &= \frac{2}{2^2 + 1} = \frac{2}{5} \\
 f(1) &= \frac{1}{2} \text{ is the abs maximum.}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad &\text{On } [-3, 3]: \\
 f(3) &= -\frac{3}{10} \\
 f(-1) &= -\frac{1}{2} \text{ is the abs minimum.} \\
 f(1) &= \frac{1}{2} \text{ is the abs maximum.} \\
 f(3) &= \frac{3}{10}
 \end{aligned}$$

$$\begin{aligned}
 34. \quad f(x) &= \frac{3x}{x^2 + 16} \\
 f'(x) &= \frac{(x^2 + 16) \cdot 3 - 3x \cdot (2x)}{(x^2 + 16)^2} \\
 &= \frac{(x^2 + 16) \cdot 3 - 3x \cdot (2x)}{(x^2 + 16)^2} = 0 \\
 &= \frac{-3x^2 + 48}{(x^2 + 16)^2} = 0 \text{ when } x = \pm 4.
 \end{aligned}$$

$x = \pm 4$ are critical numbers.

$$\begin{aligned}
 (a) \quad &\text{On } [0, 2]: \\
 f(0) &= \frac{0}{0^2 + 16} = 0 \text{ is the abs minimum.} \\
 f(2) &= \frac{2}{2^2 + 16} = \frac{3}{10} \text{ is the abs maximum.} \\
 (b) \quad &\text{on } [0, 6]: \\
 f(0) &= 0 \text{ is abs minimum.} \\
 f(4) &= \frac{3}{8} \text{ is abs maximum.} \\
 f(6) &= \frac{9}{26}
 \end{aligned}$$

$$35. \quad f'(x) = 4x^3 - 6x + 2 = 0 \text{ at about } x = 0.3660, -1.3660 \text{ and at } x = 1.$$

$$\begin{aligned}
 (a) \quad &f(-1) = -3, f(1) = 1. \\
 &\text{The absolute min is } (-1, -3) \text{ and} \\
 &\text{the absolute max is approximately} \\
 &\text{(0.3660, 1.3481).}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad &\text{The absolute min is approximately} \\
 &(-1.3660, -3.8481) \text{ and the absolute max} \\
 &\text{is } (-3, 49).
 \end{aligned}$$

$$36. \quad f'(x) = 6x^5 - 12x - 2 = 0 \text{ at about } -1.3673, -0.5860 \text{ and } 1.4522.$$

$$\begin{aligned}
 (a) \quad &f(-1) = 1, f(1) = -3. f(-0.5860) = 1.8587. \\
 &\text{The absolute min is } f(1) = -3 \\
 &\text{and the absolute max is approximately} \\
 &f(-0.5860) = 1.8587.
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad &f(-2) = 21 \text{ and } f(2) = 13. f(-1.3673) = -2.165 \\
 &\text{and } f(1.4522) = -5.8675. \\
 &\text{The absolute min is approximately} \\
 &f(1.4522) = -5.8675 \text{ and the absolute} \\
 &\text{max is } f(-2) = 21.
 \end{aligned}$$

$$37. \quad f'(x) = \sin x + x \cos x = 0 \text{ at } x = 0 \text{ and about } 2.0288 \text{ and } 4.9132.$$

$$\begin{aligned}
 (a) \quad &\text{The absolute min is } (0, 3) \text{ and the absolute} \\
 &\text{max is } (\pm\pi/2, 3 + \pi/2).
 \end{aligned}$$

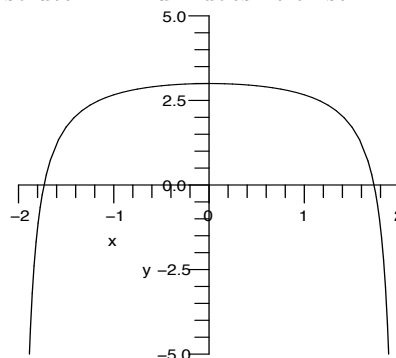
$$\begin{aligned}
 (b) \quad &\text{The absolute min is approximately} \\
 &(4.9132, -1.814) \text{ and the absolute max is} \\
 &\text{approximately } (2.0288, 4.820).
 \end{aligned}$$

$$38. \quad f'(x) = 2x + e^x = 0 \text{ at approximately } x = -0.3517.$$

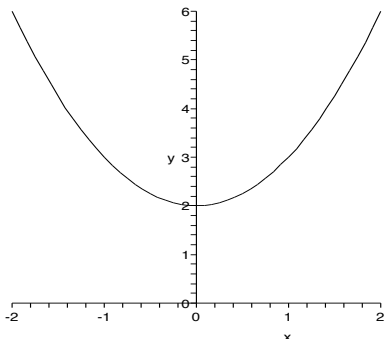
$$\begin{aligned}
 (a) \quad &f(0) = 1 \text{ and } f(1) = 1 + e \approx 3.71828. \\
 &f'(x) \neq 0 \text{ on this interval, so the absolute} \\
 &\text{min is } f(0) = 1 \text{ and the absolute max is} \\
 &f(1) = 1 + e \approx 3.71828.
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad &f(-2) \approx 4.1353 \text{ and } f(2) \approx 11.3891. \\
 &f(-0.3517) = 0.8272. \\
 &\text{The absolute min is approximately} \\
 &f(-0.3517) = 0.8272 \text{ and the absolute} \\
 &\text{max is approximately } f(2) = 11.3891.
 \end{aligned}$$

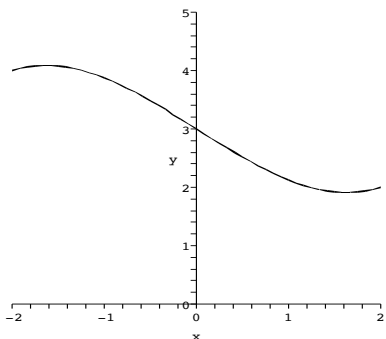
$$39. \quad \text{On } [-2, 2], \text{ the absolute maximum is } 3 \text{ and the absolute minimum doesn't exist.}$$



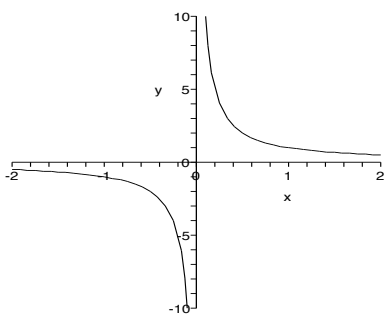
40. On $(-2, 2)$ minimum is 2 and the maximum does not exist. (The maximum would exist at the endpoints which are not included in the interval.)



41. On $(-2, 2)$ the absolute maximum is 4 and the absolute minimum is 2.



42. Absolute extrema do not exist because of the vertical asymptote.



43. $f(x) = x^3 + cx + 1$
 $f'(x) = 3x^2 + c$

We know (perhaps from a pre-calculus course) that for any cubic polynomial with positive leading coefficient, when x is large and positive the value of the polynomial is very large and positive, and when x is large and negative, the value of the polynomial is very large and negative.

Type 1: $c > 0$. There are no critical numbers. As you move from left to right, the graph of f is always rising.

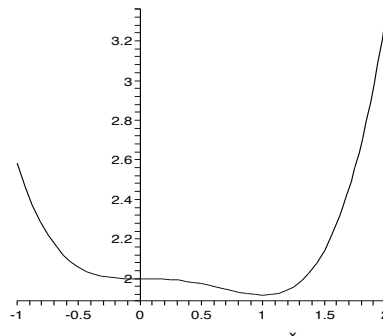
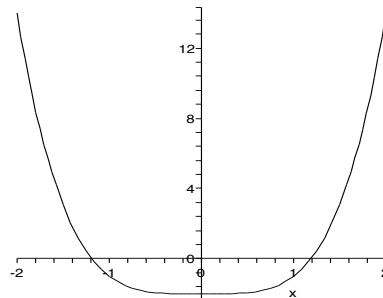
Type 2: $c < 0$. There are two critical numbers $x = \pm\sqrt{-c/3}$. As you move from left to right, the graph rises until we get to the first critical number, then the graph must fall until we get to the second critical number, and then the graph rises again. So the critical number on the left is a local maximum and the critical number on the right is a local minimum.

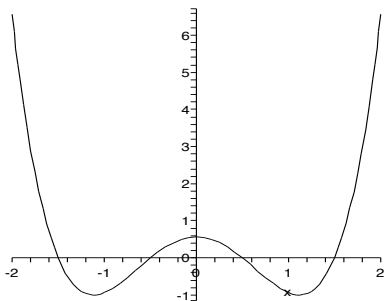
Type 3: $c = 0$. There is only one critical number, which is neither a local max nor a local min.

44. The derivative of a fourth-order polynomial is a cubic polynomial. We know that cubic polynomials must have one root, and can have up to three roots. If $p(x)$ is a fourth-order polynomial, we know that

$$\lim_{x \rightarrow \infty} p(x) = \lim_{x \rightarrow -\infty} p(x) = \infty$$

if the coefficient of x^4 is positive, and is $-\infty$ if the coefficient of x^4 is negative. This guarantees that at least one of the critical numbers will be an extremum.





45. $f(x) = x^3 + bx^2 + cx + d$
 $f'(x) = 3x^2 + 2bx + c$

The quadratic formula says that the critical numbers are

$$x = \frac{-2b \pm \sqrt{4b^2 - 12c}}{6}$$

$$= \frac{-b \pm \sqrt{b^2 - 3c}}{3}.$$

So if $c < 0$, the quantity under the square root is positive and there are two critical numbers. This is like the Type 2 cubics in Exercise 53. We know that as x goes to infinity, the polynomial $x^3 + bx^2 + cx + d$ gets very large and positive, and when x goes to minus infinity, the polynomial is very large but negative. Therefore, the critical number on the left must be a local max, and the critical number on the right must be a local min.

46. $f'(x) = 3x^2 + 2bx + c = 0$ when $x = \frac{-2b \pm \sqrt{4b^2 - 12c}}{6}$. Adding these values together yields $-2b/3$.

47. $f(x) = x^4 + cx^2 + 1$
 $f'(x) = 4x^3 + 2cx = 2x(2x^2 + c)$
 So $x = 0$ is always a critical number.

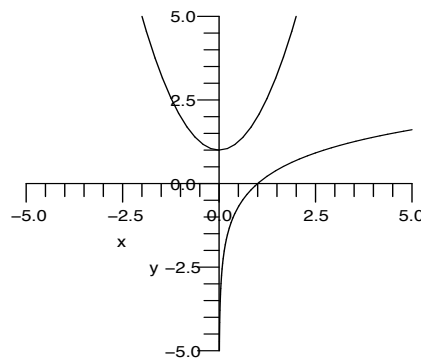
Case 1: $c \geq 0$. The only solution to $2x(2x^2 + c) = 0$ is $x = 0$, so $x = 0$ is the only critical number. This must be a minimum, since we know that the function $x^4 + cx^2 + 1$ is large and positive when $|x|$ is large (so the graph is roughly U-shaped). We could also note that $f(0) = 1$, and 1 is clearly the absolute minimum of this function if $c \geq 0$.

Case 2: $c < 0$. Then there are two other critical numbers $x = \pm\sqrt{-c/2}$. Now $f(0)$ is still equal to 1, but the value of f at both new critical numbers is less than 1. Hence $f(0)$ is a local max, and both new critical numbers are local minimums.

48. $f'(x) = 4x^3 + 3cx^2 = 0$ when $x = 0$ and $x = -3c/4$. Only $x = -3c/4$ will be an extreme point (an absolute minimum). $x = 0$ will be an inflection point.

49. Since f is differentiable on (a, b) , it is continuous on the same interval. Since f is decreasing at a and increasing at b , f must have a local minimum for some value c , where $a < c < b$. By Fermat's theorem, c is a critical number for f . Since f is differentiable at c , $f'(c)$ exists, and therefore $f'(c) = 0$.

50. Graph of $f(x) = x^2 + 1$ and $g(x) = \ln x$:



$$h(x) = f(x) - g(x) = x^2 + 1 - \ln x$$

$$h'(x) = 2x - 1/x = 0$$

$$2x^2 = 1$$

$$x = \pm\sqrt{1/2}$$

$$x = \sqrt{1/2} \text{ is min}$$

$$f'(x) = 2x$$

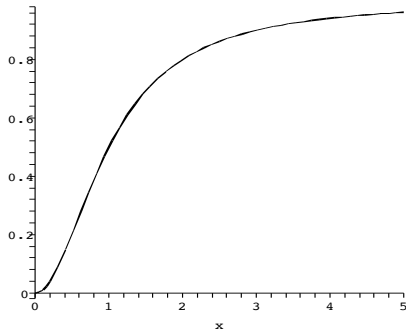
$$g'(x) = 1/x$$

$$f'(\sqrt{1/2}) = 2\sqrt{1/2} = \sqrt{2}$$

$$g'(\sqrt{1/2}) = \frac{1}{\sqrt{1/2}} = \sqrt{2}$$

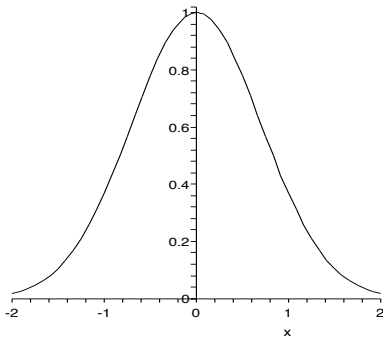
So the tangents are parallel. If the tangent lines were not parallel, then they would be getting closer together in one direction. Since the tangent lines approximate the curves, this should mean the curves are also getting closer together in that direction.

51. Graph of $f(x) = \frac{x^2}{x^2 + 1}$:



$$\begin{aligned}
 f'(x) &= \frac{2x(x^2 + 1) - x^2(2x)}{(x^2 + 1)^2} \\
 &= \frac{2x}{(x^2 + 1)^2} \\
 f''(x) &= \frac{2(x^2 + 1)^2 - 2x \cdot 2(x^2 + 1) \cdot 2x}{(x^2 + 1)^4} \\
 &= \frac{2(x^2 + 1)[(x^2 + 1) - 4x^2]}{(x^2 + 1)^4} \\
 &= \frac{2[1 - 3x^2]}{(x^2 + 1)^3} \\
 f''(x) &= 0 \text{ for } x = \pm \frac{1}{\sqrt{3}}, \\
 x &= -\frac{1}{\sqrt{3}} \notin (0, \infty) \\
 x &= \frac{1}{\sqrt{3}} \text{ is steepest point.}
 \end{aligned}$$

52. Graph of $f(x) = e^{-x^2}$:



$f(x)$ is steepest where $f'(x) = -2xe^{-x^2}$ is maximum.
 $f''(x) = -2e^{-x^2} + 4x^2e^{-x^2} = 0$ when $x = \pm\sqrt{2}/2$. This is where $f(x)$ is steepest.

53. With $t = 90$ and $r = 1/30$, we have

$P(n) = \frac{3^n}{n!}e^{-3}$. We compute P for the first few values of n :

n	P
0	e^{-3}
1	$3e^{-3}$
2	$4.5e^{-3}$
3	$4.5e^{-3}$
4	$3.375e^{-3}$

Once $n > 3$, the values of P will decrease as n increases. This is due to the fact that to get $P(n+1)$ from $P(n)$, we multiply $P(n)$ by $3/(n+1)$. Since $n > 3$, $3/(n+1) < 1$ and so $P(n+1) < P(n)$. Thus we see from the table that P is maximized at $n = 3$ (it is also maximized at $n = 2$). It makes sense that P would be maximized at $n = 3$ because $(90 \text{ mins}) \left(\frac{1}{30} \text{ goals/min} \right) = 3 \text{ goals}$.

54. $f(p) = p^m(1-p)^{n-m}$
 $f'(p) = mp^{m-1}(1-p)^{n-m} - p^m(n-m)(1-p)^{n-m-1}$

To find the critical numbers, we set $f'(p) = 0$ which gives

$$\begin{aligned}
 mp^{m-1}(1-p)^{n-m} - p^m(n-m)(1-p)^{n-m-1} &= 0 \\
 mp^{m-1}(1-p)^{n-m} &= p^m(n-m)(1-p)^{n-m-1} \\
 m(1-p) &= p(n-m) \\
 m - mp &= pn - pm \\
 p &= m/n.
 \end{aligned}$$

Since this is the only critical number, $f(p)$ is continuous, $f(0) = f(1) = 0$ and $f(m/n) > 0$, $p = m/n$ must maximize $f(p)$.

55. $y = x^5 - 4x^3 - x + 10$, $x \in [-2, 2]$
 $y' = 5x^4 - 12x^2 - 1$
 $x = -1.575, 1.575$ are critical numbers of y . There is a local max at $x = -1.575$, local min at $x = 1.575$.
 $x = -1.575$ represents the top and $x = 1.575$ represents the bottom of the roller coaster.
 $y''(x) = 20x^3 - 24x = 4x(5x^2 - 6) = 0$
 $x = 0, \pm\sqrt{6/5}$ are critical numbers of y' . We calculate y' at the critical numbers and at the endpoints $x = \pm 2$:
 $y'(0) = -1$
 $y'(\pm\sqrt{6/5}) = -41/5$
 $y'(\pm 2) = 31$
 So the points where the roller coaster is making the steepest descent are $x = \pm\sqrt{6/5}$, but the steepest part of the roller coast is during the ascents at ± 2 .

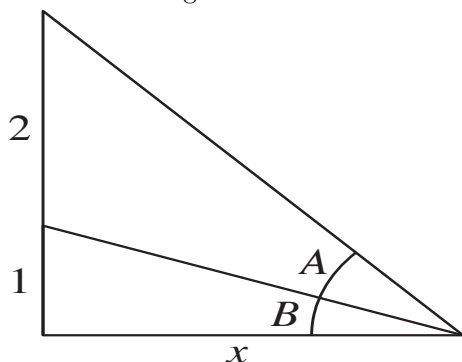
56. To maximize entropy, we find the critical numbers of H .

$H'(x) = -\ln x - 1 + \ln(1-x) + 1 = 0$
 where $\ln x = \ln(1-x)$, or where $x = 1-x$. That
 is $x = 1/2$. This maximizes unpredictability
 since for this value, errors and non-errors are
 equally likely.

57. $W(t) = a \cdot e^{-be^{-t}}$
 as $t \rightarrow \infty, -be^{-t} \rightarrow 0$, so $W(t) \rightarrow a$.
 $W'(t) = a \cdot e^{-be^{-t}} \cdot be^{-t}$
 as $t \rightarrow \infty, be^{-t} \rightarrow 0$, so $W'(t) \rightarrow 0$.
 $W''(t) = (a \cdot e^{-be^{-t}} \cdot be^{-t}) \cdot be^{-t}$
 $+ (a \cdot e^{-be^{-t}}) \cdot (-be^{-t})$
 $= a \cdot e^{-be^{-t}} \cdot be^{-t} [be^{-t} - 1]$
 $W''(t) = 0$ when $be^{-t} = 1$
 $e^{-t} = b^{-1}$
 $-t = \ln b^{-1}$
 $t = \ln b$
 $W'(\ln b) = a \cdot e^{-be^{-\ln b}} \cdot be^{-\ln b}$
 $= a \cdot e^{-b(\frac{1}{b})} \cdot b \cdot \frac{1}{b} = ae^{-1}$
 Maximum growth rate is ae^{-1} when $t = \ln b$.

58. $R'([S]) = \frac{(K_m + [S])R_m - [S]R_m}{(K_m + [S])^2} \neq 0$. The
 function doesn't have a true maximum, but
 $\lim_{[S] \rightarrow \infty} R = R_m$. The rate of reaction ap-
 proaches R_m but never reaches it.

59. Label the triangles as illustrated.



$$\begin{aligned}\tan(A+B) &= 3/x \\ A+B &= \tan^{-1}(3/x) \\ \tan B &= 1/x \\ B &= \tan^{-1}(1/x)\end{aligned}$$

Therefore,

$$\begin{aligned}A &= (A+B) - B \\ A &= \tan^{-1}(3/x) - \tan^{-1}(1/x) \\ \frac{dA}{dx} &= \frac{-3/x^2}{1 + (3/x)^2} - \frac{-1/x^2}{1 + (1/x)^2} \\ &= \frac{1}{x^2 + 1} - \frac{3}{x^2 + 9}\end{aligned}$$

The maximum viewing angle will occur at a
 critical value.

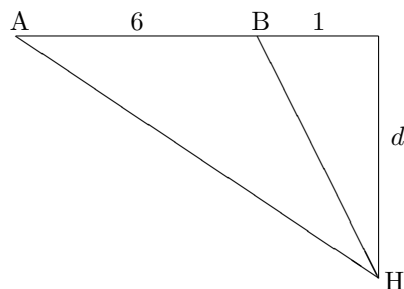
$$\frac{dA}{dx} = 0$$

$$\begin{aligned}\frac{1}{x^2 + 1} &= \frac{3}{x^2 + 9} \\ x^2 + 9 &= 3x^2 + 3 \\ 2x^2 &= 6 \\ x^2 &= 3\end{aligned}$$

$$x = \sqrt{3} \text{ ft} \approx 1.73 \text{ ft}$$

This is a maximum because when x is large
 and when x is a little bigger than 0, the angle
 is small.

60. (a) For the hockey player, $m\angle AHB$ is the
 shooting angle θ .



Therefore,

$$\theta = \tan^{-1}\left(\frac{7}{d}\right) - \tan^{-1}\left(\frac{1}{d}\right)$$

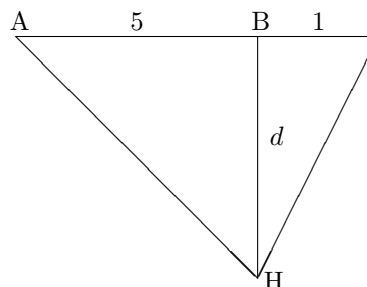
Hence,

$$\begin{aligned}\theta' &= \frac{1}{1 + (\frac{49}{d^2})} \left(\frac{-7}{d^2}\right) - \frac{1}{1 + (\frac{1}{d^2})} \left(\frac{-1}{d^2}\right) \\ &= \frac{-7}{d^2 + 49} + \frac{1}{d^2 + 1}\end{aligned}$$

To get the maximum angle,

$$\begin{aligned}\theta' &= \frac{-7}{d^2 + 49} + \frac{1}{d^2 + 1} = 0 \\ -7d^2 - 7 + d^2 + 49 &= 0 \\ 6d^2 &= 42 \\ d &= \sqrt{7}\end{aligned}$$

- (b) For the hockey player, $m\angle AHB$ is the
 shooting angle θ .



Therefore,

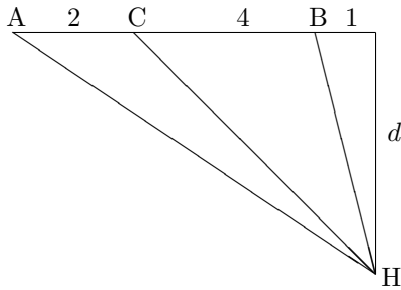
$$\theta = \tan^{-1} \left(\frac{5}{d} \right) + \tan^{-1} \left(\frac{1}{d} \right)$$

Hence,

$$\begin{aligned} \theta' &= \frac{1}{1 + \left(\frac{25}{d^2} \right)} \left(\frac{-5}{d^2} \right) + \frac{1}{1 + \left(\frac{1}{d^2} \right)} \left(\frac{-1}{d^2} \right) \\ &= -\frac{5}{d^2 + 25} - \frac{1}{d^2 + 1} \end{aligned}$$

The function is decreasing as the derivative is negative. Hence the angle is maximum when θ is minimum = 0.

- (c) For the hockey player, $m\angle AHC$ is the shooting angle, θ .



Therefore,

$$\theta = \tan^{-1} \left(\frac{7}{d} \right) - \tan^{-1} \left(\frac{5}{d} \right)$$

Hence,

$$\begin{aligned} \theta' &= \frac{1}{1 + \left(\frac{49}{d^2} \right)} \left(\frac{-7}{d^2} \right) - \frac{1}{1 + \left(\frac{25}{d^2} \right)} \left(\frac{-5}{d^2} \right) \\ &= -\frac{7}{d^2 + 49} + \frac{5}{d^2 + 25} \end{aligned}$$

To get the maximum angle,

$$\begin{aligned} \theta' &= -\frac{7}{d^2 + 49} + \frac{5}{d^2 + 25} = 0 \\ -7d^2 - 175 + 5d^2 + 245 &= 0 \\ 2d^2 &= 70 \\ d &= \sqrt{35} \end{aligned}$$

y is increasing on $(1, \infty)$ and on $(-\infty, -1)$

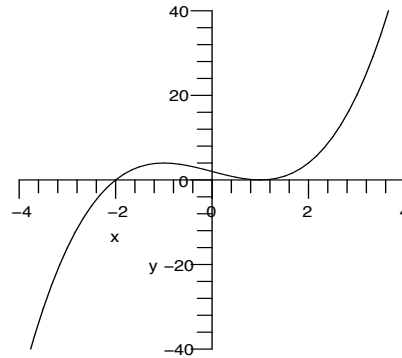
$3(x+1)(x-1) < 0$ on $(-1, 1)$, so y is decreasing on $(-1, 1)$.

$$y'' = 6x$$

$$y'' = -6 < 0 \text{ at } x = -1$$

Hence the function is a local maximum at $x = -1$.

$y'' = 6 > 0$ at $x = 1$. Hence $y(1) = 0$ is a local minimum.



2. $y = x^3 + 2x^2 + 1$

$$y' = 3x^2 + 4x = x(3x + 4)$$

The function is increasing when $x < -\frac{4}{3}$, decreasing when $-\frac{4}{3} < x < 0$, and increasing when $x > 0$.

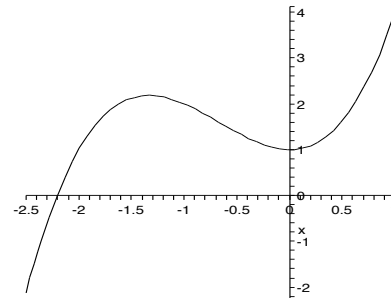
$$y'' = 6x + 4$$

$$y'' = -12 < 0 \text{ at } x = -\frac{4}{3}$$

Hence $f(-\frac{4}{3})$ is a local maximum at $x = -\frac{4}{3}$.

$$y'' = 4 > 0 \text{ at } x = 0$$

Hence $y(0)$ is a local minimum at $x = 0$.



3.4 Increasing and Decreasing Functions

1. $y = x^3 - 3x + 2$

$$\begin{aligned} y' &= 3x^2 - 3 = 3(x^2 - 1) \\ &= 3(x+1)(x-1) \end{aligned}$$

$x = \pm 1$ are critical numbers.

$(x+1) > 0$ on $(-1, \infty)$, $(x+1) < 0$ on $(-\infty, -1)$

$(x-1) > 0$ on $(1, \infty)$, $(x-1) < 0$ on $(-\infty, -1)$

$3(x+1)(x-1) > 0$ on $(1, \infty) \cup (-\infty, -1)$ so

3. $y = x^4 - 8x^2 + 1$

$$y' = 4x^3 - 16x = 4x(x^2 - 4)$$

$$= 4x(x-2)(x+2)$$

$x = 0, 2, -2$ are critical numbers.

$4x > 0$ on $(0, \infty)$, $4x < 0$ on $(-\infty, 0)$

$(x-2) > 0$ on $(2, \infty)$, $(x-2) < 0$ on $(-\infty, 2)$

$(x+2) > 0$ on $(-2, \infty)$, $(x+2) < 0$ on $(-\infty, -2)$

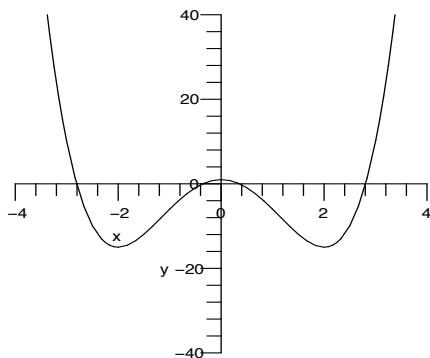
$4(x-2)(x+2) > 0$ on $(-2, 0) \cup (2, \infty)$, so the function is increasing on $(-2, 0)$ and on $(2, \infty)$.

$4(x-2)(x+2) < 0$ on $(-\infty, -2) \cup (0, 2)$, so y is decreasing on $(-\infty, -2)$ and on $(0, 2)$.

$$y'' = 12x^2 - 16$$

At $x = 0$, $y'' < 0$. Hence $y(0)$ is a local maximum at $x = 0$.

$y'' = 12(\pm 2)^2 - 16 > 0$ at $x = \pm 2$. Hence $y(\pm 2)$ are local minima at $x = \pm 2$.



4. $y = x^3 - 3x^2 - 9x + 1$

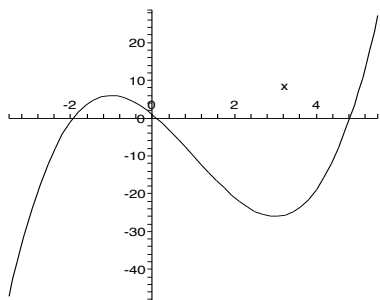
$$y' = 3x^2 - 6x - 9 = 3(x-3)(x+1).$$

The function is increasing when $x < -1$, decreasing when $-1 < x < 3$, and increasing when $x > 3$.

$$y'' = 6x - 6$$

$y'' = -12 < 0$ at $x = -1$. Hence the function is a local maximum at $x = -1$.

$y'' = 12 > 0$ at $x = 3$. Hence the function is a local minimum at $x = 3$.



5. $y = (x+1)^{2/3}$

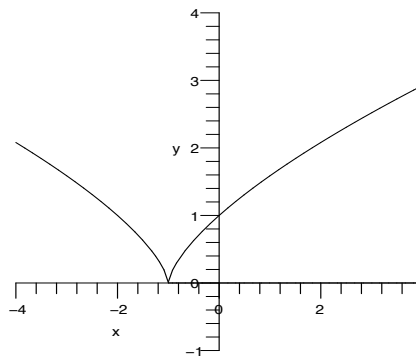
$$y' = \frac{2}{3}(x+1)^{-1/3} = \frac{2}{3\sqrt[3]{x+1}}$$

y' is not defined for $x = -1$

$\frac{2}{3\sqrt[3]{x+1}} > 0$ on $(-1, \infty)$, y is increasing

$\frac{2}{3\sqrt[3]{x+1}} < 0$ on $(-\infty, -1)$, y is decreasing

The graph has minimum at $x = -1$.

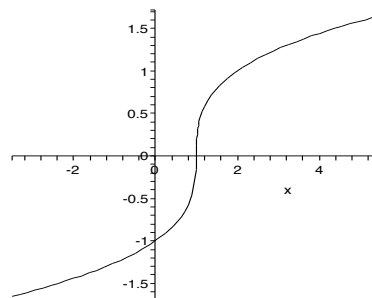


6. $y = (x-1)^{1/3}$

$$y' = \frac{1}{3}(x-1)^{-2/3}.$$

The function is increasing for all x . The slope approaches vertical as x approaches 1.

The graph has no extrema.



7. $y = \sin x + \cos x$

$$y' = \cos x - \sin x = 0$$

$$\cos x = \sin x$$

$x = \pi/4, 5\pi/4, 9\pi/4$, etc. $\cos x - \sin x > 0$ on $(-3\pi/4, \pi/4) \cup (5\pi/4, 9\pi/4) \cup \dots$

$\cos x - \sin x < 0$ on $(\pi/4, 5\pi/4) \cup (9\pi/4, 13\pi/4) \cup \dots$

So $y = \sin x + \cos x$ is decreasing on

$(\pi/4, 5\pi/4), (9\pi/4, 13\pi/4)$,

etc., and is increasing on

$(-3\pi/4, \pi/4), (5\pi/4, 9\pi/4)$, etc.

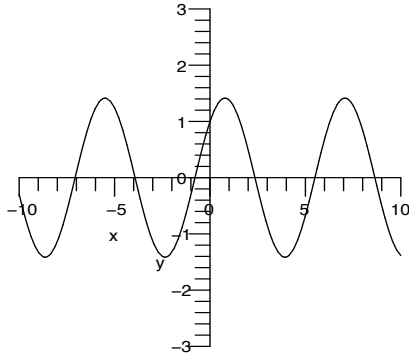
$$y'' = -\sin x - \cos x$$

$$y'' = -\frac{2}{\sqrt{2}} < 0 \text{ at } x = \pi/4, x = 9\pi/4, \text{ etc.}$$

Hence the function is local maximum at $x = \pi/4, x = 9\pi/4$, etc.

$y'' = \sqrt{2} > 0$ at $x = 5\pi/4, x = 13\pi/4$ etc.

Hence the function is local minimum at $x = 5\pi/4, x = 13\pi/4$ etc.



8. $y = \sin^2 x$

$$y' = 2 \sin x \cos x.$$

The function is increasing for $0 < x < \frac{\pi}{2}$, and decreasing for $\frac{\pi}{2} < x < \pi$, and this pattern repeats with period π .

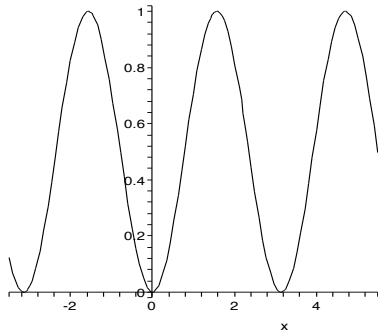
$$y'' = 2 \cos 2x$$

$$y'' = -2 < 0 \text{ at } x = \pi/2, x = 3\pi/2, \text{ etc.}$$

Hence the function is local maximum at $x = \pi/2, x = 3\pi/2$, etc.

$$y'' = \sqrt{2} > 0 \text{ at } x = 0, x = \pi, \text{ etc.}$$

Hence the function is a local minimum at $x = 0, x = \pi$, etc.



9. $y = e^{x^2-1}$

$$y' = e^{x^2-1} \cdot 2x = 2xe^{x^2-1}$$

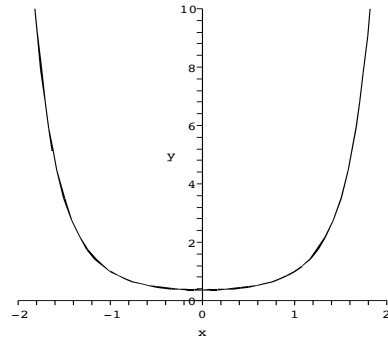
$$x = 0$$

$$2xe^{x^2-1} > 0 \text{ on } (0, \infty), y \text{ is increasing}$$

$$2xe^{x^2-1} < 0 \text{ on } (-\infty, 0), y \text{ is decreasing}$$

$$y'' = 2e^{x^2-1} [2x^2 + 1]$$

$y'' = 0.736 > 0$ at $x = 0$. Hence the function is a local minimum at $x = 0$.

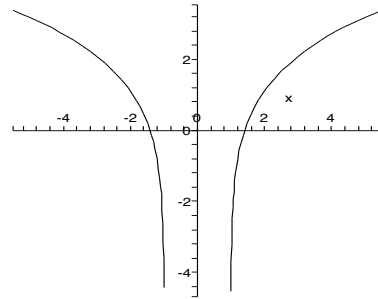


10. $y = \ln(x^2 - 1)$

$$y' = \frac{2x}{x^2 - 1}.$$

The function is defined for $|x| > 1$. The function is decreasing for $x < -1$ and increasing for $x > 1$.

The graph has no extrema.



11. $y = x^4 + 4x^3 - 2$

$$y' = 4x^3 + 12x^2 = 4x^2(x + 3)$$

Critical numbers are $x = 0, x = -3$.

$$4x^2(x + 3) > 0 \text{ on } (-3, 0) \cup (0, \infty)$$

$$4x^2(x + 3) < 0 \text{ on } (-\infty, -3)$$

Hence $x = -3$ is a local minimum and $x = 0$ is not an extremum.

12. $y = x^5 - 5x^2 + 1$

$$y' = 5x^4 - 10x = 5x(x^3 - 2).$$

At $x = 0$ the slope changes from positive to negative indicating a local maximum. At $x = \sqrt[3]{2}$ the slope changes from negative to positive indicating a local minimum.

13. $y = xe^{-2x}$

$$y' = 1 \cdot e^{-2x} + x \cdot e^{-2x}(-2)$$

$$= e^{-2x} - 2xe^{-2x}$$

$$= e^{-2x}(1 - 2x)$$

$$x = \frac{1}{2}$$

$$e^{-2x}(1 - 2x) > 0 \text{ on } (-\infty, 1/2)$$

$$e^{-2x}(1 - 2x) < 0 \text{ on } (1/2, \infty)$$

So $y = xe^{-2x}$ has a local maximum at $x = 1/2$.

14. $y = x^2 e^{-x}$

$$y' = 2xe^{-x} - x^2 e^{-x} = xe^{-x}(2 - x).$$

At $x = 0$ the slope changes from negative to positive indicating a local minimum. At $x = 2$ the slope changes from positive to negative indicating a local maximum.

15. $y = \tan^{-1}(x^2)$

$$y' = \frac{2x}{1+x^4}$$

Critical number is $x = 0$.

$$\frac{2x}{1+x^4} > 0 \text{ for } x > 0$$

$\frac{2x}{1+x^4} < 0$ for $x < 0$. Hence $x = 0$ is a local minimum.

16. $y = \sin^{-1}\left(1 - \frac{1}{x^2}\right)$

$$y' = \frac{2}{x^3} \cdot \frac{1}{\sqrt{1 - \left(1 - \frac{1}{x^2}\right)^2}}.$$

The derivative is never 0 and is defined where the function is defined, so there are no critical points.

17. $y = \frac{x}{1+x^3}$ Note that the function is not defined for $x = -1$.

$$\begin{aligned} y' &= \frac{1(1+x^3) - x(3x^2)}{(1+x^3)^2} \\ &= \frac{1+x^3-3x^3}{(1+x^3)^2} \\ &= \frac{1-2x^3}{(1+x^3)^2} \end{aligned}$$

Critical number is $x = \sqrt[3]{1/2}$

$y' > 0$ on $(-\infty, -1) \cup (-1, -\sqrt[3]{1/2})$

$y' < 0$ on $(\sqrt[3]{1/2}, \infty)$

Hence $x = \sqrt[3]{1/2}$ is a local max.

18. $y = \frac{x}{1+x^4}$

$$y' = \frac{(1+x^4) - 4x^4}{(1+x^4)^2} = \frac{1-3x^4}{(1+x^4)^2}.$$

At $x = -\sqrt[4]{1/3}$ the slope changes from negative to positive indicating a local minimum. At $x = \sqrt[4]{1/3}$ the slope changes from positive to negative indicating a local maximum.

19. $y = \sqrt{x^3 + 3x^2} = (x^3 + 3x^2)^{1/2}$

Domain is all $x \geq -3$.

$$y' = \frac{1}{2}(x^3 + 3x^2)^{-1/2}(3x^2 + 6x)$$

$$= \frac{3x^2 + 6x}{2\sqrt{x^3 + 3x^2}}$$

$$= \frac{3x(x+2)}{2\sqrt{x^3 + 3x^2}}$$

$x = 0, -2, -3$ are critical numbers.

y' undefined at $x = 0, -3$

$$y' > 0 \text{ on } (-3, -2) \cup (0, \infty)$$

$$y' < 0 \text{ on } (-2, 0)$$

So $y = \sqrt{x^3 + 3x^2}$ has local max at $x = -2$, local min at $x = 0, -3$.

20. $y = x^{4/3} + 4x^{1/3}$

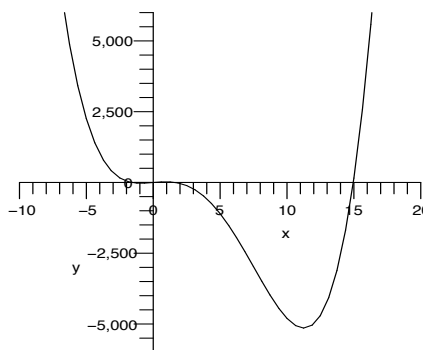
$$y' = \frac{4}{3}x^{1/3} + \frac{4}{3x^{2/3}} = \frac{4}{3} \cdot \frac{x+1}{x^{2/3}}.$$

At $x = -1$ the slope changes from negative to positive indicating a local minimum. At $x = 0$ the slope is vertical and is positive on positive side and negative on negative side, so this is neither a minimum nor a maximum.

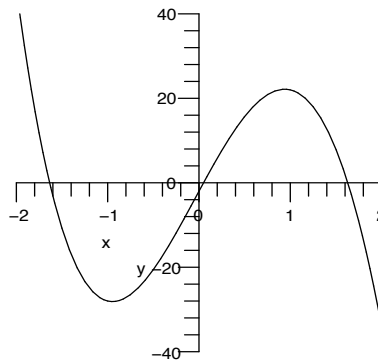
21. $y' = 4x^3 - 45x^2 - 4x + 40$

Local minima at $x = -0.9474, 11.2599$; local max at 0.9374 .

Local behavior near $x = 0$ looks like

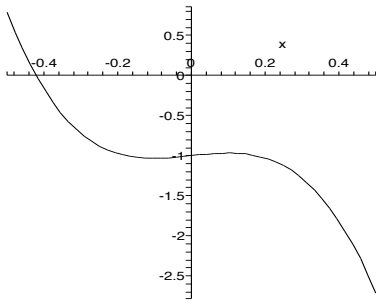


Global behavior of the function looks like

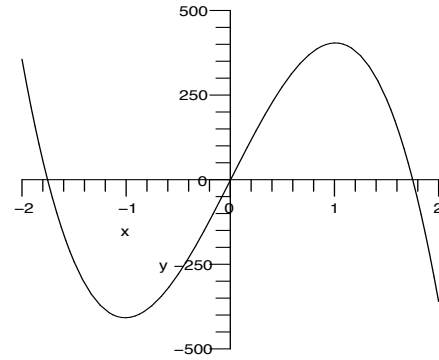
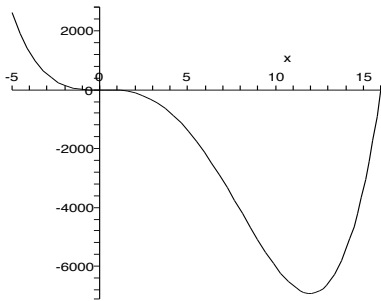


22. $y' = 4x^3 - 48x^2 - 0.2x + 0.5 = 0$ at approximately $x = -0.1037$ (local minimum), $x = 0.1004$ (local maximum), and $x = 12.003$ (local minimum).

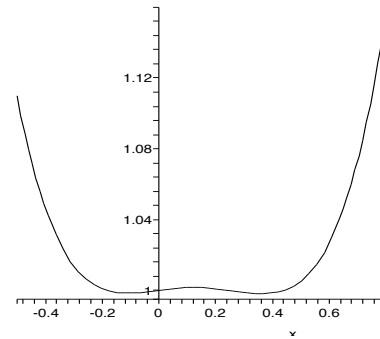
Local behavior near $x = 0$ looks like



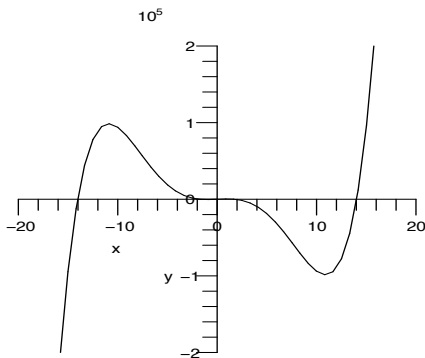
Global behavior of the function looks like



24. $y' = 4x^3 - 1.5x^2 - 0.04x + 0.02 = 0$ at approximately $x = -0.1121$ (local minimum), $x = 0.1223$ (local maximum), and $x = 0.3648$ (local minimum).

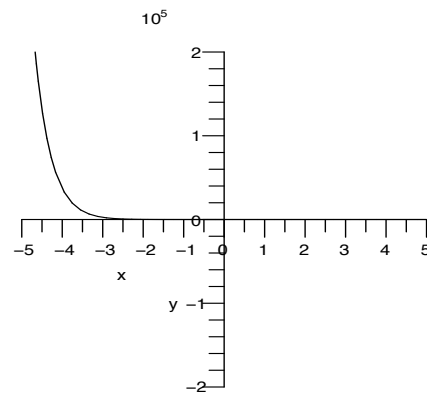


23. $y' = 5x^4 - 600x + 605$
 Local minima at $x = -1.0084, 10.9079$; local maxima at $x = -10.9079, 1.0084$.
 Local behavior near $x = 0$ looks like

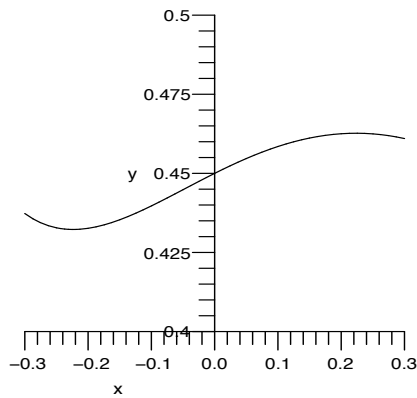


Global behavior of the function looks like

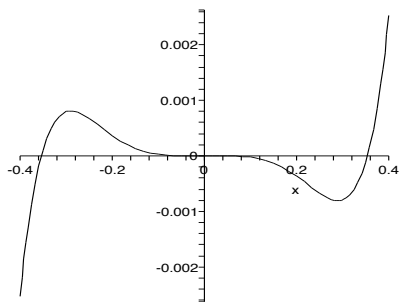
25. $y' = (2x + 1)e^{-2x} + (x^2 + x + 0.45)(-2)e^{-2x}$
 Local min at $x = -0.2236$; local max at $x = 0.2236$.
 Local behavior near $x = 0$ looks like



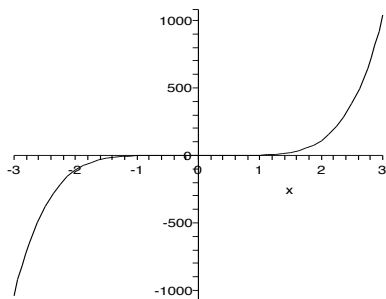
Global behavior of the function looks like



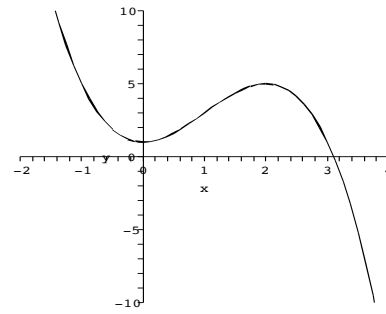
26. $y' = 5x^4 \ln(8x^2) + x^5 \frac{16x}{8x^2}$
 $= x^4(5 \ln(8x^2) + 2) = 0$ at approximately $x = \pm 0.2895$ (a local maximum and local minimum). The derivative and the function are undefined at $x = 0$, but the slope is negative on both sides (neither a minimum nor a maximum). Locally, near $x = \pm 0.2895$, the function looks like



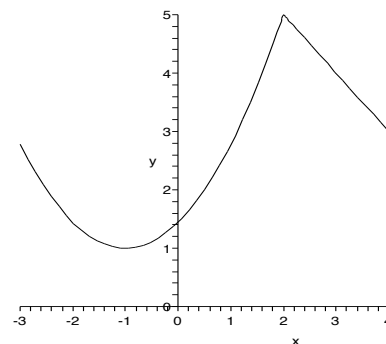
Globally, the function looks like a quintic



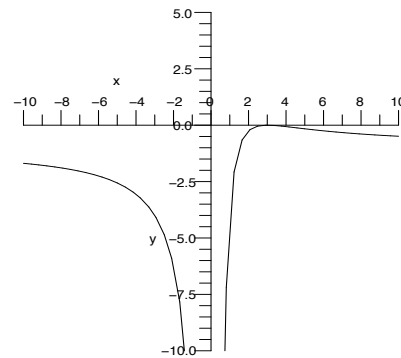
27. One possible graph:



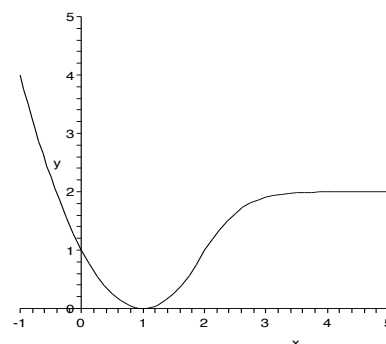
28. One possible graph:



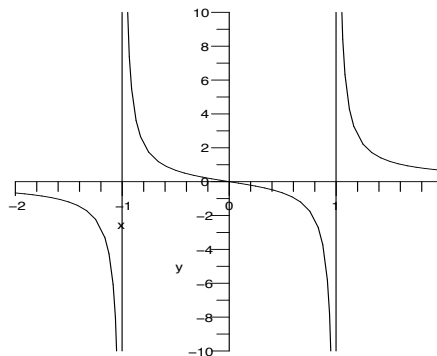
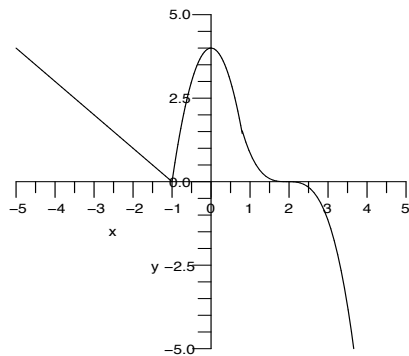
29. One possible graph:



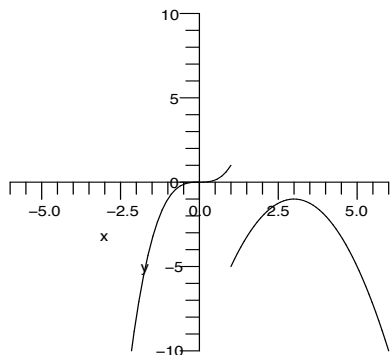
30. One possible graph:



31. One possible graph:



32. One possible graph:



$$33. \quad y = \frac{x}{x^2 - 1}$$

$$y' = \frac{x^2 - 1 - x(2x)}{(x^2 - 1)^2}$$

$$= -\frac{x^2 + 1}{(x^2 - 1)^2}$$

There are no values of x for which $y' = 0$. There are no critical points, because the values for which y' does not exist (that is, $x = \pm 1$) are not in the domain.

There are vertical asymptotes at $x = \pm 1$, and a horizontal asymptote at $y = 0$. This can be verified by calculating the following limits:

$$\lim_{x \rightarrow \pm\infty} \frac{x}{x^2 - 1} = 0$$

$$\lim_{x \rightarrow -1} \frac{x}{x^2 - 1} = \infty$$

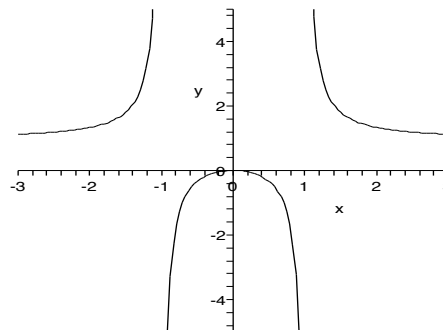
$$\lim_{x \rightarrow 1} \frac{x}{x^2 - 1} = -\infty$$

$$34. \quad y = \frac{x^2}{x^2 - 1} \text{ has vertical asymptotes at } x = \pm 1$$

and horizontal asymptote $y = 1$.

$$y' = \frac{(x^2 - 1)2x - 2x(x^2)}{(x^2 - 1)^2} = \frac{-2x}{(x^2 - 1)^2}.$$

At $x = 0$ the slope changes from positive to negative indicating a local maximum.



$$35. \quad y = \frac{x^2}{x^2 - 4x + 3} = \frac{x^2}{(x - 1)(x - 3)}$$

Vertical asymptotes $x = 1$, $x = 3$. When $|x|$ is large, the function approaches the value 1, so $y = 1$ is a horizontal asymptote.

$$y' = \frac{2x(x^2 - 4x + 3) - x^2(2x - 4)}{(x^2 - 4x + 3)^2}$$

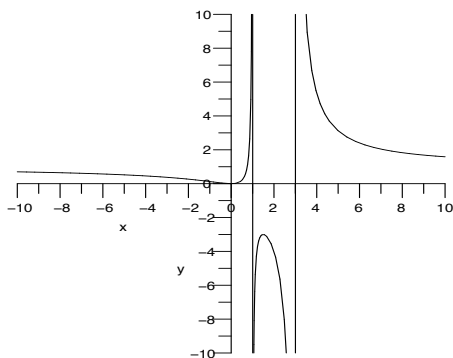
$$= \frac{2x^3 - 8x^2 + 6x - 2x^3 + 4x^2}{(x^2 - 4x + 3)^2}$$

$$= \frac{-4x^2 + 6x}{(x^2 - 4x + 3)^2}$$

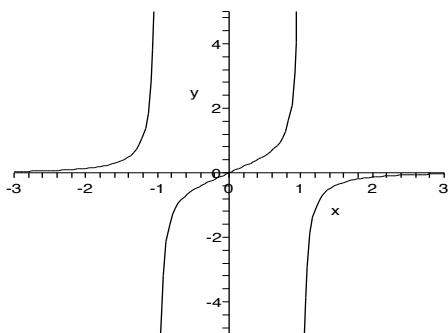
$$= \frac{2x(-2x + 3)}{(x^2 - 4x + 3)^2}$$

$$= \frac{2x(-2x + 3)}{[(x - 3)(x - 1)]^2}$$

Critical numbers are $x = 0$ (local min) and $x = 3/2$ (local max).



36. $y = \frac{x}{1-x^4}$ has vertical asymptotes at $x = \pm 1$ and horizontal asymptote $y = 0$.
 $y' = \frac{(1-x^4) + 4x^4}{(1-x^4)^2} = \frac{1+3x^4}{(1-x^4)^2} \neq 0$ for any x and is defined where the function is defined.



37. $y = \frac{x}{\sqrt{x^2+1}}$
 $y' = \frac{\sqrt{x^2+1} - x^2/\sqrt{x^2+1}}{x^2+1}$
 $= \frac{1}{(x^2+1)^{3/2}}$

The derivative is never zero, so there are no critical points. To verify that there are horizontal asymptotes at $y = \pm 1$: $y = \frac{x}{\sqrt{x^2+1}}$

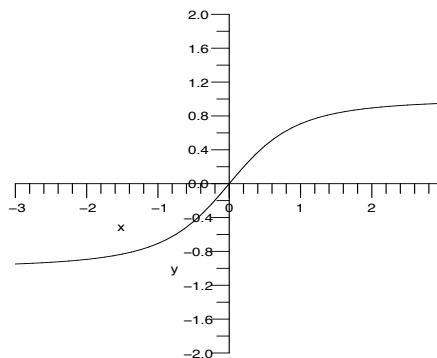
$$= \frac{x}{\sqrt{x^2} \sqrt{1 + \frac{1}{x^2}}}$$

$$= \frac{x}{|x| \sqrt{1 + \frac{1}{x^2}}}$$

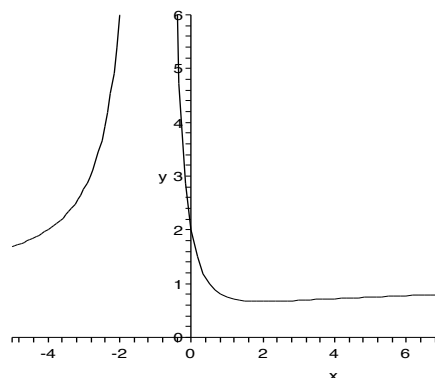
Thus,

$$\lim_{x \rightarrow \infty} \frac{x}{|x| \sqrt{1 + \frac{1}{x^2}}} = 1$$

$$\lim_{x \rightarrow -\infty} \frac{x}{|x| \sqrt{1 + \frac{1}{x^2}}} = -1$$



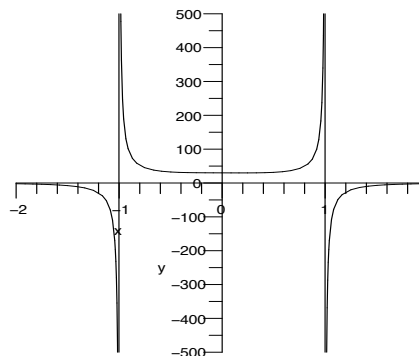
38. $y = \frac{x^2+2}{(x+1)^2}$ has a vertical asymptote at $x = -1$, and a horizontal asymptote at $y = 1$.
 $y' = \frac{2x(x+1)^2 - (x^2+2)2(x+1)}{(x+1)^4}$
 $= \frac{2(x-2)(x+1)}{(x+1)^4}$
 $x = 2$ is the only critical number. Since $f'(0) < 0$ and $f'(3) > 0$, we see that $f(2)$ is a local minimum.



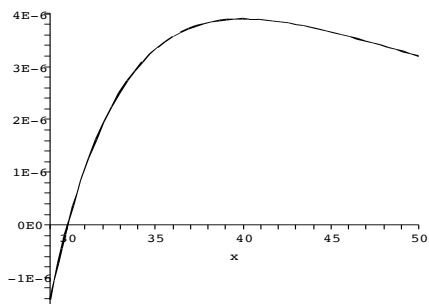
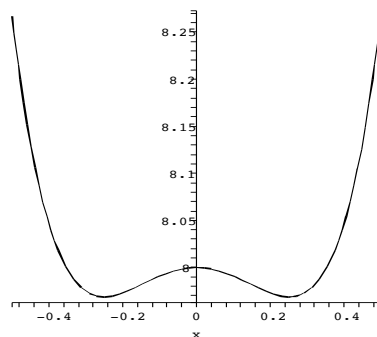
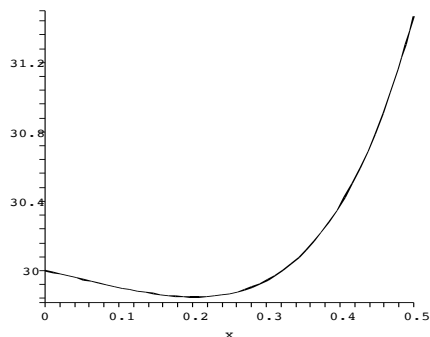
39. The derivative is
 $y' = \frac{-3x^4 + 120x^3 - 1}{(x^4 - 1)^2}$.

We estimate the critical numbers to be approximately 0.2031 and 39.999.

The following graph shows global behavior:



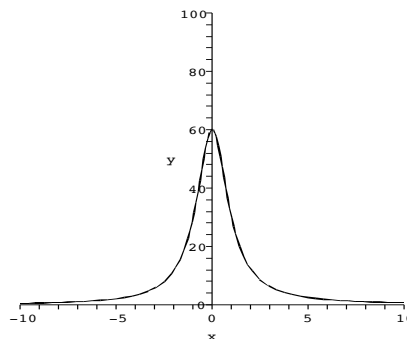
The following graphs show local behavior:



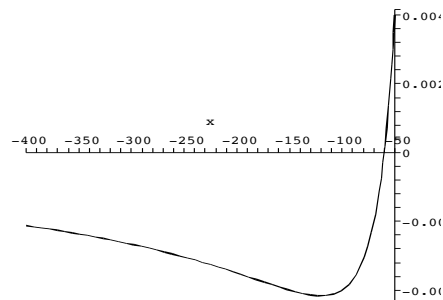
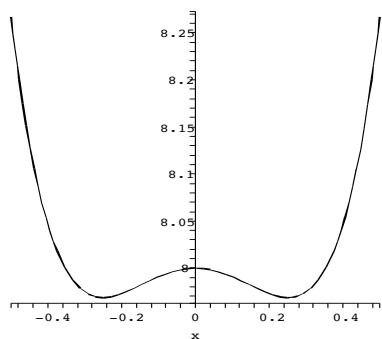
41. The derivative is $y' = \frac{-x^2 - 120x + 1}{(x^2 + 1)^2}$.

We estimate the critical numbers to be approximately 0.008 and -120.008 .

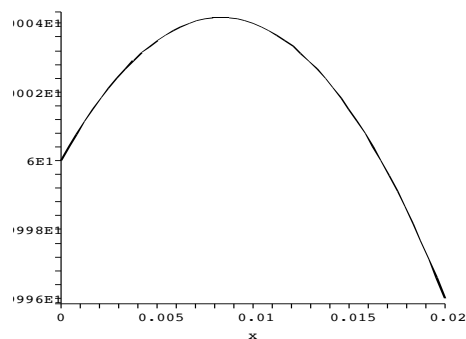
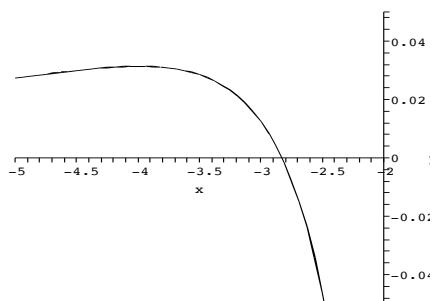
The following graph shows global behavior:



The following graphs show local behavior:



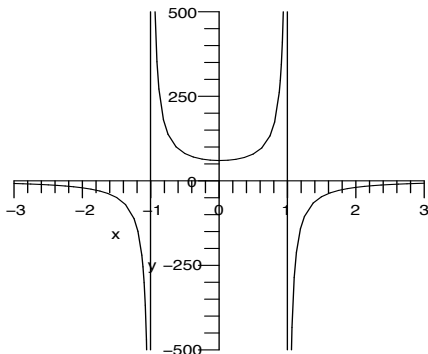
The following graphs show local behavior:



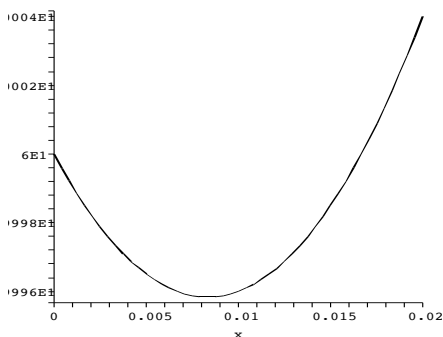
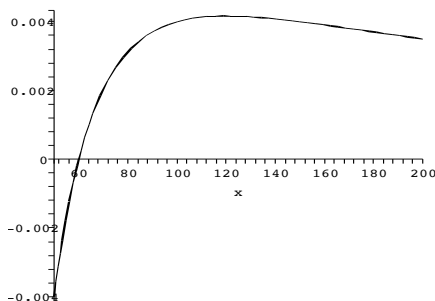
42. The derivative is $y' = \frac{-x^2 + 120x - 1}{(x^2 - 1)^2}$.

We estimate the critical numbers to be approximately 0.008 and 119.992.

The following graph shows global behavior:



The following graphs show local behavior:



43. Let $f(x) = 3 + e^{-x}$; then $f(0) = 4$, $f'(x) = -e^{-x} < 0$, so f is decreasing. But $f(x) = 3 + e^{-x} = 0$ has no solution.
44. Let y_1 and y_2 be two points in the domain of f^{-1} with $y_1 < y_2$. Let $x_1 = f^{-1}(y_1)$ and $x_2 = f^{-1}(y_2)$. We want to show $x_1 < x_2$. Suppose not. Then $x_2 \leq x_1$. But then, since f is increasing, $f(x_2) \leq f(x_1)$. That is $y_2 \leq y_1$, which contradicts our choice of y_1 and y_2 .
45. The domain of $\sin^{-1} x$ is the interval $[-1, 1]$. The function is increasing on the entire domain.

46. $\sin^{-1}\left(\frac{2}{\pi} \tan^{-1} x\right)$ is defined for all x . The derivative,

$$\frac{2}{\pi(1+x^2)\sqrt{1-\left(\frac{2}{\pi} \tan^{-1} x\right)^2}} > 0$$

for all x . The function is increasing everywhere.

47. TRUE. If $x_1 < x_2$, then $g(x_1) < g(x_2)$ since g is increasing, and then $f(g(x_1)) < f(g(x_2))$ since f is increasing.
48. We can say that $g(1) < g(4)$ and $g(f(1)) < g(f(4))$, but it is not possible to determine the maximum and minimum values without more information.

$$\begin{aligned} 49. \quad f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{f(x)}{x} \\ &= \lim_{x \rightarrow 0} \left[1 + 2x \sin\left(\frac{1}{x}\right) \right] = 1 \end{aligned}$$

For $x \neq 0$,

$$\begin{aligned} f'(x) &= 1 + 2 \left[2x \sin\left(\frac{1}{x}\right) + x^2 \left(\frac{-1}{x^2}\right) \cos\left(\frac{1}{x}\right) \right] \\ &= 1 + 4x \sin\left(\frac{1}{x}\right) - 2 \cos\left(\frac{1}{x}\right) \end{aligned}$$

For values of x close to the origin, the middle term of the derivative is small, and since the last term $-2 \cos(1/x)$ reaches its minimum value of -2 in every neighborhood of the origin, f' has negative values on every neighborhood of the origin. Thus, f is not increasing on any neighborhood of the origin. This conclusion does not contradict Theorem 4.1 because the theorem states that if a function's derivative is positive for all values in an interval, then it is increasing in that interval. In this example, the derivative is not positive throughout any interval containing the origin.

50. We have $f'(x) = 3x^2$, so $f'(x) > 0$ for all $x \neq 0$, but $f'(0) = 0$. Since $f'(x) > 0$ for all $x \neq 0$, we know $f(x)$ is increasing on any interval not containing 0. We know that if $x_1 < 0$ then $x_1^3 < 0$ and if $x_2 > 0$ then $x_2^3 > 0$. If $x_1 < 0$ and $x_2 = 0$ then $x_1^3 < 0^3 = 0$, so $f(x)$ is increasing on intervals of the form $(x_1, 0)$. Similarly, $f(x)$ is increasing on intervals of the form $(0, x_2)$. Finally, on intervals of the form (x_1, x_2) where $x_1 < 0 < x_2$, we have $x_1^3 < 0 < x_2^3$ so $f(x)$ is again increasing on these intervals. Thus $f(x)$ is increasing on any interval.

This does not contradict Theorem 4.1 because

Theorem 4.1 is not an “if and only if” statement. It says that if $f'(x) > 0$, then f is increasing (on that interval) but it does not say that if $f'(x)$ is not strictly positive that f is not increasing.

- 51.** f is continuous on $[a, b]$, and $c \in (a, b)$ is a critical number.

- (i) If $f'(x) > 0$ for all $x \in (a, c)$ and $f'(x) < 0$ for all $x \in (c, b)$, by Theorem 3.1, f is increasing on (a, c) and decreasing on (c, b) , so $f(c) > f(x)$ for all $x \in (a, c)$ and $x \in (c, b)$. Thus $f(c)$ is a local max.
- (ii) If $f'(x) < 0$ for all $x \in (a, c)$ and $f'(x) > 0$ for all $x \in (c, b)$, by Theorem 3.1, f is decreasing on (a, c) and increasing on (c, b) . So $f(c) < f(x)$ for all $x \in (a, c)$ and $x \in (c, b)$. Thus $f(c)$ is a local min.
- (iii) If $f'(x) > 0$ on (a, c) and (c, b) , then $f(c) > f(x)$ for all $x \in (a, c)$ and $f(c) < f(x)$ for all $x \in (c, b)$, so c is not a local extremum. If $f'(x) < 0$ on (a, c) and (c, b) , then $f(c) < f(x)$ for all $x \in (a, c)$ and $f(c) > f(x)$ for all $x \in (c, b)$, so c is not a local extremum.

- 52.** If $f(a) = g(a)$ and $f'(x) > g'(x)$ for all $x > a$, then $f(x) > g(x)$ for all $x > a$. Graphically, this makes sense: f and g start at the same place, but f is increasing faster, therefore f should be larger than g for all $x > a$. To prove this, apply the Mean Value Theorem to the function $f(x) - g(x)$.

If $x > a$ then there exists a number c between a and x with

$$f'(c) - g'(c) = \frac{(f(x) - g(x)) - (f(a) - g(a))}{x - a}.$$

Multiply by $(x - a)$ (and recall $f(a) = g(a)$) to get $(x - a)(f'(c) - g'(c)) = f(x) - g(x)$. The lefthand side of this equation is positive, therefore $f(x)$ is greater than $g(x)$.

- 53.** Let $f(x) = 2\sqrt{x}$, $g(x) = 3 - 1/x$.
Then $f(1) = 2\sqrt{1} = 2$, and $g(1) = 3 - 1 = 2$, so $f(1) = g(1)$.
 $f'(x) = \frac{1}{\sqrt{x}}$, $g'(x) = \frac{1}{x^2}$
So $f'(x) > g'(x)$ for all $x > 1$, and
 $f(x) = 2\sqrt{x} > 3 - \frac{1}{x} = g(x)$ for all $x > 1$.
- 54.** Let $f(x) = x$ and $g(x) = \sin x$.
Then $f(0) = g(0)$. $f'(x) = 1$. $g'(x) = \cos x$.
 $\cos x \leq 1$ for all x , therefore exercise 52 implies that $x > \sin x$ for all $x > 0$.

- 55.** Let $f(x) = e^x$, $g(x) = x + 1$.
Then $f(0) = e^0 = 1$, $g(0) = 0 + 1 = 1$, so $f(0) = g(0)$.
 $f'(x) = e^x$, $g'(x) = 1$
So $f'(x) > g'(x)$ for $x > 0$.
Thus $f(x) = e^x > x + 1 = g(x)$ for $x > 0$.

- 56.** Let $f(x) = x - 1$ and $g(x) = \ln x$.
Then $f(1) = g(1)$. $f'(x) = 1$. $g'(x) = \frac{1}{x}$.
 $1/x \leq 1$ for all $x > 1$, therefore exercise 52 implies that $x - 1 > \ln x$ for all $x > 1$.

- 57.** $f(x) = x^3 + bx^2 + cx + d$
 $f'(x) = 3x^2 + 2bx + c$
 $f'(x) \geq 0$ for all x if and only if
 $(2b)^2 - 4(3)(c) \leq 0$
if and only if $4b^2 \leq 12c$
if and only if $b^2 \leq 3c$.
Using the quadratic formula, we find
 $x^2 = \frac{-3b \pm \sqrt{9b^2 - 20c}}{10}$.
Thus, if $9b^2 < 20c$, then the roots are imaginary and so $f'(x) \geq 0$ for all x . If this is not the case, then we need to consider
 $x = \pm \sqrt{\frac{-3b \pm \sqrt{9b^2 - 20c}}{10}}$.
Now we need the expression inside the square root to be less than or equal to 0, which is the same as requiring the numerator of the expression inside the square root to be less than or equal to 0. So we need both
 $-3b < \sqrt{9b^2 - 20c}$ and
 $-3b < -\sqrt{9b^2 - 20c}$.

Of course, both are true if and only if the latter is true. In conclusion, $f(x)$ is an increasing function if $9b^2 < 20c$ or $-3b < -\sqrt{9b^2 - 20c}$.

- 58.** TRUE. $(f \circ g)'(c) = f'(g(c))g'(c) = 0$, since c is a critical number of g .
- 59.** $s(t) = \sqrt{t+4} = (t+4)^{1/2}$
 $s'(t) = \frac{1}{2}(t+4)^{-1/2} = \frac{1}{2\sqrt{t+4}} > 0$
So total sales are always increasing at the rate of $\frac{1}{2\sqrt{t+4}}$ thousand dollars per month.
- 60.** $s'(t) = \frac{1}{2\sqrt{t+4}} > 0$ for all $t > 0$. If s represents the total sales so far, then s cannot decrease. The rate of new sales can decrease, but we cannot lose sales that already have occurred.

$$\begin{aligned}
 61. \quad (a) \quad \mu'(-10) &\approx \frac{0.0048 - 0.0043}{-12 - (-8)} \\
 &= \frac{0.0005}{-4} \\
 &= -0.000125
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \mu'(-6) &\approx \frac{0.0048 - 0.0043}{-4 - (-8)} \\
 &= \frac{0.0005}{4} \\
 &= 0.000125
 \end{aligned}$$

Whether the warming of the ice due to skating makes it easier or harder depends on the current temperature of the ice. As seen from these examples, the coefficient of friction μ is decreasing when the temperature is -10° and increasing when the temperature is -6° .

62. We find the derivative of $f(t)$:

$$\begin{aligned}
 f'(t) &= \frac{a^2 + t^2 - t(2t)}{(a^2 + t^2)^2} \\
 &= \frac{a^2 - t^2}{(a^2 + t^2)^2}.
 \end{aligned}$$

The denominator is always positive, while the numerator is positive when $a^2 > t^2$, i.e., when $a > t$. We now find the derivative of $\theta(x)$:

$$\begin{aligned}
 \theta'(x) &= \frac{1}{1 + \left(\frac{29.25}{x}\right)^2} \left(\frac{-29.25}{x^2}\right) \\
 &\quad - \frac{1}{1 + \left(\frac{10.75}{x}\right)^2} \left(\frac{-10.75}{x^2}\right) \\
 &= \frac{-29.25}{x^2 + (29.25)^2} + \frac{10.75}{x^2 + (10.75)^2}.
 \end{aligned}$$

We consider each of the two terms of the last line above as instances of $f(t)$, the first as $-f(29.25)$ and the second as $f(10.75)$. Now, for any given x where $x \geq 30$, this x is our a in $f(t)$ and since $a = x$ is greater than 29.25 and greater than 10.75, $f(t)$ is increasing for these two t values and this value of a . Thus $f(29.25) > f(10.75)$. This means that

$$\theta'(x) = -f(29.25) + f(10.75) < 0$$

(where $a = x$) and so $\theta(x)$ is decreasing for $x \geq 30$. Since $\theta(x)$ is increasing for $x \geq 30$, the announcers would be wrong to suggest that the angle increases by backing up 5 yards when the team is between 50 and 60 feet away from the goal post.

3.5 Concavity and the Second Derivative Test

1. $f'(x) = 3x^2 - 6x + 4$
 $f''(x) = 6x - 6 = 6(x - 1)$
 $f''(x) > 0$ on $(1, \infty)$
 $f''(x) < 0$ on $(-\infty, 1)$
 So f is concave down on $(-\infty, 1)$ and concave up on $(1, \infty)$.
 $x = 1$ is a point of inflection.

2. $f'(x) = 4x^3 - 12x + 2$ and $f''(x) = 12x^2 - 12$.
 The graph is concave up where $f''(x)$ is positive, and concave down where $f''(x)$ is negative. Concave up for $x < -1$ and $x > 1$, and concave down for $-1 < x < 1$.
 $x = -1, 1$ are points of inflection.

3. $f(x) = x + \frac{1}{x} = x + x^{-1}$
 $f'(x) = 1 - x^{-2}$
 $f''(x) = 2x^{-3}$
 $f''(x) > 0$ on $(0, \infty)$
 $f''(x) < 0$ on $(-\infty, 0)$
 So f is concave up on $(0, \infty)$ and concave down on $(-\infty, 0)$.
 $x = 0$ is a point of inflection.

4. $y' = 1 - (1 - x)^{-2/3}$ and $y'' = \frac{-2}{3}(1 - x)^{-5/3}$.
 Concave up for $x > 1$ and concave down for $x < 1$.
 $x = 1$ is a point of inflection.

5. $f'(x) = \cos x + \sin x$
 $f''(x) = -\sin x + \cos x$
 $f''(x) < 0$ on $\dots \left(\frac{\pi}{4}, \frac{5\pi}{4}\right) \cup \left(\frac{9\pi}{4}, \frac{13\pi}{4}\right) \dots$
 $f''(x) > 0$ on $\dots \left(\frac{3\pi}{4}, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, \frac{9\pi}{4}\right) \dots$
 f is concave down on $\dots \left(\frac{\pi}{4}, \frac{5\pi}{4}\right) \cup \left(\frac{9\pi}{4}, \frac{13\pi}{4}\right) \dots$,
 concave up on $\dots \left(\frac{3\pi}{4}, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, \frac{9\pi}{4}\right) \dots$.
 $x = k\pi + \frac{\pi}{4}$ are the points of inflection for any integer k .

6. $f'(x) = \frac{2x}{1 + x^4}$ and $f''(x) = \frac{2 - 6x^4}{(1 + x^4)^2}$.
 Concave up for $-\sqrt[4]{\frac{1}{3}} < x < \sqrt[4]{\frac{1}{3}}$, and concave down for $x < -\sqrt[4]{\frac{1}{3}}$ and $x > \sqrt[4]{\frac{1}{3}}$.
 $x = -\sqrt[4]{\frac{1}{3}}, \sqrt[4]{\frac{1}{3}}$ are the points of inflection.

7. $f'(x) = \frac{4}{3}x^{1/3} + \frac{4}{3}x^{-2/3}$
 $f''(x) = \frac{4}{9}x^{-2/3} + \frac{8}{9}x^{-5/3}$
 $= \frac{4}{9x^{2/3}} \left(1 - \frac{2}{x}\right)$

The quantity $\frac{4}{9x^{2/3}}$ is never negative, so the sign of the second derivative is the same as the sign of $1 - \frac{2}{x}$. Hence the function is concave up for $x > 2$ and $x < 0$, and is concave down for $0 < x < 2$.

$x = 0, 2$ are the points of inflection.

8. $f'(x) = e^{-4x} - 4xe^{-4x}$ and $f''(x) = 8e^{-4x}(2x - 1)$.

Concave up for $x > 1/2$, and concave down for $x < 1/2$.

$x = 1/2$ is the point of inflection.

9. $f(x) = x^4 + 4x^3 - 1$
 $f'(x) = 4x^3 + 12x^2 = x^2(4x + 12)$
 So the critical numbers are $x = 0$ and $x = -3$.
 $f''(x) = 12x^2 + 24x$
 $f''(0) = 0$ so the second derivative test for $x = 0$ is inconclusive.
 $f''(-3) = 36 > 0$ so $x = -3$ is a local minimum.

10. $f(x) = x^4 + 4x^2 + 1$
 $f'(x) = 4x^3 + 8x$
 So the only critical number is $x = 0$.
 $f''(x) = 12x^2 + 8$
 $f''(0) = 8 > 0$ so $x = 0$ is a local minimum.

11. $f(x) = xe^{-x}$
 $f'(x) = e^{-x} - xe^{-x} = e^{-x}(1 - x)$
 So the only critical number is $x = 1$.
 $f''(x) = -e^{-x} - e^{-x} + xe^{-x} = e^{-x}(-2 + x)$
 $f''(1) = e^{-1}(-1) < 0$ so $x = 1$ is a local maximum.

12. $f(x) = e^{-x^2}$
 $f'(x) = -2xe^{-x^2}$
 So the only critical number is $x = 0$.
 $f''(x) = -2e^{-x^2} + 4x^2e^{-x^2}$
 $f''(0) = -2 + 0 < 0$ so $x = 0$ is a local maximum.

13. $f(x) = \frac{x^2 - 5x + 4}{x^2}$
 $f'(x) = \frac{(2x - 5)x - (x^2 - 5x + 4)(1)}{x^2}$
 $= \frac{x^2 - 4}{x^2}$

So the critical numbers are $x = \pm 2$.

$$f''(x) = \frac{(2x)(x^2) - (x^2 - 4)(2x)}{x^4} = \frac{8x}{x^4}$$

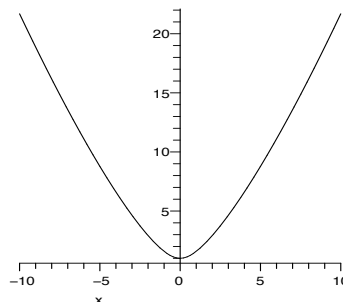
$f''(2) = 1 > 0$ so $x = 2$ is a local minimum.
 $f''(-2) = -1 < 0$ so $x = -2$ is a local maximum.

14. $f(x) = \frac{x^2 - 1}{x}$
 $f'(x) = \frac{(2x)(x) - (x^2 - 1)(1)}{x^2}$
 $= \frac{x^2 + 1}{x^2}$

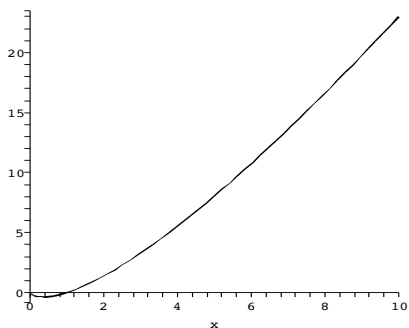
There are no critical numbers and so there are no local extrema.

15. $y = (x^2 + 1)^{2/3}$
 $y' = \frac{2}{3}(x^2 + 1)^{-1/3}(2x)$
 $f'(x) = \frac{4x(x^2 + 1)^{-1/3}}{3}$
 So the only critical number is $x = 0$.
 $y'' = \frac{4}{3} \left[(x^2 + 1)^{-1/3} + \left(\frac{-2x^2}{3} \right) (x^2 + 1)^{-4/3} \right]$
 $= \frac{4}{3} \frac{(x^2 + 1 - \frac{2x^2}{3})}{(x^2 + 1)^{4/3}} = \frac{4}{9} \frac{(3x^2 + 3 - 2x^2)}{(x^2 + 1)^{4/3}}$
 $= \frac{4}{9} \frac{(x^2 + 3)}{(x^2 + 1)^{4/3}}$

So the function is concave up everywhere, decreasing for $x < 0$, and increasing for $x > 0$. Also $x = 0$ is a local min.



16. $f(x) = x \ln x$
 $f'(x) = \ln x + 1$
 So the only critical number is e^{-1} .
 $f''(x) = 1/x$
 $f''(e^{-1}) = e > 0$ so $f(x)$ has a local minimum at $x = e^{-1}$.
 The domain of $f(x)$ is $(0, \infty)$.
 $f'(x) < 0$ on $(0, e^{-1})$ so $f(x)$ is decreasing on this interval. $f'(x) > 0$ on (e^{-1}, ∞) , so $f(x)$ is increasing on this interval.
 $f''(x) > 0$ for all x in the domain of $f(x)$, so $f(x)$ is concave up for all $x > 0$.
 Finally, $f(x)$ has a vertical asymptote at $x = 0$ such that $f(x) \rightarrow \infty$ as $x \rightarrow 0^+$.



$$\begin{aligned}
 17. \quad f(x) &= \frac{x^2}{x^2 - 9} \\
 f'(x) &= \frac{2x(x^2 - 9) - x^2(2x)}{(x^2 - 9)^2} \\
 &= \frac{-18x}{(x^2 - 9)^2} \\
 &= \frac{-18x}{\{(x+3)(x-3)\}^2} \\
 f''(x) &= \frac{-18(x^2 - 9)^2 + 18x \cdot 2(x^2 - 9) \cdot 2x}{(x^2 - 9)^4} \\
 &= \frac{54x^2 + 162}{(x^2 - 9)^3} \\
 &= \frac{54(x^2 + 3)}{(x^2 - 9)^3}
 \end{aligned}$$

$$f'(x) > 0 \text{ on } (-\infty, -3) \cup (-3, 0)$$

$$f'(x) < 0 \text{ on } (0, 3) \cup (3, \infty)$$

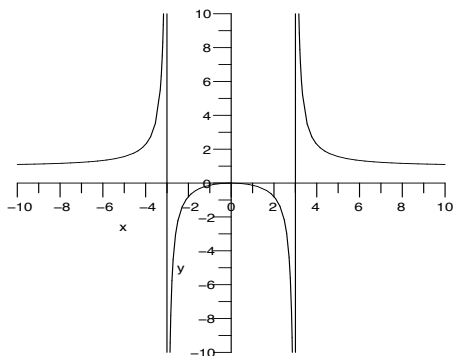
$$f''(x) > 0 \text{ on } (-\infty, -3) \cup (3, \infty)$$

$$f''(x) < 0 \text{ on } (-3, 3)$$

$$f''(0) = \frac{162}{(-9)^3}$$

f is increasing on $(-\infty, -3) \cup (-3, 0)$, decreasing on $(0, 3) \cup (3, \infty)$, concave up on $(-\infty, -3) \cup (3, \infty)$, concave down on $(-3, 3)$, $x = 0$ is a local max.

f has a horizontal asymptote of $y = 1$ and vertical asymptotes at $x = \pm 3$.



$$18. \quad f(x) = \frac{x}{x+2}$$

The domain of $f(x)$ is $\{x | x \neq -2\}$.

There is a vertical asymptote at $x = -2$ such that $f(x) \rightarrow \infty$ as $x \rightarrow -2^-$ and $f(x) \rightarrow -\infty$ as $x \rightarrow -2^+$.

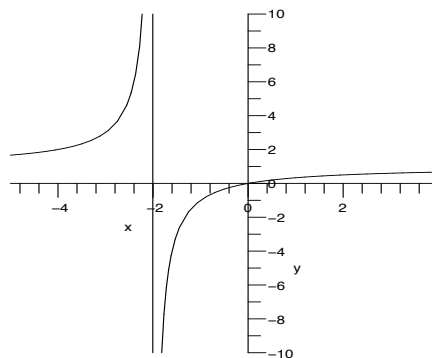
$$f'(x) = \frac{x+2-x}{(x+2)^2} = \frac{2}{(x+2)^2}$$

So there are no critical numbers. Furthermore, $f'(x) > 0$ for all $x \neq -2$, so $f(x)$ is increasing everywhere.

$$f''(x) = -4(x+2)^{-3}$$

$f''(x) > 0$ on $(-\infty, -2)$ (so $f(x)$ is concave up on this interval)

$f''(x) < 0$ on $(-2, \infty)$ (so $f(x)$ is concave down on this interval)



$$19. \quad f(x) = \sin x + \cos x$$

$$f'(x) = \cos x - \sin x$$

$$f''(x) = -\sin x - \cos x$$

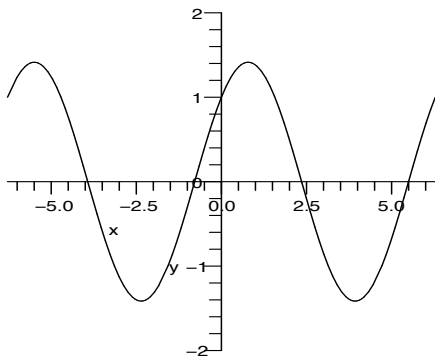
$f'(x) = 0$ when $x = \pi/4 + k\pi$ for all integers k . When k is even, $f''(\pi/4 + k\pi) = -\sqrt{2} < 0$ so $f(x)$ has a local maximum. When k is odd, $f''(\pi/4 + k\pi) = \sqrt{2} > 0$ so $f(x)$ has a local minimum.

$f'(x) < 0$ on the intervals of the form $(\pi/4 + 2k\pi, \pi/4 + (2k+1)\pi)$, so $f(x)$ is decreasing on these intervals.

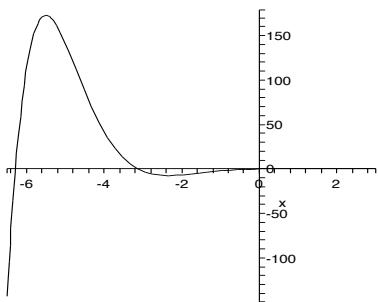
$f'(x) > 0$ on the intervals of the form $(\pi/4 + (2k+1)\pi, \pi/4 + (2k+2)\pi)$, so $f(x)$ is increasing on these intervals.

$f''(x) > 0$ on the intervals of the form $(3\pi/4 + 2k\pi, 3\pi/4 + (2k+1)\pi)$ so $f(x)$ is concave up on these intervals.

$f''(x) < 0$ on the intervals of the form $(3\pi/4 + (2k+1)\pi, 3\pi/4 + (2k+2)\pi)$ so $f(x)$ is concave down on these intervals.



20. $y = e^{-x} \sin x$
 $y' = -e^{-x} \sin x + e^{-x} \cos x = 0$ when $x = \pi/4 + k\pi$ for integers k .
 $y'' = -2e^{-x} \cos x = 0$ at $\pi/2 + 2k\pi$ for integers k . These are inflection points. The function is concave up for $-\pi/2 < x < \pi/2$ and concave down for $\pi/2 < x < 3\pi/2$, and the pattern repeats with period 2π . The critical values are all extrema, and they alternate between maxima and minima.



21. $f(x) = x^{3/4} - 4x^{1/4}$
 Domain of $f(x)$ is $\{x|x \geq 0\}$.

$$f'(x) = \frac{3}{4}x^{-1/4} - x^{-3/4} = \frac{\frac{3}{4}\sqrt{x} - 1}{x^{3/4}}$$

So $x = 0$ and $x = 16/9$ are critical points, but because of the domain we only need to really consider the latter.

$f'(1) = -1/4$ so $f(x)$ is decreasing on $(0, 16/9)$.

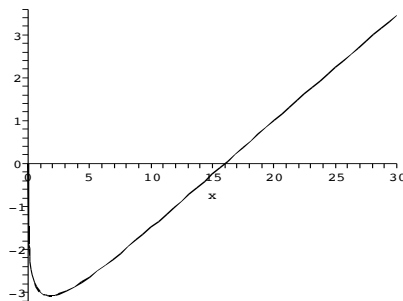
$f'(4) = \frac{0.5}{4^{3/4}} > 0$ so $f(x)$ is increasing on $(16/9, \infty)$.

Thus $x = 16/9$ is the location of a local minimum for $f(x)$.

$$\begin{aligned} f''(x) &= \frac{-3}{16}x^{-5/4} + \frac{3}{4}x^{-7/4} \\ &= \frac{-\frac{3}{16}\sqrt{x} + \frac{3}{4}}{x^{7/4}} \end{aligned}$$

The critical number here is $x = 16$. We find that $f''(x) > 0$ on the interval $(0, 16)$ (so $f(x)$ is concave up on this interval) and $f''(x) < 0$

on the interval $(16, \infty)$ (so $f(x)$ is concave down on this interval).



22. $f(x) = x^{2/3} - 4x^{1/3}$
 $f'(x) = \frac{2}{3}x^{-1/3} - \frac{4}{3}x^{-2/3}$
 $= \frac{\frac{2}{3}x^{1/3} - \frac{4}{3}}{x^{2/3}}$

So $x = 0$ and $x = 8$ are critical numbers.

$f'(-1) < 0$ so $f(x)$ is decreasing for $x < 0$.

$f'(1) < 0$ so $f(x)$ is decreasing for $0 < x < 8$.

$f'(27) > 0$ so $f(x)$ is increasing on $8 < x$.

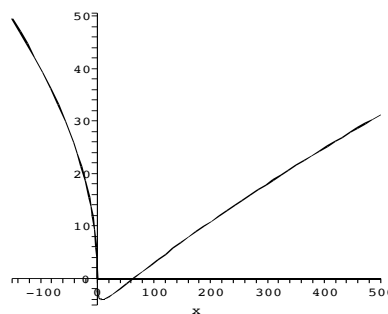
$$\begin{aligned} f''(x) &= -\frac{2}{9}x^{-4/3} + \frac{8}{9}x^{-5/3} \\ &= \frac{-\frac{2}{9}x^{1/3} + \frac{8}{9}}{x^{5/3}} \end{aligned}$$

The critical numbers here are $x = 0$ and $x = 64$.

$f''(-1) < 0$ so $f(x)$ is concave down on $(-\infty, 0)$.

$f''(1) > 0$ so $f(x)$ is concave up on $(0, 64)$.

$f''(125) < 0$ so $f(x)$ is concave down on $(64, \infty)$.



23. The easiest way to sketch this graph is to notice that

$$f(x) = x|x| = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$$

Since

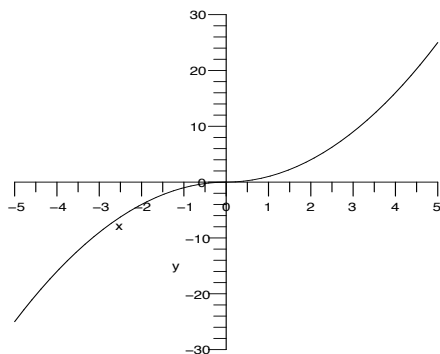
$$f'(x) = \begin{cases} 2x & x \geq 0 \\ -2x & x < 0 \end{cases}$$

there is a critical point at $x = 0$. However, it is neither a local maximum nor a local minimum.

Since

$$f''(x) = \begin{cases} 2 & x > 0 \\ -2 & x < 0 \end{cases}$$

there is an inflection point at the origin. Note that the second derivative does not exist at $x = 0$.



- 24.** The easiest way to sketch this graph is to notice that

$$f(x) = x^2|x| = \begin{cases} -x^3 & x < 0 \\ x^3 & x \geq 0 \end{cases}$$

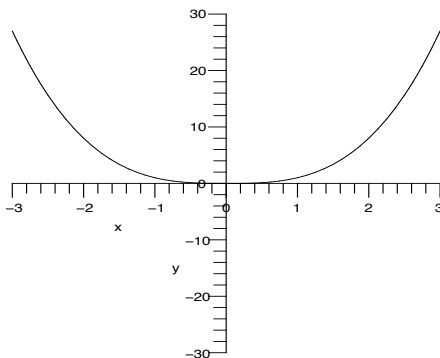
since

$$f'(x) = \begin{cases} -3x^2 & x < 0 \\ 3x^2 & x \geq 0 \end{cases}$$

there is a critical point (and local minimum) at $x = 0$. Since

$$f''(x) = \begin{cases} -6x & x < 0 \\ 6x & x \geq 0 \end{cases}$$

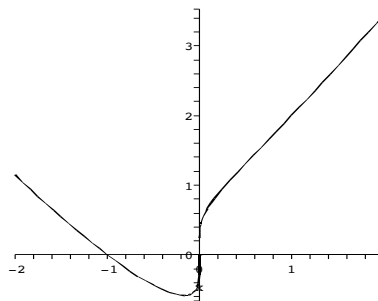
there is a critical point at the origin but this is not an inflection point.



- 25.** $f(x) = x^{1/5}(x+1) = x^{6/5} + x^{1/5}$
 $f'(x) = \frac{6}{5}x^{1/5} + \frac{1}{5}x^{-4/5}$
 $= \frac{1}{5}x^{-4/5}(6x+1)$
 $f''(x) = \frac{6}{25}x^{-4/5} - \frac{4}{25}x^{-9/5}$
 $= \frac{2}{25}x^{-9/5}(3x-2)$

Note that $f(0) = 0$, and yet the derivatives do not exist at $x = 0$. This means that there is a vertical tangent line at $x = 0$. The first derivative is negative for $x < -1/6$ and posi-

tive for $-1/6 < x < 0$ and $x > 0$. The second derivative is positive for $x < 0$ and $x > 2/3$, and negative for $0 < x < 2/3$. Thus, there is a local minimum at $x = -1/6$ and inflection points at $x = 0$ and $x = 2/3$.



26. $f(x) = \frac{\sqrt{x}}{1+\sqrt{x}}$

The domain of $f(x)$ is $\{x|x \geq 0\}$.

$$f'(x) = \frac{\frac{1}{2}x^{-1/2}(1+\sqrt{x}) - \sqrt{x}(\frac{1}{2}x^{-1/2})}{(x+\sqrt{x})^2}$$

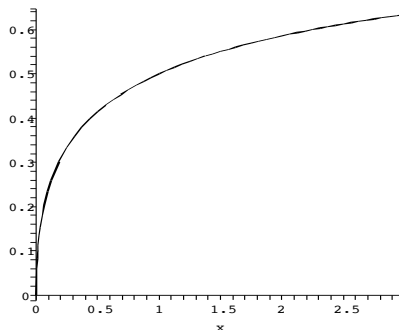
$$= \frac{x^{-1/2}}{2(1+\sqrt{x})^2}$$

The only critical point is $x = 0$, which we need not consider because of the domain. Since $f'(1) > 0$, $f(x)$ is increasing on $(0, \infty)$.

$$f''(x) = \frac{-x^{-3/2}(1+\sqrt{x})^2 - 2x^{-1/2}(1+\sqrt{x})x^{-1/2}}{4(1+\sqrt{x})^4}$$

$$= \frac{-(x^{-1/2}+3)}{4x(1+\sqrt{x})^3}$$

The critical numbers are $x = 0$ (which we again ignore) and $x = 1/9$. Since $f''(1) < 0$ and $f''(1/16) < 0$, $f(x)$ is concave down on $(0, \infty)$.



27. $f(x) = x^4 - 26x^3 + x$
 $f'(x) = 4x^3 - 78x^2 + 1$

The critical numbers are approximately -0.1129 , 0.1136 and 19.4993 .

$f'(-1) < 0$ implies $f(x)$ is decreasing on

$(-\infty, -0.1129)$.

$f'(0) > 0$ implies $f(x)$ is increasing on $(-0.1129, 0.1136)$.

$f'(1) < 0$ implies $f(x)$ is decreasing on $(0.1136, 19.4993)$.

$f'(20) > 0$ implies $f(x)$ is increasing on $(19.4993, \infty)$.

Thus $f(x)$ has local minimums at $x = -0.1129$ and $x = 19.4993$ and a local maximum at $x = 0.1136$.

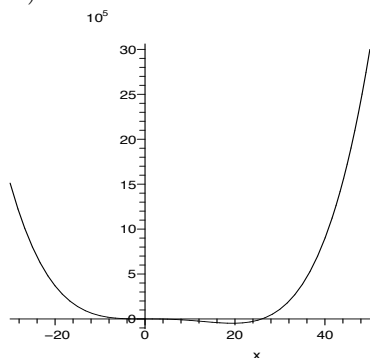
$$f''(x) = 12x^2 - 156x = x(12x - 156)$$

The critical numbers are $x = 0$ and $x = 13$.

$f''(-1) > 0$ implies $f(x)$ is concave up on $(-\infty, 0)$.

$f''(1) < 0$ implies $f(x)$ is concave down on $(0, 13)$.

$f''(20) > 0$ implies $f(x)$ is concave up on $(13, \infty)$.



28. $f(x) = 2x^4 - 11x^3 + 17x^2$

$$f'(x) = 8x^3 - 33x^2 + 34x$$

$$= x(8x - 17)(x - 2)$$

The critical numbers are $x = 0$, $x = 2$ and $x = 17/8$.

$$f''(x) = 24x^2 - 66x + 34$$

$f''(0) > 0$ implies $f(x)$ is concave up at $x = 0$ so $f(x)$ has a local minimum here and $f(x)$ is decreasing on $(-\infty, 0)$.

$f''(2) < 0$ implies $f(x)$ is concave down at $x = 2$ so $f(x)$ has a local maximum here and $f(x)$ is increasing on $(0, 2)$.

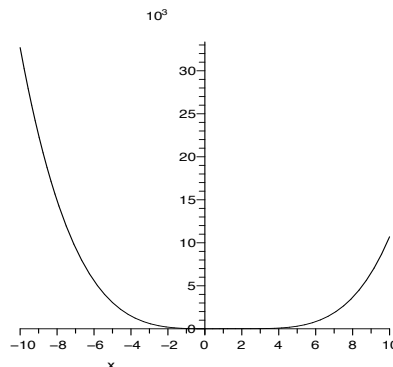
$f''(17/8) > 0$ implies $f(x)$ is concave up at $x = 17/8$ so $f(x)$ has a local minimum here and $f(x)$ is decreasing on $(2, 17/8)$ and increasing on $(17/8, \infty)$.

$$f''(x) = 2(12x^2 - 33x + 17)$$

The critical numbers are

$$x = \frac{33 \pm \sqrt{273}}{24} = 2.0635, 0.6866.$$

So $f(x)$ is concave up on $(-\infty, 0.6866)$ and $(2.0635, \infty)$ and $f(x)$ is concave down on $(0.6866, 2.0635)$.

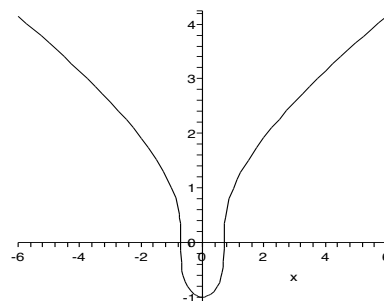


29. $y = \sqrt[3]{2x^2 - 1}$

$y' = \frac{4x}{3(2x^2 - 1)^{2/3}} = 0$ at $x = 0$ and is undefined at $x = \pm\sqrt{1/2}$.

$y'' = \frac{-4(2x^2 + 3)}{9(2x^2 - 1)^{5/3}}$ is never 0, and is undefined where y' is.

The function changes concavity at $x = \pm\sqrt{1/2}$, so these are inflection points. The slope does not change at these values, so they are not extrema. The Second Derivative Test shows that $x = 0$ is a minimum.



30. $f(x) = \sqrt{x^3 + 1}$

$f(x)$ is defined for $x \geq -1$.

$$f'(x) = \frac{1}{2}(x^3 + 1)^{-1/2}(3x^2)$$

The critical numbers are $x = -1$ (which we ignore because of the domain) and $x = 0$.

$f'(-1/2) > 0$ so $f(x)$ is increasing on $(-1, 0)$.

$f'(1) > 0$ so $f(x)$ is also increasing on $(0, \infty)$ so $f(x)$ has no relative extrema.

$$f''(x) =$$

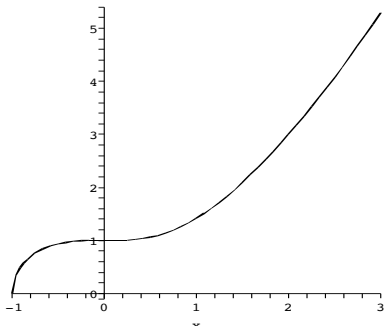
$$\frac{3}{2} \cdot \frac{2x(x^3 + 1)^{1/2} - x^2 \frac{1}{2}(x^3 + 1)^{-1/2} 3x^2}{x^3 + 1}$$

$$= \frac{2x(x^3 + 1) - \frac{3}{2}x^4}{(x^3 + 1)^{3/2}}$$

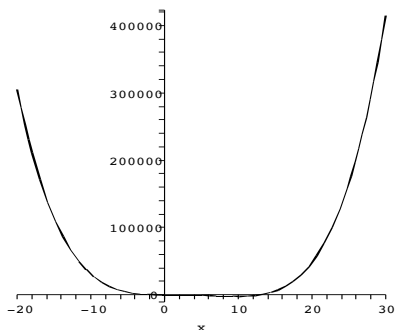
$$= \frac{-\frac{1}{2}x^4 + 2x}{(x^3 + 1)^{3/2}}$$

The critical numbers are $x = 0$ and $x = 4^{1/3}$ (and $x = -1$, which we need not consider).

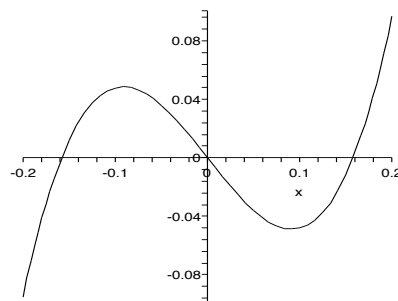
$f''(-1/2) < 0$ so $f(x)$ is concave down on $(-1, 0)$. $f''(1) > 0$ so $f(x)$ is concave up on $(0, 4^{1/3})$. $f''(2) > 0$ so $f(x)$ is concave up on $(4^{1/3}, \infty)$.



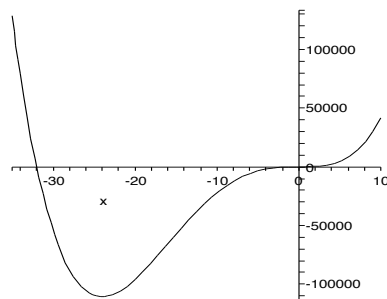
31. $f(x) = x^4 - 16x^3 + 42x^2 - 39.6x + 14$
 $f'(x) = 4x^3 - 48x^2 + 84x - 39.6$
 $f''(x) = 12x^2 - 96x + 84$
 $= 12(x^2 - 8x + 7)$
 $= 12(x - 7)(x - 1)$
 $f'(x) > 0$ on $(.8952, 1.106) \cup (9.9987, \infty)$
 $f'(x) < 0$ on $(-\infty, .8952) \cup (1.106, 9.9987)$
 $f''(x) > 0$ on $(-\infty, 1) \cup (7, \infty)$
 $f''(x) < 0$ on $(1, 7)$
 f is increasing on $(.8952, 1.106)$ and on $(9.9987, \infty)$, decreasing on $(-\infty, .8952)$ and on $(1.106, 9.9987)$, concave up on $(-\infty, 1) \cup (7, \infty)$, concave down on $(1, 7)$, $x = .8952, 9.9987$ are local min, $x = 1.106$ is local max, $x = 1, 7$ are inflection points.



32. $y = x^4 + 32x^3 - 0.02x^2 - 0.8x$
 $y' = 4x^3 + 96x^2 - 0.04x - 0.8 = 0$ at approximately $x = -24, -0.09125$, and 0.09132 .
 $y'' = 12x^2 + 192x - 0.04 = 0$ at approximately $x = 16.0002$ and 0.0002 , and changes sign at these values, so these are inflection points. The Second Derivative Test shows that $x = -24$ and 0.09132 are minima, and that $x = -0.09125$ is a maxima. The extrema near $x = 0$ look like this:



The global behavior looks like this:



33. $f(x) = x\sqrt{x^2 - 4}$; f undefined on $(-2, 2)$
 $f'(x) = \sqrt{x^2 - 4}$

$$+ x \left(\frac{1}{2}\right) (x^2 - 4)^{-1/2} (2x)$$

$$= \sqrt{x^2 - 4} + \frac{x^2}{\sqrt{x^2 - 4}}$$

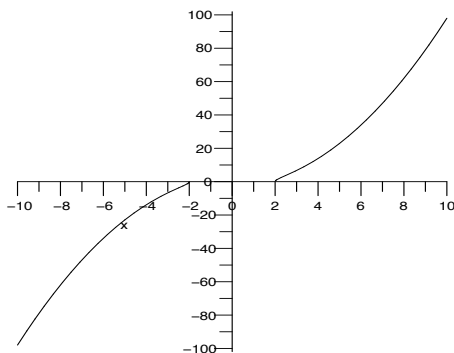
$$= \frac{2x^2 - 4}{\sqrt{x^2 - 4}}$$

$$f''(x) = \frac{4x\sqrt{x^2 - 4} - (2x^2 - 4)\frac{1}{2}(x^2 - 4)^{-1/2}(2x)}{(x^2 - 4)^{3/2}}$$

$$= \frac{4x(x^2 - 4) - (2x^2 - 4)x}{(x^2 - 4)^{3/2}}$$

$$= \frac{2x^3 - 12x}{(x^2 - 4)^{3/2}} = \frac{2x(x^2 - 6)}{(x^2 - 4)^{3/2}}$$

$f'(x) > 0$ on $(-\infty, -2) \cup (2, \infty)$
 $f''(x) > 0$ on $(-\sqrt{6}, 2) \cup (\sqrt{6}, \infty)$
 $f''(x) < 0$ on $(-\infty, -\sqrt{6}) \cup (2, \sqrt{6})$
 f is increasing on $(-\infty, -2)$ and on $(2, \infty)$, concave up on $(-\sqrt{6}, -2) \cup (\sqrt{6}, \infty)$, concave down on $(-\infty, -\sqrt{6}) \cup (2, \sqrt{6})$, $x = \pm\sqrt{6}$ are inflection points.



$$34. f(x) = \frac{2x}{\sqrt{x^2 + 4}}$$

$$f'(x) = \frac{2\sqrt{x^2 + 4} - 2x(\frac{1}{2})(x^2 + 4)^{-1/2}2x}{(x^2 + 4)}$$

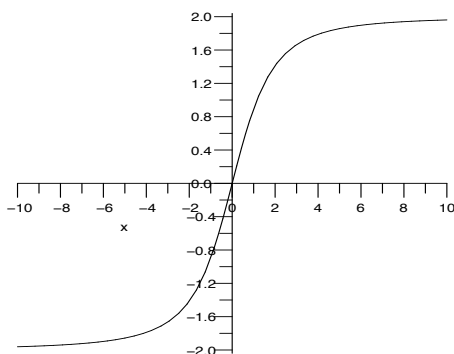
$$= \frac{8}{(x^2 + 4)^{3/2}}$$

$f'(x)$ is always positive, so there are no critical points and $f(x)$ is always increasing.

$$f''(x) = \frac{8(-\frac{3}{2})(x^2 + 4)^{-5/2}(2x)}{-24x}$$

$$= \frac{-24x}{(x^2 + 4)^{5/2}}$$

The only critical point is $x = 0$. Since $f''(-1) > 0$, $f(x)$ is concave up on $(-\infty, 0)$. Also $f''(1) < 0$, so $f(x)$ is concave down on $(0, \infty)$ and $x = 0$ is an inflection point for f .



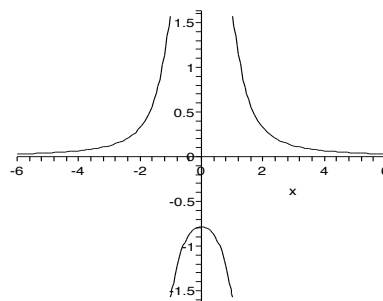
35. The function has horizontal asymptote $y = 0$, and is undefined at $x = \pm 1$.

$$y' = \frac{-2x}{x^4 - 2x^2 + 2} = 0$$

only when $x = 0$.

$$y'' = \frac{2(3x^4 - 2x^2 - 2)}{(x^4 - 2x^2 + 2)^2} = 0$$

at approximately $x = \pm 1.1024$ and changes sign there, so these are inflection points (very easy to miss by looking at the graph). The Second Derivative Test shows that $x = 0$ is a local maximum.



$$36. f(x) = e^{-2x} \cos x$$

$$f'(x) = -2e^{-2x} \cos x - e^{-2x} \sin x$$

$$= e^{-x}(-2 \cos x - \sin x)$$

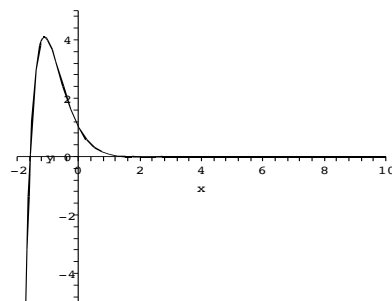
$$f''(x) = -2e^{-2x}(-2 \cos x - \sin x)$$

$$+ e^{-2x}(2 \sin x - \cos x)$$

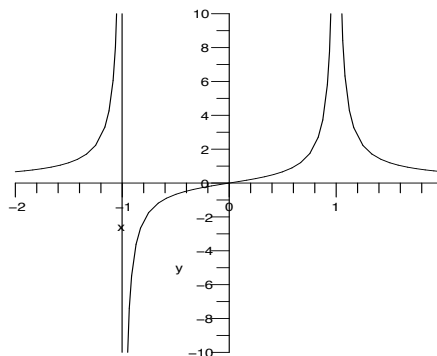
$$= e^{-2x}(4 \sin x + 3 \cos x)$$

$f'(x) = 0$ when $\sin x = -2 \cos x$ so when $x = k\pi + \tan^{-1}(-2)$ for any integer k .

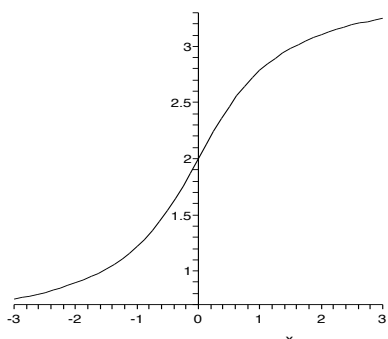
$f''(2k\pi + \tan^{-1}(-2)) < 0$ so there are local maxima at all such points, while $f''((2k+1)\pi + \tan^{-1}(-2)) > 0$, so there are local minima at all such points. $f''(x) = 0$ when $4 \sin x = -3 \cos x$ or $x = k\pi + \tan^{-1}(-3/4)$ for any integer k . All such points x are inflection points.



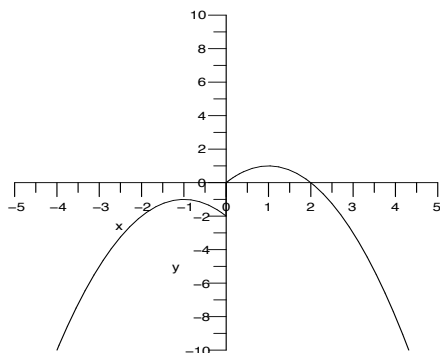
37. One possible graph:



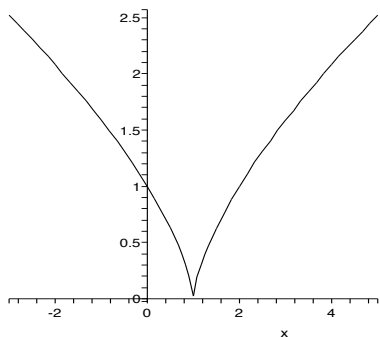
38. One possible graph:



39. One possible graph:



40. One possible graph:



41. $f(x) = ax^3 + bx^2 + cx + d$

$$f'(x) = 3ax^2 + 2bx + c$$

$$f''(x) = 6ax + 2b$$

Thus, $f''(x) = 0$ for $x = -b/3a$. Since f'' changes sign at this point, f has an inflection point at $x = -b/3a$. Note that $a \neq 0$.

For the quartic function (where again $a \neq 0$),

$$f(x) = ax^4 + bx^3 + cx^2 + dx + e$$

$$f'(x) = 4ax^3 + 3bx^2 + 2cx + d$$

$$f''(x) = 12ax^2 + 6bx + 2c \\ = 2(6ax^2 + 3bx + c)$$

The second derivative is zero when

$$x = \frac{-3b \pm \sqrt{9b^2 - 24ac}}{12a} \\ = \frac{-3b \pm \sqrt{3(3b^2 - 8ac)}}{12a}$$

There are two distinct solutions to the previous equation (and therefore two inflection points) if and only if $3b^2 - 8ac > 0$.

42. Since $f'(0) = 0$ and $f''(0) > 0$, $f(x)$ must have a local minimum at $x = 0$. Since we also know that $f(0) = 0$, this means that there is some neighborhood (possibly very small) of 0 such that for all x in this neighborhood (excluding $x = 0$), $f(x) > 0$.

Similarly, $g'(0) = 0$ and $g''(0) < 0$ implies that $g(x)$ must have a local maximum at $x = 0$. Again we know that $g(0) = 0$, so there is some neighborhood of 0 such that for all x in this neighborhood (excluding $x = 0$), $g(x) < 0$.

On the smaller of these two neighborhoods, we know that $g(x) < 0 < f(x)$.

43. Let $f(x) = -1 - x^2$. Then

$$f'(x) = -2x$$

$$f''(x) = -2$$

so f is concave down for all x , but $-1 - x^2 = 0$ has no solution.

44. The statement is true.

45. $f(x)$ is concave up on $(-\infty, -0.5)$ and $(0.5, \infty)$; $f(x)$ is concave down on $(-0.5, 0.5)$. $f(x)$ is decreasing on the intervals $(-\infty, 1)$ and $(0, 1)$; increasing on the intervals $(-1, 0)$ and $(1, \infty)$. $f(x)$ has local maxima at 0 and minima at -1 and 1. Inflection points of $f(x)$ are -0.5 and 0.5.

46. $f(x)$ is concave up on $(1, \infty)$; $f(x)$ is concave down on $(-\infty, 1)$. $f(x)$ is increasing on the intervals $(-\infty, 0)$ and $(2, \infty)$; decreasing on the intervals $(0, 2)$. Inflection point of $f(x)$ is 1.

47. (a) **For #45 :**

The interval of increase is $(-\infty, -1.5)$ and $(1.5, \infty)$. The interval of decrease is $(-1.5, 1.5)$. Minima at $x = 1.5$ and Maxima at $x = -1.5$. It is concave up for $(-1, 0) \cup (1, \infty)$. It is concave down for $(-\infty, -1) \cup (0, 1)$. The points of inflection are $x = 0$ and ± 1 .

For #46:

The interval of increase is $(-\frac{1}{2}, \frac{1}{2}) \cup (3, \infty)$. The interval of decrease is $(-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, 3)$. Minima at $x = -\frac{1}{2}$ and Maxima at $x = \frac{1}{2}, 3$. It is concave up for $(-\infty, 0) \cup (2, \infty)$. It is concave down for $(0, 2)$. The points of inflection

are $x = 0, 2..$

(b) **For #45:**

It is concave up for $(-\infty, -1.5) \cup (1.5, \infty)$. It is concave down for $(-1.5, 1.5)$. The points of inflection are $x = 0$ and ± 1.5 .

For #46:

It is concave up for $(-\frac{1}{2}, \frac{1}{2}) \cup (3, \infty)$. It is concave down for $(-\infty, -\frac{1}{2}) \cup (\frac{1}{2}, 3)$. The points of inflection are $x = \pm\frac{1}{2}, 3..$

48. If $f''(c) < 0$, then f' is decreasing at c . Because $f'(c) = 0$, this means that $f' > 0$ to the left of c and $f' < 0$ to the right of c . Therefore, by the First Derivative Test, $f(c)$ is a local maximum. The proof of the second claim is similar.

49. Add and subtract 16 to complete square.

$$x^4 - 8x^2 + 10$$

$$= x^4 - 8x^2 + 16 + 10 - 16$$

$$= (x^2 - 4)^2 - 6$$

Therefore, absolute minimum occurs when $(x^2 - 4)^2 = 0$. That is absolute minimum is -6 and occurs when $x = \pm 2$.

Similarly, add and subtract 9 to $x^4 - 6x^2 + 1$.

$$x^4 - 6x^2 + 1$$

$$= x^4 - 6x^2 + 9 + 1 - 9$$

$$= (x^2 - 3)^2 - 8$$

Therefore, absolute minimum occurs when $(x^2 - 3)^2 = 0$. That is absolute minimum is -8 and occurs when $x = \pm\sqrt{3}$.

50. $f(x) = x^4 + bx^3 + cx^2 + dx + 2$
 $f'(x) = 4x^3 + 3bx^2 + 2cx + d$
 $f''(x) = 12x^2 + 6bx + 2c$

To find inflection points, solve $f''(x) = 0$.

$$x = \frac{-6b \pm \sqrt{36b^2 - 96c}}{24}$$

x is real only if $36b^2 - 96c > 0$

$$\Rightarrow c < \frac{3}{8}b^2$$

The critical numbers are

$$x = \frac{-6b + \sqrt{36b^2 - 96c}}{24}$$

and

$$x = \frac{-6b - \sqrt{36b^2 - 96c}}{24}$$

Therefore sum of x -coordinates

$$\begin{aligned} &= \frac{-6b + \sqrt{36b^2 - 96c}}{24} + \frac{-6b - \sqrt{36b^2 - 96c}}{24} \\ &= \frac{-6b + \sqrt{36b^2 - 96c} - 6b - \sqrt{36b^2 - 96c}}{24} \\ &= \frac{-12b}{24} = -\frac{b}{2} \end{aligned}$$

51. We need to know $w'(0)$ to know if the depth is increasing.

52. We assume the sick person's temperature is too high, and not too low. We do need to know $T'(0)$ in order to tell which is better.

If $T''(0) = 2$ and $T' > 0$, the person's temperature is rising alarmingly.

If $T''(0) = -2$ and $T' > 0$, the person's temperature is increasing, but leveling off. Negative T'' is better if $T' > 0$.

If $T''(0) = 2$ and $T' < 0$, the person's temperature is decreasing and leveling off.

If $T''(0) = -2$ and $T' < 0$, the person's temperature is dropping too steeply to be safe. Positive T'' is probably better if $T' < 0$.

53. $s(x) = -3x^3 + 270x^2 - 3600x + 18000$

$$s'(x) = -9x^2 + 540x - 3600$$

$$s''(x) = -18x + 540 = 0$$

$x = 30$. This is a max because the graph of $s'(x)$ is a parabola opening down. So spend \$30,000 on advertising to maximize the rate of change of sales. This is also the inflection point of $s(x)$.

54. $Q'(t)$ measures the number of units produced per hour. If this number is larger, the worker is more efficient.

$Q'(t) = -3t^2 + 12t + 12$ will be maximized where

$Q'' = -6t + 12 = 0$, or $t = 2$ hours. (This is a maximum by the First Derivative Test.)

It is reasonable to call this inflection point the point of diminishing returns, because after this point, the efficiency of the worker decreases.

55. $C(x) = .01x^2 + 40x + 3600$

$$\bar{C}(x) = \frac{C(x)}{x} = .01x + 40 + 3600x^{-1}$$

$$\bar{C}'(x) = .01 - 3600x^{-2} = 0$$

$x = 600$. This is a min because $\bar{C}''(x) = 7200x^{-3} > 0$ for $x > 0$, so the graph is concave up. So manufacture 600 units to minimize average cost.

56. Solving $c' = 0$ yields $t = 19.8616$. The Second Derivative Test shows this is a maximum.

Solving $c'' = 0$ yields $t = 41.8362$. Suppose a second drug produced a similar plasma concentration graph, with the same maximum, but a later inflection point. Then the plasma concentration decays faster for the second drug, since it takes longer for the rate of decay to level off.

57. Since the tangent line points above the sun, the sun appears higher in the sky than it really is.

3.6 Overview of Curve Sketching

1. $f(x) = x^3 - 3x^2 + 3x$
 $= x(x^2 - 3x + 3)$

The only x -intercept is $x = 0$; the y -intercept is $(0, 0)$.

$$f'(x) = 3x^2 - 6x + 3$$

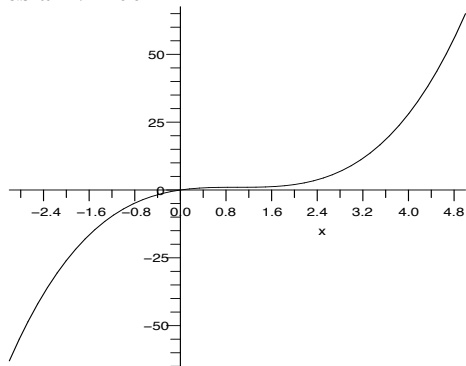
$$= 3(x^2 - 2x + 1) = 3(x - 1)^2$$

$f'(x) > 0$ for all x , so $f(x)$ is increasing for all x and has no local extrema.

$$f''(x) = 6x - 6 = 6(x - 1)$$

There is an inflection point at $x = 1$: $f(x)$ is concave down on $(-\infty, 1)$ and concave up on $(1, \infty)$.

Finally, $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.



2. $f(x) = x^4 - 3x^2 + 2x$
 $= x(x^3 - 3x + 2)$

The x -intercepts are $x = -2$, $x = 1$ and $x = 0$; the y -intercept is $(0, 0)$.

$$f'(x) = 4x^3 - 6x + 2$$

$$= 2(2x^3 - 3x + 1)$$

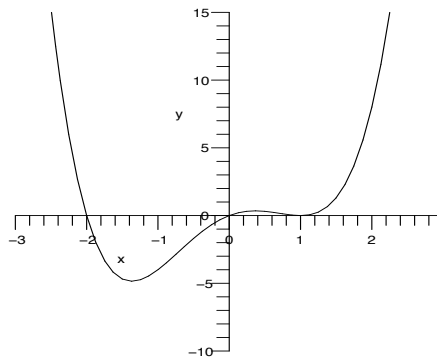
The critical numbers are $x = -1.366$, 0.366 and 1 .

$f'(x) > 0$ on $(-1.366, 0.366)$ and $(1, \infty)$, so $f(x)$ is increasing on these intervals. $f'(x) < 0$ on $(-\infty, -1.366)$ and $(0.366, 1)$, so $f(x)$ is decreasing on these intervals. Thus $f(x)$ has local minima at $x = -1.366$ and $x = 1$ and a local maximum at $x = 0.366$.

$$f''(x) = 12x^2 - 6 = 6(2x^2 - 1)$$

The critical numbers here are $x = \pm 1/\sqrt{2}$. $f''(x) > 0$ on $(-\infty, -1/\sqrt{2})$ and $(1/\sqrt{2}, \infty)$ so $f(x)$ is concave up on these intervals. $f''(x) < 0$ on $(-1/\sqrt{2}, 1/\sqrt{2})$ so $f(x)$ is concave down on this interval. Thus $f(x)$ has inflection points at $x = \pm 1/\sqrt{2}$.

Finally, $f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.



3. $f(x) = x^5 - 2x^3 + 1$

The x -intercepts are $x = 1$ and $x \approx -1.5129$; the y -intercept is $(0, 1)$.

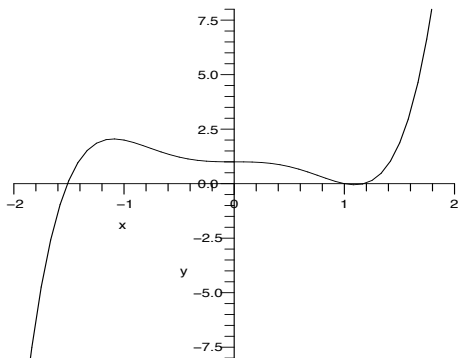
$$f'(x) = 5x^4 - 6x^2 = x^2(5x^2 - 6)$$

The critical numbers are $x = 0$ and $x = \pm\sqrt{6/5}$. Plugging values from each of the intervals into $f'(x)$, we find that $f'(x) > 0$ on $(-\infty, -\sqrt{6/5})$ and $(\sqrt{6/5}, \infty)$ so $f(x)$ is increasing on these intervals. $f'(x) < 0$ on $(-\sqrt{6/5}, 0)$ and $(0, \sqrt{6/5})$ so $f(x)$ is decreasing on these intervals. Thus $f(x)$ has a local maximum at $-\sqrt{6/5}$ and a local minimum at $\sqrt{6/5}$.

$$f''(x) = 20x^3 - 12x = 4x(5x^2 - 3)$$

The critical numbers are $x = 0$ and $x = \pm\sqrt{3/5}$. Plugging values from each of the intervals into $f''(x)$, we find that $f''(x) > 0$ on $(-\sqrt{3/5}, 0)$ and $(\sqrt{3/5}, \infty)$ so $f(x)$ is concave up on these intervals. $f''(x) < 0$ on $(-\infty, -\sqrt{3/5})$ and $(0, \sqrt{3/5})$ so $f(x)$ is concave down on these intervals. Thus $f(x)$ has inflection points at all three of these critical numbers.

Finally, $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.



4. $f(x) = x^4 + 4x^3 - 1$

The x -intercepts are $x \approx -4.01541$ and $x \approx 0.6012$; the y -intercept is $(0, -1)$.

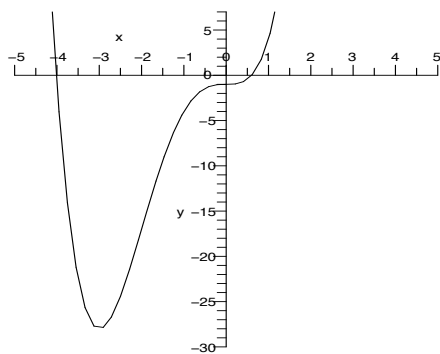
$$f'(x) = 4x^3 + 12x^2 = 4x^2(x + 3)$$

The critical numbers are $x = 0$ and $x = -3$. Plugging values from each of the intervals into $f'(x)$, we find that $f'(x) > 0$ on $(-3, 0)$ and $(0, \infty)$ so $f(x)$ is increasing on these intervals. $f'(x) < 0$ on $(-\infty, -3)$ so $f(x)$ is decreasing on these intervals. Thus $f(x)$ has a local minimum at -3 .

$$f''(x) = 12x^2 + 24x = 12x(x + 2)$$

The critical numbers are $x = 0$ and $x = -2$. Plugging values from each of the intervals into $f''(x)$, we find that $f''(x) > 0$ on $(-\infty, -2)$ and $(0, \infty)$ so $f(x)$ is concave up on $(-\infty, -2)$ and $(0, \infty)$. $f''(x) < 0$ on $(-2, 0)$ so $f(x)$ is concave down on $(-2, 0)$. The graph has inflection points at -2 and 0 .

Finally, $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow -\infty$.



5. $f(x) = x + \frac{4}{x} = \frac{x^2 + 4}{x}$

This function has no x - or y -intercepts. The domain is $\{x|x \neq 0\}$.

$f(x)$ has a vertical asymptote at $x = 0$ such that $f(x) \rightarrow -\infty$ as $x \rightarrow 0^-$ and $f(x) \rightarrow \infty$ as $x \rightarrow 0^+$.

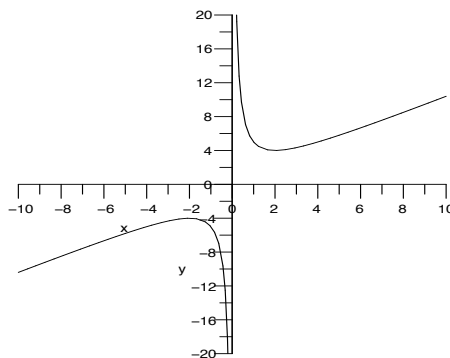
$$f'(x) = 1 - 4x^{-2} = \frac{x^2 - 4}{x^2}$$

The critical numbers are $x = \pm 2$. We find that $f'(x) > 0$ on $(-\infty, -2)$ and $(2, \infty)$ so $f(x)$ is increasing on these intervals. $f'(x) < 0$ on $(-2, 0)$ and $(0, 2)$, so $f(x)$ is decreasing on these intervals. Thus $f(x)$ has a local maximum at $x = -2$ and a local minimum at $x = 2$.

$$f''(x) = 8x^{-3}$$

$f''(x) < 0$ on $(-\infty, 0)$ so $f(x)$ is concave down on this interval and $f''(x) > 0$ on $(0, \infty)$ so $f(x)$ is concave up on this interval, but $f(x)$ has an asymptote (not an inflection point) at $x = 0$.

Finally, $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.



6. $f(x) = \frac{x^2 - 1}{x} = x - \frac{1}{x}$

There are x -intercepts at $x = \pm 1$, but no y -intercepts. The domain is $\{x|x \neq 0\}$.

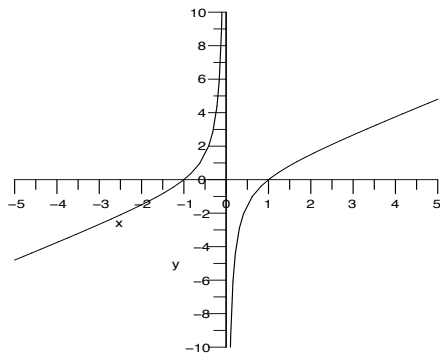
$f(x)$ has a vertical asymptote at $x = 0$ such that $f(x) \rightarrow \infty$ as $x \rightarrow 0^-$ and $f(x) \rightarrow -\infty$ as $x \rightarrow 0^+$.

$f'(x) = 1 + x^{-2} > 0$, So there is no critical numbers. $f(x)$ is increasing function.

$$f''(x) = -2x^{-3}$$

$f''(x) > 0$ on $(-\infty, 0)$ so $f(x)$ is concave up on this interval and $f''(x) < 0$ on $(0, \infty)$ so $f(x)$ is concave down on this interval, but $f(x)$ has an vertical asymptote (not an inflection point) at $x = 0$.

Finally, $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.



7. $f(x) = \frac{x^2 + 4}{x^3}$ has no x -intercept and no y -intercept. The domain of f includes all real numbers $x \neq 0$. $f(x)$ has a vertical asymptote at $x = 0$

$$f'(x) = \frac{2x(x^3) - (x^2 + 4)(3x^2)}{(x^3)^2}$$

$$= \frac{-(x^2 + 12)}{x^4}$$

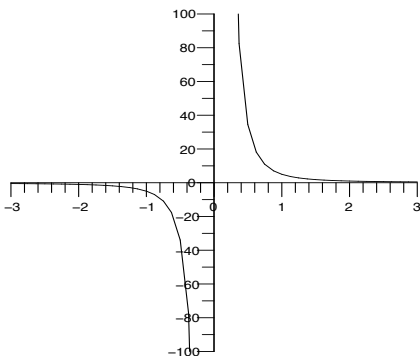
Since $f'(x) = 0$ has no real roots, the graph has no extrema. $f'(x) < 0$ on $(-\infty, 0)$ and $(0, \infty)$ so $f(x)$ is decreasing on these intervals.

$$f''(x) = -\left[\frac{x^4(2x) - (x^2 + 12)(4x^3)}{(x^4)^2} \right]$$

$$= \frac{2[x^2 + 24]}{x^5}$$

$f''(x) < 0$ on $(-\infty, 0)$ so $f(x)$ is concave down on this interval and $f''(x) > 0$ on $(0, \infty)$ so $f(x)$ is concave up on this interval, but $f(x)$ has an asymptote (not an inflection point) at $x = 0$.

Finally, $f(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore, the graph has horizontal asymptote $y = 0$.



8. $f(x) = \frac{x-4}{x^3}$

The graph has x -intercepts at $x = 4$, but no y -intercepts. The domain of f includes all real numbers $x \neq 0$. $f(x)$ has a vertical asymptote at $x = 0$

$$f'(x) = \frac{x^3 - (x-4)(3x^2)}{(x^3)^2}$$

$$= \frac{-2x + 12}{x^4}$$

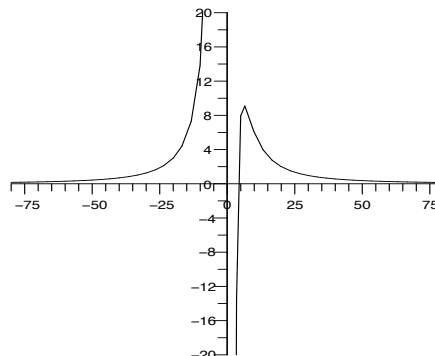
The critical numbers is $x = 6$. We find that $f'(x) > 0$ on $(-\infty, 0)$ and $(0, 6)$ so $f(x)$ is increasing on these intervals. $f'(x) < 0$ on $(6, \infty)$, so $f(x)$ is decreasing on these intervals. Therefore, the graph has a local maximum at $x = 6$.

$$f''(x) = \frac{(x^4)(-2) - (-2x + 12)(4x^3)}{(x^4)^2}$$

$$= \frac{6x - 48}{x^5}$$

$f''(x) > 0$ on $(-\infty, 0)$ and $(8, \infty)$ so $f(x)$ is concave up on this interval and $f''(x) < 0$ on $(0, 8)$ so $f(x)$ is concave down on this interval, but $f(x)$ has an inflection point at $x = 8$.

Finally, $f(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore, the graph has horizontal asymptote $y = 0$.



9. $f(x) = \frac{2x}{x^2 - 1}$

The graph has x -intercept and y -intercept at $(0, 0)$. The domain of f includes all real numbers $x \neq \pm 1$. $f(x)$ has vertical asymptotes at $x = \pm 1$.

$$f'(x) = \frac{2(x^2 - 1) - (2x)(2x)}{(x^2 - 1)^2}$$

$$= \frac{-2(x^2 + 1)}{(x^2 - 1)^2}$$

Since $f'(x) = 0$ has no real roots, the graph has no extrema. $f'(x) < 0$ on $(-\infty, -1)$, $(-1, 0)$, $(0, 1)$ and $(1, \infty)$ so $f(x)$ is decreasing on these intervals.

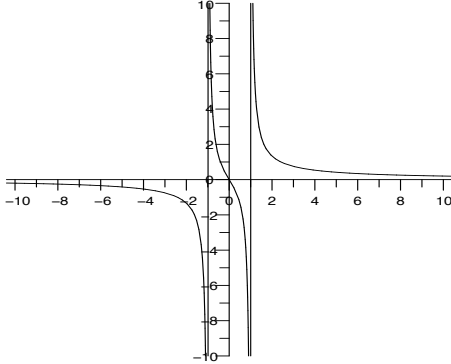
$$f''(x) = -2 \left[\frac{2x(x^2 - 1)[x^2 - 1 - 2x^2 - 2]}{(x^2 - 1)^4} \right]$$

$$= \frac{4x[x^2 + 3]}{(x^2 - 1)^3}$$

$f''(x) > 0$ on $(-1, 0)$ and $(1, \infty)$ so $f(x)$ is concave up on this interval and $f''(x) < 0$ on

$(-\infty, -1)$ and $(0, 1)$ so $f(x)$ is concave down on this interval, but $f(x)$ has an inflection point at $x = 0$.

Finally, $f(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore, the graph has horizontal asymptote $y = 0$.



10. $f(x) = \frac{3x^2}{x^2 + 1}$

The graph has x -intercept and y -intercept at $(0, 0)$. The domain of f includes all real numbers.

$$f'(x) = \frac{(x^2 + 1)(6x) - (3x^2)(2x)}{(x^2 + 1)^2}$$

$$= \frac{6x}{(x^2 + 1)^2}$$

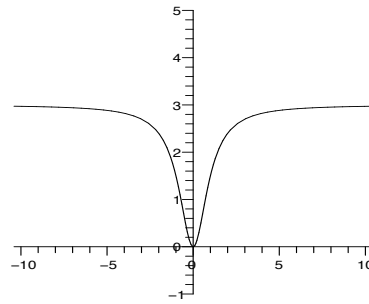
$f'(x) < 0$ on $(-\infty, 0)$ so $f(x)$ is decreasing on these intervals and $f'(x) > 0$ on $(0, \infty)$ so $f(x)$ is increasing on these interval.

$$f''(x) = \frac{(x^2 + 1)[6(x^2 + 1) - 24x^2]}{(x^2 + 1)^4}$$

$$= \frac{6 - 18x^2}{(x^2 + 1)^3}$$

The critical numbers are $x = \pm\sqrt{\frac{1}{3}}$. We find that $f''(x) > 0$ on $(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}})$ so $f(x)$ is concave up on this interval and we find that $f''(x) < 0$ on $(-\infty, -\sqrt{\frac{1}{3}})$ and $(\sqrt{\frac{1}{3}}, \infty)$ so $f(x)$ is concave down on this interval, but the graph has inflection points at $x = \pm\sqrt{\frac{1}{3}}$.

Finally, $f(x) \rightarrow 3$ as $x \rightarrow -\infty$ and $f(x) \rightarrow 3$ as $x \rightarrow \infty$. Therefore, the graph has horizontal asymptote at $y = 3$.



11. $f(x) = (x + \sin x)$

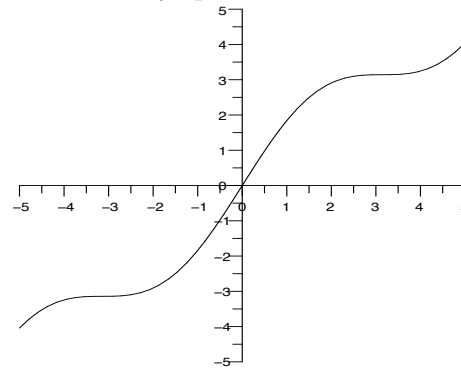
The graph has x -intercepts and y -intercepts at $(0, 0)$. The domain of f includes all real numbers.

$f'(x) = 1 + \cos x \geq 0$, therefore the graph has no extrema and $f(x)$ is a increasing function.

$$f''(x) = -\sin x$$

$f''(x) < 0$ on $(2n\pi, (2n+1)\pi)$ so $f(x)$ is concave down on this interval and we find that $f''(x) > 0$ on $((2n+1)\pi, 2(n+1)\pi)$ so $f(x)$ is concave up on this interval, but the graph has inflection points at $x = n\pi$.

Finally, $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$. Therefore, the graph has no horizontal asymptote.



12. $f(x) = \sin x - \cos x$

$f'(x) = \cos x + \sin x$ is zero for $x = n\pi - \frac{\pi}{4}$.

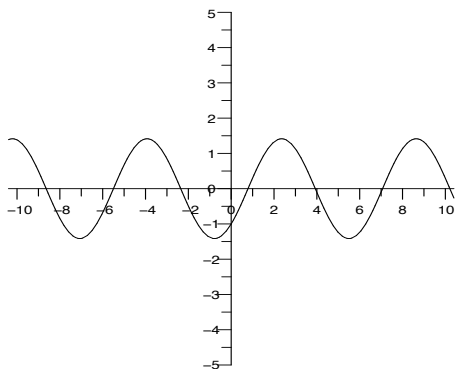
$$f''(x) = -\sin x + \cos x$$

When n is even, $f''(x) > 0$ and so f is minimum at $x = n\pi - \frac{\pi}{4}$.

When n is odd, $f''(x) < 0$ and so f is maximum at $x = n\pi - \frac{\pi}{4}$.

$f''(x) = 0$ for $x = n\pi + \frac{\pi}{4}$. So inflection points are $n\pi + \frac{\pi}{4}$.

$f''(x) < 0$ on $(\frac{\pi}{4} + n\pi, \frac{5\pi}{4} + n\pi)$ so $f(x)$ is concave down on this interval and we find that $f''(x) > 0$ on $(\frac{5\pi}{4} + n\pi, \frac{9\pi}{4} + n\pi)$ so $f(x)$ is concave up on this interval.

13. $f(x) = x \ln x$

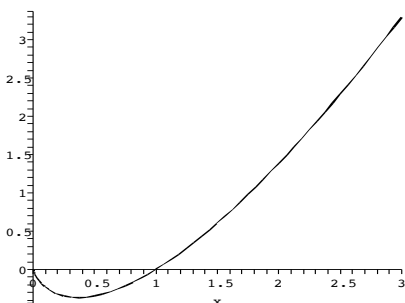
The domain is $\{x|x > 0\}$. There is an x -intercept at $x = 1$ and no y -intercept.

$$f'(x) = \ln x + 1$$

The only critical number is $x = e^{-1}$. $f'(x) < 0$ on $(0, e^{-1})$ and $f'(x) > 0$ on (e^{-1}, ∞) so $f(x)$ is decreasing on $(0, e^{-1})$ and increasing on (e^{-1}, ∞) . Thus $f(x)$ has a local minimum at $x = e^{-1}$.

$f''(x) = 1/x$, which is positive for all x in the domain of f , so $f(x)$ is always concave up.

$f(x) \rightarrow \infty$ as $x \rightarrow \infty$.

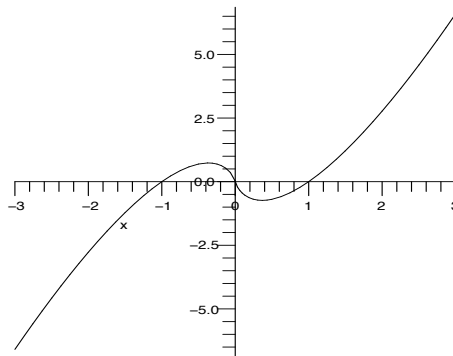
14. $f(x) = x \ln x^2$

The domain is $\{x|x \neq 0\}$. There are x -intercepts at $x = \pm 1$ but no y -intercept.

$$f'(x) = \ln x^2 + 2$$

The critical numbers are $x = \pm e^{-1}$. $f''(x) = 2/x$, so $x = -e^{-1}$ is a local maximum and $x = e^{-1}$ is a local minimum. $f(x)$ is increasing on $(-\infty, -e^{-1})$ and (e^{-1}, ∞) ; $f(x)$ is decreasing on $(-e^{-1}, 0)$ and $(0, e^{-1})$. $f(x)$ is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$.

$f(x) \rightarrow -\infty$ as $x \rightarrow \infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.

15. $f(x) = \sqrt{x^2 + 1}$

The y -intercept is $(0, 1)$. There are no x -intercepts.

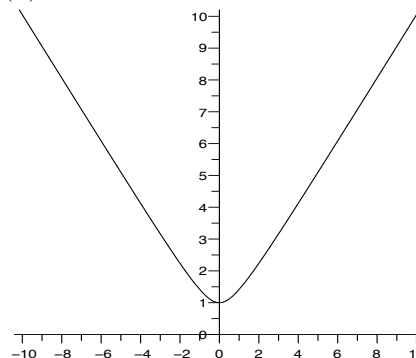
$f'(x) = \frac{1}{2}(x^2 + 1)^{-1/2}2x = \frac{x}{\sqrt{x^2 + 1}}$ The only critical number is $x = 0$. $f'(x) < 0$ when $x < 0$ and $f'(x) > 0$ when $x > 0$ so $f(x)$ is increasing on $(0, \infty)$ and decreasing on $(-\infty, 0)$. Thus $f(x)$ has a local minimum at $x = 0$.

$$f''(x) = \frac{\sqrt{x^2 + 1} - x \cdot \frac{1}{2}(x^2 + 1)^{-1/2}2x}{x^2 + 1}$$

$$= \frac{1}{(x^2 + 1)^{3/2}}$$

Since $f''(x) > 0$ for all x , we see that $f(x)$ is concave up for all x .

$f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.

16. $f(x) = \sqrt{2x - 1}$

The domain is $\{x|x \geq 1/2\}$. There is an x -intercept at $x = 1/2$.

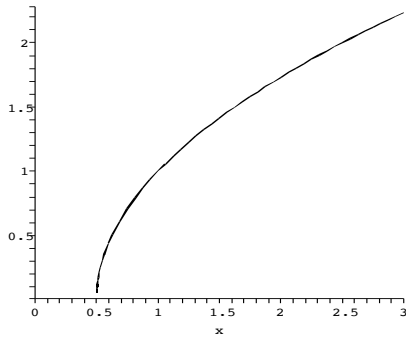
$$f'(x) = \frac{1}{2}(2x - 1)^{-1/2}2 = \frac{1}{\sqrt{2x - 1}}$$

$f'(x)$ is undefined at $x = 1/2$, but this is an endpoint of $f(x)$ and there are no other critical points. Since $f'(x)$ is positive for all x in the domain of f , we see that $f(x)$ is increasing for all x in the domain.

$$f''(x) = -\frac{1}{2}(2x - 1)^{-3/2}2 = \frac{-1}{(2x - 1)^{3/2}}$$

$f''(x) < 0$ for all x in the domain of f , so f is concave down for all x for which it is defined.

$f(x) \rightarrow \infty$ as $x \rightarrow \infty$.



$$17. f(x) = (x^3 - 3x^2 + 2x)^{1/3}$$

$$f'(x) = \frac{3x^2 - 6x + 2}{3(x^3 - 3x^2 + 2x)^{2/3}}$$

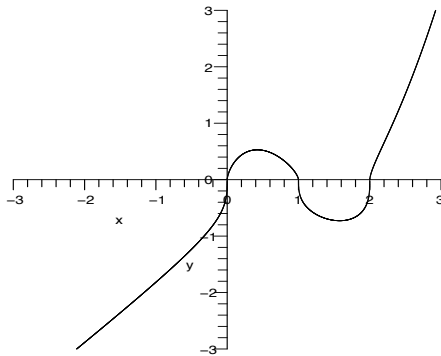
There are critical numbers at $x = \frac{3 \pm \sqrt{3}}{3}$, 0, 1 and 2.

$$f''(x) = \frac{-6x^2 + 12x - 8}{9(x^3 - 3x^2 + 2x)^{5/3}}$$

with critical numbers $x = 0, 1$ and 2 . $f''(x)$ changes sign at these values, so these are inflection points. The Second Derivative test shows that $x = \frac{3 + \sqrt{3}}{3}$ is a minimum, and

$x = \frac{3 - \sqrt{3}}{3}$ is a maximum.

$f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.



$$18. f(x) = (x^3 - 3x^2 + 2x)^{1/2}$$

$f(x)$ is defined for $0 \leq x \leq 1$ and $x \geq 2$.
 $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.

$$f'(x) = \frac{3x^2 - 6x + 2}{2(x^3 - 3x^2 + 2x)^{1/2}}$$

There are critical numbers at $x = \frac{3 \pm \sqrt{3}}{3}$, 0, 1 and 2.

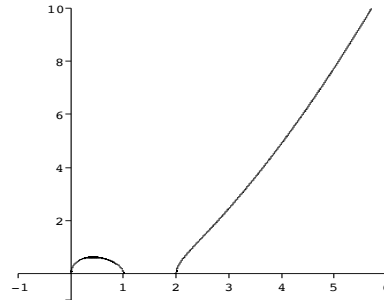
$$f''(x) = \frac{3x^4 - 12x^3 + 12x^2 - 4}{4(x^3 - 3x^2 + 2x)^{3/2}}$$

with critical numbers $x = 0, 1$ and 2 and $x \approx -0.4679$ and 2.4679 . $f(x)$ is undefined at $x = -0.4679$, so we do not consider this point.

$f''(x)$ changes sign at $x = 2.4679$, so this is an inflection point. The Second Derivative test

shows that $x = \frac{3 - \sqrt{3}}{3}$ is a maximum.

At $x = 0, 1, 2$, $f(x)$ is minimum.



$$19. f(x) = x^{5/3} - 5x^{2/3}$$

The domain of f includes all real numbers.

$$f'(x) = \frac{5}{3}x^{2/3} - \frac{10}{3}x^{-1/3}$$

$$= \frac{5}{3} \left(x^{2/3} - 2x^{-1/3} \right)$$

$$= \frac{5}{3} \left(\frac{x - 2}{x^{1/3}} \right)$$

Critical number is $x = 2$.

$f'(x) > 0$ on $(-\infty, 0)$ and $(2, \infty)$. So $f(x)$ is increasing on these intervals.

$f'(x) < 0$ on $(0, 2)$ and so $f(x)$ is decreasing on this interval.

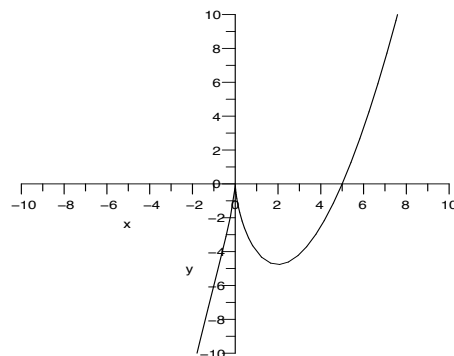
Therefore $f(x)$ is maximum at $x = 0$ and minimum at $x = 2$.

$$f''(x) = \frac{5}{3} \left(\frac{2}{3}x^{-1/3} + \frac{2}{3}x^{-4/3} \right)$$

$$= \frac{10}{9} \left(x^{-1/3} + x^{-4/3} \right)$$

$$= \frac{10}{9} \left(\frac{x + 1}{x^{4/3}} \right)$$

The critical number is at $x = 0, -1$. $f''(x)$ changes sign at these values, so these are inflection points. $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.



20. $f(x) = x^3 - \frac{3}{400}x = x(x^2 - \frac{3}{400})$

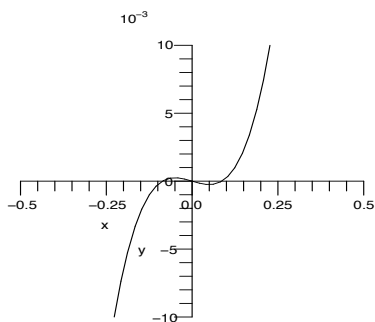
The y -intercept (also an x -intercept) is $(0, 0)$ and there are also x -intercepts at $x = \pm\sqrt{3}/20$.

$$f'(x) = 3x^2 - \frac{3}{400}$$

The critical numbers are $x = \pm 1/20$.

$f''(x) = 6x$, so $x = -1/20$ is a local maximum and $x = 1/20$ is a local minimum. $f(x)$ is increasing on $(-\infty, -1/20)$ and $(1/20, \infty)$ and decreasing on $(-1/20, 1/20)$. It is concave up on $(0, \infty)$ and concave down on $(-\infty, 0)$, with an inflection point at $x = 0$.

$f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.



21. $f(x) = e^{-2/x}$

$$f'(x) = e^{-2/x} \left(\frac{2}{x^2} \right) = \frac{2}{x^2} e^{-2/x}$$

$$f''(x) = \frac{-4}{x^3} e^{-2/x} + \frac{2}{x^2} e^{-2/x} \left(\frac{2}{x^2} \right)$$

$$= \frac{4}{x^4} e^{-2/x} - \frac{4}{x^3} e^{-2/x}$$

$$f'(x) > 0 \text{ on } (-\infty, 0) \cup (0, \infty)$$

$$f''(x) > 0 \text{ on } (-\infty, 0) \cup (0, 1)$$

$$f''(x) < 0 \text{ on } (1, \infty)$$

f increasing on $(-\infty, 0)$ and on $(0, \infty)$, concave up on $(-\infty, 0) \cup (0, 1)$, concave down on $(1, \infty)$, inflection point at $x = 1$. f is undefined at $x = 0$.

$$\lim_{x \rightarrow 0^+} e^{-2/x} = \lim_{x \rightarrow 0^+} \frac{1}{e^{2/x}} = 0 \text{ and}$$

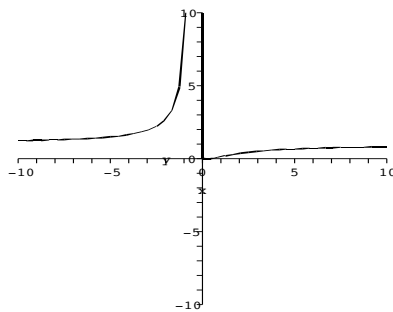
$$\lim_{x \rightarrow 0^-} e^{-2/x} = \infty$$

So f has a vertical asymptote at $x = 0$.

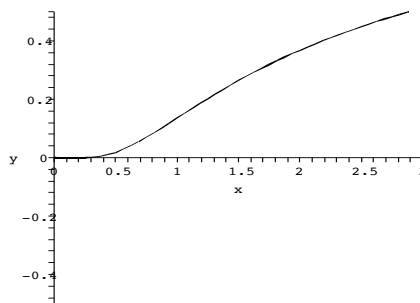
$$\lim_{x \rightarrow \infty} e^{-2/x} = \lim_{x \rightarrow -\infty} e^{-2/x} = 1$$

So f has a horizontal asymptote at $y = 1$.

Global graph of $f(x)$:



Local graph of $f(x)$:



22. $f(x) = e^{1/x^2}$

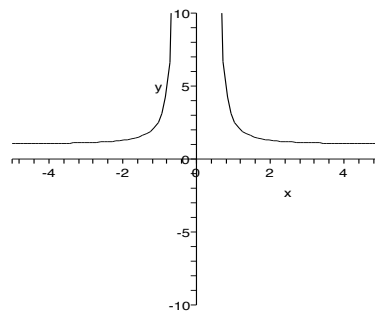
The function has a vertical asymptote at $x = 0$ such that $f(x) \rightarrow \infty$ as x approaches 0 from the right or left. There is a horizontal asymptote of $y = 1$ as $x \rightarrow \pm\infty$.

$$f'(x) = \frac{-2}{x^3} \cdot e^{1/x^2}$$

$f'(x) > 0$ for $x < 0$, so $f(x)$ is increasing on $(-\infty, 0)$ and $f'(x) < 0$ for $x > 0$, so $f(x)$ is decreasing on $(0, \infty)$.

$$f''(x) = \frac{2e^{1/x^2}(3x^2 + 2)}{x^6}$$

is positive for all $x \neq 0$, so $f(x)$ is concave up for all $x \neq 0$.



23. $f(x) = \frac{1}{x^3 - 3x^2 - 9x + 1}$

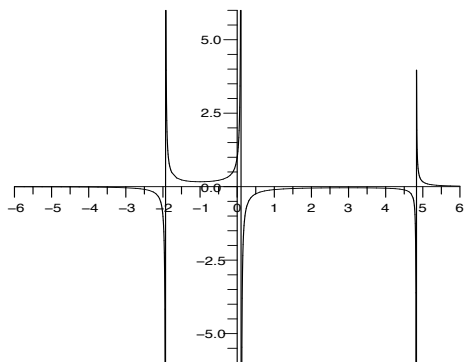
$$f'(x) = -\frac{3x^2 - 6x - 9}{(x^3 - 3x^2 - 9x + 1)^2}$$

The critical numbers are $x = -1, 3$.

$$f''(x) = \frac{6(6x^4 - 4x^3 - 7x^2 + 12x + 2)}{(x^3 - 3x^2 - 9x + 1)^3}$$

The Second Derivative test shows that the graph has a local minimum at $x = -1$ and a local maximum at $x = 3$. The graph has a vertical asymptote at $x = -1.9304$. Similarly, the graph has vertical asymptotes at $x = 0.1074$ and 4.8231 .

$f(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore, the graph has horizontal asymptote $y = 0$.



24. $f(x) = \frac{1}{x^3 + 3x^2 + 4x + 1}$

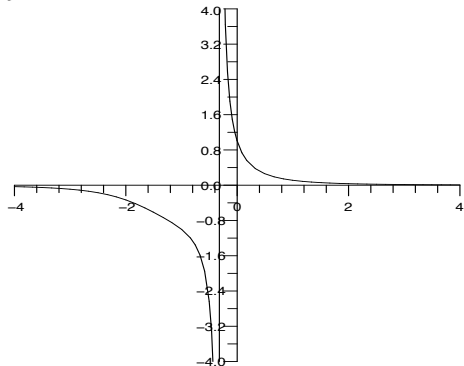
$$f'(x) = -\frac{3x^2 + 6x + 4}{(x^3 + 3x^2 + 4x + 1)^2}$$

Since $f'(x) = 0$ has no real roots, the graph has no extrema.

$$f''(x) = \frac{12x^4 + 48x^3 + 78x^2 + 66x + 26}{(x^3 + 3x^2 + 4x + 1)^3}$$

The Critical number is $x = -0.316722$. $f''(x) > 0$ on $(-0.316722, \infty)$ so the graph is concave up on this interval. $f''(x) < 0$ on $(-\infty, -0.316722)$ so the graph is concave down on this interval. the graph has a vertical asymptote at $x = -0.316722$. $f(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $f(x) \rightarrow 0$ as $x \rightarrow \infty$.

Therefore, the graph has horizontal asymptote $y = 0$.



25. $f(x) = (x^3 - 3x^2 + 2x)^{2/3}$

$$f'(x) = \frac{2(3x^2 - 6x + 2)}{3(x^3 - 3x^2 + 2x)^{1/3}}$$

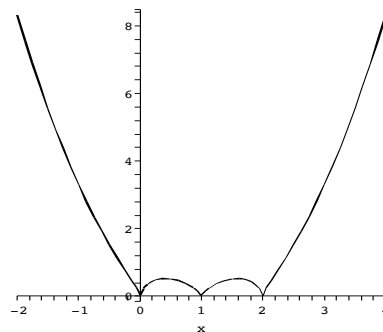
There are critical numbers at $x = \frac{3 \pm \sqrt{3}}{3}$, 0, 1 and 2.

$$f''(x) = \frac{18x^4 - 72x^3 + 84x^2 - 24x - 8}{9(x^3 - 3x^2 + 2x)^{4/3}}$$

with critical numbers $x = 0, 1$ and 2 and $x \approx -0.1883$ and 2.1883 . $f''(x)$ changes sign at these last two values, so these are inflection points. The Second Derivative test shows that

$x = \frac{3 \pm \sqrt{3}}{3}$ are both maxima. Local minima occur at $x = 0, 1$ and 2 .

$f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.

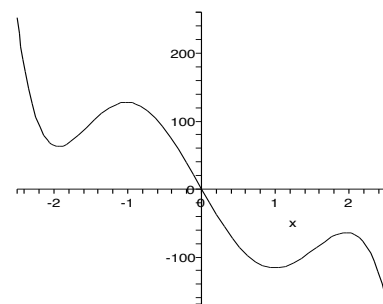


26. $f(x) = x^6 - 10x^5 - 7x^4 + 80x^3 + 12x^2 - 192x$
 $f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.

$$f'(x) = 6x^5 - 50x^4 - 28x^3 + 240x^2 + 24x - 192$$

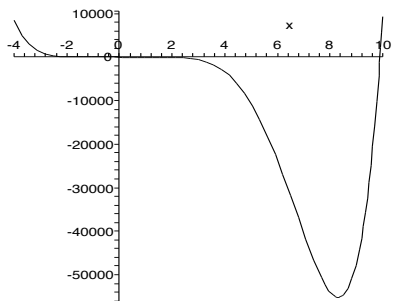
$$f''(x) = 30x^4 - 200x^3 - 84x^2 + 480x + 24$$

Critical numbers at approximately $x = -1.9339, -1.0129, 1, 1.9644$, and 8.3158 . Critical numbers at approximately $x = -1.5534, -0.0496, 1.5430$, and 6.7267 , and changes sign at each of these values, so these are inflection points. The Second Derivative Test shows that $x = -1.9339, 1$, and 8.3158 are local minima, and $x = -1.0129$ and 1.9644 are local maxima. The extrema near $x = 0$ look like this:



The inflection points, and the global behavior of the function can be seen on the following

graph.



27. $f(x) = \frac{x^2 + 1}{3x^2 - 1}$

Note that $x = \pm\sqrt{1/3}$ are not in the domain of the function, but yield vertical asymptotes.

$$\begin{aligned} f'(x) &= \frac{2x(3x^2 - 1) - (x^2 + 1)(6x)}{(3x^2 - 1)^2} \\ &= \frac{(6x^3 - 2x) - (6x^3 + 6x)}{(3x^2 - 1)^2} \\ &= \frac{-8x}{(3x^2 - 1)^2} \end{aligned}$$

So the only critical point is $x = 0$.

$$f'(x) > 0 \text{ for } x < 0$$

$$f'(x) < 0 \text{ for } x > 0$$

so f is increasing on $(-\infty, -\sqrt{1/3})$ and on $(-\sqrt{1/3}, 0)$; decreasing on $(0, \sqrt{1/3})$ and on $(\sqrt{1/3}, \infty)$. Thus there is a local max at $x = 0$.

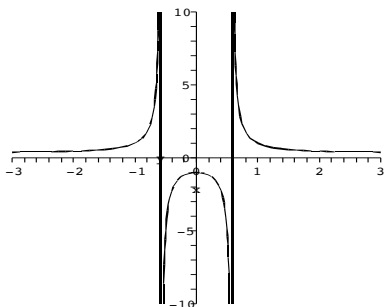
$$f''(x) = 8 \cdot \frac{9x^2 + 1}{(3x^2 - 1)^3}$$

$$f''(x) > 0 \text{ on } (-\infty, -\sqrt{1/3}) \cup (\sqrt{1/3}, \infty)$$

$$f''(x) < 0 \text{ on } (-\sqrt{1/3}, \sqrt{1/3})$$

Hence f is concave up on $(-\infty, -\sqrt{1/3})$ and on $(\sqrt{1/3}, \infty)$; concave down on $(-\sqrt{1/3}, \sqrt{1/3})$.

Finally, when $|x|$ is large, the function approached $1/3$, so $y = 1/3$ is a horizontal asymptote.



28. $f(x) = \frac{5x}{x^3 - x + 1}$

Looking at the graph of $x^3 - x + 1$, we see

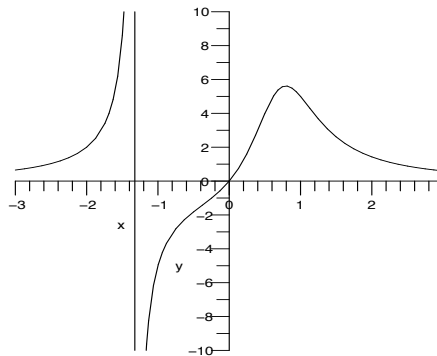
that there is one real root, at approximately -1.325 ; so the domain of the function is all x except for this one point, and $x = -1.325$ will be a vertical asymptote. There is a horizontal asymptote of $y = 0$.

$$f'(x) = 5 \frac{1 - 2x^3}{(x^3 - x + 1)^2}$$

The only critical point is $x = \sqrt[3]{1/2}$. By the first derivative test, this is a local max.

$$f''(x) = 10 \frac{3x^5 + x^3 - 6x^2 + 1}{(x^3 - x + 1)^3}$$

The numerator of f'' has three real roots, which are approximately $x = -.39018$, $x = .43347$, and $x = 1.1077$. $f''(x) > 0$ on $(-\infty, -1.325) \cup (-.390, .433) \cup (1.108, \infty)$ $f''(x) < 0$ on $(-1.325, -.390) \cup (.433, 1.108)$. So f is concave up on $(-\infty, -1.325) \cup (-.390, .433) \cup (1.108, \infty)$ and concave down on $(-1.325, -.390) \cup (.433, 1.108)$. Hence $x = -.39018$, $x = .43347$, and $x = 1.1077$ are inflection points.



29. $f(x) = x^2\sqrt{x^2 - 9}$

f is undefined on $(-3, 3)$.

$$\begin{aligned} f'(x) &= 2x\sqrt{x^2 - 9} + x^2 \left(\frac{1}{2}(x^2 - 9)^{-1/2} \cdot 2x \right) \\ &= 2x\sqrt{x^2 - 9} + \frac{x^3}{\sqrt{x^2 - 9}} \\ &= \frac{2x(x^2 - 9) + x^3}{\sqrt{x^2 - 9}} \\ &= \frac{3x^3 - 18x}{\sqrt{x^2 - 9}} = \frac{3x(x^2 - 6)}{\sqrt{x^2 - 9}} \\ &= \frac{3x(x + \sqrt{6})(x - \sqrt{6})}{\sqrt{x^2 - 9}} \end{aligned}$$

Critical points ± 3 . (Note that f is undefined at $x = 0, \pm\sqrt{6}$.)

$$\begin{aligned}
 f''(x) &= \frac{(9x^2 - 18)\sqrt{x^2 - 9}}{x^2 - 9} \\
 &\quad - \frac{(3x^3 - 18x) \cdot \frac{1}{2}(x^2 - 9)^{-1/2} \cdot 2x}{x^2 - 9} \\
 &= \frac{(9x^2 - 18)(x^2 - 9) - x(3x^3 - 18x)}{(x^2 - 9)^{3/2}} \\
 &= \frac{(6x^4 - 81x^2 + 162)}{(x^2 - 9)^{3/2}} \\
 f''(x) &= 0 \text{ when} \\
 x^2 &= \frac{81 \pm \sqrt{81^2 - 4(6)(162)}}{2(6)} \\
 &= \frac{81 \pm \sqrt{2673}}{12} = \frac{1}{4}(27 \pm \sqrt{297})
 \end{aligned}$$

So $x \approx \pm 3.325$ or $x \approx \pm 1.562$, but these latter values are not in the same domain. So only ± 3.325 are potential inflection points.

$$f'(x) > 0 \text{ on } (3, \infty)$$

$$f'(x) < 0 \text{ on } (-\infty, -3)$$

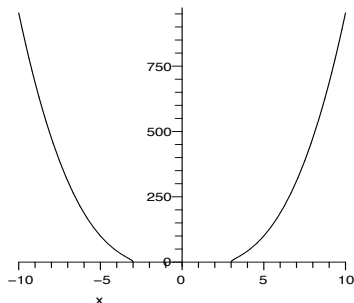
$$f''(x) > 0 \text{ on } (-\infty, -3.3) \cup (3.3, \infty)$$

$$f''(x) < 0 \text{ on } (-3.3, -3) \cup (3, 3.3)$$

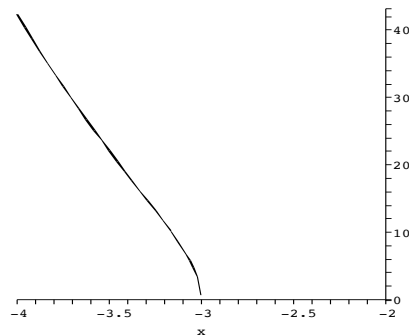
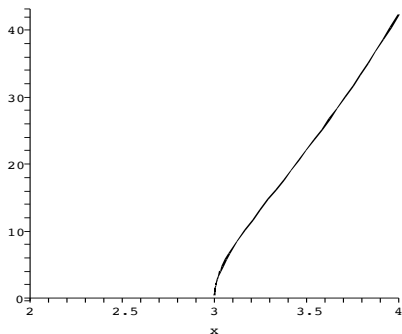
f is increasing on $(3, \infty)$, decreasing on $(-\infty, -3)$, concave up on $(-\infty, -3.3) \cup (3.3, \infty)$, concave down on $(-3.3, -3) \cup (3, 3.3)$.

$x = \pm 3.3$ are inflection points.

Global graph of $f(x)$:



Local graphs of $f(x)$:

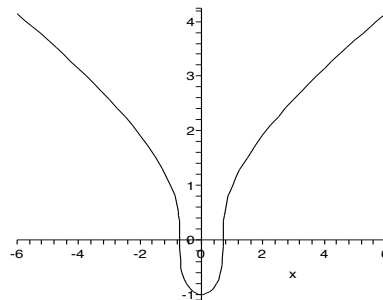


$$\begin{aligned}
 30. \quad f(x) &= \sqrt[3]{2x^2 - 1} \\
 f'(x) &= \frac{4x}{3(2x^2 - 1)^{2/3}} \\
 f'(x) &= 0 \text{ at } x = 0 \text{ and is undefined at } x = \pm \sqrt{1/2}.
 \end{aligned}$$

$$f''(x) = \frac{-4(2x^2 + 3)}{9(2x^2 - 1)^{5/3}}$$

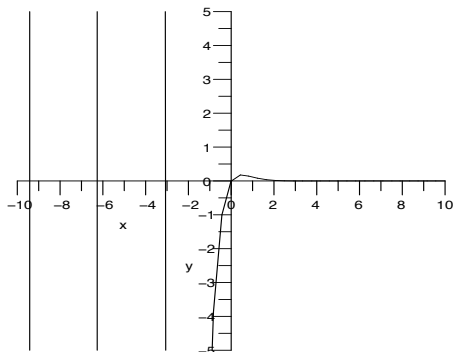
$f''(x)$ is never 0, and is undefined where f' is. The function changes concavity at $x = \pm \sqrt{1/2}$, so these are inflection points. The slope does not change at these values, so they are not extrema. The Second Derivative Test shows that $x = 0$ is a minimum.

$$f(x) \rightarrow \infty \text{ as } x \rightarrow \pm \infty.$$

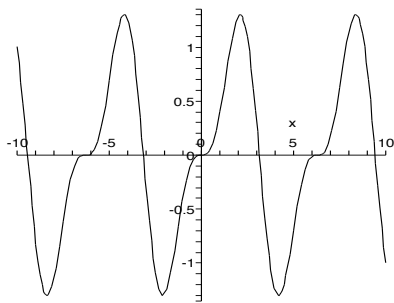


$$\begin{aligned}
 31. \quad f(x) &= e^{-2x} \sin x \\
 f'(x) &= e^{-2x} (\cos x - 2 \sin x) \\
 f''(x) &= e^{-2x} (3 \sin x - 4 \cos x) \\
 f'(x) &= 0 \text{ when } \cos x = 2 \sin x; \text{ that is,} \\
 &\text{when } \tan x = 1/2; \text{ that is, when } x = \\
 &k\pi + \tan^{-1}(1/2), \text{ where } k \text{ is any integer.} \\
 f'(x) &< 0, \text{ and } f \text{ is decreasing, on intervals of} \\
 &\text{the form } (2k\pi + \tan^{-1}(\frac{1}{2}), (2k+1)\pi + \tan^{-1}(\frac{1}{2})) \\
 f'(x) &> 0 \text{ and } f \text{ is increasing, on intervals of} \\
 &\text{the form } ((2k-1)\pi + \tan^{-1}(\frac{1}{2}), 2k\pi + \tan^{-1}(\frac{1}{2})) \\
 \text{Hence } f &\text{ has a local max at } x = 2k\pi + \\
 &\tan^{-1}(1/2) \text{ and a local min at } x = (2k+1)\pi + \\
 &\tan^{-1}(1/2). \\
 f''(x) &= 0 \text{ when } 3 \sin x = 4 \cos x; \text{ that} \\
 &\text{is, when } \tan x = 4/3; \text{ that is, when } x = \\
 &k\pi + \tan^{-1}(4/3). \text{ The sign of } f'' \text{ changes at}
 \end{aligned}$$

each of these points, so all of them are inflection points.



- 32.** $f(x) = \sin x - \frac{1}{2} \sin 2x$
 $f'(x) = \cos x - \cos 2x$
 $f'(x) = 0$ when $x = 2k\pi$, $2\pi/3 + 2k\pi$, or $4\pi/3 + 2k\pi$.
 $f''(x) = -\sin x + 2 \sin 2x$
 $f''(x) = 0$ when $x = 0, \pi$ and approximately ± 1.3181 , and the pattern repeats with period 2π . f'' changes sign at each of these values, so these are inflection points. The First Derivative Test shows that $x = 2k\pi$ is neither a minimum nor a maximum. The Second Derivative Test shows that the other critical numbers are extrema that alternate between minima and maxima.

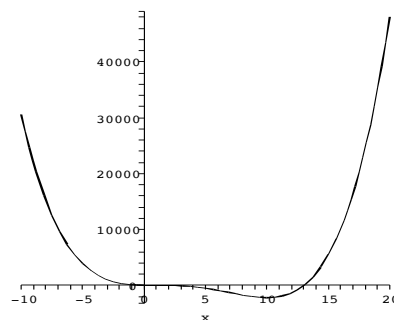


- 33.** $f(x) = x^4 - 16x^3 + 42x^2 - 39.6x + 14$
 $f'(x) = 4x^3 - 48x^2 + 84x - 39.6$
 $f''(x) = 12x^2 - 96x + 84$
 $= 12(x^2 - 8x + 7)$
 $= 12(x - 7)(x - 1)$
 $f'(x) > 0$ on $(.8952, 1.106) \cup (9.9987, \infty)$
 $f'(x) < 0$ on $(-\infty, .8952) \cup (1.106, 9.9987)$
 $f''(x) > 0$ on $(-\infty, 1) \cup (7, \infty)$
 $f''(x) < 0$ on $(1, 7)$
 f is increasing on $(.8952, 1.106)$ and on $(9.9987, \infty)$, decreasing on $(-\infty, .8952)$ and on $(1.106, 9.9987)$, concave up on $(-\infty, 1) \cup (7, \infty)$, concave down on $(1, 7)$, $x = .8952$, 9.9987 are local min, $x = 1.106$ is local max, $x = 1, 7$ are

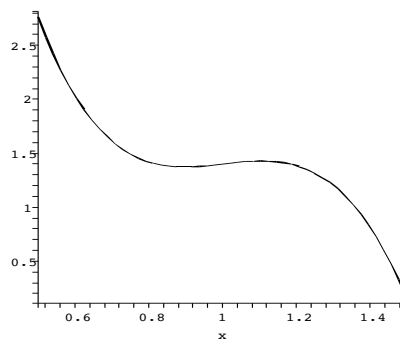
inflection points.

$f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.

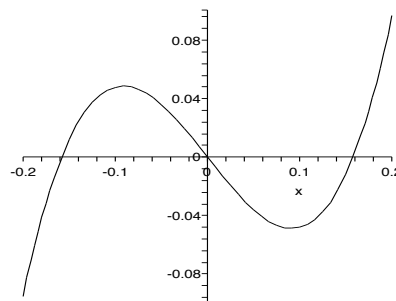
Global graph of $f(x)$:



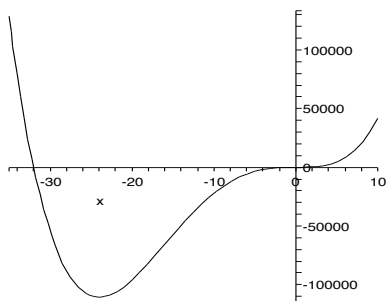
Local graph of $f(x)$:



- 34.** $f(x) = x^4 + 32x^3 - 0.02x^2 - 0.8x$
 $f'(x) = 4x^3 + 96x^2 - 0.04x - 0.8$
 $f'(x) = 0$ at approximately $x = -24$, -0.09125 , and 0.09132 .
 $f''(x) = 12x^2 + 192x - 0.04$
 $f''(x) = 0$ at approximately $x = 16.0002$ and 0.0002 , and changes sign at these values, so these are inflection points. The Second Derivative Test shows that $x = -24$ and 0.09132 are minima, and that $x = -0.09125$ is a maxima. The extrema near $x = 0$ look like this:



The global behavior looks like this:



$$\begin{aligned}
 35. \quad f(x) &= \frac{25 - 50\sqrt{x^2 + 0.25}}{x} \\
 &= 25 \left(\frac{1 - 2\sqrt{x^2 + 0.25}}{x} \right) \\
 &= 25 \left(\frac{1 - \sqrt{4x^2 + 1}}{x} \right)
 \end{aligned}$$

Note that $x = 0$ is not in the domain of the function.

$$f'(x) = 25 \left(\frac{1 - \sqrt{4x^2 + 1}}{x^2 \sqrt{4x^2 + 1}} \right)$$

We see that there are no critical points. Indeed, $f' < 0$ wherever f is defined. One can verify that

$$f''(x) > 0 \text{ on } (0, \infty)$$

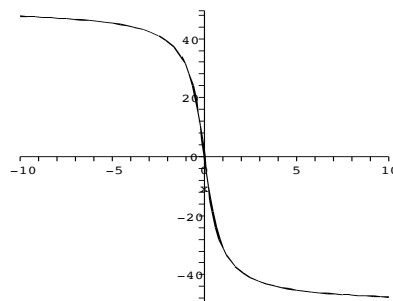
$$f''(x) < 0 \text{ on } (-\infty, 0)$$

Hence the function is concave up on $(0, \infty)$ and concave down on $(-\infty, 0)$.

$$\begin{aligned}
 &\lim_{x \rightarrow \infty} \frac{25 - 50\sqrt{x^2 + 0.25}}{x} \\
 &= \lim_{x \rightarrow \infty} \frac{25}{x} - \frac{50\sqrt{x^2 + 0.25}}{x} \\
 &= \lim_{x \rightarrow \infty} 0 - 50 \frac{x\sqrt{1 + \frac{0.25}{x^2}}}{x} \\
 &= \lim_{x \rightarrow \infty} -50\sqrt{1 + \frac{0.25}{x^2}} = -50
 \end{aligned}$$

$$\begin{aligned}
 &\lim_{x \rightarrow -\infty} \frac{25 - 50\sqrt{x^2 + 0.25}}{x} \\
 &= \lim_{x \rightarrow \infty} \frac{25}{x} - \frac{50\sqrt{x^2 + 0.25}}{x} \\
 &= \lim_{x \rightarrow -\infty} 0 - 50 \frac{(-x)\sqrt{1 + \frac{0.25}{x^2}}}{x} \\
 &= \lim_{x \rightarrow -\infty} 50\sqrt{1 + \frac{0.25}{x^2}} = 50
 \end{aligned}$$

So f has horizontal asymptotes at $y = 50$ and $y = -50$.



$$36. \quad f(x) = \tan^{-1} \left(\frac{1}{x^2 - 1} \right)$$

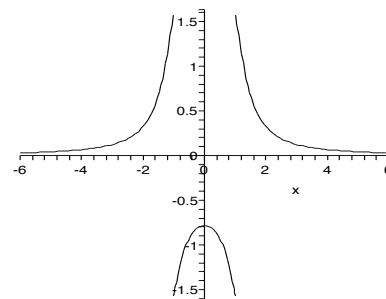
The function has horizontal asymptote $y = 0$, and is undefined at $x = \pm 1$.

$$f'(x) = \frac{-2x}{x^4 - 2x^2 + 2}$$

$$f'(x) = 0 \text{ only when } x = 0.$$

$$f''(x) = \frac{2(3x^4 - 2x^2 - 2)}{(x^4 - 2x^2 + 2)^2}$$

$f''(x) = 0$ at approximately $x = \pm 1.1024$ and changes sign there, so these are inflection points (very easy to miss by looking at the graph). The Second Derivative Test shows that $x = 0$ is a local maximum.



$$37. \quad f(x) = x^4 + cx^2$$

$$f'(x) = 4x^3 + 2cx$$

$$f''(x) = 12x^2 + 2c$$

$c = 0$: 1 extremum, 0 inflection points

$c < 0$: 3 extrema, 2 inflection points

$c > 0$: 1 extremum, 0 inflection points

$c \rightarrow -\infty$: the graph widens and lowers

$c \rightarrow +\infty$: the graph narrows

$$38. \quad f(x) = x^4 + cx^2 + x$$

$$f'(x) = 4x^3 + 2cx + 1$$

$$f''(x) = 12x^2 + 2c$$

If c is negative, there will be two solutions to $f'' = 0$, and these will be inflection points. For $c > 0$ there will be no solutions to $f'' = 0$, and no inflection points. For $c = 0$, $f'' = 0$

when $x = 0$, but does not change sign there, so this is not an inflection point. $f' = 0$ has one solution, corresponding to a minimum, for all $c > -1.5$. For $c = -1.5$, there is a second critical point which is neither a minimum nor a maximum. For $c < -1.5$ there are three critical points, two minima and a maximum. As $c \rightarrow \infty$ the curve has one minimum, and narrows. As $c \rightarrow -\infty$, the two minima get farther apart and drop lower. The local maximum approaches $(0, 0)$.

$$\begin{aligned} 39. \quad f(x) &= \frac{x^2}{x^2 + c^2} \\ f'(x) &= \frac{2c^2 x}{(x^2 + c^2)^2} \\ f''(x) &= \frac{2c^4 - 6c^2 x^2}{(x^2 + c^2)^3} \end{aligned}$$

If $c = 0$: $f(x) = 1$, except that f is undefined at $x = 0$. $c < 0$, $c > 0$: horizontal asymptote at $y = 1$, local min at $x = 0$, since the derivative changes sign from negative to positive at $x = 0$; also there are inflection points at $x = \pm c/\sqrt{3}$. As $c \rightarrow -\infty$, $c \rightarrow +\infty$: the graph widens.

$$\begin{aligned} 40. \quad f(x) &= e^{-x^2/c} \\ f'(x) &= \frac{-2x}{c} \cdot e^{-x^2/c} \\ f''(x) &= \frac{-2c + 4x^2}{c^2} \cdot e^{-x^2/c} \end{aligned}$$

For $c > 0$ the graph is a bell curve centered at its maximum point $(0, 1)$, and the inflection points are at $x = \pm\sqrt{c/2}$. As $c \rightarrow \infty$, the curve widens.

The function is not defined for $c = 0$.

For $c < 0$, there are no inflection points, and $x = 0$ is a minimum. The graph is cup shaped and widens as $c \rightarrow -\infty$.

$$41. \text{ When } c = 0, f(x) = \sin(0) = 0.$$

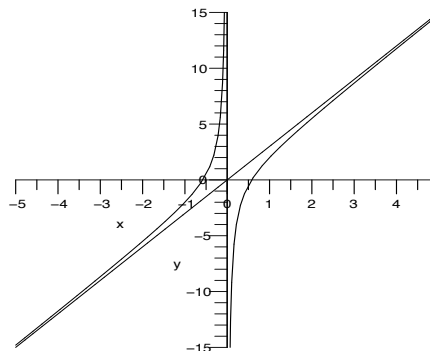
Since $\sin x$ is an odd function, $\sin(-cx) = -\sin(cx)$. Thus negative values of c give the reflection through the x -axis of their positive counterparts. For large values of c , the graph looks just like $\sin x$, but with a very small period.

$$42. \text{ When } c = 0, \text{ we have } f(x) = x^2\sqrt{-x^2}, \text{ which is undefined.}$$

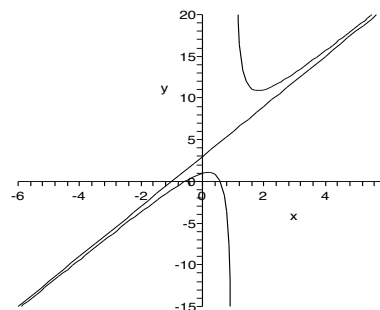
Since $x^2\sqrt{c^2 - x^2} = x^2\sqrt{(-c)^2 - x^2}$, the function is the same regardless of whether c is negative or positive. The function is always 0 at $x = 0$ and undefined for $|x| > |c|$. Where it

is defined, $f(x) \geq 0$, attaining its minimum at $x = 0$. It reaches its maximum value at $x = \pm\sqrt{2c^2/3}$. At these points, f attains the value $2\sqrt{3}|c|^3/9$. The function looks generally the same as $|c|$ gets large, with the domain and range increasing as $|c|$ does.

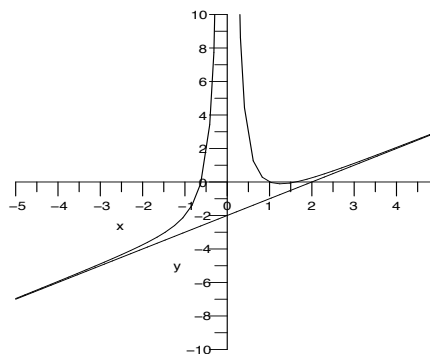
$$\begin{aligned} 43. \quad f(x) &= \frac{3x^2 - 1}{x} = 3x - \frac{1}{x} \\ y &= 3x \text{ is a slant asymptote.} \end{aligned}$$



$$\begin{aligned} 44. \quad f(x) &= \frac{3x^2 - 1}{x - 1} = 3x + 3 + \frac{2}{x - 1}, \\ \text{so the slant asymptote is } y &= 3x + 3. \end{aligned}$$

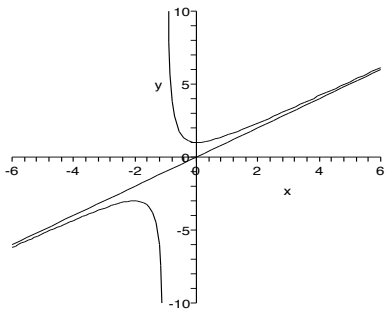


$$\begin{aligned} 45. \quad f(x) &= \frac{x^3 - 2x^2 + 1}{x^2} = x - 2 + \frac{1}{x^2} \\ y &= x - 2 \text{ is a slant asymptote.} \end{aligned}$$

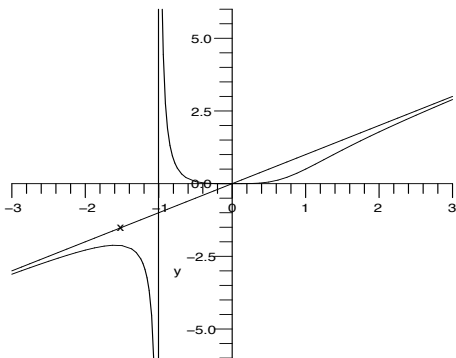


$$46. \quad f(x) = \frac{x^3 - 1}{x^2 - 1} = x + \frac{x - 1}{x^2 - 1},$$

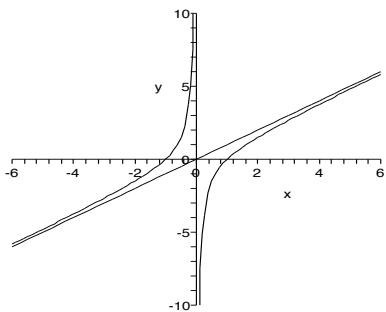
so the slant asymptote is $y = x$.



47. $f(x) = \frac{x^4}{x^3 + 1} = x - \frac{x}{x^3 + 1}$
 $y = x$ is a slant asymptote.



48. $f(x) = \frac{x^4 - 1}{x^3 + x} = x + \frac{-x^2 - 1}{x^3 + x}$,
 so the slant asymptote is $y = x$.



49. One possibility:

$$f(x) = \frac{3x^2}{(x-1)(x-2)}$$

50. One possibility:

$$f(x) = \frac{x}{x^2 - 1}$$

51. One possibility:

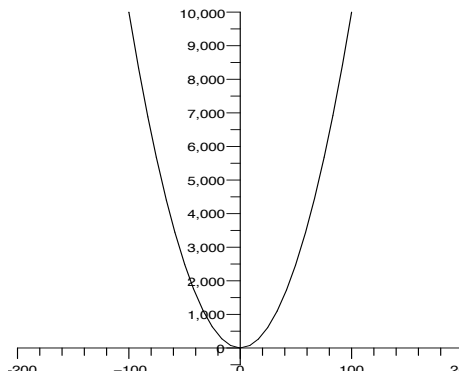
$$f(x) = \frac{2x}{\sqrt{(x-1)(x+1)}}$$

52. One possibility:

$$f(x) = \frac{2x^2}{(x-1)(x-3)}$$

53. $\lim_{x \rightarrow \infty} \left[\frac{x^4 - x^2 + 1}{x^2 - 1} - x^2 \right]$
 $= \lim_{x \rightarrow \infty} \left[\frac{x^4 - x^2 + 1 - x^2(x^2 - 1)}{x^2 - 1} \right]$
 $= \lim_{x \rightarrow \infty} \left[\frac{1}{x^2 - 1} \right] = 0$

Thus $f(x) = \frac{x^4 - x^2 + 1}{x^2 - 1}$ has x^2 as an asymptote.



54. (a) $f(x) = \frac{x^4}{x+1}$
 $= \frac{x^4 - 1 + 1}{x+1}$
 $= \frac{(x^2 + 1)(x+1)(x-1)}{x+1} + \frac{1}{x+1}$
 $= (x^2 + 1)(x+1) + \frac{1}{x+1}$

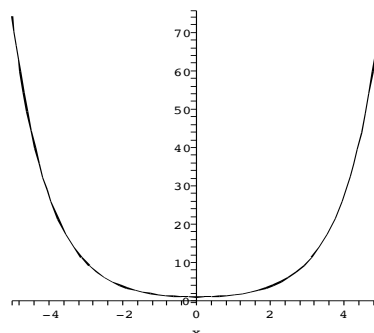
One possible polynomial is $p(x) = (x^2 + 1)(x+1)$. Then $|f(x) - p(x)| = \left| \frac{1}{x+1} \right| \rightarrow 0$ as $x \rightarrow \infty$.

(b) $f(x) = \frac{x^5 - 1}{x+1}$
 $= \frac{x^5 + 1 - 2}{x+1}$
 $= x^4 - x^3 + x^2 - x + 1 - \frac{2}{x+1}$

One possible polynomial is $p(x) = x^4 - x^3 + x^2 - x + 1$. Then $|f(x) - p(x)| = \left| \frac{2}{x+1} \right| \rightarrow 0$ as $x \rightarrow \infty$.

$$\begin{aligned}
 \text{(c) } f(x) &= \frac{x^6 - 2}{x + 1} \\
 &= \frac{x^6 - 1 - 1}{x + 1} \\
 &= \frac{(x^3 - 1)(x^3 + 1) - 1}{x + 1} \\
 &= \frac{(x^3 - 1)(x + 1)(x^2 - x + 1) - 1}{x + 1} \\
 &= (x^3 - 1)(x^2 - x + 1) - \frac{1}{x + 1}
 \end{aligned}$$

One possible polynomial is $p(x) = (x^3 - 1)(x^2 - x + 1)$. Then $|f(x) - p(x)| = \left| \frac{1}{x+1} \right| \rightarrow 0$ as $x \rightarrow \infty$.



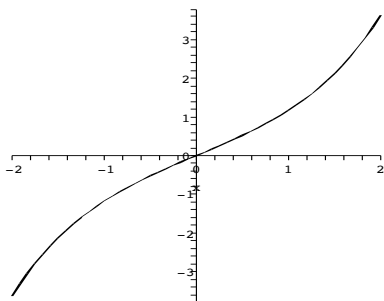
$$55. f(x) = \sinh x = \frac{e^x - e^{-x}}{2}$$

$$f'(x) = \frac{e^x + e^{-x}}{2}$$

$f'(x) > 0$ for all x so $f(x)$ is always increasing and has no extrema.

$$f''(x) = \frac{e^x - e^{-x}}{2}$$

$f''(x) = 0$ only when $x = 0$ and changes sign here, so $f(x)$ has an inflection point at $x = 0$.



$$f(x) = \cosh x = \frac{e^x + e^{-x}}{2}$$

$$f'(x) = \frac{e^x - e^{-x}}{2}$$

$f'(x) = 0$ only when $x = 0$.

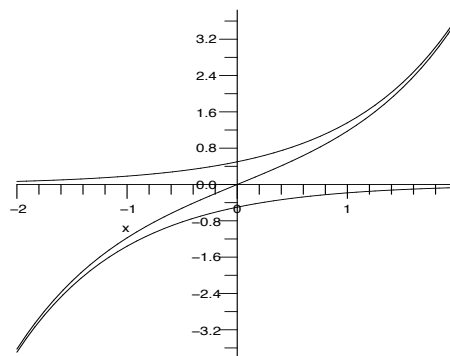
$$f''(x) = \frac{e^x + e^{-x}}{2}$$

$f''(x) > 0$ for all x , so $f(x)$ has no inflection points, but $x = 0$ is a minimum.

56. For $y = \sinh x$ we need to use $-\frac{1}{2}e^{-x}$ instead of $\frac{1}{2}e^{-x}$. To explain the enveloping behavior, note that:

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} \sinh x &= \lim_{x \rightarrow -\infty} \frac{e^x - e^{-x}}{2} \\
 &= \lim_{x \rightarrow -\infty} -\frac{e^{-x}}{2}
 \end{aligned}$$

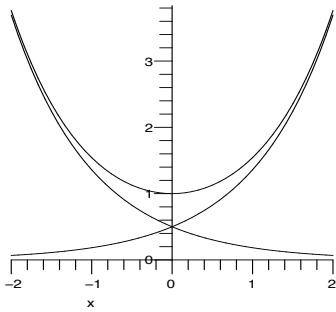
$$\begin{aligned}
 \lim_{x \rightarrow \infty} \sinh x &= \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{2} \\
 &= \lim_{x \rightarrow \infty} \frac{e^x}{2}
 \end{aligned}$$



To explain the enveloping behavior for $y = \cosh x$, note that:

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} \cosh x &= \lim_{x \rightarrow -\infty} \frac{e^x + e^{-x}}{2} \\
 &= \lim_{x \rightarrow -\infty} \frac{e^{-x}}{2}
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \cosh x &= \lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{2} \\
 &= \lim_{x \rightarrow \infty} \frac{e^x}{2}
 \end{aligned}$$



57. $f(x) = xe^{-bx}$

$$f(0) = 0$$

$$f(x) > 0 \text{ for } x > 0$$

$$\lim_{x \rightarrow \infty} xe^{-bx} = \lim_{x \rightarrow \infty} \frac{x}{e^{bx}} = \lim_{x \rightarrow \infty} \frac{1}{be^{bx}} = 0$$

(by L'Hôpital's rule)

$f'(x) = e^{-bx}(1 - bx)$, so there is a unique critical point at $x = 1/b$, which must be the maximum. The bigger b is, the closer the max is to the origin. For time since conception, $1/b$ represents the most common gestation time. For survival time, $1/b$ represents the most common life span.

58. From the graph we can count 15 maxima and 16 minima in the range $0 \leq x \leq 10$. Using a CAS to solve

$$f'(x) = -\sin(10x + 2\cos x)(10 - 2\sin x) = 0,$$

we find the following values of x at the extrema.

Minima	Maxima
0.11549	0.44806
0.80366	1.18055
1.57080	1.96104
2.33793	2.69353
3.02610	3.33776
3.63216	3.91326
4.18477	4.45009
4.71239	7.97469
5.24001	5.51152
5.79261	6.08702
6.39868	6.73125
7.08685	7.46374
7.85398	8.24422
8.62112	8.97672
9.30929	9.62094
9.91535	

59. $f(x) = A = \tan^{-1}\left(\frac{29.25}{x}\right) - \tan^{-1}\left(\frac{10.75}{x}\right)$

$$f'(x) = \left[\frac{x^2}{x^2 + (29.25)^2} \left(\frac{-29.25}{x^2} \right) \right] - \left[\frac{x^2}{x^2 + (10.75)^2} \left(\frac{-10.75}{x^2} \right) \right]$$

$$f'(x) = 0 \Rightarrow \frac{-29.25}{x^2 + (29.25)^2} + \frac{10.75}{x^2 + (10.75)^2} = 0$$

$$x = 17.73 \text{ ft.}$$

Substitute $x = 17.73$ in $f(x)$.

$$A = \tan^{-1}\left(\frac{29.25}{17.73}\right) - \tan^{-1}\left(\frac{10.75}{17.73}\right)$$

$$= 58.78 - 31.23$$

$$= 27.55^\circ$$

Now x is increased to $(x + 15)$.

$$f(x + 15) = A$$

$$= \tan^{-1}\left(\frac{29.25}{x + 15}\right) - \tan^{-1}\left(\frac{10.75}{x + 15}\right)$$

$$f'(x + 15)$$

$$= \left[\frac{1}{1 + \left(\frac{29.25}{x+15}\right)^2} \left(\frac{-29.25}{(x+15)^2} \right) \right] - \left[\frac{1}{1 + \left(\frac{10.75}{x+15}\right)^2} \left(\frac{-10.75}{(x+15)^2} \right) \right]$$

$$= \frac{-29.25}{(x+15)^2 + (29.25)^2} + \frac{10.75}{(x+15)^2 + (10.75)^2}$$

$$f'(x) = 0 \Rightarrow x = 2.73 \text{ ft.}$$

Substitute $x = 2.73$ in $f(x)$.

$$A = \tan^{-1}\left(\frac{29.25}{2.73}\right) - \tan^{-1}\left(\frac{10.75}{2.73}\right)$$

$$= 84.67 - 75.75$$

$$= 8.92^\circ$$

Therefore, A decreases by 18.63° .

60. $x(t) = \frac{2.5}{w}t - \frac{2.5}{4w^2} \sin 4wt$

Since $0 \leq t \leq 0.68$. Hence

$$0 \leq x(t) \leq \left(\frac{(2.5)(0.68)}{w} - \frac{2.5}{4w^2} \sin(4w(0.68)) \right)$$

$$\leq \left(\frac{1.7}{w} - \frac{2.5}{4w^2} \sin(2.72w) \right)$$

$$\leq \left(\frac{6.8w - 2.5\sin(2.72w)}{4w^2} \right)$$

Taking limit as $w \rightarrow 0$

$$\lim_{w \rightarrow 0} 0 \leq \lim_{w \rightarrow 0} x(t) \leq \lim_{w \rightarrow 0} \left(\frac{6.8w - 2.5\sin(2.72w)}{4w^2} \right)$$

(by L'Hôpital's rule)

$$\lim_{w \rightarrow 0} \left(\frac{6.8w - 2.5 \sin(2.72w)}{4w^2} \right)$$

$$= \lim_{w \rightarrow 0} \frac{6.8 - 6.8 \cos(2.72w)}{8w}$$

(by L'Hôpital's rule)

$$\lim_{w \rightarrow 0} \frac{18.496 \sin(2.72w)}{8} = 0$$

Hence

$$\lim_{w \rightarrow 0} 0 \leq \lim_{w \rightarrow 0} x(t) \leq 0.$$

As $w \rightarrow 0$, $x(t) \rightarrow 0$ that is the knuckleball will move in a straight direction.

$$A(x) = x \left(40 - \frac{2}{3}x \right)$$

$$A'(x) = 1 \left(40 - \frac{2}{3}x \right) + x \left(-\frac{2}{3} \right)$$

$$= 40 - \frac{4}{3}x = 0$$

$$40 = \frac{4}{3}x$$

$$x = 30$$

$$A'(x) > 0 \text{ for } 0 < x < 30$$

$$A'(x) < 0 \text{ for } x > 30.$$

$$\text{So } x = 30 \text{ is max, } y = 40 - \frac{2}{3} \cdot 30 = 20.$$

So the dimensions are $20' \times 30'$.

3.7 Optimization

1. $A = xy = 1800$

$$y = \frac{1800}{x}$$

$$P = 2x + y = 2x + \frac{1800}{x}$$

$$P' = 2 - \frac{1800}{x^2} = 0$$

$$2x^2 = 1800$$

$$x = 30$$

$$P'(x) > 0 \text{ for } x > 30$$

$$P'(x) < 0 \text{ for } 0 < x < 30$$

So $x = 30$ is min.

$$y = \frac{1800}{x} = \frac{1800}{30} = 60$$

So the dimensions are $30' \times 60'$ and the minimum perimeter is 120 ft.

2. If y is the length of fence opposite the river, and x is the length of the other two sides, then we have the constraint $2x + y = 96$. We wish to maximize

$$A = xy = x(96 - 2x).$$

$$A' = 96 - 4x = 0 \text{ when } x = 24.$$

$A'' = -4 < 0$ so this gives a maximum. Reasonable possible values of x range from 0 to 48, and the area is 0 at these extremes. The maximum area is $A = 1152$, and the dimensions are $x = 24$, $y = 48$.

3. $P = 2x + 3y = 120$

$$3y = 120 - 2x$$

$$y = 40 - \frac{2}{3}x$$

$$A = xy$$

4. Let x be the length of the sides facing each other and y be the length of the third side. We have the constraint that $xy = 800$, or $y = 800/x$. We also know that $x > 6$ and $y > 10$. The function we wish to minimize is the length of walls needed, or the side length minus the width of the doors.

$$L = (y - 10) + 2(x - 6) = 800/x + 2x - 22.$$

$$L' = -800/x^2 + 2 = 0 \text{ when } x = 20.$$

$L'' = 1600/x^3 > 0$ when $x = 20$ so this is a minimum. Possible values of x range from 6 to 80. $L(6) = 123.3$, $L(80) = 148$, and $L(20) = 58$. To minimize the length of wall, the facing sides should be 20 feet, and the third side should be 40 feet.

5. $A = xy$

$$P = 2x + 2y$$

$$2y = P - 2x$$

$$y = \frac{P}{2} - x$$

$$A(x) = x \left(\frac{P}{2} - x \right)$$

$$A'(x) = 1 \cdot \left(\frac{P}{2} - x \right) + x(-1)$$

$$= \frac{P}{2} - 2x = 0$$

$$P = 4x$$

$$x = \frac{P}{4}$$

$$A'(x) > 0 \text{ for } 0 < x < P/4$$

$$A'(x) < 0 \text{ for } x > P/4$$

So $x = P/4$ is max,

$$y = \frac{P}{2} - x = \frac{P}{2} - \frac{P}{4} = \frac{P}{4}$$

So the dimensions are $\frac{P}{4} \times \frac{P}{4}$. Thus we have a square.

6. We have a rectangle with sides x and y and area $A = xy$, and that we wish to minimize the perimeter,

$$P = 2x + 2y = 2x + 2 \cdot \frac{A}{x}.$$

$$P' = 2 - \frac{2A}{x^2} = 0 \text{ when } x = \sqrt{A}.$$

$P'' = 4A/x^3 > 0$ here, so this is a minimum. Possible values of x range from 0 to ∞ . As x approaches these values the perimeter grows without bound. For fixed area, the rectangle with minimum perimeter has dimensions $x = y = \sqrt{A}$, a square.

7. $V = l \cdot w \cdot h$

$$V(x) = (10 - 2x)(6 - 2x) \cdot x, \quad 0 \leq x \leq 3$$

$$V'(x) = -2(6 - 2x) \cdot x + (10 - 2x)(-2) \cdot x$$

$$+ (10 - 2x)(6 - 2x)$$

$$= 60 - 64x + 12x^2$$

$$= 4(3x^2 - 16x + 15)$$

$$= 0$$

$$x = \frac{16 \pm \sqrt{(-16)^2 - 4 \cdot 3 \cdot 15}}{6}$$

$$= \frac{8}{3} \pm \frac{\sqrt{19}}{3}$$

$$x = \frac{8}{3} + \frac{\sqrt{19}}{3} > 3.$$

$$V'(x) > 0 \text{ for } x < 8/3 - \sqrt{19}/3$$

$$V'(x) < 0 \text{ for } x > 8/3 - \sqrt{19}/3$$

$$\text{So } x = \frac{8}{3} - \frac{\sqrt{19}}{3} \text{ is a max.}$$

8. If we cut squares out of the corners of a 12" by 16" sheet and fold it into a box, the volume of the resulting box will be

$$V = x(12 - 2x)(16 - 2x)$$

$$= 4x^3 - 56x^2 + 192x,$$

where the value of x must be between 0 and 6.

$$V' = 12x^2 - 112x + 192 = 0$$

when $x = \frac{14 \pm 2\sqrt{13}}{3} \approx 7.07$ and 2.26. The critical value $x = \frac{14 + 2\sqrt{13}}{3}$ is outside of the reasonable range. The volume is 0 when x is 0 or 6. The First Derivative Test shows that

$x = \frac{14 - 2\sqrt{13}}{3}$ gives the maximum volume.

9. (a) $V = l \cdot w \cdot h$

The volume of the first box (without top) is

$$V_1 = V_1(x) = (6 - 2x)^2(x) = 4x(3 - x)^2$$

where $0 < x < 3$. The volume of the second box (without top and bottom) is

$$V_2 = V_2(x) = x^3.$$

Thus, we find the absolute maximum of

the continuous function

$$V = V(x) = V_1(x) + V_2(x) = 4x(3 - x)^2 + x^3$$

on the interval $0 < x < 3$.

$$V'(x) = 4(3 - x)^2 + 4x(2(3 - x)(-1)) + 3x^2$$

$$= 4(9 - 6x + x^2) - 8x(3 - x) + 3x^2$$

$$= 15x^2 - 48x + 36$$

$$= (x - 2)(15x - 18)$$

Now compare the value of the function at the critical points.

$$V(1.2) = 17.28$$

$$V(2) = 16$$

Therefore, the value $x = 1.2$ maximizes the sum of volumes of the boxes.

- (b) The volume of the first box (without top) is

$$V_1 = V_1(x) = (6 - 2x)(4 - 2x)(x)$$

$$= 4x(3 - x)(2 - x), \text{ where } 0 < x < 2$$

The volume of the second box (without top and bottom) is

$$V_2 = V_2(x) = x^3.$$

Thus, we find the absolute maximum of the continuous function

$$V = V(x) = V_1(x) + V_2(x)$$

$$= 4x(3 - x)(2 - x) + x^3, \text{ on the interval } 0 < x < 2.$$

We have,

$$V'(x) = 4(3 - x)(2 - x) + 4x(2 - x)(-1)$$

$$+ 4x(3 - x)(-1) + 3x^2$$

$$= 4(6 - 5x + x^2) - 4x(2 - x)$$

$$- 4x(3 - x) + 3x^2$$

$$= 15x^2 - 40x + 24$$

Now compare the value of the function at the critical points.

$$V(0.91169) = 9.0$$

$$V(1.75496) = 5.4$$

Therefore, $x = 0.91169$ maximizes the sum of volumes of the boxes.

10. The volume of the first box (without top) is

$$V_1 = V_1(x) = (6 - 2x)(d - 2x)(x),$$

$$\text{where } 0 < x < \min\{\frac{d}{2}, 3\}$$

The volume of the second box (without top and bottom) is

$$V_2 = V_2(x) = x^3.$$

Thus, we find the absolute maximum of the continuous function

$$V = V(x) = V_1(x) + V_2(x)$$

$$= x(6 - 2x)(d - 2x) + x^3, \text{ on the interval } 0 < x < \min\{\frac{d}{2}, 3\}$$

$$0 < x < \min\{\frac{d}{2}, 3\}$$

We have,

$$\begin{aligned}
 V'(x) &= (-2x)(d-2x) - 2x(6-2x) \\
 &\quad + (d-2x)(6-2x) + 3x^2 \\
 &= 15x^2 - 24x - 4dx + 6d \\
 &= 15x^2 - x(24+4d) + 6d = 0
 \end{aligned}$$

To get real values of x , $(24+4d)^2 - 4 \cdot 15 \cdot 6d \geq 0$ and this is true for all $d > 0$.

Therefore for each $d > 0$, we can find x such that V is maximum.

11. $d = \sqrt{(x-0)^2 + (y-1)^2}$

$$y = x^2$$

$$d = \sqrt{x^2 + (x^2-1)^2}$$

$$= (x^4 - x^2 + 1)^{1/2}$$

$$d'(x) = \frac{1}{2}(x^4 - x^2 + 1)^{-1/2}(4x^3 - 2x)$$

$$= \frac{2x(2x^2 - 1)}{2\sqrt{x^4 - x^2 + 1}} = 0$$

$$x = 0, \pm\sqrt{1/2};$$

$$f(0) = 1, f(\sqrt{1/2}) = 3/4, f(-\sqrt{1/2}) = 3/4;$$

Thus $x = \pm\sqrt{1/2}$ are min, and the points on $y = x^2$ closest to $(0, 1)$ are $(\sqrt{1/2}, 1/2)$ and $(-\sqrt{1/2}, 1/2)$.

12. Points on the curve $y = x^2$ can be written (x, x^2) . The distance from such a point to $(3, 4)$ is

$$D = \sqrt{(x-3)^2 + (x^2-4)^2}$$

$$= \sqrt{x^4 - 7x^2 - 6x + 25}.$$

We numerically approximate the solution of

$$D' = \frac{2x^3 - 7x - 3}{\sqrt{x^4 - 7x^2 - 6x + 25}} = 0 \text{ to be } x \approx$$

2.05655, and two negative solutions. The negative critical numbers clearly do not minimize the distance. The closest point is approximately $(2.05655, 4.22940)$.

13. $d = \sqrt{(x-0)^2 + (y-0)^2}$

$$y = \cos x$$

$$d = \sqrt{x^2 + \cos^2 x}$$

$$d'(x) = \frac{2x - 2\cos x \sin x}{2\sqrt{x^2 + \cos^2 x}} = 0$$

$$x = \cos x \sin x$$

$$x = 0$$

So $x = 0$ is min and the point on $y = \cos x$ closest to $(0, 0)$ is $(0, 1)$.

14. Points on the curve $y = \cos x$ can be written $(x, \cos x)$. The distance from such a point to $(1, 1)$ is

$$D = \sqrt{(x-1)^2 + (\cos x - 1)^2}$$

$$= \sqrt{x^2 - 2x + \cos^2 x - 2\cos x + 2}$$

We numerically approximate the solution of

$$D' = \frac{x - 1 - \cos x \sin x + \sin x}{\sqrt{x^2 - 2x + \cos^2 x - 2\cos x + 2}} = 0$$

to be $x \approx 0.789781$. The First or Second Derivative Test shows that this is a minimum distance. The closest point is approximately $(0.789781, 0.704001)$.

15. For $(0, 1)$, $(\sqrt{1/2}, 1/2)$ on $y = x^2$, we have

$$y' = 2x, y'(\sqrt{1/2}) = 2 \cdot \sqrt{1/2} = \sqrt{2} \text{ and}$$

$$m = \frac{\frac{1}{2} - 1}{-\sqrt{\frac{1}{2}} - 0} = \frac{1}{\sqrt{2}}.$$

For $(0, 1)$, $(-\sqrt{1/2}, 1/2)$ on $y = x^2$, we have

$$y'(-\sqrt{1/2}) = 2(-\sqrt{1/2}) = -\sqrt{2} \text{ and}$$

For $(3, 4)$, $(2.06, 4.2436)$ on $y = x^2$, we have $y'(2.06) = 2(2.06) = 4.12$ and

$$m = \frac{4.2436 - 4}{2.06 - 3} = -0.2591 \approx -\frac{1}{4.12}.$$

16. For $(3, 9)$, $(1, 8)$ on $y = 9 - x^2$, we have

$$y' = -2x, y'(1) = -2 \cdot 1 = -2 \text{ and}$$

$$m = \frac{8 - 9}{1 - 3} = \frac{1}{2}.$$

For $(5, 11)$, $(0.79728, 8.364)$ on $y = 9 - x^2$, we have $y'(0.79728) = -2(0.79728) = -1.59456$ and

$$m = \frac{8.364 - 11}{0.79728 - 5} = 0.6272 \approx \frac{1}{1.59456}.$$

17. Cost: $C = 2(2\pi r^2) + 2\pi rh$

Convert from fluid ounces to cubic inches:

$$12 \text{ fl oz} = 12 \text{ fl oz} \cdot 1.80469 \text{ in}^3/\text{fl oz} = 21.65628 \text{ in}^3$$

Volume: $V = \pi r^2 h$ so

$$h = \frac{V}{\pi r^2} = \frac{21.65628}{\pi r^2}$$

$$C = 4\pi r^2 + 2\pi r \left(\frac{21.65628}{\pi r^2} \right)$$

$$C(r) = 4\pi r^2 + 43.31256r^{-1}$$

$$C'(r) = 8\pi r - 43.31256r^{-2}$$

$$= \frac{8\pi r^3 - 43.31256}{r^2}$$

$$r = \sqrt[3]{\frac{43.31256}{8\pi}} = 1.1989''$$

when $C'(r) = 0$.

$$C'(r) < 0 \text{ on } (0, 1.1989)$$

$$C'(r) > 0 \text{ on } (1.1989, \infty)$$

Thus $r = 1.1989$ minimizes the cost and

$$h = \frac{21.65628}{\pi(1.1989)^2} = 4.7957''.$$

18. If the top and bottom of the cans are 2.23 times as thick as the sides, then the new cost func-

tion will be

$$C(r) = 2\pi \left(2.23r^2 + \frac{21.65628}{\pi r} \right).$$

$$\text{Then } C'(r) = 2\pi \left(4.46r - \frac{21.65628}{\pi r^2} \right) = 0$$

$$\text{when } r = \sqrt[3]{\frac{21.65628}{4.46\pi}} \approx 1.156.$$

The First Derivative Test shows this is a minimum, and we can verify that the cost increases without bound as $r \rightarrow 0$ and $r \rightarrow \infty$.

19. Let x be the distance from the connection point to the easternmost development. Then $0 \leq x \leq 5$.

$$f(x) = \sqrt{3^2 + (5-x)^2} + \sqrt{4^2 + x^2},$$

$$0 \leq x \leq 5$$

$$f'(x) = -(9 + (5-x)^2)^{-1/2}(5-x) + \frac{1}{2}(16 + x^2)^{-1/2}(2x)$$

$$= \frac{x-5}{\sqrt{9 + (5-x)^2}} + \frac{x}{\sqrt{16 + x^2}}$$

$$= 0$$

$$x = \frac{20}{7} \approx 2.857$$

$$f(0) = 4 + \sqrt{34} \approx 9.831$$

$$f\left(\frac{20}{7}\right) = \sqrt{74} \approx 8.602$$

$$f(5) = 3 + \sqrt{41} \approx 9.403$$

So $x = 20/7$ is minimum. The length of new line at this point is approximately 8.6 miles. Since $f(0) \approx 9.8$ and $f(5) \approx 9.4$, the water line should be $20/7$ miles west of the second development.

20. Say the pipeline intersects the shore at a distance x from the closest point on the shore to the oil rig. Then x will be between 0 and 8. The length of underwater pipe is then $W = \sqrt{x^2 + 25^2}$, and the length of pipe constructed on land will be $L = \sqrt{(8-x)^2 + 5^2}$. The total cost will be $C = 50W + 20L$.

We numerically solve

$$C' = \frac{50x}{\sqrt{625 + x^2}} + \frac{10(2x-16)}{\sqrt{x^2 - 16x + 89}} = 0$$

to find $x \approx 5.108987$. The first derivative test shows that this gives a minimum. The cost at this value is \$1391 thousand. The cost when $x = 0$ is \$1439 thousand, and the cost when $x = 8$ is \$1412 thousand, so $x = 5.108987$ gives the absolute minimum cost.

$$21. \quad (a) \quad C(x) = 5\sqrt{16 + x^2} + 2\sqrt{36 + (8-x)^2}$$

$$0 \leq x \leq 8$$

$$C(x) = 5\sqrt{16 + x^2} + 2\sqrt{100 - 16x + x^2}$$

$$C'(x) = 5 \left(\frac{1}{2} \right) (16 + x^2)^{-1/2} \cdot 2x$$

$$+ 2 \left(\frac{1}{2} \right) (100 - 16x + x^2)^{-1/2} (2x - 16)$$

$$= \frac{5x}{\sqrt{16 + x^2}} + \frac{2x - 16}{\sqrt{100 - 16x + x^2}}$$

$$= 0$$

$$x \approx 1.2529$$

$$C(0) = 40$$

$$C(1.2529) \approx 39.0162$$

$$C(8) \approx 56.7214$$

The highway should emerge from the marsh 1.2529 miles east of the bridge.

- (b) If we build a straight line to the interchange, we have $x = (3.2)$.

Since $C(3.2) - C(1.2529) \approx 1.963$, we save \$1.963 million.

22. (a) Say the road intersects the edge of the marsh at a distance x from the closest point on the edge to the bridge. Then x will be between 0 and 8. The length of road over marsh is now $M = \sqrt{x^2 + 4^2}$, and the length of road constructed on dry land will be $L = \sqrt{(8-x)^2 + 6^2}$. The total cost will be $C = 6M + 2L$.

We numerically solve

$$C' = \frac{6x}{\sqrt{16 + x^2}} + \frac{2x - 16}{\sqrt{x^2 - 16x + 100}} = 0$$

to find $x \approx 1.04345$. The first derivative test shows that this gives a minimum. The cost at this value is \$43.1763 million. The cost when we use the solution $x = 1.2529$ from exercise 19 is \$43.2078 million, so the increase is \$31,500.

$$(b) \quad C(x) = 5\sqrt{16 + x^2} + 3\sqrt{36 + (8-x)^2}$$

$$0 \leq x \leq 8$$

$$C'(x) = \frac{5x}{\sqrt{16 + x^2}} + \frac{3x - 24}{\sqrt{100 - 16x + x^2}}$$

Setting $C'(x) = 0$ yields

$$x \approx 1.8941$$

$$C(0) = 50$$

$$C(1.8941) \approx 47.8104$$

$$C(8) \approx 62.7214$$

The highway should emerge from the marsh 1.8941 miles east of the bridge. So if we must use the path from exercise 21, the extra cost is

$$\begin{aligned} C(1.2529) - C(1.8941) \\ = 48.0452 - 47.8104 = 0.2348 \\ \text{or about \$234.8 thousand.} \end{aligned}$$

- 23.** Let r denote the running speed, and s be the swimming speed and let $f(y)$ be the time to get to the ball. Since time = distance/speed, we have

$$f(y) = \frac{z-y}{r} + \frac{\sqrt{x^2+y^2}}{s}$$

$$\text{Solving } f'(y) = 0 \text{ for } y, \text{ we get } y = \frac{sx}{\sqrt{r^2-s^2}}.$$

Substitute $x = 4\text{m}$ and $z = 8\text{m}$. Therefore, we have

$$\begin{aligned} y &= \frac{(0.9)(4)}{\sqrt{(6.4)^2 - (0.9)^2}} \\ &\approx 0.56815. \end{aligned}$$

Therefore, Elvis should enter into the water at $y \approx 0.56815$.

From the equation $y = \frac{x}{\sqrt{r/s+1}\sqrt{r/s-1}}$, we get that the value of y is independent of $z > 1$.

- 24.** Let r denote the running speed, and s be the swimming speed and let $f(y)$ be the time to get to the ball. Since time = distance/speed, we have

$$f(y) = \frac{z-y}{r} + \frac{\sqrt{x^2+y^2}}{s}$$

$$\text{Solving } f'(y) = 0 \text{ for } y, \text{ we get } y = \frac{sx}{\sqrt{r^2-s^2}}.$$

Substitute $r = 6.4$ and $s = 0.9$.

$$y = \frac{0.9x}{\sqrt{(6.4)^2 - (0.9)^2}} \approx 0.144x$$

Therefore, for any x the optimal entry point is approximately $y = 0.144x$.

$$\begin{aligned} \mathbf{25.} \quad T(x) &= \frac{\sqrt{1+x^2}}{v_1} + \frac{\sqrt{1+(2-x)^2}}{v_2} \\ T'(x) &= \frac{1}{v_1} \cdot \frac{1}{2}(1+x^2)^{-1/2} \cdot 2x \\ &\quad + \frac{1}{v_2}(1+(2-x)^2)^{-1/2} \cdot (2-x)(-1) \\ &= \frac{x}{v_1\sqrt{1+x^2}} + \frac{x-2}{v_2\sqrt{1+(2-x)^2}} \end{aligned}$$

Note that

$$\begin{aligned} T'(x) &= \frac{1}{v_1} \cdot \frac{x}{\sqrt{1+x^2}} \\ &\quad - \frac{1}{v_2} \cdot \frac{(2-x)}{\sqrt{1+(2-x)^2}} \\ &= \frac{1}{v_1} \sin \theta_1 - \frac{1}{v_2} \sin \theta_2 \end{aligned}$$

When $T'(x) = 0$, we have

$$\begin{aligned} \frac{1}{v_1} \sin \theta_1 &= \frac{1}{v_2} \sin \theta_2 \\ \frac{\sin \theta_1}{\sin \theta_2} &= \frac{v_1}{v_2} \end{aligned}$$

- 26.** The distance light travels is

$$D = \sqrt{2^2+x^2} + \sqrt{1^2+(4-x)^2}.$$

We maximize this by solving

$$D' = \frac{x}{\sqrt{4+x^2}} + \frac{2x-8}{2\sqrt{x^2-8x+17}} = 0$$

to find $x = 8/3$. For this value of x , $\theta_1 = \theta_2 = \tan^{-1}(3/4)$. (Or simply note similar triangles.)

- 27.** $V(r) = cr^2(r_0 - r)$

$$V'(r) = 2cr(r_0 - r) + cr^2(-1)$$

$$= 2crr_0 - 3cr^2$$

$$= cr(2r_0 - 3r)$$

$$V'(r) = 0 \text{ when } r = 2r_0/3$$

$$V'(r) > 0 \text{ on } (0, 2r_0/3)$$

$$V'(r) < 0 \text{ on } (2r_0/3, \infty)$$

Thus $r = 2r_0/3$ maximizes the velocity.

$r = 2r_0/3 < r_0$, so the windpipe contracts.

- 28.** We wish to minimize

$$E(\theta) = \frac{\csc \theta}{r^4} + \frac{1 - \cot \theta}{R^4}.$$

We find

$$\begin{aligned} E'(\theta) &= -\frac{\csc \theta \cot \theta}{r^4} + \frac{1 + \cot^2 \theta}{R^4} \\ &= \frac{-\cos \theta R^4 + r^4}{r^4 R^4 \sin^2 \theta}. \end{aligned}$$

This is zero when $\cos \theta = r^4/R^4$, so $\theta = \cos^{-1}(r^4/R^4)$. The derivative changes from negative to positive here, so this gives a minimum as desired.

29. $p(x) = \frac{V^2 x}{(R+x)^2}$
 $p'(x) = \frac{V^2(R+x)^2 - V^2 x \cdot 2(R+x)}{(R+x)^4}$
 $= \frac{V^2 R^2 - V^2 x^2}{(R+x)^4}$
 $p'(x) = 0$ when $x = R$
 $p'(x) > 0$ on $(0, R)$
 $p'(x) < 0$ on (R, ∞)
 Thus $x = R$ maximizes the power absorbed.
30. If the meter registers 115 volts, then $v = 115\sqrt{2}$. The function $V(t) = v \sin(2\pi ft)$ has amplitude v , so the maximum value of the voltage is $115\sqrt{2}$.
31. $\pi r + 4r + 2w = 8 + \pi$
 $w = \frac{8 + \pi - r(\pi + 4)}{2}$
 $A(r) = \frac{\pi r^2}{2} + 2rw$
 $= \frac{\pi r^2}{2} + r(8 + \pi - r(\pi + 4))$
 $= r^2 \left(-4 - \frac{\pi}{2}\right) + r(8 + \pi)$
 $A'(r) = -2r \left(4 + \frac{\pi}{2}\right) + (8 + \pi) = 0$
 $A'(r) = 0$ when $r = 1$
 $A'(r) > 0$ on $(0, 1)$
 $A'(r) < 0$ on $(1, \infty)$
 Thus $r = 1$ maximizes the area so
 $w = \frac{8 + \pi - (\pi + 4)}{2} = 2$.
 The dimensions of the rectangle are 2×2 .
32. Let x be the distance from the end at which the wire is cut. Due to symmetry, we may consider $0 \leq x \leq 1$. We wish to minimize the area of the squares formed by the two pieces. The total area is
 $A(x) = \left(\frac{x}{4}\right)^2 + \left(\frac{2-x}{4}\right)^2$
 $= \frac{2x^2 - 4x + 4}{16}$.
 We compute
 $A'(x) = \frac{x}{4} - \frac{1}{4} = 0$ when $x = 1$.
 $A'' = \frac{1}{4} > 0$, so this is a minimum.
 We check $A(0) = 1/4$ and $A(1) = 1/8$ and see that cutting the wire in half minimizes the area of the two squares.
33. $l \times w = 92$, $w = 92/l$
 $A(l) = (l+4)(w+2)$
 $= (l+4)(92/l+2)$
 $= 92 + 368/l + 2l + 8$
 $= 100 + 368l^{-1} + 2l$

$$A'(l) = -368l^{-2} + 2$$

$$= \frac{2l^2 - 368}{l^2}$$

$$A'(l) = 0 \text{ when } l = \sqrt{184} = 2\sqrt{46}$$

$$A'(l) < 0 \text{ on } (0, 2\sqrt{46})$$

$$A'(l) > 0 \text{ on } (2\sqrt{46}, \infty)$$

So $l = 2\sqrt{46}$ minimizes the total area. When $l = 2\sqrt{46}$, $w = \frac{92}{2\sqrt{46}} = \sqrt{46}$.

For the minimum total area, the printed area has width $\sqrt{46}$ in. and length $2\sqrt{46}$ in., and the advertisement has overall width $\sqrt{46} + 2$ in. and overall length $2\sqrt{46} + 4$ in.

34. Let x and y be the width and height of the advertisement. Then $xy = 120$ and $y = 120/x$. We wish to maximize the printed area

$$A = (x-2)(y-3) = (x-2)\left(\frac{120}{x} - 3\right)$$

$$= 126 - 3x - \frac{240}{x}.$$

We find $A' = -3 + \frac{240}{x^2} = 0$ when $x = 4\sqrt{5}$.

The first Derivative Test shows that this is a maximum. The smallest x could be is 2, and this gives $A(2) = 0$. The largest x could be is 40, and this also gives $A(40) = 0$. Thus, we see that the dimensions which maximize the printed area are $x = 4\sqrt{5}$ and $y = 6\sqrt{5}$.

35. (a) Let L represent the length of the ladder. Then from the diagram, it follows that $L = a \sec \theta + b \csc \theta$.

Therefore,

$$\frac{dL}{d\theta} = a \sec \theta \tan \theta - b \csc \theta \cot \theta$$

$$0 = a \sec \theta \tan \theta - b \csc \theta \cot \theta$$

$$a \sec \theta \tan \theta = b \csc \theta \cot \theta$$

$$\frac{b}{a} = \frac{\sec \theta \tan \theta}{\csc \theta \cot \theta}$$

$$= \frac{1}{\cos \theta} \frac{\sin \theta}{\cos \theta} \frac{\sin \theta}{1} \frac{\sin \theta}{\cos \theta}$$

$$= \tan^3 \theta$$

Thus,

$$\tan \theta = \sqrt[3]{b/a}$$

$$\theta = \tan^{-1} \left(\sqrt[3]{b/a} \right)$$

$$= \tan^{-1} \left(\sqrt[3]{4/5} \right)$$

$$\approx 0.748 \text{ rad or } 42.87 \text{ degrees}$$

Thus, the length of the longest ladder that can fit around the corner is approximately

$$L = a \sec \theta + b \csc \theta$$

$$= 5 \sec(0.748) + 4 \csc(0.748)$$

$$\approx 12.7 \text{ ft}$$

- (b) From part (a), we have that $\theta = \tan^{-1}(\sqrt[3]{b/a})$ is the critical number limiting the length of the ladder. Thus $\tan \theta = b^{1/3}/a^{1/3}$. We can then draw a right triangle with θ as one angle and the length of the side opposite θ equal to $b^{1/3}$ and the length of the side adjacent to θ equal to $a^{1/3}$. By the Pythagorean Theorem, the hypotenuse of this triangle is $(a^{2/3} + b^{2/3})^{1/2}$. From this triangle, we find

$$\sin \theta = \frac{b^{1/3}}{(a^{2/3} + b^{2/3})^{1/2}} \text{ and}$$

$$\cos \theta = \frac{a^{1/3}}{(a^{2/3} + b^{2/3})^{1/2}}$$

so

$$\csc \theta = \frac{(a^{2/3} + b^{2/3})^{1/2}}{b^{1/3}} \text{ and}$$

$$\sec \theta = \frac{(a^{2/3} + b^{2/3})^{1/2}}{a^{1/3}}.$$

Thus

$$\begin{aligned} L &= a \sec \theta + b \csc \theta \\ &= a \cdot \frac{(a^{2/3} + b^{2/3})^{1/2}}{a^{1/3}} + b \cdot \frac{(a^{2/3} + b^{2/3})^{1/2}}{b^{1/3}} \\ &= a^{2/3}(a^{2/3} + b^{2/3})^{1/2} + b^{2/3}(a^{2/3} + b^{2/3})^{1/2} \\ &= (a^{2/3} + b^{2/3})(a^{2/3} + b^{2/3})^{1/2} \\ &= (a^{2/3} + b^{2/3})^{3/2}. \end{aligned}$$

- (c) Using the result of part (b) and solving for b :

$$\begin{aligned} L &= (a^{2/3} + b^{2/3})^{3/2} \\ L^{2/3} &= a^{2/3} + b^{2/3} \\ b^{2/3} &= L^{2/3} - a^{2/3} \\ b &= (L^{2/3} - a^{2/3})^{3/2} \\ &= (8^{2/3} - 5^{2/3})^{3/2} \\ &\approx 1.16 \text{ ft} \end{aligned}$$

- (d) This was already done in part (c) while solving for b :

$$b = (L^{2/3} - a^{2/3})^{3/2}.$$

36. (a) $R(x) = \frac{35x - x^2}{x^2 + 35}$

$$R'(x) = -35 \frac{x^2 + 2x - 35}{(x^2 + 35)^2}$$

$$= -35 \frac{(x-5)(x+7)}{(x^2 + 35)^2}$$

Hence the only critical number for $x \geq 0$ is $x = 5$ (that is, 5000 items). This must correspond to the absolute maximum, since $R(0) = 0$ and $R(x)$ is nega-

tive for large x . So maximum revenue is $R(5) = 2.5$ (that is, \$2500).

- (b) To maximize

$$R(x) = \frac{cx - x^2}{x^2 + c},$$

we compute

$$R'(x) = \frac{c(c - 2x - x^2)}{(x^2 + c)^2}.$$

This is zero when $x^2 + 2x - c = 0$, so

$$x = \frac{-2 \pm \sqrt{4 + 4c}}{2}.$$

The First Derivative Test shows that

$$x = \frac{-2 + \sqrt{4 + 4c}}{2}$$

is a maximum.

- 37.** (a) $Q'(t)$ is efficiency because it represents the number of additional items produced per unit time.

$$Q(t) = -t^3 + 12t^2 + 60t$$

$$Q'(t) = -3t^2 + 24t + 60$$

$$= 3(-t^2 + 8t + 20)$$

This is the quantity we want to maximize.

$Q''(t) = 3(-2t + 8)$ so the only critical number is $t = 4$ hours. This must be the maximum since the function $Q'(t)$ is a parabola opening down.

- (b) The worker's efficiency, Q' is maximized at the point of diminishing returns because at this point Q'' changes from positive to negative. The First Derivative Test applied to Q' shows that Q' has a local maximum at this point. (This assumes that the graph of Q changes from concave up to concave down at the inflection point. If this was reversed, the inflection point would not be a point of diminishing returns, and the efficiency would be minimized at such a point.)

- 38.** (a) Let $C(t)$ be the total cost of the tickets.

Then

$$C(t) = (\text{price per ticket})(\# \text{ of tickets})$$

$$C(t) = (40 - (t - 20))(t)$$

$$= (60 - t)(t) = 60t - t^2$$

for $20 < t < 50$. Then $C'(t) = 60 - 2t$, so $t = 30$ is the only critical number. This must correspond to the maximum since $C(t)$ is a parabola opening down.

- (b) If each additional ticket over 20 reduces the cost-per-ticket by c dollars, then the total cost for ordering x tickets (with x between 20 and 50) is

$$C(x) = (40 - c(x - 20))x$$

$$= (40 + 20c)x - cx^2.$$

This is a downward facing parabola with one maximum at $x = \frac{20+10c}{c}$. If we want the maximum cost to be at $x = 50$, we must choose c so that the peak of the parabola is at or to the right of 50. The value of $x = \frac{20+10c}{c}$ increases as c decreases, and equals 50 when $c = \frac{1}{2}$. Any discount of 50 cents or less will cause the maximum cost to occur when the group orders 50 tickets.

$$\begin{aligned}
 39. \quad R &= \frac{2v^2 \cos^2 \theta}{g} (\tan \theta - \tan \beta) \\
 R'(\theta) &= \frac{2v^2}{g} [2 \cos \theta (-\sin \theta) (\tan \theta - \tan \beta) \\
 &\quad + \cos^2 \theta \cdot \sec^2 \theta] \\
 &= \frac{2v^2}{g} \left[-2 \cos \theta \sin \theta \cdot \frac{\sin \theta}{\cos \theta} \right. \\
 &\quad \left. + 2 \cos \theta \sin \theta \tan \beta \right. \\
 &\quad \left. + \cos^2 \theta \cdot \frac{1}{\cos^2 \theta} \right] \\
 &= \frac{2v^2}{g} [-2 \sin^2 \theta + \sin(2\theta) \tan \beta + 1] \\
 &= \frac{2v^2}{g} [-2 \sin^2 \theta + \sin(2\theta) \tan \beta \\
 &\quad + (\sin^2 \theta + \cos^2 \theta)] \\
 &= \frac{2v^2}{g} [\sin(2\theta) \tan \beta \\
 &\quad + (\cos^2 \theta - \sin^2 \theta)] \\
 &= \frac{2v^2}{g} [\sin(2\theta) \tan \beta + \cos(2\theta)]
 \end{aligned}$$

$$R'(\theta) = 0 \text{ when}$$

$$\begin{aligned}
 \tan \beta &= \frac{-\cos(2\theta)}{\sin(2\theta)} = -\cot(2\theta) \\
 &= -\tan\left(\frac{\pi}{2} - 2\theta\right) \\
 &= \tan\left(2\theta - \frac{\pi}{2}\right)
 \end{aligned}$$

Hence $\beta = 2\theta - \pi/2$, so

$$\begin{aligned}
 \theta &= \frac{1}{2} \left(\beta + \frac{\pi}{2} \right) \\
 &= \frac{\beta}{2} + \frac{\pi}{4} = \frac{\beta^\circ}{2} + 45^\circ
 \end{aligned}$$

$$\text{i. } \beta = 10^\circ, \theta = 50^\circ$$

$$\text{ii. } \beta = 0^\circ, \theta = 45^\circ$$

$$\text{iii. } \beta = -10^\circ, \theta = 40^\circ$$

$$40. \quad A = 4xy$$

$$\frac{dA}{dx} = 4(xy' + y)$$

To determine $y' = \frac{dy}{dx}$, use the equation for the ellipse:

$$\begin{aligned}
 1 &= \frac{x^2}{a^2} + \frac{y^2}{b^2} \\
 0 &= \frac{2x}{a^2} + \frac{2yy'}{b^2} \\
 \frac{2yy'}{b^2} &= -\frac{2x}{a^2} \\
 y' &= -\frac{b^2}{a^2} \frac{x}{y}
 \end{aligned}$$

Substituting this expression for y' into the expression for $\frac{dA}{dx}$, we get

$$\begin{aligned}
 \frac{dA}{dx} &= xy' + y \\
 &= x \left(-\frac{b^2}{a^2} \frac{x}{y} \right) + y \\
 &= -\frac{b^2}{a^2} \frac{x^2}{y} + y
 \end{aligned}$$

The area is maximized when its derivative is zero:

$$0 = -\frac{b^2}{a^2} \frac{x^2}{y} + y$$

$$\frac{b^2}{a^2} \frac{x^2}{y} = y$$

$$\frac{x^2}{a^2} = \frac{y^2}{b^2}$$

Substituting the previous relationship into the equation for the ellipse, we get

$$\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{1}{2}$$

and therefore,

$$x = \frac{a}{\sqrt{2}} \quad \text{and} \quad y = \frac{b}{\sqrt{2}}$$

Thus, the maximum area is

$$A = 4 \frac{a}{\sqrt{2}} \frac{b}{\sqrt{2}} = 2ab$$

Since the area of the circumscribed rectangle is $4ab$, the required ratio is

$$2ab : \pi ab : 4ab = 1 : \frac{\pi}{2} : 2$$

41. Let V_c be the volume of the cylinder, h be the height of the cylinder and r the radius of the cylinder so that

$$V_c = h\pi r^2.$$

Let V_s be the volume of the sphere and R be the radius of the sphere so that

$$V_s = \frac{4}{3}\pi R^3.$$

Draw the sphere on coordinate axes with center $(0, 0)$ and inscribe the cylinder. Then draw a right triangle as follows: draw a straight line from the origin to the side of the cylinder (this line has length r , the radius of the cylinder); draw a line from this point to the point where the cylinder meets the sphere (this line has length $h/2$, half the height of the cylinder); connect this point with the origin to create the

hypotenuse of the triangle (this line has length R , the radius of the sphere). Thus we see that

$$R^2 = r^2 + \left(\frac{h}{2}\right)^2.$$

Now we have

$$V_s = \frac{4}{3}\pi \left(r^2 + \frac{h^2}{4}\right)^{3/2}.$$

Taking the derivative of both sides with respect to h gives

$$0 = 2\pi \left(r^2 + \frac{h^2}{4}\right)^{1/2} \left(2rr' + \frac{h}{2}\right).$$

Solving for r' , we find $r' = -h/4r$. Taking the derivative with respect to h of both sides of the formula for the volume for the cylinder yields

$$\frac{dV_c}{dh} = \pi r^2 + 2h\pi r r'.$$

Plugging in the formula we found for r' gives

$$\begin{aligned} \frac{dV_c}{dh} &= \pi r^2 + 2h\pi r \left(-\frac{h}{4r}\right) \\ &= \pi r^2 - \frac{h^2\pi}{2}. \end{aligned}$$

To maximize the volume of the cylinder, we set this equal to 0 and find that the volume of the cylinder is maximized when $h^2 = 2r^2$. In this case, the formula relating R , r and h above gives

$$h = \sqrt{\frac{4}{3}R^2} = \frac{2R}{\sqrt{3}}.$$

The maximum volume of the cylinder is then

$$\begin{aligned} V_c &= h\pi r^2 \\ &= \frac{\pi h^3}{2} = \frac{\pi \left(\frac{2R}{\sqrt{3}}\right)^3}{2} \\ &= \frac{1}{\sqrt{3}} \left(\frac{4}{3}\pi R^3\right) \\ &= \frac{1}{\sqrt{3}} V_s. \end{aligned}$$

42. Suppose that $a = b$ in the isoscles triangle, so that

$$A^2 = s(s-a)(s-b)(s-c) = s(s-a)^2(s-c)$$

Since $s = \frac{1}{2}(a+b+c)$, it follows that

$s = \frac{1}{2}(2a+c) = a + \frac{c}{2}$, so that $s-a = \frac{c}{2}$. Thus,

$$\begin{aligned} A^2 &= s \left(\frac{c^2}{4}\right) (s-c) \\ &= \frac{s}{4} (sc^2 - c^3) \end{aligned}$$

Since s is a constant (it's half of the perimeter), we can now differentiate to get

$$\begin{aligned} 2A \frac{dA}{dc} &= \frac{s}{4} (2sc - 3c^2) \\ 0 &= c(2s - 3c) \end{aligned}$$

Thus, the area is maximized when $2s - 3c = 0$, which means $c = \frac{2}{3}s$. Solving for a , we get

$$a = s - \frac{c}{2} = s - \frac{s}{3} = \frac{2}{3}s.$$

Thus, the area is maximized when $a = b = c$; in other words the area is maximized when the triangle is equilateral.

The maximum area is

$$\begin{aligned} A &= \sqrt{s(s-c)^3} = \sqrt{s \left(\frac{s}{3}\right)^3} \\ &= \frac{s^2}{9} \sqrt{3} = \frac{p^2}{36} \sqrt{3} \end{aligned}$$

3.8 Related Rates

1. $V(t) = (\text{depth})(\text{area}) = \frac{\pi}{48} [r(t)]^2$
(units in cubic feet per min)
 $V'(t) = \frac{\pi}{48} 2r(t)r'(t) = \frac{\pi}{24} r(t)r'(t)$
We are given $V'(t) = \frac{120}{7.5} = 16$.
Hence $16 = \frac{\pi}{24} r(t)r'(t)$ so
 $r'(t) = \frac{(16)(24)}{\pi r(t)}.$

(a) When $r = 100$,

$$r'(t) = \frac{(16)(24)}{100\pi} = \frac{96}{25\pi} \approx 1.2223 \text{ ft/min},$$

(b) When $r = 200$,

$$r'(t) = \frac{(16)(24)}{200\pi} = \frac{48}{25\pi} \approx 0.61115 \text{ ft/min}$$

2. $V = (\text{depth})(\text{area}). \quad \frac{1''}{8} = \frac{1'}{96}$, so
 $V(t) = \frac{1}{96}\pi r(t)^2.$

Differentiating we find $\frac{dV}{dt} = \frac{2\pi}{96} r(t) \frac{dr}{dt}.$

Using $1 \text{ ft}^3 = 7.5 \text{ gal}$, the rate of change of volume is $\frac{90}{7.5} = 12$. So when $r(t) = 100$,

$$12 = \frac{2\pi}{96} 100 \frac{dr}{dt}, \text{ and}$$

$$\frac{dr}{dt} = \frac{144}{25\pi} \text{ feet per minute.}$$

3. (a) From #1,
 $V'(t) = \frac{\pi}{48} 2r(t)r'(t) = \frac{\pi}{24} r(t)r'(t),$
so $\frac{g}{7.5} = \frac{\pi}{24} (100)(.6) = 2.5\pi,$
so $g = (7.5)(2.5)\pi$
 $= 18.75\pi \approx 58.905 \text{ gal/min.}$

(b) If the thickness is doubled, then the rate of change of the radius is halved.

4. (a) t = hours elapsed since injury
 r = radius of the infected area
 A = area of the infection
 $A = \pi r^2$
 $A'(t) = 2\pi r(t) \cdot r'(t)$
 When $r = 3$ mm, $r' = 1$ mm/hr,
 $A' = 2\pi(3)(1) = 6\pi$ mm²/hr
- (b) We have $A'(t) = 2\pi r r'(t)$, and $r'(t) = 1$ mm/hr, so when the radius is 6 mm we have
 $A'(t) = 2\pi \cdot 6 \cdot 1 = 12\pi$ mm²/hr.
 This rate is larger when the radius is larger because the area is changing by the same amount along the entire circumference of the circle. When the radius is larger, there is more circumference, so the same change in radius causes a larger change in area.

5. $V(t) = \frac{4}{3}\pi[r(t)]^3$
 $V'(t) = 4\pi[r(t)]^2 r'(t) = Ar'(t)$
 If $V'(t) = kA(t)$, then
 $r'(t) = \frac{V'(t)}{A(t)} = \frac{kA(t)}{A(t)} = k.$

6. We have $A'(t) = 2\pi r r'(t)$, and $r'(t) = 5$ ft/min, so when the radius is 200 ft we have
 $A'(t) = 2\pi \cdot 200 \cdot 5 = 2,000\pi$ ft²/min.

7. (a) $10^2 = x^2 + y^2$
 $0 = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$
 $\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}$
 $= -\frac{6}{8}(3)$
 $= -2.25$ ft/s

- (b) We have
 $\cos \theta(t) = \frac{x(t)}{10}.$
 Differentiating with respect to t gives
 $-\sin \theta(t) \cdot \theta'(t) = \frac{x'(t)}{10}.$
 When the bottom is 6 feet from the wall, the top of the ladder is 8 feet from the floor and this distance is the opposite side of the triangle from θ . Thus, at this point, $\sin \theta = 8/10$. So
 $-\frac{8}{10} \theta'(t) = \frac{3}{10}$
 $\theta'(t) = -\frac{3}{8}$ rad/s.

8. (a) $\theta = \pi - \tan^{-1} \left(\frac{40}{60-x} \right) - \tan^{-1} \left(\frac{20}{x} \right)$

$$\frac{d\theta}{dx} = -\frac{40 \left(\frac{1}{60-x} \right)^2}{1 + \left(\frac{40}{60-x} \right)^2} + \frac{\frac{20}{x^2}}{1 + \left(\frac{20}{x} \right)^2}$$

When $x = 30$, this becomes

$$\begin{aligned} \frac{d\theta}{dx} &= -\frac{40 \left(\frac{1}{30} \right)^2}{1 + \left(\frac{40}{30} \right)^2} + \frac{\frac{20}{900}}{1 + \left(\frac{20}{30} \right)^2} \\ &= -\frac{1}{1625} \text{ rad/ft} \end{aligned}$$

$$\begin{aligned} \frac{d\theta}{dt} &= \frac{d\theta}{dx} \frac{dx}{dt} \\ &= \left(-\frac{1}{1625} \right) (4) \\ &\approx -0.00246 \text{ rad/s} \end{aligned}$$

- (b) As in the solution to #8(a), let x be the distance from the 20' building to the person. To find the maximum θ , we set $\frac{d\theta}{dx} = 0$ and solve for x :

$$\begin{aligned} 0 &= -\frac{40 \left(\frac{1}{60-x} \right)^2}{1 + \left(\frac{40}{60-x} \right)^2} + \frac{\frac{20}{x^2}}{1 + \left(\frac{20}{x} \right)^2} \\ \frac{20}{x^2 + 40} &= \frac{40}{(60-x)^2 + 1} \\ 0 &= 20x^2 + 2400x - 56000 \\ 0 &= x^2 + 120x - 2800 \end{aligned}$$

Using the quadratic formula, we find two roots:

$$x = -60 \pm 80$$

We discard the x value obtained from the minus sign as it is negative and does not make sense for our problem. The other value is $x = 20$. We find $\theta'(10) > 0$ and $\theta'(30) < 0$, so $x = 20$ must be a maximum as desired.

9. (a) We know $[x(t)]^2 + 4^2 = [s(t)]^2$. Hence
 $2x(t)x'(t) = 2s(t)s'(t)$, so
 $x'(t) = \frac{s(t)s'(t)}{x(t)} = \frac{-240s(t)}{x(t)}$. When $x = 40$, $s = \sqrt{40^2 + 4^2} = 4\sqrt{101}$, so at that moment
 $x'(t) = \frac{(-240)(4\sqrt{101})}{40} = -24\sqrt{101}$.
 So the speed is $24\sqrt{101} \approx 241.2$ mph.

- (b) From #9(a), we have
 $x'(t) = \frac{s(t)s'(t)}{x(t)} = \frac{-240s(t)}{x(t)}$.
 This time the height is 6 miles, so $s = \sqrt{40^2 + 6^2} = 2\sqrt{409}$, so at that moment
 $x'(t) = \frac{(-240)(2\sqrt{409})}{40} = -12\sqrt{409}$.

So the speed is $12\sqrt{409} \approx 242.7$ mph. The difference in height does not make a large difference in the speed of the plane.

10. (a) If the police car is not moving, then $x'(t) = 0$, but all the other data are unchanged. So

$$\begin{aligned} d'(t) &= \frac{x(t)x'(t) + y(t)y'(t)}{\sqrt{[x(t)]^2 + [y(t)]^2}} \\ &= \frac{-(1/2)(50)}{\sqrt{1/4 + 1/16}} \\ &= \frac{-100}{\sqrt{5}} \approx -44.721. \end{aligned}$$

This is more accurate.

- (b) If the police car is at the intersection, then the rate of change the police car measures is

$$\frac{0 \cdot (-40) + \frac{1}{2} \cdot (-50)}{\sqrt{\frac{1}{4} + 0}} = -50,$$

the true speed of the car.

$$\begin{aligned} 11. \quad d'(t) &= \frac{x(t)x'(t) + y(t)y'(t)}{\sqrt{[x(t)]^2 + [y(t)]^2}} \\ &= \frac{-(1/2)(\sqrt{2} - 1)(50) - (1/2)(50)}{\sqrt{1/4 + 1/4}} \\ &= -50. \end{aligned}$$

12. The radar gun will read less than the actual speed if the police car is not at the intersection, and is travelling away from the intersection.

13. From the table, we see that the recent trend is for advertising to increase by \$2000 per year. A good estimate is then $x'(2) \approx 2$ (in units of thousands). Starting with the sales equation $s(t) = 60 - 40e^{-0.05x(t)}$, we use the chain rule to obtain
- $$\begin{aligned} s'(t) &= -40e^{-0.05x(t)}[-0.05x'(t)] \\ &= 2x'(t)e^{-0.05x(t)}. \end{aligned}$$

Using our estimate that $x'(2) \approx 2$ and since $x(2) = 20$, we get $s'(2) \approx 2(2)e^{-1} \approx 1.471$. Thus, sales are increasing at the rate of approximately \$1471 per year.

14. The year 2 rate of change for the average cost is given by $\overline{C}'(t) = \frac{-94}{x^2} \cdot x'(t)$.

From the table we see that in year two $x = 9.4$ and $x' = 0.6$, so

$$\overline{C}'(t) = \frac{-94}{9.4^2} \cdot 0.6 = -0.6383 \text{ per year.}$$

$$15. \quad \overline{C}(x) = 10 + \frac{100}{x}$$

$$\overline{C}'(x(t)) = \frac{-100}{x^2} \cdot x'(t)$$

$\overline{C}'(10) = -1(2) = -2$ dollars per item, so average cost is decreasing at the rate of \$2 per year.

16. The rate of change of sales is

$$s' = 0.8e^{-0.04x}x'(t).$$

We are given $x = 40$ and $x'(t) = 1.5$, so

$$s' = 0.8e^{-0.04 \cdot 40} \cdot 1.5 = 0.242 \text{ thousand dollars per year.}$$

17. (a) We have $\tan \theta = \frac{x}{2}$, so

$$\frac{d}{dt}(\tan \theta) = \frac{d}{dt}\left(\frac{x}{2}\right)$$

$$\sec^2 \theta \cdot \theta' = \frac{1}{2}x'$$

$$\theta' = \frac{1}{2\sec^2 \theta} \cdot x' = \frac{x' \cos^2 \theta}{2}$$

at $x = 0$, we have $\tan \theta = \frac{x}{2} = \frac{0}{2}$ so $\theta = 0$

and we have $x' = -130$ ft/s so

$$\theta' = \frac{(-130) \cdot \cos^2 0}{2} = -65 \text{ rad/s.}$$

- (b) $x = 2 \tan \theta$, so $\frac{dx}{dt} = 2 \sec^2 \theta \frac{d\theta}{dt}$. $\theta = 0$ (and $\sec \theta = 1$) as the ball crosses home plate, so $\frac{d\theta}{dt} = \frac{1}{2} \frac{dx}{dt}$. For this to be less than 3 radians per sec, the pitch must be less than 6 ft/sec.

18. (a) t = number of seconds since launch
 x = height of rocket in miles after t seconds

θ = camera angle in radians after t seconds

$$\tan \theta = \frac{x}{2}$$

$$\frac{d}{dx}(\tan \theta) = \frac{d}{dx}\left(\frac{x}{2}\right)$$

$$\sec^2 \theta \cdot \theta' = \frac{1}{2}x'$$

$$\theta' = \frac{\cos^2 \theta \cdot x'}{2}$$

When $x = 3$, $\tan \theta = 3/2$, so $\cos \theta = 2/\sqrt{13}$.

$$\theta' = \frac{\left(\frac{2}{\sqrt{13}}\right)^2 (2)}{2} \approx .03 \text{ rad/s}$$

- (b) If the height of the rocket is x , then $x = 2 \tan \theta$, and $\frac{dx}{dt} = 2 \sec^2 \theta \frac{d\theta}{dt}$.

When $x = 1$ and $\frac{dx}{dt} = 0.2$, we have

$0.2 = 2 \cdot \frac{5}{4} \cdot \frac{d\theta}{dt}$ and $\frac{d\theta}{dt} = 0.08$ radians per sec. This is larger because the angle

changes more quickly when the rocket is close to the ground. When the rocket is far away, large changes in height result in small changes in the angle, since the angle is approaching a limit of $\pi/2$.

19. (a) Let θ be the angle between the end of the shadow and the top of the lamppost.

Then $\tan \theta = \frac{6}{s}$ and $\tan \theta = \frac{18}{s+x}$, so

$$\begin{aligned}\frac{x+s}{18} &= \frac{s}{6} \\ \frac{d}{dx} \left(\frac{x+s}{18} \right) &= \frac{d}{dx} \left(\frac{s}{6} \right) \\ \frac{x'+s'}{18} &= \frac{s'}{6} \\ x'+s' &= 3s' \\ s' &= \frac{x'}{2}\end{aligned}$$

Since $x' = 2$, $s' = 2/2 = 1$ ft/s.

- (b) From #19(a), $s' = x'/2$. Since $x' = -3$, $s' = -3/2$ ft/s.

20. (a) $P(t) \cdot V'(t) + P'(t)V(t) = 0$

$$\frac{P'(t)}{V'(t)} = -\frac{P(t)}{V(t)} = -\frac{c}{V(t)^2}$$

- (b) Solving Boyle's Law for P gives $P = \frac{c}{V}$.

Then differentiating gives

$$P'(V) = \frac{-c}{V^2}, \text{ the same as } P'(t)/V'(t).$$

21. Let $r(t)$ be the length of the rope at time t and $x(t)$ be the distance (along the water) between the boat and the dock.

$$r(t)^2 = 36 + x(t)^2$$

$$2r(t)r'(t) = 2x(t)x'(t)$$

$$\begin{aligned}x'(t) &= \frac{r(t)r'(t)}{x(t)} = \frac{-2r(t)}{x(t)} \\ &= \frac{-2\sqrt{36+x^2}}{x}\end{aligned}$$

When $x = 20$, $x' = -2.088$; when $x = 10$, $x' = -2.332$.

22. The volume of a cone is $V = \frac{1}{3}\pi r^2 h$, and we know that this cone has $r = \frac{h}{2}$, so we have

$V = \frac{\pi}{12}h^3$. Differentiating gives

$$\frac{dV}{dt} = \frac{\pi h^2}{4} \cdot \frac{dh}{dt}.$$

We are given that $\frac{dV}{dt} = 5 \text{ m}^3/\text{s}$, so when $h = 2$ meters, we have

$$\begin{aligned}5 &= \frac{\pi 2^2}{4} \cdot \frac{dh}{dt}, \\ \text{so } \frac{dh}{dt} &= \frac{5}{\pi} \text{ meters per second.}\end{aligned}$$

$$\begin{aligned}23. f(t) &= \frac{1}{2L(t)} \sqrt{\frac{T}{\rho}} = \frac{110}{L(t)}. \\ f'(t) &= \frac{-110}{L(t)^2} L'(t).\end{aligned}$$

When $L = 1/2$, $f(t) = 220$ cycles per second.

If $L' = -4$ at this time, then $f'(t) = 1760$ cycles per second per second. It will only take $1/8$ second at this rate for the frequency to go from 220 to 440, and raise the pitch one octave.

$$\begin{aligned}24. V &= \frac{4}{3}\pi r^3 \\ \frac{dV}{dt} &= \frac{4}{3}\pi(3r^2) \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt} \\ 1 &= 4\pi r^2 \frac{dr}{dt} \\ \frac{dr}{dt} &= \frac{1}{4\pi r^2}\end{aligned}$$

$$\text{When } r = .01, \frac{dr}{dt} = \frac{2500}{\pi}$$

$$\text{When } r = .1, \frac{dr}{dt} = \frac{25}{\pi}.$$

At first, the radius expands rapidly; later it expands more slowly.

25. (a) Let R represent the radius of the circular surface of the water in the tank.

$$V(R) = \pi \left[60^2(60^2 - R^2)^{1/2} - \right.$$

$$\left. \frac{1}{3}(60^2 - R^2)^{3/2} + \frac{2}{3}60^3 \right]$$

$$\frac{dV}{dR} = \pi \left[60^2 \left(\frac{1}{2} \right) (60^2 - R^2)^{-1/2} (-2R) - \right.$$

$$\left. \frac{1}{3} \left(\frac{3}{2} \right) (60^2 - R^2)^{1/2} (-2R) \right]$$

$$= \pi \left[\frac{-60^2 R}{\sqrt{60^2 - R^2}} + R \sqrt{60^2 - R^2} \right]$$

$$= \pi R \left[\frac{-60^2 + 60^2 - R^2}{\sqrt{60^2 - R^2}} \right]$$

$$= \frac{-\pi R^3}{\sqrt{60^2 - R^2}}$$

$$\frac{dR}{dt} = \frac{dV/dt}{dV/dR}$$

$$= \frac{10}{dV/dR}$$

$$= \frac{-10\sqrt{60^2 - R^2}}{\pi R^3}$$

- i. Substituting $R = 60$ into the previous equation, we get $\frac{dR}{dt} = 0$.

- ii. We need to determine the value of R when the tank is three-quarters full. The volume of the spherical tank is $\frac{4}{3}\pi 60^3$, so when the tank is three-quarters full, $V(R) = \pi 60^3$. Substituting this value into the formula for $V(R)$ and solving for R (using a CAS, for example) we get $R \approx 56.265$. Substituting this value into the formula for dR/dt , we get

$$\begin{aligned}\frac{dR}{dt} &= \frac{-10\sqrt{60^2 - R^2}}{\pi R^3} \\ &\approx \frac{-10\sqrt{60^2 - 56.265^2}}{\pi 56.265^3} \\ &\approx -0.00037 \text{ ft/s}\end{aligned}$$

- (b) Assuming the tank is at least half full, we can represent the height of the water in the tank by $h(t) = \sqrt{60^2 - R^2} + 60$. Differentiating gives

$$\begin{aligned}h'(t) &= \frac{1}{2}(60^2 - R^2)^{-1/2}(-2R)R'(t) \\ &= -(60^2 - R^2)^{-1/2}R \cdot R'(t) \\ &= \frac{-(60^2 - R^2)^{-1/2}R \cdot (-10\sqrt{60^2 - R^2})}{\pi R^3}.\end{aligned}$$

Here we have used the expression for $R'(t)$ found in exercise 35.

- Substituting $R = 60$ into the previous equation, we get $h'(t) = 0$.
- Substituting $R \approx 56.265$ into the formula for $h'(t)$ gives $h'(t) \approx 0.001006$ ft/s.

26. (a) The volume of the conical pile is $V = \frac{1}{3}\pi r^2 h$. Since $h = 2r$, we can write the volume as

$$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{1}{12}\pi h^3 \text{ Thus,}$$

$$\frac{dV}{dt} = \frac{\pi h^2}{4} \cdot \frac{dh}{dt}$$

$$20 = \frac{\pi 6^2}{4} \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{20}{9\pi}$$

$$\frac{dr}{dt} = \frac{10}{9\pi}$$

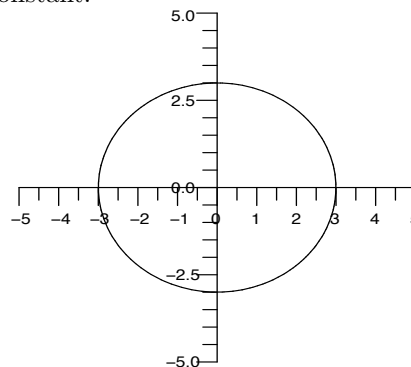
- (b) In this case, we have $r = h$ so

$$V = \frac{1}{3}\pi h^2 h = \frac{\pi h^3}{3}$$

Thus $V'(t) = \pi h^2 h'(t)$ so when the height is 6 feet,

$$h'(t) = r'(t) = \frac{20}{36\pi} = \frac{5}{9\pi}.$$

27. (a) Let an object move around the circle $x^2(t) + y^2(t) = r^2$. Both x and y coordinates are the functions of t and r is a constant.



Therefore, on differentiating w.r.t. t , we get

$$2x(t)x'(t) + 2y(t)y'(t) = 0$$

$$x(t)x'(t) + y(t)y'(t) = 0$$

Therefore, $y'(t) = \frac{-x(t)x'(t)}{y(t)}$ and

$$x'(t) = -\frac{y(t)y'(t)}{x(t)}$$

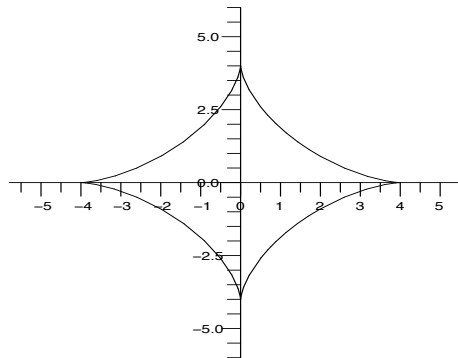
Thus, if $x(t) = 0$, then $y'(t) = 0$ and if $y(t) = 0$, then $x'(t) = 0$

From the graph it can be observed that:

At $x(t) = 0$ the tangent is horizontal which means $\frac{y'(t)}{x'(t)} = 0 \Rightarrow y'(t) = 0$ and

At $y(t) = 0$ the tangent is vertical which means $\frac{x'(t)}{y'(t)} = 0 \Rightarrow x'(t) = 0$

- (b) An object move around the asteroid $x^{2/3}(t) + y^{2/3}(t) = 1$. Both x and y coordinates are the functions of time.



Therefore, on differentiating w.r.t. t , we get

$$\frac{2}{3}x^{-1/3}(t)x'(t) + \frac{2}{3}y^{-1/3}(t)y'(t) = 0$$

$$x(t)[y'(t)]^3 + y(t)[x'(t)]^3 = 0$$

$$y'(t) = \left(\frac{-y(t)}{x(t)} \right)^{1/3} x'(t) \text{ and}$$

$$x'(t) = \left(\frac{-x(t)}{y(t)} \right)^{1/3} y'(t)$$

thus, if $x(t) = 0$, then $y(t) = 1$, $x'(t) = 0$

and if $y(t) = 0$, then $x(t) = 1$, $y'(t) = 0$

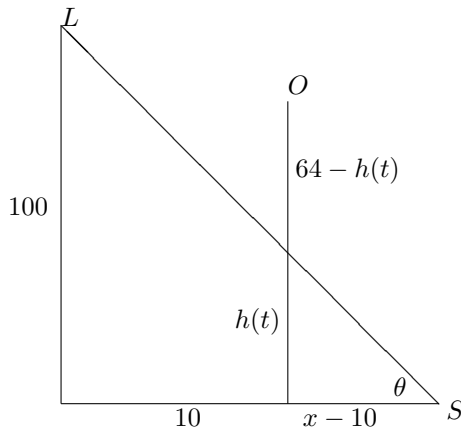
From the graph it can be observed that,
at $x(t) = 0$ the tangent is vertical which

means $\frac{x'(t)}{y'(t)} = 0 \Rightarrow x'(t) = 0$ and

at $y(t) = 0$ the tangent is horizontal which

means $\frac{y'(t)}{x'(t)} = 0 \Rightarrow y'(t) = 0$

28. (a) Let θ be the angle of the light at the shadow as shown in figure below:



Then,

$$\tan \theta = \frac{h(t)}{x(t) - 10} = \frac{100}{x(t)}$$

$$\Rightarrow x(t) = \frac{1000}{100 - h(t)}$$

$$x'(t) = \frac{1000h'(t)}{[100 - h(t)]^2}$$

$$= -\frac{8000\sqrt{(64 - h(t))}}{[100 - h(t)]^2}$$

At $h = 0$,

$$x'(t) = -\frac{8000\sqrt{(64 - 0)}}{[100 - 0]^2}$$

$$= -\frac{64000}{10000}$$

$$= -6.4$$

(b) $|x'(t)| = 8000 \frac{\sqrt{(64 - h(t))}}{[100 - h(t)]^2}$

At maxima or minima of $|x'(t)|$,

$$\frac{d}{dx} |x'(t)| = 0$$

$$\Rightarrow \frac{-\frac{1}{2}(64 - h(t))^{-1/2}h'(t)}{[100 - h(t)]^2}$$

$$+ \frac{\sqrt{(64 - h(t))}}{[100 - h(t)]^3} \cdot 2h'(t) = 0$$

$$\Rightarrow -h'(t)\{(100 - h(t)) - 4(64 - h(t))\} = 0$$

$$\Rightarrow h'(t) = 0 \text{ or } 100 - 256 - h(t) + 4h(t) = 0$$

$$\Rightarrow h'(t) = 0 \text{ or } h(t) = 52$$

$$\text{At } h'(t) = 0 : |x'(t)| = 0$$

$$\text{At } h(t) = 52 :$$

$$|x'(t)| = 8000 \frac{\sqrt{64 - h(t)}}{[100 - h(t)]^2}$$

$$= 8000 \frac{\sqrt{(64 - 52)}}{(100 - 52)^2} = 12.02$$

Therefore, $h(t) = 52$ is the height in which $|x'(t)|$ is maximum.

29. (a) $d(t) = \sqrt{(x(t) - 8)^2 + (0 - 4)^2}$

$$\text{therefore } d'(t) = \frac{(x(t) - 8)x'(t)}{\sqrt{(x(t) - 8)^2 + 16}}$$

$$\text{Now } d'(t) = 0.9 \text{ and } x'(t) = 6.4$$

$$\text{gives } x(t) = 8.5681$$

$$\text{hence } t = \frac{x(t) - x(0)}{x'(t)} = \frac{8.5681}{6.4} = 1.3388$$

- (b) Thus the location at this moment is $(8.5681, 0)$

30. $\theta = \tan^{-1} \left(\frac{2s}{vT} \right)$

$$\frac{d\theta}{dt} = \frac{\left(-\frac{2s}{T}\right)v^{-2}v'(t)}{1 + \left(\frac{2s}{vT}\right)^2}$$

$$= \frac{-2sv'(t)}{Tv^2 \left[1 + \frac{4s^2}{v^2T^2}\right]}$$

$$= \frac{-2sTv'(t)}{T^2v^2 + 4s^2}$$

$$\text{For } T = 1, s = 0.6 \text{ and } v'(t) = 1,$$

$$\frac{d\theta}{dT} = \frac{-1.2}{v^2 + 1.44}$$

(a)

$$\frac{d\theta}{dT} = \frac{-1.2}{2.44} \approx -0.4918 \text{ rad/s}$$

(b)

$$\frac{d\theta}{dT} = \frac{-1.2}{5.44} \approx -0.2206 \text{ rad/s}$$

3.9 Rates of Change in Economics and the Sciences

1. The marginal cost function is

$$C'(x) = 3x^2 + 40x + 90.$$

The marginal cost at $x = 50$ is $C'(50) = 9590$. The cost of producing the 50th item is $C(50) - C(49) = 9421$.

2. The marginal cost function is

$$C'(x) = 4x^3 + 28x + 60.$$

The marginal cost at $x = 50$ is $C'(50) = 501460$. The cost of producing the 50th item is $C(50) - C(49) = 486645$.

3. The marginal cost function is

$$C'(x) = 3x^2 + 42x + 110.$$

The marginal cost at $x = 100$ is $C'(100) = 34310$. The cost of producing the 100th item is $C(100) - C(99) = 33990$.

4. The marginal cost function is

$$C'(x) = 3x^2 + 22x + 40.$$

The marginal cost at $x = 100$ is $C'(100) = 32240$. The cost of producing the 100th item is $C(100) - C(99) = 31930$.

5. $C'(x) = 3x^2 - 60x + 300$

$$C''(x) = 6x - 60 = 0$$

$x = 10$ is the inflection point because $C'''(x)$ changes from negative to positive at this value. After this point, cost rises more sharply.

6. A linear model doesn't reflect the capacity of the stadium, or the presence of a certain number of fans who would attend no matter what the price, but away from the extremes a linear model might serve adequately. For ticket price x , the revenue function is

$$\begin{aligned} R(x) &= x(-3,000x + 57,000) \\ &= -3,000x^2 + 57,000x. \end{aligned}$$

We solve

$$R'(x) = -6,000x + 57,000 = 0$$

and find that $x = 9.5$ dollars per ticket is the critical number. Since $R'' = -6,000 < 0$, this is a maximum.

7. $\bar{C}(x) = C(x)/x = 0.1x + 3 + \frac{2000}{x}$

$$\bar{C}'(x) = 0.1 - \frac{2000}{x^2}$$

Critical number is $x = 100\sqrt{2} \approx 141.4$.

$\bar{C}'(x)$ is negative to the left of the critical number and positive to the right, so this must be the minimum.

8. The average cost function is

$$\begin{aligned} \bar{C}(x) &= \frac{0.2x^3 + 4x + 4000}{x} \\ &= 0.2x^2 + 4 + \frac{4000}{x}. \end{aligned}$$

$$\bar{C}'(x) = 0.4x - \frac{4000}{x^2} = 0$$

when $x \approx 21.54$. This is a minimum because $\bar{C}'' = 0.4 + \frac{4000}{x^3} > 0$ at this x .

9. $\bar{C}(x) = C(x)/x = 10\frac{e^{0.02x}}{x}$

$$\bar{C}'(x) = 10e^{.02x} \left(\frac{.02x - 1}{x^2} \right)$$

Critical number is $x = 50$. $\bar{C}'(x)$ is negative to the left of the critical number and positive to the right, so this must be the minimum.

10. The average cost function is

$$\bar{C}(x) = \frac{\sqrt{x^3 + 800}}{x} \text{ and}$$

$$\bar{C}'(x) = \frac{x^3 - 1600}{2x^2\sqrt{x^3 + 800}}.$$

This is zero when $x = \sqrt[3]{1600}$. This is a minimum because

$$\bar{C}'' = \frac{5,120,000 + 12,800x^3 - x^6}{4x^3(x^3 + 800)^{3/2}} > 0 \text{ at this } x.$$

11. (a) $C(x) = 0.01x^2 + 40x + 3600$

$$C'(x) = 0.02x + 40$$

$$\bar{C}(x) = \frac{C(x)}{x} = 0.01x + 40 + \frac{3600}{x}$$

$$C'(100) = 42$$

$$\bar{C}(100) = 77$$

$$\text{so } C'(100) < \bar{C}(100)$$

$$\bar{C}(101) = 76.65 < \bar{C}(100)$$

- (b) $C'(x) = 0.02x + 40$

$$C'(1000) = 60$$

$$\bar{C}(x) = \frac{0.01x^2 + 40x + 3600}{x}$$

$$\bar{C}(1000) = 53.6$$

$$\bar{C}(1001) = 53.6064$$

- (c) $\bar{C}'(x) = 0.01 - \frac{3600}{x^2} = 0$

so $x = 600$ is min and

$$C'(600) = 52$$

$$\bar{C}(600) = 52$$

12. (a) $P(x) = R(x) - C(x)$

$$P'(x) = R'(x) - C'(x) = 0$$

$$R'(x) = C'(x)$$

- (b) $P(x) = (10x - 0.001x^2) - (2x + 5,000)$.

$$P'(x) = 8 - 0.002x = 0 \text{ if } x = 4,000.$$

This is a maximum because $P''(x) = -0.002 < 0$.

13. $E = \frac{p}{f(p)} f'(p)$

$$= \frac{p}{200(30 - p)} (-200) = \frac{p}{p - 30}$$

To solve $\frac{p}{p - 30} < -1$, multiply both sides

by the negative quantity $p - 30$, to get $p > (-1)(p - 30)$ or $p > 30 - p$, so $2p > 30$, so $15 < p < 30$.

$$14. E = \frac{pf'(p)}{f(p)} = \frac{p(-200)}{200(20-p)} = \frac{p}{p-20}$$

$\frac{p}{p-20} < -1$ when $p > 20 - p$, so demand is elastic when $10 < p < 20$.

$$15. f(p) = 100p(20-p) = 100(20p-p^2)$$

$$E = \frac{p}{f(p)} f'(p)$$

$$= \frac{p}{100p(20-p)} (100)(20-2p)$$

$$= \frac{20-2p}{20-p}$$

To solve $\frac{20-2p}{20-p} < -1$, multiply both sides by the positive quantity $20-p$ to get $20-2p < (-1)(20-p)$, or $20-2p < p-20$, so $40 < 3p$, so $40/3 < p < 20$.

$$16. E = \frac{pf'(p)}{f(p)}$$

$$= \frac{p(600-120p)}{60p(10-p)} = \frac{2p-10}{p-10}$$

If $\frac{2p-10}{p-10} < -1$ for positive p , then $p-10$ must be negative. this means $\frac{2p-10}{p-10} < -1$ when $2p-10 > 10-p$, so demand is elastic when $\frac{20}{3} < p < 10$.

$$17. [pf(p)]' < 0$$

if and only if $p'f(p) + pf'(p) < 0$

if and only if $f(p) + pf'(p) < 0$

if and only if $pf'(p) < -f(p)$

if and only if $\frac{pf'(p)}{f(p)} < -1$

$$18. \text{The percentage change in quantity purchased (using the chain rule) is } \frac{Q'(I) \cdot I'}{Q(I)}.$$

The percentage change in income is $\frac{I'}{I}$.

The income elasticity of demand is then $\frac{Q'(I) \cdot I'}{Q(I)} \cdot \frac{I}{I'}$ or $\frac{Q'(I) \cdot I}{Q(I)}$.

$$19. (a) \text{ Rewrite } x'(t) \text{ as } f(x) = 2x[4-x].$$

$$f'(x) = 2(4-x) + 2x(-1)$$

$$= 8-4x$$

$$f'(x) = 0 \Rightarrow x = 2 \text{ where the } f(x) \text{ is maximum}$$

(b) The critical points of $x'(t) = 2x[4-x]$ are $x = 0$ and $x = 4$.

$$x'(t) > 0, 0 < x(t) < 4$$

$$x'(t) < 0, x > 4 \text{ or } x < 0$$

Therefore, the limiting concentration is 4.

$$20. (a) \text{ Rewrite } x'(t) \text{ as } f(x) = 0.5x[5-x].$$

$$f'(x) = 0.5(5-x) + 0.5x(-1)$$

$$= 2.5-x$$

$$f'(x) = 0 \Rightarrow x = 2.5 \text{ where the } f(x) \text{ is maximum.}$$

$$(b) \text{ The critical points of } x'(t) = 0.5x[5-x] \text{ are } x = 0 \text{ and } x = 5.$$

$$x'(t) > 0, 0 < x(t) < 5$$

$$x'(t) < 0, x > 5 \text{ or } x < 0$$

Therefore, the limiting concentration is 5.

$$21. y'(t) = c \cdot y(t)[K - y(t)]$$

$$y(t) = Kx(t)$$

$$y'(t) = Kx'(t)$$

$$Kx'(t) = c \cdot Kx(t)[K - Kx(t)]$$

$$x'(t) = c \cdot Kx(t)[1 - x(t)]$$

$$= rx(t)[1 - x(t)]$$

$$r = cK$$

$$22. \text{The given conditions translate into equations } 3 = c \cdot 2(K-2) \text{ and } 4 = c \cdot 4(K-4).$$

Solving the first equation for c and substituting into the second equation gives

$$4 = \frac{4 \cdot 3(K-4)}{2(K-2)} \Rightarrow K = 8 \text{ and } c = 1/4.$$

$$23. x'(t) = [a - x(t)][b - x(t)]$$

for $x(t) = a$,

$$x'(t) = [a - a][b - a] = 0$$

So the concentration of product is staying the same.

If $a < b$ and $x(0) = 0$ then $x'(t) > 0$ for $0 < x < a < b$

$$x'(t) < 0 \text{ for } a < x < b$$

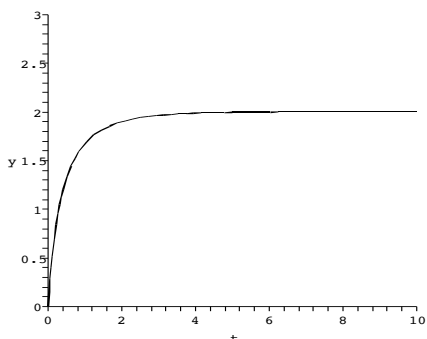
Thus $x(t) = a$ is a maximum.

$$24. x(0) = \frac{a[1 - e^{-(b-a) \cdot 0}]}{1 - \left(\frac{a}{b}\right)e^{-(b-a) \cdot 0}}$$

$$= \frac{a[1-1]}{1 - \left(\frac{a}{b}\right)} = 0$$

$$\lim_{t \rightarrow \infty} x(t) = \frac{a[1-0]}{1-0} = a$$

For $a = 2$ and $b = 3$ the graph looks like this:



25. The first inflection point occurs around $f = 1/3$, before the step up. The second occurs at the far right of the graph. The equivalence point is presumably more stable. The first inflection point would be hard to measure, since the pH takes drastic leap right after the inflection point occurs.

26. Recall that we are assuming $0 < f < 1$. As $f \rightarrow 1^-$,

$$p'(f) = \frac{1}{f(1-f)} \rightarrow \infty$$

27. $R(x) = \frac{rx}{k+x}$, $x \geq 0$

$$R'(x) = \frac{rk}{(k+x)^2}$$

There are no critical numbers. Any possible maximum would have to be at the endpoint $x = 0$, but in fact R is increasing on $[0, \infty)$, so there is no maximum (although as x goes to infinity, R approaches r).

28. $PV^{7/5} = c$

$$\frac{d}{dP} (PV^{7/5}) = \frac{d}{dP} (c) = 0$$

$$V^{7/5} + \frac{7}{5}PV^{2/5}\frac{dV}{dP} = 0$$

$$V + \frac{7}{5}P\frac{dV}{dP} = 0$$

$$\frac{dV}{dP} = \frac{-5}{7} \frac{V}{P}$$

 But $V^{7/5} = c/P$, so $V = (c/P)^{5/7}$. Hence

$$\frac{dV}{dP} = \frac{-5}{7} \frac{V}{P} = \frac{-5}{7} \frac{(c/P)^{5/7}}{P} = \frac{-5c^{5/7}}{7P^{12/7}}$$

As pressure increases, volume decreases.

29. $m'(x) = 4 - \cos x$, so the rod is less dense at the ends.
30. $m'(x) = 3(x-1)^2 + 6$.
 Density is maximum at the ends and at a minimum in the middle.

31. $m'(x) = 4$, so the rod is homogeneous.

32. $m'(x) = 8x$.

Density increases from 0 at the left end to a maximum at the right end.

33.
$$\begin{aligned} Q'(t) &= e^{-2t} \cdot (-2)(\cos 3t - 2 \sin 3t) \\ &\quad + e^{-2t} ((-\sin 3t \cdot 3) - 2 \cos 3t \cdot 3) \\ &= e^{-2t} (-8 \cos 3t + \sin 3t) \text{ amps} \end{aligned}$$

34.
$$\begin{aligned} Q'(t) &= e^t (3 \cos 2t + \sin 2t) \\ &\quad + e^t (-6 \sin 2t + 2 \cos 2t) \\ &= 5e^t (\cos 2t - \sin 2t) \text{ amps} \end{aligned}$$

35. As $t \rightarrow \infty$, $Q(t) \rightarrow 4 \sin 3t$, so $e^{-3t} \cos 2t$ is called the transient term and $4 \sin 3t$ is called the steady-state value.

$$\begin{aligned} Q'(t) &= e^{-3t} \cdot (-3) \cos 2t \\ &\quad + e^{-3t} (-\sin 2t \cdot 2) + 4 \cos 3t \cdot 3 \\ &= e^{-3t} (-3 \cos 2t - 2 \sin 2t) \\ &\quad + 12 \cos 3t \end{aligned}$$

The transient term is $e^{-3t}(-3 \cos 2t - 2 \sin 2t)$ and the steady-state value is $12 \cos 3t$.

36.
$$\begin{aligned} Q'(t) &= -2e^{-2t}(\cos t - 2 \sin t) \\ &\quad + e^{-2t}(-\sin t - 2 \cos t) \\ &\quad + e^{-3t} - 3te^{-3t} - 8 \sin 4t \\ Q'(t) &= e^{-2t}(-4 \cos t + 3 \sin t) \\ &\quad + e^{-3t}(1 - 3t) - 8 \sin 4t \end{aligned}$$

The transient term is $e^{-2t}(-4 \cos t + 3 \sin t) + e^{-3t}(1 - 3t)$ and the steady-state value is $-8 \sin 4t$.

37. The rate of population growth is given by
 $f(p) = 4p(5-p) = 4(5p-p^2)$
 $f'(p) = 4(5-2p)$,
 so the only critical number is $p = 2.5$. Since the graph of f is a parabola opening down, this must be a max.

38. The rate of growth $R = 2p(7-2p)$, so $R' = 14-8p = 0$ when $p = 7/4$. This is a maximum because $R'' = -8 < 0$.

39.
$$\begin{aligned} p'(t) &= \frac{-B(1+Ae^{-kt})'}{(1+Ae^{-kt})^2} \\ &= \frac{-B(-kAe^{-kt})}{(1+Ae^{-kt})^2} \\ &= \frac{kABe^{-kt}}{(1+Ae^{-kt})^2} \\ &= \frac{kABe^{-kt}}{1+2Ae^{-kt}+A^2e^{-2kt}} \\ &= \frac{kAB}{e^{kt}+2A+A^2e^{-kt}} \end{aligned}$$

As t goes to infinity, the exponential term goes

to 0, and so the limiting population is

$$\frac{B}{1 + A(0)} = B.$$

40. If the inflection point is $p = 120$, then the maximum population is $B = 240$. If the initial population is $p(0) = 40$, then

$$40 = \frac{240}{1 + A}.$$

We solve to get $A = 5$. If then $p(12) = 160$, we have the equation

$$160 = \frac{240}{1 + 5e^{-12k}}$$

which we can solve to get

$$k = \frac{\ln 10}{12}.$$

41. For $a = 70$, $b = 0.2$, $f(t) = \frac{70}{1 + 3e^{-0.2t}} =$

$$70(1 + 3e^{-0.2t})^{-1}$$

$$f(2) = \frac{70}{1 + 3e^{-0.2 \cdot 2}} \approx 23$$

$$f'(t) = -70(1 + 3e^{-0.2t})^{-2}(3e^{-0.2t})(-0.2)$$

$$= \frac{42e^{-0.2t}}{(1 + 3e^{-0.2t})^2}$$

$$f'(2) = \frac{42e^{0.2 \cdot 2}}{(1 + 3e^{-0.2 \cdot 2})^2} \approx 3.105$$

This says that at time $t = 2$ hours, the rate at which the spread of the rumor is increasing is about 3% of the population per hour.

$$\lim_{t \rightarrow \infty} f(t) = \frac{70}{1 + 0} = 70$$

so 70% of the population will eventually hear the rumor.

42. $f'(t) = -0.02e^{-0.02t} + 0.42e^{-0.42t}$
 $f'(t) = 0$ when $0.42e^{-0.42t} = 0.02e^{-0.02t}$, or
 $e^{-0.4t} = 0.02/0.42$. So we see that

$$t = -\frac{\ln 0.047619}{0.4} \approx 7.6113$$

is the critical value. The Second Derivative Test shows that it is a maximum.

43. $f'(x) = \frac{-64x^{-1.4}(4x^{-0.4} + 15)}{(4x^{-0.4} + 15)^2}$
 $- \frac{(160x^{-0.4} + 90)(-1.6x^{-1.4})}{(4x^{-0.4} + 15)^2}$
 $= \frac{-816x^{-1.4}}{(4x^{-0.4} + 15)^2} < 0$

So $f(x)$ is decreasing. This shows that pupils shrink as light increases.

44. $T(x) = 102 - \frac{1}{6}x^2 + \frac{1}{54}x^3$.

To maximize $|T'(x)|$, we find all extrema of $T'(x)$ and compare their magnitudes.

$$T'(x) = \frac{-1}{3}x + \frac{1}{18}x^2.$$

$$T''(x) = \frac{-1}{3} + \frac{1}{9}x = 0 \text{ when } x = 3.$$

We test the critical numbers and the endpoints:

$$T'(0) = 0, T'(6) = 0, \text{ and } T'(3) = \frac{-1}{2}.$$

The dosage that maximizes sensitivity is 3 mg.

45. If v is not greater than c , the fish will never make any headway. $E'(v) = \frac{v(v-2c)}{(v-c)^2}$ so the only critical number is $v = 2c$. When v is large, $E(v)$ is large, and when v is just a little bigger than c , $E(v)$ is large, so we must have a minimum.

46. We wish to minimize $P = \frac{1}{v} + cv^3$.

$$P' = \frac{-1}{v^2} + 3cv^2 = 0 \text{ when } v = \sqrt[4]{\frac{1}{3c}}.$$

$P'' = \frac{2}{v^3} + 6cv > 0$ at this velocity, so this gives the minimum power.

47. (a) $xy = c$

$$y = \frac{c}{x}$$

$$\text{Time spent to cover } y \text{ miles} = \frac{y}{r_1}$$

$$\text{Time spent to cover } x \text{ miles} = \frac{x}{r_2}$$

$$\text{So, the total time spent } (T) = \frac{y}{r_1} + \frac{x}{r_2}$$

Now by taking $f(x) = T$ we get:

$$f(x) = \left(\frac{y}{r_1} + \frac{x}{r_2} \right)$$

$$= \left(\frac{c}{r_1 x} + \frac{x}{r_2} \right)$$

$$f'(x) = \frac{-c}{r_1} \cdot \frac{1}{x^2} + \frac{1}{r_2}$$

$$f'(x) = 0 \Rightarrow$$

$$-cr_2 + r_1x^2 = 0$$

$$r_1x^2 = cr_2$$

$$x^2 = \frac{cr_2}{r_1}$$

$$x = \sqrt{\frac{cr_2}{r_1}}$$

$$\text{Substitute } x = \sqrt{\frac{cr_2}{r_1}} \text{ in } y = \frac{c}{x}.$$

$$y = \frac{c}{\sqrt{\frac{cr_2}{r_1}}}$$

$$= \sqrt{\frac{r_1c}{r_2}}$$

$$\text{Therefore, when } x = \sqrt{\frac{cr_2}{r_1}} \text{ and } y =$$

$\sqrt{\frac{r_1c}{r_2}}$, the time spent by the commuter is minimum.

- (b) Time spent driving at $r_1 = \frac{y}{r_1}$

$$= \frac{\sqrt{\frac{r_1 c}{r_2}}}{r_1} = \sqrt{\frac{c}{r_1 r_2}}$$

Time spent driving at $r_2 = \frac{x}{r_2}$

$$= \frac{\sqrt{\frac{r_2 c}{r_1}}}{r_2} = \sqrt{\frac{c}{r_1 r_2}}$$

Therefore, equal time is spent driving at r_1 and r_2 .

48. (a) $C(v) = avp + b\frac{p}{v}$
 $C'(v) = ap + \left(\frac{-bp}{v^2}\right)$
 $C'(v) = 0 \Rightarrow ap + \left(\frac{-bp}{v^2}\right) = 0$

$$a + \left(\frac{-b}{v^2}\right) = 0$$

$$a = \frac{b}{v^2}$$

$$v^2 = \frac{b}{a}$$

$$v = \pm \sqrt{\frac{b}{a}}$$

$$C''(v) = \frac{bp}{v^3}$$

$$C''(v) > 0 \text{ at } v = \sqrt{\frac{b}{a}}.$$

Therefore, $v = \sqrt{\frac{b}{a}}$ to minimize $C(v)$.

(b) $C(v) = ap\frac{v^2}{v-v_c} + b\frac{p}{v-v_c}$
 $C'(v) = ap\left[\frac{(v-v_c)(2v)-v^2}{(v-v_c)^2}\right]$
 $+ \left[\frac{-bp}{(v-v_c)^2}\right]$
 $= \frac{1}{(v-v_c)^2} [2apv(v-v_c) - apv^2 - bp]$
 $= \frac{1}{(v-v_c)^2} [apv^2 - 2apv_c v - bp]$

$$C'(v) = 0 \Rightarrow$$

$$apv^2 - 2apv_c v - bp = 0$$

$$v = \frac{2apv_c \pm \sqrt{(2apv_c)^2 + 4abp^2}}{2ap}$$

$$v = v_c \pm \sqrt{v_c^2 + \frac{b}{a}}$$

Therefore, $v = v_c \pm \sqrt{v_c^2 + \frac{b}{a}}$ minimizes

$$C(v)$$

Ch. 3 Review Exercises

1. $f(x) = e^{3x}$, $x_0 = 0$,
 $f'(x) = 3e^{3x}$
 $L(x) = f(x_0) + f'(x_0)(x - x_0)$
 $= f(0) + f'(0)(x - 0)$
 $= e^{3 \cdot 0} + 3e^{3 \cdot 0}x$
 $= 1 + 3x$

2. $f'(x) = \frac{2x}{2\sqrt{x^2+3}}$.
 $f(1) = 2$, and $f'(1) = 1/2$.
 $L(x) = \frac{1}{2}(x - 1) + 2$.

3. $f(x) = \sqrt[3]{x} = x^{1/3}$, $x_0 = 8$
 $f'(x) = \frac{1}{3}x^{-2/3}$
 $L(x) = f(x_0) + f'(x_0)(x - x_0)$
 $= f(8) + f'(8)(x - 8)$
 $= \sqrt[3]{8} + \frac{1}{3}(8)^{-2/3}(x - 8)$
 $= 2 + \frac{1}{12}(x - 8)$
 $L(7.96) = 2 + \frac{1}{12}(7.96 - 8) \approx 1.99666$

4. $\sin 3$ is close to $\sin \pi$. If $y = \sin x$, $y' = \cos x$.
The point is $(\pi, 0)$ and the slope is -1 . The
linear approximation of $\sin x$ at $x = \pi$ is
 $L(x) = -(x - \pi)$, so
 $\sin 3 \approx -(3 - \pi) \approx 0.14159$.

5. From the graph of $f(x) = x^3 + 5x - 1$, there is
one root.
 $f'(x) = 3x^2 + 5$
Starting with $x_0 = 0$, Newton's method gives
 $x_1 = 0.2$, $x_2 = 0.198437$, and $x_3 = 0.198437$.

6. From the graph of $f(x) = x^3 - e^{-x}$, there is
one root.
 $f'(x) = 3x^2 + e^{-x}$
Starting with $x_0 = 1$, Newton's method gives
 $x_1 = 0.8123$, $x_2 = 0.7743$, and $x_3 = 0.7729$,
which is accurate to 4 decimal places.

7. Near an inflection point, the rate of change
of the rate of change of $f(x)$ is very small so
there aren't any big dropoffs or sharp increases
nearby to make the linear approximation inac-
curate.

8. If $y = \frac{1}{1-x}$, then $y' = \frac{1}{(1-x)^2}$.

For "small" x , x is near 0. The point on the
curve when $x = 0$ is $(0, 1)$, and the slope is 1,
so the linear approximation is $L(x) = x + 1$,
and this is valid for "small" x .

9. $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1}$ is type $\frac{0}{0}$;

L'Hôpital's Rule gives

$$\lim_{x \rightarrow 1} \frac{3x^2}{2x} = \frac{3}{2}.$$

10. $\lim_{x \rightarrow 0} \frac{\sin x}{x^2 + 3x}$ is type $\frac{0}{0}$;

L'Hôpital's Rule gives

$$\lim_{x \rightarrow 0} \frac{\cos x}{2x + 3} = \frac{1}{3}.$$

11. $\lim_{x \rightarrow 0} \frac{e^{2x}}{x^4 + 2}$ is type $\frac{\infty}{\infty}$;

applying L'Hôpital's Rule twice gives:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2e^{2x}}{4x^3} &= \lim_{x \rightarrow \infty} \frac{4e^{2x}}{12x^2} = \lim_{x \rightarrow \infty} \frac{8e^{2x}}{24x} \\ &= \lim_{x \rightarrow \infty} \frac{16e^{2x}}{24} = \infty \end{aligned}$$

12. $\lim_{x \rightarrow \infty} (x^2 e^{-3x}) = \lim_{x \rightarrow \infty} \frac{x^2}{e^{3x}}$ is type $\frac{\infty}{\infty}$;

applying L'Hôpital's Rule twice gives:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2x}{3e^{3x}} &= \lim_{x \rightarrow \infty} \frac{2}{9e^{3x}} = 0 \end{aligned}$$

13. $L = \lim_{x \rightarrow 2^+} \left| \frac{x+1}{x-2} \right|^{\sqrt{x^2-4}}$

$$\ln L = \lim_{x \rightarrow 2^+} \left(\sqrt{x^2-4} \ln \left| \frac{x+1}{x-2} \right| \right)$$

$$= \lim_{x \rightarrow 2^+} \left(\frac{\ln \left| \frac{x+1}{x-2} \right|}{(x^2-4)^{-1/2}} \right)$$

$$= \lim_{x \rightarrow 2^+} \left(\frac{\left| \frac{x-2}{x+1} \right| \frac{-3}{(x-2)^2}}{-x(x^2-4)^{-3/2}} \right)$$

$$= \lim_{x \rightarrow 2^+} \left(\frac{3(x^2-4)^{3/2}}{x(x+1)(x-2)} \right)$$

$$= \lim_{x \rightarrow 2^+} \left(\frac{3(x-2)^{1/2}(x+2)^{3/2}}{x(x+1)} \right)$$

$$\ln L = 0$$

$$L = 1$$

14. $\lim_{x \rightarrow \infty} x \ln \left(1 + \frac{1}{x} \right) = \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{x} \right)}{\frac{1}{x}}$

is type $\frac{0}{0}$ so we can apply L'Hôpital's Rule:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\frac{1}{(1+\frac{1}{x})}(-x^{-2})}{-x^{-2}} &= \lim_{x \rightarrow \infty} \frac{1}{(1+\frac{1}{x})} = 1 \end{aligned}$$

15. $\lim_{x \rightarrow 0^+} (\tan x \ln x) = \lim_{x \rightarrow 0^+} \left(\frac{\ln x}{\cot x} \right)$
 $= \lim_{x \rightarrow 0^+} \left(\frac{1/x}{-\csc^2 x} \right)$
 $= \lim_{x \rightarrow 0^+} - \left(\frac{\sin^2 x}{x} \right)$
 $= - \lim_{x \rightarrow 0^+} \left(\frac{\sin x}{x} \sin x \right)$
 $= (-1)(0) = 0$

16. $\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{\sin^{-1} x}$ is type $\frac{0}{0}$;

we can apply L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{\frac{1}{1+x^2}}{\frac{1}{\sqrt{1-x^2}}} = \lim_{x \rightarrow 0} \frac{\sqrt{1-x^2}}{1+x^2} = 1$$

17. $f'(x) = 3x^2 + 6x - 9 = 3(x^2 + 2x - 3)$
 $= 3(x+3)(x-1)$

So the critical numbers are $x = 1$ and $x = -3$.

$f'(x) > 0$ on $(-\infty, -3) \cup (1, \infty)$

$f'(x) < 0$ on $(-3, 1)$

Hence f is increasing on $(-\infty, -3)$ and on $(1, \infty)$ and f is decreasing on $(-3, 1)$. Thus there is a local max at $x = -3$ and a local min at $x = 1$.

$$f''(x) = 3(2x+2) = 6(x+1)$$

$f''(x) > 0$ on $(-1, \infty)$

$f''(x) < 0$ on $(-\infty, -1)$

Hence f is concave up on $(-1, \infty)$ and concave down on $(-\infty, -1)$, and there is an inflection point at $x = -1$.

18. $f'(x) = 4x^3 - 4$

$f'(x) = 0$ when $x = 1$, and this is the only critical number. The function is decreasing for $x < 1$ and increasing for $x > 1$.

$f'' = 12x^2 > 0$ when $x = 1$, so this is a local minimum. $f'' = 0$ when $x = 0$, but does not change sign there, so there are no inflection points. The function is concave up everywhere.

19. $f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$

$x = 0, 3$ are critical numbers.

$f'(x) > 0$ on $(3, \infty)$

$f'(x) < 0$ on $(-\infty, 0) \cup (0, 3)$

f increasing on $(3, \infty)$, decreasing on $(-\infty, 3)$ so $x = 3$ is a local min.

$$f''(x) = 12x^2 - 24x = 12x(x-2)$$

$f''(x) > 0$ on $(-\infty, 0) \cup (2, \infty)$
 $f''(x) < 0$ on $(0, 2)$
 f is concave up on $(-\infty, 0) \cup (2, \infty)$, concave down on $(0, 2)$ so $x = 0, 2$ are inflection points.

20. $f'(x) = 3x^2 - 6x - 24 = 3(x-4)(x+2)$
 $f'(x) = 0$ when $x = 4$ and $x = -2$. The function is increasing for $x < -2$, then decreasing for $-2 < x < 4$, and increasing for $x > 4$. $x = -2$ represents a local maximum, and $x = 4$ represents a local minimum.

$f''(x) = 6x - 6$
 $f''(x) = 0$ when $x = 1$, and changes sign there, so $x = 1$ is an inflection point. The function is concave down for $x < 1$ and concave up for $x > 1$.

21. $f'(x) = e^{-4x} + xe^{-4x}(-4) = e^{-4x}(1 - 4x)$
 $x = 1/4$ is a critical number.

$f'(x) > 0$ on $(-\infty, \frac{1}{4})$
 $f'(x) < 0$ on $(\frac{1}{4}, \infty)$
 f increasing on $(-\infty, \frac{1}{4})$, decreasing on $(\frac{1}{4}, \infty)$ so $x = 1/4$ is a local max.
 $f''(x) = e^{-4x}(-4)(1 - 4x) + e^{-4x}(-4)$
 $= -4e^{-4x}(2 - 4x)$
 $f''(x) > 0$ on $(\frac{1}{2}, \infty)$
 $f''(x) < 0$ on $(-\infty, \frac{1}{2})$
 f is concave up on $(\frac{1}{2}, \infty)$, concave down on $(-\infty, \frac{1}{2})$ so $x = 1/2$ is inflection point.

22. $f'(x) = 2x \ln x + x = x(2 \ln x + 1)$
 $f'(x) = 0$ when $\ln x = -1/2$, so $x = e^{-1/2}$. ($x = 0$ is not a critical number because it is not in the domain of the function.) The function is decreasing for $0 < x < e^{-1/2}$, and increasing for $x > e^{-1/2}$. The critical number $x = e^{-1/2}$ represents a minimum.

$f''(x) = 2 \ln x + 3$
 $f''(x) = 0$ when $x = e^{-3/2}$ and the sign changes from negative to positive there, so this is an inflection point. The function is concave down for $0 < x < e^{-3/2}$ and concave up for $x > e^{-3/2}$.

23. $f'(x) = \frac{x^2 - (x-90)(2x)}{x^4}$
 $= \frac{-(x-180)}{x^3}$

$x = 180$ is the only critical number.
 $f'(x) < 0$ on $(-\infty, 0) \cup (180, \infty)$
 $f'(x) > 0$ on $(0, 180)$
 $f(x)$ is decreasing on $(-\infty, 0) \cup (180, \infty)$ and increasing on $(0, 180)$ so $f(x)$ has a local maximum at $x = 180$.
 $f''(x) = -\frac{x^3 - (x-180)(3x^2)}{x^6}$

$$= -\frac{-2x + 540}{x^4}$$

$f''(x) < 0$ on $(-\infty, 0) \cup (0, 270)$
 $f''(x) > 0$ on $(270, \infty)$ so $x = 90$ is an inflection point.

24. $f'(x) = \frac{4x}{3(x^2 - 1)^{1/3}}$
 $f'(x) = 0$ at $x = 0$ and is undefined at $x = \pm 1$. The function is decreasing for $x < -1$, increasing for $-1 < x < 0$, decreasing for $0 < x < 1$, and increasing for $1 < x$. Critical numbers $x = \pm 1$ are minima, and $x = 0$ is a maximum.

$f''(x) = \frac{4(x^2 - 3)}{9(x^2 - 1)^{4/3}}$
 $f''(x) = 0$ when $x = \pm\sqrt{3}$, and undefined for $x = \pm 1$. The function is concave up for $x < -\sqrt{3}$, concave down for $-\sqrt{3} < x < -1$, concave down for $-1 < x < 1$, concave down for $1 < x < \sqrt{3}$, and concave up for $\sqrt{3} < x$. The inflection points are $x = \pm\sqrt{3}$.

25. $f'(x) = \frac{x^2 + 4 - x(2x)}{(x^2 + 4)^2}$
 $= \frac{4 - x^2}{(x^2 + 4)^2}$

$x = \pm 2$ are critical numbers.
 $f'(x) > 0$ on $(-2, 2)$
 $f'(x) < 0$ on $(-\infty, -2) \cup (2, \infty)$
 f increasing on $(-2, 2)$, decreasing on $(-\infty, -2)$ and on $(2, \infty)$ so f had a local min at $x = -2$ and a local max at $x = 2$.

$f''(x) = \frac{-2x(x^2 + 4)^2 - (4 - x^2)[2(x^2 + 4) \cdot 2x]}{(x^2 + 4)^4}$
 $= \frac{2x^3 - 24x}{(x^2 + 4)^3}$
 $f''(x) > 0$ on $(-\sqrt{12}, 0) \cup (\sqrt{12}, \infty)$
 $f''(x) < 0$ on $(-\infty, -\sqrt{12}) \cup (0, \sqrt{12})$
 f is concave up on $(-\sqrt{12}, 0) \cup (\sqrt{12}, \infty)$, concave down on $(-\infty, -\sqrt{12}) \cup (0, \sqrt{12})$ so $x = \pm\sqrt{12}, 0$ are inflection points.

26. $f'(x) = \frac{2}{(x^2 + 4)^{3/2}}$

$f'(x)$ is never zero and is defined for all x , so there are no critical numbers. The function is increasing for all x .

$f''(x) = \frac{-6x}{(x^2 + 4)^{5/2}}$
 $f''(x) = 0$ when $x = 0$. The function is concave up for $x < 0$, concave down for $x > 0$, and the inflection point is $x = 0$.

27. $f'(x) = 3x^2 + 6x - 9$
 $= 3(x+3)(x-1)$

$x = -3$, $x = 1$ are critical numbers, but $x = -3 \notin [0, 4]$.
 $f(0) = 0^3 + 3 \cdot 0^2 - 9 \cdot 0 = 0$
 $f(4) = 4^3 + 3 \cdot 4^2 - 9 \cdot 4 = 76$
 $f(1) = 1^3 + 3 \cdot 1^2 - 9 \cdot 1 = -5$
 So $f(4) = 76$ is absolute max on $[0, 4]$, $f(1) = -5$ is absolute min.

28. First note that $f(x) = \sqrt{x(x-1)(x-2)}$ is only defined on $[0, 1] \cup [2, \infty)$. So we are looking at the intervals $[0, 1] \cup [2, 3]$.

$$f'(x) = \frac{3x^2 - 6x + 2}{2\sqrt{x^3 - 3x^2 + 2x}}$$

The numerator has roots $x = \frac{3 \pm \sqrt{3}}{3}$, but $f(x)$ is only defined at $\frac{3 - \sqrt{3}}{3}$. The denominator has zeros at $x = 0, 1$ and 2 . Plus we have to check the values of f at the endpoint $x = 3$. We find:
 $f(0) = 0$
 $f(\frac{3 - \sqrt{3}}{3}) \approx 0.6204$
 $f(1) = 0$
 $f(2) = 0$
 $f(3) = \sqrt{6} \approx 2.4495$
 Thus $f(x)$ has an absolute maximum on this interval at $x = 3$ and absolute minimums at $x = 0$, $x = 1$ and $x = 2$.

29. $f'(x) = \frac{4}{5}x^{-1/5}$
 $x = 0$ is critical number.
 $f(-2) = (-2)^{4/5} \approx 1.74$
 $f(3) = (3)^{4/5} \approx 2.41$
 $f(0) = (0)^{4/5} = 0$
 $f(0) = 0$ is absolute min, $f(3) = 3^{4/5}$ is absolute max.
30. $f'(x) = 2xe^{-x} - x^2e^{-x} = xe^{-x}(2 - x)$
 $f'(x) = 0$ when $x = 0$ and $x = 2$. We test $f(x)$ at the critical numbers in the interval $[-1, 4]$, and the endpoints.
 $f(-1) = e \approx 2.718$
 $f(0) = 0$
 $f(2) = 4/e^2 \approx 0.541$
 $f(4) = 16/e^4 \approx 0.293$
 The absolute maximum is $f(-1) = e$, and the absolute minimum is $f(0) = 0$.

31. $f'(x) = 3x^2 + 8x + 2$
 $f'(x) = 0$ when

$$x = \frac{-8 \pm \sqrt{64 - 24}}{6} = -\frac{4}{3} \pm \frac{\sqrt{10}}{3}$$

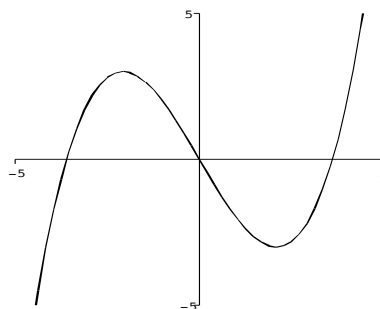
 $x = -\frac{4}{3} - \frac{\sqrt{10}}{3}$ is local max, $x = -\frac{4}{3} + \frac{\sqrt{10}}{3}$ is local min.

32. $f'(x) = 4x^3 - 6x + 2$
 $= 2(x - 1)(2x^2 + 2x - 1)$
 $f'(x) = 0$ when $x = 1$ and $x = \frac{-1 \pm \sqrt{3}}{2}$, and the derivative changes sign at these values, so these critical numbers are all extrema.

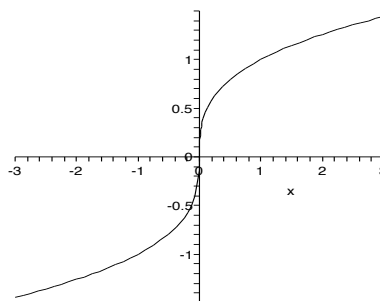
33. $f'(x) = 5x^4 - 4x + 1 = 0$
 $x \approx 0.2553, 0.8227$
 local min at $x \approx 0.8227$,
 local max at $x \approx 0.2553$.

34. $f'(x) = 5x^4 + 8x - 4$
 $f'(x) = 0$ at approximately $x = -1.3033$ and $x = 0.4696$ (found using Newton's method, or a CAS numerical solver). The derivative changes sign at these values so they correspond to extrema: $x = -1.3033$ is a local max and $x = 0.4696$ is a local min.

35. One possible graph:



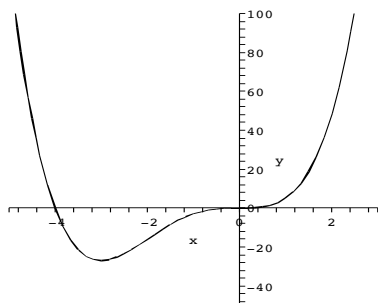
36. One possible graph:



37. $f'(x) = 4x^3 + 12x^2 = 4x^2(4x + 3)$
 $f''(x) = 12x^2 + 24x = 12x(x + 2)$
 $f'(x) > 0$ on $(-3, 0) \cup (0, \infty)$
 $f'(x) < 0$ on $(-\infty, -3)$
 $f''(x) > 0$ on $(-\infty, -2) \cup (0, \infty)$
 $f''(x) < 0$ on $(-2, 0)$
 f increasing on $(-3, \infty)$, decreasing on $(-\infty, -3)$, concave up on $(-\infty, -2) \cup (0, \infty)$, concave down on $(-2, 0)$, local min at $x = -3$,

inflection points at $x = -2, 0$.

$f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.

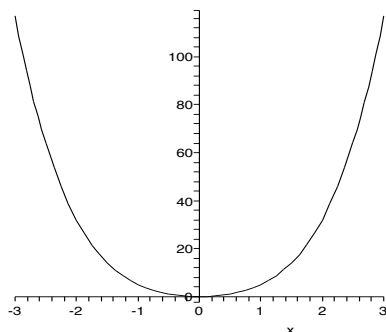


38. $f'(x) = 4x^3 + 8x$

$f'(x) = 0$ when $x = 0$.

$f'' = 12x^2 + 8 > 0$ at $x = 0$, so this is a minimum. $f''(x) > 0$ for all x so there are no inflection points.

$f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.



39. $f'(x) = 4x^3 + 4 = 4(x^3 + 1)$

$f''(x) = 12x^2$

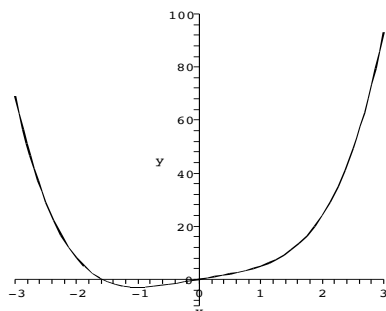
$f'(x) > 0$ on $(-1, \infty)$

$f'(x) < 0$ on $(-\infty, -1)$

$f''(x) > 0$ on $(-\infty, 0) \cup (0, \infty)$

f increasing on $(-1, \infty)$, decreasing on $(-\infty, -1)$, concave up on $(-\infty, \infty)$, local min at $x = -1$.

$f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.



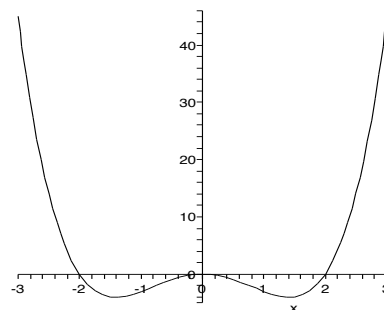
40. $f'(x) = 4x^3 - 8x$

$f'(x) = 0$ when $x = 0$ and $x = \pm\sqrt{2}$.

$f'' = 12x^2 - 8 < 0$ at $x = 0$, so this is a maximum. $f''(x) > 0$ for $x = \pm\sqrt{2}$, so these are minima.

$f''(x) = 0$ when $x = \pm\sqrt{2/3}$, and changes sign there, so these are inflection points.

$f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.



$$\begin{aligned} 41. \quad f'(x) &= \frac{x^2 + 1 - x(2x)}{(x^2 + 1)^2} \\ &= \frac{1 - x^2}{(x^2 + 1)^2} \\ f''(x) &= \frac{-2x(x^2 + 1)^2 - (1 - x^2)2(x^2 + 1)2x}{(x^2 + 1)^4} \\ &= \frac{2x(x^2 - 3)}{(x^2 + 1)^4} \end{aligned}$$

$f'(x) > 0$ on $(-1, 1)$

$f'(x) < 0$ on $(-\infty, -1) \cup (1, \infty)$

$f''(x) > 0$ on $(-\sqrt{3}, 0) \cup (\sqrt{3}, \infty)$

$f''(x) < 0$ on $(-\infty, -\sqrt{3}) \cup (0, \sqrt{3})$

f increasing on $(-1, 1)$, decreasing on $(-\infty, -1)$ and on $(1, \infty)$, concave up on

$$(-\sqrt{3}, 0) \cup (\sqrt{3}, \infty),$$

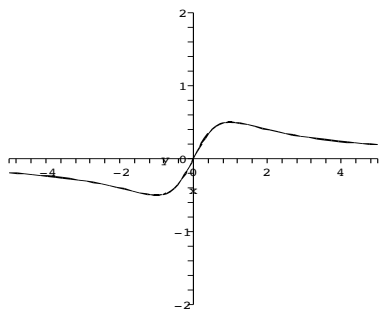
concave down on

$$(-\infty, -\sqrt{3}) \cup (0, \sqrt{3}),$$

local min at $x = -1$, local max at $x = 1$, inflection points at $0, \pm\sqrt{3}$.

$$\lim_{x \rightarrow \infty} \frac{x}{x^2 + 1} = \lim_{x \rightarrow -\infty} \frac{x}{x^2 + 1} = 0$$

So f has a horizontal asymptote at $y = 0$.

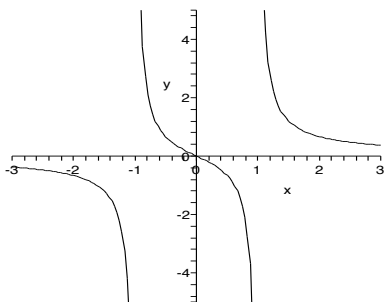


42. $f'(x) = -\frac{x^2 + 1}{(x^2 - 1)^2}$

is undefined when $f(x)$ is undefined, and is never zero. There are no extrema. There are vertical asymptotes at $x = \pm 1$, and horizontal asymptote $y = 0$.

$$f''(x) = \frac{2x(x^2 + 3)}{(x^2 - 1)^3}$$

$f''(x) = 0$ when $x = 0$, and this is the inflection point: $f(x)$ is concave down on $(-\infty, -1)$ and $(0, 1)$; $f(x)$ is concave up on $(-1, 0)$ and $(1, \infty)$.



43. $f'(x) = \frac{(2x)(x^2 + 1) - x^2(2x)}{(x^2 + 1)^2}$

$$= \frac{2x}{(x^2 + 1)^2}$$

$$f''(x) = \frac{2(x^2 + 1)^2 - 2x \cdot 2(x^2 + 1)2x}{(x^2 + 1)^4}$$

$$= \frac{2 - 6x^2}{(x^2 + 1)^3}$$

$f'(x) > 0$ on $(0, \infty)$
 $f'(x) < 0$ on $(-\infty, 0)$
 $f''(x) > 0$ on $(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}})$
 $f''(x) < 0$ on $(-\infty, -\sqrt{\frac{1}{3}}) \cup (\sqrt{\frac{1}{3}}, \infty)$
 f increasing on $(0, \infty)$ decreasing on $(-\infty, 0)$,

concave up on

$$\left(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}\right),$$

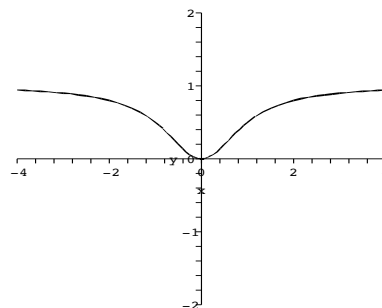
concave down on

$$\left(-\infty, -\sqrt{\frac{1}{3}}\right) \cup \left(\sqrt{\frac{1}{3}}, \infty\right),$$

local min at $x = 0$, inflection points at $x = \pm\sqrt{1/3}$.

$$\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 1} = \lim_{x \rightarrow -\infty} \frac{x^2}{x^2 + 1} = 1$$

So f has a horizontal asymptote at $y = 1$.

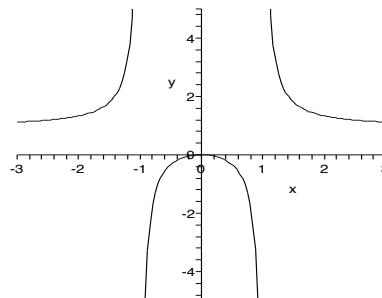


44. $f'(x) = -\frac{2x}{(x^2 - 1)^2}$

$f'(x) = 0$ when $x = 0$, and is undefined when $f(x)$ is undefined. There is a local maximum at $x = 0$. There are vertical asymptotes at $x = \pm 1$, and horizontal asymptote $y = 1$.

$$f''(x) = \frac{2(3x^2 + 1)}{(x^2 - 1)^3}$$

$f''(x) \neq 0$ for any x , and there are no inflection points: $f(x)$ is concave up on $(-\infty, -1) \cup (1, \infty)$ and concave down on $(-1, 1)$.



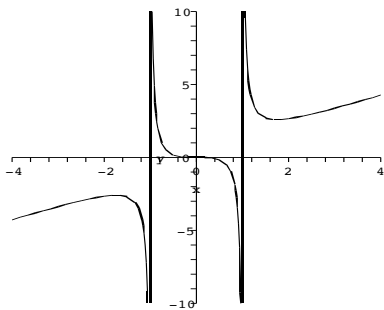
$$\begin{aligned}
 45. \quad f'(x) &= \frac{3x^2(x^2 - 1) - x^3(2x)}{(x^2 - 1)^2} \\
 &= \frac{x^4 - 3x^2}{(x^2 - 1)^2} \\
 f''(x) &= \frac{(4x^3 - 6x)(x^2 - 1)^2}{(x^2 - 1)^4} \\
 &\quad - \frac{(x^4 - 3x^2)2(x^2 - 1)2x}{(x^2 - 1)^4} \\
 &= \frac{2x^3 + 6x}{(x^2 - 1)^4}
 \end{aligned}$$

$f'(x) > 0$ on $(-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$
 $f'(x) < 0$ on $(-\sqrt{3}, -1) \cup (-1, 0) \cup (0, 1) \cup (1, \sqrt{3})$
 $f''(x) > 0$ on $(-1, 0) \cup (1, \infty)$
 $f''(x) < 0$ on $(-\infty, -1) \cup (0, 1)$
 f increasing on $(-\infty, -\sqrt{3})$ and on $(\sqrt{3}, \infty)$;
decreasing on $(-\sqrt{3}, -1)$ and on $(-1, 1)$ and
on $(1, \sqrt{3})$; concave up on $(-1, 0) \cup (1, \infty)$, con-
cave down on $(-\infty, -1) \cup (0, 1)$; $x = -\sqrt{3}$ local
max; $x = \sqrt{3}$ local min; $x = 0$ inflection point.
 f is undefined at $x = -1$ and $x = 1$.

$$\lim_{x \rightarrow 1^+} \frac{x^3}{x^2 - 1} = \infty, \text{ and}$$

$$\lim_{x \rightarrow 1^-} \frac{x^3}{x^2 - 1} = -\infty$$

So f has vertical asymptotes at $x = 1$ and $x = -1$.



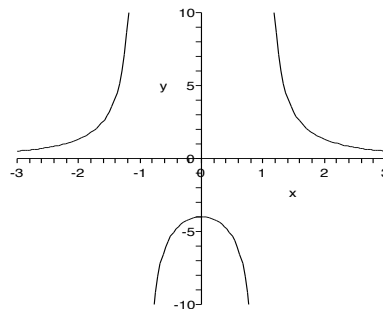
$$46. \quad f'(x) = -\frac{8x}{(x^2 - 1)^2}$$

$f'(x) = 0$ when $x = 0$, and is undefined
when $f(x)$ is undefined. $f(x)$ is increasing
on $(-\infty, -1)$ and $(-1, 0)$; $f(x)$ is decreasing
on $(0, 1)$ and $(1, \infty)$. There is a local maxi-
mum at $x = 0$. There are vertical asymptotes
at $x = \pm 1$, and horizontal asymptote $y = 0$.

$$f''(x) = \frac{8(3x^2 + 1)}{(x^2 - 1)^3}$$

$f''(x) \neq 0$ for any x , and there are no inflec-
tion points. $f(x)$ is concave up on $(-\infty, -1)$

and $(1, \infty)$; $f(x)$ is concave down on $(-1, 1)$.



$$\begin{aligned}
 47. \quad d &= \sqrt{(x - 2)^2 + (y - 1)^2} \\
 &= \sqrt{(x - 2)^2 + (2x^2 - 1)^2} \\
 f(x) &= (x - 2)^2 + (2x^2 - 1)^2 \\
 f'(x) &= 2(x - 2) + 2(2x^2 - 1)4x \\
 &= 16x^3 - 6x - 4 \\
 f'(x) &= 0 \text{ when } x \approx 0.8237 \\
 f'(x) &< 0 \text{ on } (-\infty, 0.8237) \\
 f'(x) &> 0 \text{ on } (0.8237, \infty) \\
 \text{So } x &\approx 0.8237 \text{ corresponds to the closest point.} \\
 y &= 2x^2 = 2(0.8237)^2 = 1.3570 \\
 (0.8237, 1.3570) &\text{ is closest to } (2, 1).
 \end{aligned}$$

48. We compute the slope of the tangent line to
 $y = 2x^2$ at the closest point $(0.8237, 1.3570)$.
 When $x = 0.8237$, we get $y' = 3.2948$.
 The slope of the line between $(2, 1)$ and
 $(0.8237, 1.3570)$ is

$$\frac{1 - 1.3570}{2 - 0.8237} = -0.3035 = \frac{-1}{3.2948},$$

so the lines are perpendicular.

$$\begin{aligned}
 49. \quad C(x) &= 6\sqrt{4^2 + (4 - x)^2} + 2\sqrt{2^2 + x^2} \\
 C'(x) &= \\
 6 \cdot \frac{1}{2} [16 + (4 - x)^2]^{-1/2} \cdot 2(4 - x)(-1) \\
 &\quad + 2 \cdot \frac{1}{2} (4 + x^2)^{-1/2} \cdot 2x \\
 &= \frac{6(x - 4)}{\sqrt{16 + (4 - x)^2}} + \frac{2x}{\sqrt{4 + x^2}}
 \end{aligned}$$

$$C'(x) = 0 \text{ when } x \approx 2.864$$

$$C'(x) < 0 \text{ on } (0, 2.864)$$

$$C'(x) > 0 \text{ on } (2.864, 4)$$

So $x \approx 2.864$ gives the minimum cost. Locate
 highway corner $4 - 2.864 = 1.136$ miles east of
 point A.

50. Let $F(v) = e^{-v/2}$. Then $F'(v) = -0.5e^{-v/2}$,
 so $F'(v) < 0$ for all v . Thus $F(v)$ is decreasing
 for all v . This says that as the speed of contrac-
 tion increases, the force produced decreases.

Let $P(v) = ve^{-v/2}$. Then

$$P'(v) = e^{-v/2}(1 - \frac{1}{2}v).$$

$P'(v) = 0$ when $v = 2$. We check that $P'(0) > 0$ and $P'(4) < 0$ so $v = 2$ is in fact a maximum.

51. Area: $A = 2\pi r^2 + 2\pi rh$

Convert to in^3 :

$$16 \text{ fl oz} = 16 \text{ fl oz} \cdot 1.80469 \text{ in}^3/\text{fl oz} \\ = 28.87504 \text{ in}^3$$

Volume: $V = \pi r^2 h$

$$h = \frac{\text{Vol}}{\pi r^2} = \frac{28.87504}{\pi r^2}$$

$$A(r) = 2\pi \left(r^2 + \frac{28.87504}{\pi r} \right)$$

$$A'(r) = 2\pi \left(2r - \frac{28.87504}{\pi r^2} \right)$$

$$2\pi r^3 = 28.87504$$

$$r = \sqrt[3]{\frac{28.87504}{2\pi}} \approx 1.663$$

$$A'(r) < 0 \text{ on } (0, 1.663)$$

$$A'(r) > 0 \text{ on } (1.663, \infty)$$

So $r \approx 1.663$ gives the minimum surface area.

$$h = \frac{28.87504}{\pi(1.663)^2} \approx 3.325$$

52. If $C(x) = 0.02x^2 + 4x + 1200$, then $C'(x) = 0.04x + 4 > 0$ for positive values of x (number of items manufactured). This must be positive because the cost function must be increasing. It must cost more to manufacture more items.
 $C''(x) = 0.04 > 0$. This means that the cost per item is rising as the number of items produced increases. (For an efficient process, the cost per item should decrease as the number of items increases.)

53. Let θ_1 be the angle from the horizontal to the upper line segment defining θ and let θ_2 be the angle from the horizontal to the lower line segment defining θ . Then the length of the side opposite θ_2 is $\frac{H-P}{2}$ while the length of the side opposite θ_1 is $\frac{H+P}{2}$. Then

$$\begin{aligned} \theta(x) &= \theta_1 - \theta_2 \\ &= \tan^{-1} \left(\frac{H+P}{2x} \right) \\ &\quad - \tan^{-1} \left(\frac{H-P}{2x} \right) \end{aligned}$$

and so

$$\begin{aligned} \theta'(x) &= \frac{1}{1 + \left(\frac{H+P}{2x} \right)^2} \left(-\frac{H+P}{2x^2} \right) \\ &\quad - \frac{1}{1 + \left(\frac{H-P}{2x} \right)^2} \left(-\frac{H-P}{2x^2} \right). \end{aligned}$$

We set this equal to 0:

$$0 = \frac{-2(H+P)}{4x^2 + (H+P)^2} + \frac{2(H-P)}{4x^2 + (H-P)^2}$$

and solve for x :

$$\frac{2(H+P)}{4x^2 + (H+P)^2} = \frac{2(H-P)}{4x^2 + (H-P)^2}$$

$$8x^2(H+P) - 8x^2(H-P)$$

$$= 2(H-P)(H+P)^2$$

$$- 2(H+P)(H-P)^2$$

$$8x^2(2P) = 2(H-P)(H+P)(2P)$$

$$x^2 = \frac{H^2 - P^2}{4}$$

$$x = \frac{\sqrt{H^2 - P^2}}{2}.$$

54. From exercise 53 we know that

$$\theta'(x) = \frac{-2(H+P)}{4x^2 + (H+P)^2} + \frac{2(H-P)}{4x^2 + (H-P)^2}$$

and that the function $\theta(x)$ is maximized at

$$x = \frac{\sqrt{H^2 - P^2}}{2}.$$

Plugging in the appropriate H and P values for high school shows that $\theta(x)$ is maximized by $x \approx 23.9792$. This is not in the range specified. In order to find out whether $\theta(x)$ is increasing or decreasing in the interval specified we plug the H and P values into the expression for $\theta'(x)$ and then plug in a value in our interval, say 55. We find that $\theta'(55) \approx -0.00392$. Since this is negative, $\theta(x)$ is decreasing on this interval, so the announcers must be wrong.

Following the same procedure for college, we find that $\theta(x)$ is maximized by $x \approx 17.7324$ and $\theta'(55) \approx -0.00412$ so again the announcers would be wrong.

Finally, for pros we see that $\theta(x)$ is maximized at $x = 0$ and $\theta'(55) \approx -0.0055$ so the announcers would be wrong once again. In this situation there is no x value for which the announcers would be correct, but in the high school and college situations, if the field goal is taken from some x less than the x which maximized $\theta(x)$, the announcers would be correct.

$$\begin{aligned} 55. \quad Q'(t) &= -3e^{-3t} \sin 2t + e^{-3t} \cos 2t \cdot 2 \\ &= e^{-3t}(2 \cos 2t - 3 \sin 2t) \text{ amps} \end{aligned}$$

$$56. \quad f(x) = 0.3x(4-x), \quad f'(x) = 1.2 - 0.6x = 0 \text{ when } x = 2, \text{ and changes from positive to negative there, so this represents a maximum.}$$

$$57. \quad \rho(x) = m'(x) = 2x$$

As you move along the rod to the right, its density increases.

$$58. \quad \text{With no studying, the person scores } f(0) = \frac{90}{1+4} = 18.$$

$$f'(x) = \frac{144e^{-0.4t}}{(1+4e^{-0.4t})^2}.$$

If the student were to study one hour, the score will increase by approximately

$$f'(0) = \frac{144}{25} = 5.76 \text{ points.}$$

$$\begin{aligned} 59. \quad C'(x) &= 0.04x + 20 \\ C'(20) &= 0.04(20) + 20 = 20.8 \\ C(20) - C(19) &= \\ 0.02(20)^2 + 20(20) + 1800 & \\ - [0.02(19)^2 + 20(19) + 1800] & \\ &= 20.78 \end{aligned}$$

$$60. \quad \overline{C}(x) = \frac{0.02x^2 + 20x + 1800}{x}$$

$$= 0.02x + 20 + \frac{1800}{x},$$

$$\overline{C}'(x) = 0.02 - \frac{1800}{x^2}$$

$\overline{C}'(x) = 0$ when $x = 300$, and the derivative changes from negative to positive here, so $x = 300$ gives the minimum average cost.